Pragmatics & Beyond New Series

Editor:
Andreas H. Jucker
(Justus Liebig University, Giessen)

Associate Editors:
Jacob L. Mey
(Odense University)

Herman Parret
(Belgian National Science Foundation, Universities of Louvain and Antwerp)

Jef Verschueren
(Belgian National Science Foundation, University of Antwerp)

Editorial Address:
Justus Liebig University Giessen, English Department
Otto-Behaghel-Strasse 10, D-35394 Giessen, Germany
e-mail: andreas.jucker@anglistik.uni-giessen.de

Editorial Board:
Shoshana Blum-Kulka (*Hebrew University of Jerusalem*)
Chris Butler (*University College of Ripon and York*)
Jean Caron (*Université de Poitiers*); Robyn Carston (*University College London*)
Bruce Fraser (*Boston University*); John Heritage (*University of California at Los Angeles*)
David Holdcroft (*University of Leeds*); Sachiko Ide (*Japan Women's University*)
Catherine Kerbrat-Orecchioni (*University of Lyon 2*)
Claudia de Lemos (*University of Campinas, Brasil*); Marina Sbisà (*University of Trieste*)
Emanuel Schegloff (*University of California at Los Angeles*)
Paul O. Takahara (*Kobe City University of Foreign Studies*)
Sandra Thompson (*University of California at Santa Barbara*)
Teun A. Van Dijk (*University of Amsterdam*); Richard Watts (*University of Bern*)

54

Ken Hyland

Hedging in Scientific Research Articles

HEDGING IN SCIENTIFIC RESEARCH ARTICLES

KEN HYLAND
City University of Hong Kong

JOHN BENJAMINS PUBLISHING COMPANY
AMSTERDAM/PHILADELPHIA

Contents

Preface	viii
Acknowledgments	x
1 On Hedging and Hedges	1
What is hedging?	1
What is a hedge?	3
Why study scientific hedging?	6
A brief background to hedging	8
Purpose and methodology	10
2 Towards a context of scientific hedging	13
Empirical and rhetorical models of scientific writing	14
Research writing as social action	22
Rhetorical features of research articles	25
Disciplinary culture and rhetorical context	34
3 Perspective and hedging in discourse	38
Linguistic approaches to hedging	39
Hedging in academic writing	51
Knowledge claims and scientific truth	56
Knowledge claims and audience design	62
Knowledge claims and writer commitment	69
Conclusion	79

4 Theoretical and methodological considerations 81
Scientific claims in the construction of knowledge 81
A characterisation of hedging in scientific RAs 90
Data selection and description of the corpora 94
Method of analysis 98

5 Surface features of hedging 102
Lexical and strategic hedges 103
Modal auxiliaries 105
Epistemic lexical verbs 119
Epistemic adjectives, adverbs and nouns 130
Hedging numerical data 139
Non-lexical hedges 141
Distributional information 148
Conclusion 155

6 A pragmatic analysis of hedging 156
A polypragmatic model 156
Content-oriented hedges 162
a) Accuracy-oriented hedges 162
Attribute hedges 163
Reliability hedges 166
b) Writer-oriented hedges 170
Reader-oriented hedges 177
Hedging in scientific writing 185

7 Examining hedges in scientific discourse 189
Extract 1: Hedging in an introduction 190
Extract 2: Hedging results 193
Extract 3: Hedging a controversial claim 196
Extract 4: Hedging a research article 200
Some comments on the interpretations 214

8 Hedging and second language learners 217
Hedging in L2 students' academic writing 218 218
Advice from the style guides 221 221
Hedging in ESP textbooks 223 223
Teaching ESP students to hedge 232 232
Conclusion 242 242

9 Conclusions and Implications
Some principal issues of hedging 244 244
Hedging in science: proposing some answers 252 252
Some implications 258 258
Conclusions and directions 261 261

Notes 265

Appendix: corpus of journal articles 267

References 271

Index 304

Preface

There is a widely held belief that scientific writing is purely objective, impersonal and informational, designed to disguise the author and deal directly with facts and the search for an independent truth. But effective academic writing is like any other kind of discourse in that it is interactive, it involves writers trying to influence their readers by persuading them of the correctness of their claims. This means that research papers do not only carry the writer's point of view, but that arguments have to be expressed in ways that are acceptable, meaningful and plausible to colleagues. A growing literature in both sociology and applied linguistics has therefore begun to turn its attention to the social contexts of writing and the ways that knowledge is a negotiated accomplishment. Persuading readers to accept a particular observation as a fact involves relying on disciplinary assumptions about how best to represent experiments, contextualise results and present oneself. One important means of achieving this is the use of hedges to present claims with caution, precision and humility.

Hedging is central to academic writing as it expresses possibility rather than certainty and collegiality rather than presumption. Scientific claims are rarely made without interpretive statements and these involve both assessments of probability and judgements concerning the impact of linguistic choices on readers. In other words, academic writing involves cognition and cognition almost invariably involves discretion. Hedging is one crucial means by which writers manage this pragmatic dimension of discourse and in this study I attempt to throw some light on their use in published scientific papers. In particular, I examine why scientists hedge and set out to describe the ways they do this, identifying the forms they use, how they are distributed and their specific functions in this genre.

There are four central arguments in this book. The first is that hedges are employed to achieve a single primary objective: to overcome the inherent negatability of statements and gain the reader's acceptance of a knowledge

claim. Second, I suggest that in achieving this objective hedges can emphasise an orientation to either the proposition or the reader, and that the former may focus primarily on achieving propositional accuracy or minimising writer accountability. Third, I show that in actual use the epistemic and affective functions of hedges are often conveyed simultaneously and that this indeterminacy prevents the formation of discrete descriptive categories. My fourth main argument is that hedging in scientific research writing is the product of informational, rhetorical and personal choices and cannot be fully understood in isolation from its social and institutional contexts. The pragmatic framework therefore seeks to accommodate both a complex interpretive environment and a perspective which accepts imprecision as an important characteristic of language use.

Given this broad context, the book ranges over a number of areas and critically examines their contributions to the field before presenting the descriptive and analytical model. Chapter 1 provides some background and an overview of the topic. In chapters 2 and 3 I seek to provide a rationale for the use of hedging by examining a variety of perspectives in the literature. My focus in chapter 2 is the broad socio-linguistic institutional contexts within which hedging occurs and has meaning. Here I explore the practices, beliefs, assumptions and goals which constitute biologists culturally and historically organised social worlds, providing a characterisation of scientific discourse from sociological, speech act and applied linguistic approaches. In chapter 3 I review the work on hedging more specifically.

In chapter 4 I discuss some central assumptions of the study and outline the research design and methodology. Chapters 5, 6 and 7 are the core of the book. Chapter 5 describes the surface features of hedging in the corpora and Chapter 6 presents a framework to account for their pragmatic functions in this genre. In chapter 7 I apply this framework to a number of extended text fragments and a complete research article. Having established the significance of hedging in scientific writing, I investigate its importance to second language students in chapter 8, exploring the coverage it receives in pedagogic methods and materials, and outlining some teaching approaches. In the concluding chapter I consider some implications of the study, and suggest future directions for research.

Acknowledgements

My intellectual debts to scholars in various disciplines will I hope be clear from the text that follows, but additional influences have been significant. I am particularly grateful for the advice and encouragement of Lynn Wales and David Lee at the University of Queensland, where I begun this project. Lynn's interest in modality inspired my own, and her insights into texts were invaluable.

I am also grateful to a number of specialist informants for perspectives on their discourse practices and commentary on various texts. Among these are Bruce Richardson, Desmond O'Toole, and Peter Tanner of the Department of Biology and Chemistry at City University of Hong Kong. My thanks are particularly due to Crista Chrichley of the Department of Botany, University of Queensland, whose strong interest in the language of science not only made her an interesting and invaluable informant, but led me to several years immersion in the discourse of cell biology.

I would also like to thank two reviewers at John Benjamins for their comments, and the anonymous readers of the journals cited below where earlier versions of particular chapters appeared.

An earlier version of chapter 5 appeared as "Talking to the academy: forms of hedging in science research articles" in *Written Communication*, vol. 13, no. 2, pp. 251-281, copyright © 1996 by *Written Communication*. Reprinted by permission of Sage Publications, Inc. An earlier version of chapter 6 appeared as "Writing without conviction? Hedging in science research articles" in *Applied Linguistics*, vol. 17, no. 4, pp. 433-454 (1996). Reprinted by permission of Oxford University Press. Parts of chapter 8 appeared as "Hedging in academic textbooks and EAP", *English for Specific Purposes* (1994), 13, 3: 239-256, and "Nurturing hedges in the ESP curriculum", *System* (1996), 24, 4: 477-490. Reprinted in revised form with kind permission of Elsevier Science Ltd., The boulevard, Langford Lane, Kindlington OX5 1GB, UK.

1 On Hedging and Hedges

> Since ... perception only gives information of this external world or of "physical reality" indirectly, we can only grasp the latter by speculative means. It follows from this that our notions of physical reality can never be final. We must always be ready to change these notions ...
>
> (Albert Einstein)

What is hedging?

The notion of hedging has been in the linguistic vocabulary since the term was introduced by Lakoff (1972) to describe "words whose job it is to make things more or less fuzzy". It has subsequently been applied to the linguistic devices used to qualify a speaker's confidence in the truth of a proposition, the kind of caveats like *I think, perhaps, might* and *maybe* which we routinely add to our statements to avoid commitment to categorical assertions. Hedges therefore express tentativeness and possibility in communication, and their appropriate use in scientific discourse is critical.

Definitions are surprisingly rare in the literature. Zuck and Zuck (1986) refer to hedging as the process whereby the author reduces the strength of a statement, while for Markkanen and Schröder (1989) it is any manipulative, non-direct sentence strategy of saying less than one means. My sense of the term is closer to the first of these definitions, as I wish to include in it statements that express exactly what the author means, saying no more than is warranted by available evidence. Hedging is not a strategy to obfuscate or confuse, any more than it is simply a convention of academic style.

In this study then, "hedging" refers to any linguistic means used to indicate either a) a lack of complete commitment to the truth value of an accompanying proposition, or b) a desire not to express that commitment categorically.

In essence, the purpose of this book is to elucidate this definition, locate it within linguistic theory and demonstrate its relevance to scientific discourse. The value of such an enterprise has been emphasised by Holmes (1984: 364) who states that:

> Identifying and describing the linguistic devices which may be used to modify illocutionary force constitutes a rich research field for those interested in pragmatics. And for the sociolinguist there is the challenge of investigating the differential use made of such pragmatic resources by different categories of speakers, to different addressees, in different social contexts.

The writer or speaker's judgements about statements and their possible effects on interlocutors is the essence of hedging, and this clearly places epistemic modality at the centre of our interest. Coates (1987: 112) describes epistemic modality as concerned with "the speaker's assumptions, or assessment of possibilities, and, in most cases, it indicates the speaker's confidence, or lack of confidence in the truth of the proposition expressed." Lyons' (1977: 797) definition of epistemic modality is well known: "any utterance in which the speaker explicitly qualifies his commitment to the truth of the proposition expressed by the sentence he utters ... is an epistemically modal, or modalised sentence."

Most commentators therefore refer to the epistemic system as realising a continuum of "commitment" on the part of the writer to the truth value of his or her statements, ranging from uncertain possibility to confident necessity. Its importance lies in the fact that such tentativeness avoids personal accountability for statements, reducing the author's "degree of liability" (Huebler, 1983:18), while expressing caution and facilitating open discussion. Beyond this is the declarative sentence. Epistemic comment is seen by many writers as being at the heart of language use and a principal means by which speakers can use language flexibly to adopt positions, express a point of view and display allegiances (eg Lyons, 1977; Stubbs, 1986). It is one of the principal resources we have for conveying choice and meaning (Skelton, 1997) and without it, language would be purely propositional and rigid.

Writers cannot avoid encoding their point of view towards whatever they say, commenting on and committing or distancing themselves from their propositions. As Stubbs (1986: 4) observes: the expression of commitment and

detachment, or of modality in all its senses, can be seen as a central organising principle in language. Where this involves the writer withholding full commitment to statements, I have called it hedging.

What is a hedge?

Halliday (1994:356) refers to modality as "the area of meaning that lies between yes and no", taking in "either yes or no" and "both yes and no". Because judgements about truth and falsehood, certainty and doubt, probability and possibility play such an important role in our lives, they allow a wide range of lexical, grammatical and strategic realisations. There are an enormous number of ways writers can signal tentative assessments of referential information, and a variety of reasons for doing so, therefore it is worth briefly clarifying what 'counts' as a hedge in the sense used in this book. It is also worth making the obvious point that hedges are represented by surface phenomena of texts and we require their presence to detect the function. The most apparently categorical statement may draw on participant understandings to convey attitudes about its content or readers, but we are able to say little about this without linguistic confirmation.

As noted above, hedging is one part of epistemic modality; it indicates an unwillingness to make an explicit and complete commitment to the truth of propositions. In everyday conversation hedges are commonly expressed through auxiliary verbs and by epistemic adjectives, adverbs and lexical verbs. Common conversational forms include words like *Perhaps, I guess, may be, quite* and *sort of,* while prosody, tag questions, verbal fillers and hesitation markers also function as hedges when they are used to attenuate the strength of an utterance (Coates & Cameron, 1988; Holmes, 1995). Academic research writing also includes diverse expressions of hedging which make use of the full range of epistemic lexical resources[1]:

(1) ... it seems that this group plays a critical role in orienting the carboxyl function. (E1:294)

(2) It might be speculated that the lack of crDNA methylation in cv Platenese could result in enhanced amounts of mRNA in choroplasts. (F3:17)

(3) ..a model implying almost complete lateral heterogeneity in the thylakoid membrane could be suggested, nearly all Photosystem II being situated in the appressed granal domain. (A5:172)

Hedges can thus be employed either singly or in combination. But in addition to lexical items, scientific statements can be weakened using a variety of strategies which limit the confidence invested in the claims made for the research. These can include drawing attention to the limitations of results or the experimental conditions in which they were obtained, and this is often achieved by commenting on difficulties encountered, shortcomings of findings or the possibility of alternative interpretations:

(4) Crystallographic investigation of membrane proteins is difficult because of the difficulty of obtaining suitable diffraction quality crystals. (C3:281)

(5) ...implies that phytochrome A is also not necessary for normal photomorphogenesis, at least under these irradiation conditions. (D4:46)

(6) ...could be explained by the fact that in these fractions the majority of PMB and PBF results from functional infected cells that do not contain the 21-kD protein, and/or by the fact that senescent vesicules are more fragile than ..... (D2:266)

Thus hedging is not limited to lexical expression and the data reveals questions, IF-clauses and contrastive markers that can qualify certainty and indicate a measure of the writer's doubt by making the truth of a proposition relative to the current state of (limited) knowledge:

(7) How is it, then, that this gene from a species that never forms nodules has the capacity to be expressed at elevated levels in nodules? The answer presumably has implications for the evolutionary origin of nodulation ability and nodule-specific gene expression .. (D1:637)

(8) If correct, this prediction might explain why previous exhaustive screens have .. (D4:40)

(9) Although a causal relationship between the latter processes remains to be verified, the correlation may not reflect mere coincidence. (B1:192)

Writers have also noted that use of the passive voice and tense can serve to hedge statements by distancing the writer from his or her assertions. However, the difficulty of determining the precise motivation and effect of these syntactic strategies with any assurance places them outside the scope of this study.

It should also be noted that many statements express opinions, viewpoints, attitudes and feelings, but not all are hedges. Commentary on propositional content can also signal the writer's personal feelings about a topic (*Surprisingly..., it is terrifying to think that...*) or intensify one's conviction of its truth (*It is obviously the case that ..., the results clearly show..*). However, the term *hedging* does not include all devices used to express attitudes or to comment metadiscoursally on the text, nor, of course, does it comprise epistemic devices which convey the writer's *conviction* in the truth of a statement. Sometimes it may be difficult distinguishing truth judgements from value judgements (Skelton, 1997:125), particularly in a context as rhetorically charged as scientific argument where the precise appeal to the audience is not always clear. However, my definition does not require this and allows any statement of reservation to be examined for its hedging value.

Similarly, while hedges have been seen as one device often used to mitigate threats to the 'face' or self-image of interlocutors (Brown & Levinson, 1987; Myers, 1989), not all expressions of politeness act to qualify writer commitment. One function of hedges is to contribute to a relationship by alerting readers to the writer's perspective towards both propositional information and to the readers themselves. "Facework" however is accomplished by a range of strategies which also include markers of obligation, politeness and reluctance, as well as swear words and title forms (Turnbull & Saxton, 1997; Wood & Kroger, 1994). A range of interactional and evaluative features are subsumed under Halliday's (1994) interpersonal macro function of language, which is contrasted with the textual, or enabling, and the ideational, or informational, functions. Thus while hedges constitute one aspect of the interpersonal domain of language, they are distinct from other elements of it.

In sum, in this study hedges are the means by which writers can present a proposition as an opinion rather than a fact: items are only hedges in their epistemic sense, and only then when they mark uncertainty.

Why study scientific hedging?

Despite a widely held belief that professional scientific writing is a series of impersonal statements of fact which add up to the truth, hedges are abundant in science and play a critical role in academic writing. Hedges are important to scientists because, as the quotation from Einstein at the head of this chapter emphasises, even the most assured scientific propositions have an inherently limited period of acceptance. Categorical assertions of truth in these circumstances are decidedly hazardous. Science writing involves weighing evidence, drawing conclusions from data, and stating circumstances which allow these conclusions to be accepted; it assesses the claims it makes. Nash states this position succinctly:

> The writer .... currently evaluates and criticises the information and the propositions he or she tries to set down as fully, accurately. and objectively as possible. For centuries this dialectical processing of objective fact and subjective evaluation has been the goal of academic writing and of the training that leads to academic writing. (Nash, 1990: 10).

Academic discourse invariably involves interpretative statements because academics are crucially concerned with varieties of cognition, and cognition is inevitably hedged, with writers offering an assessment of the referential information they provide. Rather than being factual and impersonal, effective scientific writing actually depends on means of augmenting propositional information in order to alert readers to the writer's opinion.

A good reason for studying scientific hedging is therefore to reveal an essential element of academic argument. Hedges are a crucial means of presenting new claims for ratification and are among the primary features which shape the research article (RA) as the principal vehicle for new knowledge. Hedging enables writers to express a perspective on their statements, to present unproven claims with caution and to enter into a dialogue with their audiences. It is therefore a substantial means by which scientists confirm their membership of the scientific community: "hedging is the mark of a professional scientist, one who acknowledges the caution with which he or she does science and writes on science" (Crismore & Farnsworth, 1990: 135). Its study therefore has important implications for a number of areas.

Most importantly, a better understanding of how hedges are used in scientific research articles can contribute to the growing literature on the rhetoric of science, providing insights into how science establishes its claims to knowledge, and how scientists carry out their work. The research article is a genre clearly distinguished by its role in reporting new experiments intended to add to the existing knowledge of the discipline (cf. Swales, 1990). It is the major vehicle for knowledge in academic cultures and an artefact central to the legitimation of a discipline and the reputation of its practitioners. Its importance has been confirmed in recent years by studies which have critically debated the epistemological foundation of scientific knowledge and questioned the role of the RA as simply a transmitter of facts. These studies have established that the RA is a significant rhetorical artefact, which acts to both conceal the contingency of knowledge and persuade readers to accept claims (eg Latour & Woolgar, 1979; Gilbert & Mulkay, 1984; Bazerman, 1988).

Essentially, scientific knowledge is seen as less a coherent body of objective truth about the world than a set of justifiable beliefs reached by the scientific discourse community and constructed through interactions among individuals. In this view the accreditation of knowledge is a social process and research is perceived as a quest for collective agreement rather than a search for truth (eg Kuhn, 1970; Ziman, 1984; Longino, 1990). Consequently, scientific argument is part of a rhetoric employed to convince readers of the validity of claims. This search for consensus provides the motivation for the expression of doubt and tentativeness in science for such interpretations can help secure readers' acceptance. Statements require audience ratification and this requires the reader's active role in how the writer constructs claims. In other words, hedges are an important rhetorical device in academic writing as they signal the writer's anticipation of the negatability of his/her propositions.

Information about hedging can also contribute to our understanding of the practice of evidential reasoning and the structure of argumentation as discussed by Toulmin (1958), Longino (1990) and others. Toulmin's widely known system defines the elements of reasoning to be the grounds (established data), the warrant (facts supporting the hypothesis), the qualifier (degree of uncertainty concerning the connection between grounds and claim), the claim (proposed conclusion) and the rebuttal (elements of doubt concerning the claim). Hedges clearly contribute to the repertoire of devices used to anticipate possible rebuttals, and their study can help reveal how writers move between grounds and claims in the process of gaining reader ratification for statements.

This study also has practical implications for the teaching of research writing in both L1 and L2 contexts, where the advice frequently given to students is to avoid hedging altogether (eg Winkler & McCuen, 1989: 97; Strunk & White, 1959:59). Many textbooks emphasise the impersonality of scientific discourse and encourage the idea that the goals, methods and topics of academic discourse transcend personality. The demands imposed by linguistic and rhetorical fluency in a second language, however, represent a major encumbrance for non-native English speakers studying science or seeking publication for their research (Gosden, 1992). Such demands may be responsible for the low level of NNS contributions to the scientific literature in English (Swales, 1985:42).

English is now the established lingua franca of scientific communication with perhaps 80% of the world's research output in English (Garfield, 1983). There can be little doubt that the ability to write effectively in English is a prerequisite for full participation in international research communities and of professional success. Scholars all over the world realise that publication in English guarantees the widest circulation for their ideas and the results of their research. However, the inadequacies of ESP materials on this topic (Hyland, 1994) underline why even otherwise proficient second language students find hedging their propositions notoriously problematic (Skelton, 1988a; Bloor & Bloor, 1991). Clearly, the more we understand such features of academic writing, the more we can assist NNS to participate fully and successfully in the research world.

The research may also provide analytic and descriptive insights that will enable teachers of English for Science and Technology (EST) to assist their students to write more effectively. The implications for English language teaching are therefore briefly discussed and suggestions made for teaching materials and approaches. This information may therefore be of relevance to teachers producing study materials to assist students with the pragmatic demands of hedging in scientific discourse.

A brief background to hedging

The concept of hedging is an area of some neglect (Crystal, 1995: 120) although the literature contains references to various aspects of hedging over the past 20 years. In fact, a plethora of labels demonstrates a continuing interest in

the topic. Hedges have been referred to as compromisers (James, 1983), downtoners (Quirk et al, 1972), weakeners (Brown & Levinson, 1987), downgraders (House & Kasper, 1981), softeners (Crystal & Davy, 1975), backgrounding terms (Low, 1996) and pragmatic devices (Stubbe & Holmes, 1995).

Hedging has received most attention in casual conversation where it is extremely common and represents a significant interpersonal and facilitative communicative resource for speakers (eg Stubbs, 1986; Coates, 1987). Speakers use hedges when they wish to create an informal or congenial atmosphere, facilitate turn-taking, sustain discussion, show politeness, mitigate face-threats or disguise deficient knowledge or vocabulary (Brown & Levinson, 1978; Coates, 1987 & 1988; Holmes, 1984; Turnbull & Saxton, 1997). Some writers also follow Lakoff (1972) in associating hedges with conveying purposive vagueness. Stubbs (1986) and Myers (1989) have argued that claiming precision is only appropriate in some situations and Latour & Woolgar (1979: 75-87), Powell (1985), Dubois (1987) and Channell (1990 & 1994) all claim that deliberate imprecision is a characteristic of academic discourse. Hedging has also been seen as a means of signalling the distance between a speaker and what is said (Prince et al, 1982; Rounds, 1982; Nash, 1990; Skelton, 1988b), and as a means to structure organisational roles and negotiate discourse statuses (He, 1993).

From the readers' perspective, hedges have been found to be an important element of persuasive writing (Conner & Lauer, 1988), although the extent to which it is attended to in different genres is uncertain. Low's (1996) evidence suggests that hedges may be 'lexically invisible' to respondents when included in questionnaire statements, while Crismore and Vande Kopple (1988) found that hedges presented in personal voice and in low intensity enhanced students' memorisation of a text.

Hedges have also been treated as a form of 'metadiscourse' by some Applied Linguists who have included it in the elements of a text used to explicitly organise the discourse, engage the audience and signal the writer's attitude (Crismore & Farnsworth, 1990; Crismore et al, 1993; Hyland, 1998 & in press a; Vande Kopple, 1985).

Metadiscourse refers to non-propositional aspects of texts which "bracket the discourse organisation and the expressive implication of what is being said" (Schiffrin, 1980: 231). Together with textual and attitude markers, hedges enable authors to project themselves into their work to signal their communicative intentions. They are therefore seen as a means of indicating the writer's

stance towards both the material and the reader. Hedges constitute an important feature of the metadiscourse in various kinds of persuasive prose of various kinds (Crismore & Farnsworth, 1989; Hyland, 1998), and are the most significant aspect of interpersonal metadiscourse in academic articles in a number of disciplines and in undergraduate textbooks (Hyland, in press a & b). This view of 'metadiscourse' therefore acknowledges a critical aspect of interactive discourse and alerts us to an important rhetorical feature of science RAs.

Analyses of written academic corpora have revealed some of the characteristics of hedging in text-books (Hyland, in press b; Myers, 1992), economic forecasting (Pindi & Bloor, 1987; Bloor & Bloor, 1993), science digests (Fahnestock, 1986), abstracts (Rounds, 1982), medical discourse (Salager-Meyer, 1994; Skelton, 1997) and molecular genetics articles (Myers, 1989 & 1990). Studies have also shown the importance to academic discourse in general of modal verbs (Hanania & Akhtar, 1985; Butler, 1990), imprecise numeric expressions (Channell, 1990; Dubois, 1987) and "commentative" items (Adams-Smith, 1984; Skelton, 1988 & 1997). More generally, hedges have been seen as an important means by which the values of scientific cultures are articulated in discourse meanings (Hyland, 1997). Together this literature has demonstrated the clear pragmatic importance of hedging as a discoursal resource for expressing uncertainty, scepticism and displaying an open mind about one's propositions.

Purpose and methodology

Unfortunately, while the literature places the importance of hedging beyond dispute, we know little about its frequency, distribution or use in different disciplines or genres. Almost no work has been carried out into the expression and functions of mitigation in academic contexts based on analyses of adequate corpora and no convincing account of its use in scientific writing exists. Much of the attention it has attracted has been of a theoretical nature, focusing on intuitive and decontextualised examples (Klinge, 1993; Lyons, 1977). Coates (1983) and Palmer (1990) focus only on modal verbs and draw on non-academic sources while Holmes (1988b) examines expressions of certainty as well as mitigation. Hedging in scientific research writing therefore represents a little studied area of pragmatic competence. In particular, greater attention needs to be paid to the fact that hedging represents a writer's *attitude* within a

particular context. There is, therefore, a need for an explanatory framework which accounts for its pervasiveness in academic discourse by situating hedging in its socio-pragmatic contexts.

The study was therefore undertaken to provide a comprehensive description of this feature in one particular context, the discourse of a major scientific discipline, by relating a systematic analysis of linguistic forms to a pragmatic explanation for their use. The research therefore helps situate the concept of hedging within wider argumentative strategies based on an analysis of authentic sources. In particular, the study seeks to identify the major hedging devices used in cell and molecular biology articles, determine their distribution across different rhetorical sections, explore their pragmatic functions, and to briefly compare the overall findings with other forms of academic writing.

To achieve these objectives, I will seek answers to the following questions:

1. Which lexico-grammatical forms and strategies are principally used to hedge in the corpus and do these differ from other forms of discourse?
2. Are hedges equally distributed across the rhetorical structure of the genre?
3. Are there clear pragmatic functions which hedges perform in science research articles and do these differ from other kinds of discourse?
4. Can particular hedging forms and strategies be related to particular rhetorical functions?
5. Do these findings suggest any particular teaching strategies and materials ?

In exploring these issues I draw on theoretical work on epistemic modality and extends studies in pragmatics, composition, applied linguistics and the sociology of science. It examines the rhetorical and professional contexts in which scientists work and which motivate the use of hedging, and analyses 26 research articles in cell and molecular biology. In addition, data from three large computer corpora of academic English, equivalent to over 2, 000 pages of text, are examined to determine differences in the frequencies and range of hedging expressions used. The book therefore moves from a discussion of the social and discursive practices of biologists to focus more explicitly on hedging. A detailed description of surface features of hedging in the corpus is then followed by a functional rationale for their selection and distribution. This procedure avoids a reliance on predetermined notional categories, so that an ex-

planatory framework can emerge from an analysis of actual language behaviour.

The study shows that the core function of hedging in science is a negotiation of meaning which seeks to balance reader and writer perspectives in gaining accreditation for knowledge claims. So, while the study of modality traditionally falls within the domains of linguistics and semantics, linguistic knowledge alone is insufficient to understand how discourse is rhetorically and stylistically organised. Texts must be seen as constituting a social dialogue between writers and readers within the wider context of social and discursive practices of the scientific discourse community. Together, the analyses demonstrate that hedging is a major aspect of academic argument and confirms its centrality to both individual scientists and to science itself. Ultimately, research articles reveal a relationship between a discourse community, standards of knowledge and textual representations and it is these in combination which clarify the use of hedges.

2. Towards a context of scientific hedging

Any attempt to provide a detailed analysis of how hedging is deployed as a constitutive feature of scientific writing needs to offer a characterisation of the settings within which it is used. Hedging cannot be properly understood, appropriately interpreted or relevantly described without going beyond the event itself. Language is a type of social practice and texts result from the actions of socially situated writers operating within particular institutional constraints. In other words, attention to hedging requires some understanding of the cultural practices of scientists and their epistemological assumptions and values, together with the specific genre within which the feature is situated and has significance. Context is thus seen here as an interactive and dialogic notion, both sustaining and sustained by the discourse of social actors at a given point of time. This chapter therefore turns briefly from a focus on hedging to look at the social institution of science and the particular discoursal activities and purposes of its practitioners. It seeks to provide an overview of relevant research to locate hedging in the wider context of our knowledge of scientific discourse and of scientific communities

The links between academic communities and their discourses have mainly been studied from two perspectives. Applied linguists have largely focused on the informational, rhetorical and stylistic organisation of written texts for pedagogic purposes, while sociological studies have sought to explore the interactions between scientists for evidence of the processes which maintain social order.

A major focus of the sociological work has been to examine the practices by which knowledge is socially constructed. This has revealed the influence of contingent factors in scientific research, and the rhetorical nature of the "empiricist repertoire" by which scientists persuade colleagues of their claims. This work has illuminated the central roles of power, allegiance, and self-

esteem in scientific cultures, and shown how researchers selectively draw on a range of potential accounts to explain their results. Linguistic approaches, on the other hand, have long recognised that texts are both interactive and situated within particular social contexts which constrain discourse choices. Recent research into scientific texts has focused on the rhetorical function of surface features or generic structure and has revealed a great deal about the interactions between text producers and their discourse communities.

The sociological accounts have therefore emphasised that linguistic conventions help to generate a social world of science without describing the systematic means by which this is accomplished, while the linguistic work has largely neglected the significance of the wider context in which linguistic features occur. Together, however, the two approaches offer considerable insights into the context of scientific research writing and describe how discursive practices are derived from and constrained by scientific cultures. The analysis of discourse invites questions about what certain choices accomplish in a text, and the purposes achieved by selecting particular options over others, although textual analyses alone cannot provide a rationale for such choices. Equally, particular representations of the nature of institutional and social relations require discoursal evidence, as the socio-rhetorical functions of language must be systematically grounded in linguistic description. An adequate account of an academic hedging thus requires an understanding of how the features of scientific discourse represent and produce disciplinary practices, while contributing to situated forms of argument.

In this chapter, then, I will first explore the discursive practices of text production and interpretation and the institutional and social circumstances which surround it. Then I will look more closely at scientific argumentation and the genre of the research article itself. Finally I will briefly characterise salient aspects of the context in which hedging needs to be understood.

Empirical and rhetorical models of scientific writing

It is a common belief that scientific discourse simply reports or describes natural facts with almost no human intervention. In a strictly empiricist view, a scientific text is regarded as a neutral descriptive medium which allows a scientist to act simply as "a messenger relaying the truth from nature" (Gilbert, 1976: 285). Research findings in the natural sciences are generally communicated in

research papers which follow a conventional style and 4- part IMRD format. This format now represents "the standard product of the knowledge manufacturing industry" (Swales, 1987: 42) and typically omits descriptions of wrong turns, false leads, unsuccessful procedures, and often the factors that resulted in the choice of problem or methods. It also ostensibly excludes reference to the author's opinions, doubts and interests and removes any allusion to the author's identity.

The effect of these stylistic devices is to maintain the assumption of an objective reality with an existence quite separate from the investigating scientist. The production of knowledge and its communication are presented as different activities. In following the correct procedures for revealing the truth, the researcher can appear to allow the truth to speak for itself. This involves presenting a research narrative, sometimes referred to as a 'phony story' (Berkenkotter & Huckin, 1995), which reinforces a view of scientific activity as collective, inductive and cumulative. Sir Peter Medawar, who shared the Nobel prize for physiology in 1960, has observed that RAs do "not merely conceal but misrepresent the reasoning that goes into the work they describe" (in Judson, 1995:34). The view of the scientist-researcher completing lab experiments then writing up the research according to a strictly circumscribed format is a modern myth which acts to bolster this normative view of scientific activity, and underline the objectivity of scientific knowledge.

In reporting new experiments and seeking endorsement of claims, RAs also demonstrate how disciplines exercise control over approved knowledge. Although it suggests an objective reality independent of the researcher, the research article reflects forms of culturally determined and socially agreed knowledge. A growing number of sociological studies have shown that scientific discourse is both socially situated and structured to accomplish rhetorical objectives (Bazerman & Paradis, 1991; Fahnestock, 1986; Gilbert & Mulkay, 1984; Myers, 1990). Persuading readers to accept a particular observation as a fact involves relying on a social consensus about how best to represent experiments, contextualise results, and present claims. Writers have to consider an expected audience and anticipate its background knowledge, processing problems and reactions to the text. Simultaneously, readers are trying to predict lines of thought and interrogate authors on their positions. Bazerman (1985), for example, has shown how physicists' reading processes reveal personal schema based on their active research purposes. All articles are evaluated for usefulness and importance and the paper that is unused and uncited receives

harsh judgement. In such circumstances the ability to anticipate the reader's needs, interests and schema are critical.

In fact, scientific language serves two main functions. It both assists the interests of competing scientists and research groups in procuring assent to propositions about the world, and it establishes and maintains the authority of science itself. Principally, rhetorical objectives involve persuading readers that a particular observation actually lies beyond questions of persuasion and is situated within the realm of fact. What counts as a fact however is grounded in disciplinary values and practices transmitted through socialisation and secured via a system of peer review. In other words, scientific knowledge is seen to be socially contingent and scientific discourse to be a rhetorical artifact.

Studies of research practices tell us a great deal about the importance of writing among the scientific community and about the conventions that constrain its patterns of discourse. An important aspect of this is the tension between the private aspects of experimental practices and science as legitimate public knowledge, discussed as different registers, or "interpretative repertoires" (Latour & Woolgar, 1979; McKinlay and Potter, 1987). Gilbert & Mulkay show that scientists routinely use two alternative ways of accounting for the same phenomena, referring to an "empiricist repertoire", which emphasises impersonal statements of evidence and experimental practices, and the "contingent repertoire", which acknowledges personal or social factors. James Watson's (1968) discussion of the events leading to the discovery of the DNA structure is a good account of how chance, competition and deception can play a role in research. In more recent times, one of the informants interviewed for this study cited the role of self-interest in maintaining the massively funded research "bandwagon" into the greenhouse gasses effects.

It is unsurprising that the choice of register often varies systematically with context, but while the former is common in the research literature and the latter in informal conversation, writers can shift strategically between the two. The account that is selected is directly employed to bolster scientific claims and so scientists will typically employ an empiricist repertoire to account for their own work and give versions of their opponents' research in contingent terms, drawing attention to the potential influence of distorting social and personal factors. But while this strategy has been observed in forums such as conference presentations (McKinlay & Potter, 1987), and published papers (Yearley, 1981), it rarely involves specific attacks accusing particular individuals of bias or incompetence. Explicit refutation is rare, and writers typically hedge their

criticisms and manage conflict through acceptable channels (Bazerman, 1988, chap. 5). This kind of hedging is central to the norms of rhetoric advocated by Boyle 300 years ago. Boyle argued that disputes should refer to findings rather than their supporters "for I love to speak of persons with civility, though of things with freedom" (quoted in Shapin, 1984: 502). Such appropriate scientific rules of discourse allow disputants to express divergent (appropriately hedged) views while contributing to the growth of knowledge.

How the empiricist repertoire is used to characterise scientific practices is illustrated by Knorr-Cetina's (1981) description of the way lab work is differently represented in published accounts. The sequence of Introduction, Method, Results and Discussion (IMRD) falsely suggests that laboratory activity proceeds in a unidirectional, systematic fashion of experiment- result- interpretation- conclusion. Science rarely proceeds like that however. One informant put it thus:

> The whole business is to do with serendipity in a sense, because while you set out expecting to find something, in many cases discoveries are made by chance, just where you're not looking. You have to be aware of this, of course, to find something when it occurs.

In Knorr-Cetina's study the research team responded to a propitious laboratory finding rather than consciously solved a stated problem. By reversing the research events in terms of a problem-solution dynamic however, the writers were able to reformulate the activity as extracting potato protein by an efficient new method and depict their results as a contribution to experimental technique.

This conventional IMRD organisation of the research article has been discussed by Gross (1985) as a rhetorical device which helps justify the enterprise of experimental science. The form of the research paper, moving "from the contingency of laboratory events to the necessity of natural processes" (ibid. p 20), is designed less to explore research issues than to perpetuate the myth that the process of induction, upon which science relies for the certainty of its knowledge, is unproblematic. The Introduction places the experiment in the context of a research programme rather than laboratory events, thus creating a theoretical rather than contingent interpretive world, and this process is continued in Methods and Results where the objectivity of laboratory practices is emphasised over contingencies of human activities. Finally, the Discussion frames the relevance of these practices in the theoretical perspective of the Introduction, emphasising a correspondence of the data to the paper's claim. The

research article thus leads from sensory experience to reliable knowledge, creating a uniform interpretive context.

Moreover, not only are a paper's proposed conclusions made to appear as if they followed unproblematically from empirical evidence, but rhetorical decisions also intervene when scientists revise successive drafts of a paper for publication. Myers' (1990) case studies of two biologists attempting to get research funded and published, shows how significant alterations are made in the conceptual organisation and macro-structure of papers. Claims become more cautious, speculations and proposals more restricted and a discursive text replaces a more goal-directed one. There is also a considerable increase in hedging in the final versions as authors anticipate the text's reception in the discourse community and try and place their claims appropriately within it. The writing of a research article thus involves selective representation and rhetorical reconstruction as a means of anticipating negative responses to claims. Such processes as an after-the-fact reconstruction of a goal-directed purpose in the literature review, a procedural account of the methodology and a selective narrative of results are all means of persuasively presenting the writer's knowledge claims as unproblematically related to observed data.

While scientific discourse reflects disciplinary needs for a convincing rhetorical style, we cannot deny the importance of an epistemological system founded on the possibility of direct and reliable knowledge of the natural world. Experimental science has evolved at least partly as a result of beliefs that correct inductive procedures can reveal apodictic truths about reality, and all the biologists interviewed for this research strongly subscribed to the view that there was an external world which could be understood using scientifically appropriate means. However, an empiricist position cannot convincingly explain the evolution of the formal and terminological stability of scientific prose, nor the basic contradictions of inductive procedures in its production.

Importantly, these familiar ways of discussing nature, which seem a timeless and self evident means of establishing and defending knowledge, have in actuality been consciously developed historically in response to particular rhetorical situations. The conventional linguistic practices for expounding and securing support for scientific knowledge are historical artifacts which date from the 1600s (Shapin, 1984). Establishing rules of discourse to generate and validate authentic facts and to create a "public" for experimental research was crucial when science was in its infancy. The literary technology of Robert Boyle and his colleagues served to both proselytise and create linguistic

boundaries which segregated authenticated facts from man-made hypotheses, producing the "illusion that scientists' speech about natural reality is simply a reflection of that reality" (Shapin, 1984: 510).

By avoiding unfounded speculation, providing elaborate accounts of both successful and failed experiments, giving detailed illustrations, and employing devices which distinguished "matters of fact" from speculations, Boyle helped create a rhetoric which laid the foundations for scientific communication. One of the most important linguistic devices of this rhetoric of science was the use of what we now call 'hedges'. As Boyle states:

> in almost every one of the following essays I ... speak so doubtingly, and use so often, 'perhaps', 'it seems', it is not improbable' and other such expressions, as argue a diffidence of the truth of the opinions I incline to ... (in Shapin, 1984: 495)

Only facts, as discovered rather than invented, could be spoken of assuredly, "...I dare speak confidently and positively of very few things, except of matters of fact" (ibid p 496). A clear distinction between facts and interpretation was recommended, both in terms of a moral stance and manner of speech to provide the basis of an embryonic scientific philosophy and academic discourse.

The development of scientific prose to the present has been traced by Atkinson's (1996) survey of *The Philosophical Transactions of the Royal Society of London* between 1675 and 1975. This shows how linguistic practices, particularly those relating to changes in the presentation of experiments, were increasingly refined. This enabled contributors to gain publicity and acceptance for their work and for figures like Boyle, Hooke and Newton to become public figures through regular publication. As publication became essential to science and audiences became increasingly professional, critical and research-motivated, a communication network of scientists evolved which required institutionalised standards of public argument, moving from author-centred norms of genteel conduct and deductive reasoning to a greater attention to methodology and experimental research. Such changes were in response to evolving research communities and represented a movement away from the observations of gentlemen to the evolution of specialist disciplines and research problems.

The subsequent de-emphasising of methods and the substantial expansion of theoretical discussions in this century further point to changes in rhetorical and social practices (Atkinson, 1996; Bazerman, 1988; Berkenkotter & Huckin, 1995). The reasons for the emergence of Discussion to replace Methods as the

dominant basis of persuasion is uncertain, but is probably related to the standardisation of experimental procedures and the greater need to contextualise scientific work in terms of discipline recognised problems and a community generated literature. Critically, scientific communication builds on prior texts, and "common theory has become an extremely strong force in structuring articles and binding articles to each other" (Bazerman, 1988: 157). The explosive growth in scientific research to address these problems and the increasing pressure to publish findings has also resulted in the development of the RA as a 'promotional' genre. This discourse emphasises features such as titles, abstracts and graphics to foreground novelty and significance, to accommodate the scanning reading patterns of information saturated readers searching rapidly for relevance and novelty (Bazerman, 1988; Berkenkotter & Huckin, 1995: Ch 2).

The changes noted in these studies suggest that RAs are strongly responsive to changing disciplinary norms and the ways community members consume texts. Thus the means by which arguments are presented, procedures enumerated, literature cited, theory and data discussed, can only be seen as effectively persuasive against a backdrop of disciplinary practices and rhetorical expectations. The development of a scientific community has helped codify the language of scientific presentation we see today as a result of a need to socially validate knowledge. This discourse community has achieved a relatively stable rhetorical situation in order to discuss similar topics under similar conditions, for similar purposes. Readers therefore not only share an acceptance of a great deal of knowledge of content and methods, they also share an acceptance of a pattern of discourse which has evolved in response to particular rhetorical contexts.

Texts play many roles in the scientific community but the norms of discourse and writers' conformity to them suggest a strong common goal. Each new paper is based on the existing consensus among scientists and seeks to become part of that consensus. The strong continuity of language and discourse forms and the heavy citation of previous work demonstrate the dependency of argument on this process, while the impersonality of voice and the use of technical lexis illustrate attempts by the new contribution to insinuate itself into this body of agreed knowledge. Formal text features represent fundamental relations among scientists and are the result of a number of considerations, which are commonly cited as respect for reason, accuracy, truthfulness, humility, universalism, efficient ways of representing nature, and so on. The importance of persuasion in scientific discourse, however, also indicates that language serves

the competitive interests of separate individuals and research groups, and is partisan and argumentative. Scientific language cannot, as a human artifact, be a dispassionate, objective representation of nature. Rather, it serves personal goals for authors, institutions and the discipline itself.

One reason given for the use of persuasive rhetoric is that scientific publication is a route to a range of tangible rewards which satisfy the specific wants of individuals in the scientific community:

> The academic science community operates an institutionalised system which is simultaneously a communication system and a reward system and what might be called a system for 'distributing property'. This system rests on peer judgement. (Barnes & Edge, 1982: 15)

A scientist's contribution is judged by colleagues who are potentially able to make use of it. If they judge it to be significant, grant it publication, borrow and cite it, then the scientist achieves recognition. This may lead to tenure, promotion, research funding or a Nobel Prize. Latour and Woolgar (1979) argue that a scientist's goal is to maximise credibility in the discourse community. Credibility can then be used as credit in a series of moves which allows them, through grants, equipment, more data and fresh publications, to increase their credibility further. The apparent objectivity of scientific papers is therefore seen as compromised by the fact that they are principally contributions to professional reputations in a forum of competitive publication. As Gaston (1971) writes:

> In their attempts to gain recognition for their accomplishments scientists work in a competitive situation which is quite different from that of other professions ... Competition in science is more like a race between runners in the same track and over the same distance at the same time.

Communication facilitates the social interactions which assist in the production of knowledge and, as Berkenkotter and Huckin (1995: 1) observe, this is vital to institutions and individuals:

> ....maintaining the production of knowledge is crucial for institutional recognition, the development of sub-specialties, and the advancement of scientists' and scholars' research programs. Scientific and scholarly productivity are also the criteria by which careers are assessed, tenure given and grants awarded.

Authors want to see their papers published in prestigious journals that testify to the importance of their claims, help promote their reputations among peers, and provide support for their related proposals for research funding (Myers, 1990). The need for recognition and reward among active researchers then, is a principle motivation for publishing research and a principle reason for the essentially rhetorical character of the research article.

Research writing as social action

The scientific paper emerges from the research reported above as a rhetorically competent product through which scientific knowledge is negotiated and ratified. Language is seen as central to members of scientific communities in gaining acceptance for their claims, as it is the means by which arguments are presented and opposition anticipated. A successful claim thus depends on linguistic choices which appropriately convey the writer's intention, facilitate the smooth exchange of information and create conditions for persuading readers of a point of view. Speech act theory offers some useful insights into how this may be accomplished, as it challenges a view that comprehension is simply dependent on an interpretation of words on the page. It both confirms our conviction that purposes underpin actions, and suggests a means of analysing rhetorical acts as social accomplishments which involve shared writer-reader knowledge of disciplinary discourse conventions.

The translation of words into acts is only possible within the structure of a particular discourse since, the decisive function of any sentence depends on a framework of understandings about the text to which it contributes, together with the expectations established by a particular discourse community. Austin's (1962) distinction between propositional (locutionary) meanings, functional (illocutionary) meanings and the interlocutor's response to these acts (perlocutionary effects) is well known. For our purposes its relevance lies in the fact it facilitate a principled division between meanings which convey information clearly and those which are effectively persuasive.

For speech act theorists, argumentation can be seen as an "illocutionary act complex composed of elementary illocutions" (Van Eemeren and Grootendorst, 1984: 39). These elementary illocutions are to be found within Searle's (1979) category of *Assertions*[1] which commit the writer to the truth of the expressed proposition and include acts such as asserting, suggesting, claiming,

reporting and so on. Essentially then, speech act theorists might see a scientific text as a constellation of illocutionary acts which constitutes an argument by justifying or refuting a given opinion. The purpose of such argumentation is to elicit from readers a particular response: writers hope they will be convinced or persuaded. For Van Eemeren and Grootendorst, this means the language used must serve both a communicative and an interactional purpose, a writer not only wants the words to be understood (an illocutionary effect), but also to be accepted (a perlocutionary effect). The reader is expected to rationally decide whether to accept the perlocutionary effect desired by the writer, but the illocutionary effect of understanding an argument is not a sufficient condition to achieve this.

The principal insight of speech act theory therefore is that linguistic communication involves not only surface forms but also the reader's recognition of the writer's intention in the context of the discourse. In writing up their research, scientists convey propositional content and perform illocutionary acts by expressing at least one attitude which reflexively intends the reader to understand that he or she has that attitude. These examples from my corpus help make this clear:

(1) The proposed model reconciles a number of observations and data already in the literature for which a cohesive theory was previously lacking. (A1:274)

(2) These results demonstrate that Arabidopsis produces at least two classes of screening pigments that reduce UV-B damage: flavonoids and sinapate esters. (D3:176)

In performing the illocutionary act of asserting by making a claim for a model (1), or stating a result (2), these writers have firmly expressed their intent in performing such an act by using appropriate syntactic forms (declaratives). This signals both their belief in the truth of the propositions and their intention that readers should believe that they believe the propositions are true. The illocutionary act is accomplished by readers recognising that intention, and communication is achieved by them experiencing this "illocutionary effect" (Searle) or "uptake" (Austin).

The accomplishment of social acts in scientific writing therefore concerns epistemic change; the intention of the writer is to alter the knowledge set

of the reader according to the writer's purpose. While this change is consequent upon the reader's identification of a semantic act of meaning and reference, it also involves pragmatic interpretation. A scientific assertion thus has as part of its essential force the intent to persuade an audience, changing "a context in which the speaker is not committed to justifiable true belief in x, into a context in which he is so committed" (Gazdar, 1981:69).

The relation between illocution and perlocution however has never been entirely clear, and neither Austin nor Searle discuss it in any detail. Essentially the difference between the two is expressed as being the result of the fact they have different felicity conditions, i.e. arguing is felicitous if the reader has understood, and convincing is felicitous if the reader accepts the writer's view. Thus we have to regard illocution and perlocution as two distinct aspects of the same speech act. Van Eemeren and Grootendorst believe that there is a conventional relationship between the illocution *argument* and the perlocution *convince,* such that *convince* is an "associated perlocution", regularly, but not necessarily, brought about by means of the particular illocution *argument.* To explain this they posit the existence of "argumentation schema" which represent regularities in the language use of particular discourse communities. Without providing descriptions of such institutional schemata, they insist that discourse community members expect and prefer their occurrence, such that if they are employed, there is at least a possibility of the desired perlocutionary act occurring.

While this position appears to have received little support, the introduction of schemata begins to clarify some of the uncertainty of the illocution/perlocution distinction. But although Van Eemeren and Grootendorst point to the genres and institutional units of language as linked to socio-cultural contexts, they still view illocutions as functional units grounded in the unobservable intentions of language users. The uncertainty of this distinction also remains because it is difficult to precisely analyse the kinds of acts such as *asserting, stating, questioning* and so on, which characterise RAs. There are serious difficulties in specifying either the conditions required for the performance of such diffuse acts (de Beaugrande and Dressler, 1981:117), or the actual length of speech act units. It may be that in expository writing an entire text might perform an act of argumentation. In the next section I shall examine this view, and show how greater descriptive detail has been proposed for subcategorising acts in research genres by focusing on rhetorical objectives within particular genre frameworks.

Rhetorical features of research articles

While the sociological literature has shown that scientific writing is a socially based form of action, linguistic accounts have sought to relate linguistic features to rhetorical functions, demonstrating how academic writers effectively argue their claims for a particular disciplinary audience. This interest in pragmatic aspects of texts has been fuelled by the increasingly influential view that scientific writing is defined less by surface forms than by particular meanings and text stages (eg de Beaugrande, 1989; Paltridge, 1993). Many studies have therefore sought to analyse the rhetorical and discoursal subdivisions in academic texts and the differential use of features which both define and signal the rhetorical goals of those sections.

A writer's ability to simultaneously promote empirical facticity and gain the community's certification of knowledge claims is largely accomplished through careful linguistic choices which help structure appropriate social interactions and rhetorical purposes within the RA's IMRD macrostructure. A number of studies suggest that forms indicating greater writer intrusion occur in Introduction and Discussion sections, which are points in the discourse where argument is emphasised and where decisions, claims and justifications most commonly occur. Thus pronominals, verbs of reasoning, "*that* - nominals", and adverbs, adjectives and modals qualifying assertions tend to cluster here (Bernhardt, 1985: Butler, 1990; West, 1980). Moreover, the more discursive I and D sections contain a heavier use of present forms, and a greater variety of tense and modality usage than the procedural M and R sections (Heslot, 1982; Hanania & Akhtar, 1985). They also show a marked increase in authorial comment, where themes generally foreground overt author visibility and viewpoint rather than research-based entities as in Methods (Adams Smith, 1984; Gosden, 1993).

While these observations indicate important "grammatical-rhetorical dependencies", the division into a rhetorically simple, objective and detached M and R, and a complex, subjective and author centred I and D is less straightforward. Even the most rhetorically innocent sections reveal the concerted attempts by writers to persuade their audience of the veracity of their claims. The primary objective of the research article for those who write them is not simply to present new claims, but to ensure those claims are accepted by their peers and ratified as new knowledge by the disciplinary community. The familiar

IMRD structure is a major means by which these rhetorical goals are pursued and the following sections briefly describe how this is achieved.

Introduction Sections

The considerable work that has been carried out into the rhetorical structure of Introductions clearly demonstrates its rhetorical role (Crookes, 1986; Swales & Najjar, 1987; Swales, 1984 & 1990; Paltridge, 1993). These analyses have attempted to describe the discourse structure of Introductions in terms of authors' rhetorical purposes rather than the orthodox interpretive practices of the empiricist repertoire of objectivity and reason (eg Zappen, 1983).

The most familiar of these analyses is Swale's (1990) 'Create a Research Space' (CARS) model which seeks to capture generalisable elements of introductions. This model suggests a conventional rhetorical structure of three predictably sequenced moves involving establishing a territory, establishing a niche, and then occupying that niche. These moves are motivated by the writer's

> need to reestablish in the eyes of the discourse community the significance of the research field itself; the need to 'situate' the actual research in terms of that significance; and the need to show how this niche in the wider ecosystem will be occupied and defended. (Swales, 1990: 142)

Applying this model to the brief introduction in (3), we can see that the text opens with the author claiming centrality (step 1A) by appealing to the discourse community to accept the study as related to a significant research area. This is typical of such claims in being a single, text initial sentence. "It is well known .." (line 3) introduces a topic generalisation (step 1B) concerning the current state of knowledge in the area and this is supported by a brief review of recent research developments (step 1C) (lines 5-10). The adversative sentence connector "However" (line 11) signals a Move 2 opening indicating a gap in previous research (step 2B). The role of Move 3 is to turn the niche just established into "the research space that justifies the present article" (ibid: 159) and this connection is made here by the writer, in paragraph two, proposing to fill the created gap.

(3) The reversible phosphorylation of protein translation factors is
2 considered as one of the main mechanisms of regulation of protein synthesis in eukaryotic cell[1]. It is well known that phosphorylation of

only 25-30% of s subunit of the rabbit reticulocyte translation initiation factor 2 (eIF-2) results in a strong inhibition of protein synthesis [1-4]. Recently it was shown that the elongation factor 2 (EF-2) from animal cells can be phosphorylated in vitro and in vivo by specific Ca^{2+}/ calmodulin-dependent EF-2 kinase and phosphorylation makes it inactive in protein synthesis [5-8]. It was assumed that the phosphorylation of EF-2 may play a role in the regulation of protein biosynthesis in animal cells [1-9]. However the existence of such a possible mechanism of translation regulation in plant cells was not yet investigated.

The present work was undertaken to study the ability of plant (wheat germ) EF-2 to serve as substrate for EF-2 kinase from rabbit reticulocytes. Also an attempt was made to detect endogenous EF-2 kinase activity in plant cells. (B3:219)

Clearly a great deal of rhetorical effort is being invested in this short text. The author is not simply stating a problem he intends to solve, but also establishing both the importance of his area and his contribution to the field. It employs all rhetorical Move 1 options to emphasise the importance of the area, while avoiding any hostile depiction of earlier work. Move 2 also subtly avoids imputing oversight or neglect to other researchers, and merely refers to work not yet undertaken, while Move 3 states the writer's purpose very early in the paper and in direct co-occurrence with the specified 'gap'. Swales (and Crookes, 1986) found a considerable amount of step and move cycling in long introductions which suggests that the rhetorical effort involved in any text varies according to specific features of the discourse context, such as the competition, the importance of the niche to be established, and the writer's reputation (Swales, 1990: 142). The work on introductions thus shows that writers are responsive to rhetorical pressures, rarely commencing with a simple purpose statement, but typically displaying an attempt to situate a claim in an important area of scientific endeavour.

Methods Sections

Methods sections are often believed to be straightforward reports of the procedures and materials employed in the lab in order to facilitate replication. This

is clear in the advice given to undergraduate biologists in this university laboratory manual:

> This section should be written so completely and explicitly that a competent investigator could repeat your experiments. In the preparation of an actual scientific paper, this section would be written in the past tense and would describe in detail the materials and equipment employed and the methods followed in the investigation. (Botany Dept., U of Q, 1996)

The injunction to report in such a way as to permit replication originates with Bacon's advice at the origins of modern science that anyone "may be free to judge for themselves whether the information obtained ...be trustworthy or fallacious" (Bacon, 1960: 282).

Methods sections are characterised by formulaic procedures and methodological rules. The extract here shows that Method reporting is highly formulaic, with a restricted use of lexis, and an extensive reliance on 'inferential bridging' and repetition replacing explicit cohesion (Myers, 1991; Weissberg, 1984). Texts assume a great deal of domain knowledge from the reader. This not only involves knowledge of lexical relations, but also a tacit understanding of many craft and procedural skills of laboratory work and of the argumentative function of the text itself.

> (4)[2] The viable embryos were isolated from wheat grains (*Tritium aestivum*, Kazakhstanskaya 4 varieties) by the method of Johnston and Stern (1957). The cytoplasmic extract of wheat germ (S23) and the ribosome-free cytoplasmic fraction (S100) were obtained as described earlier (Shaikhin et al, 1992). To isolate EF-2, S100 was fractionated by ammonium sulphate as described in Walthall et al, (1979). [....]
>
> Phosphorylation in vitro was performed in the reaction mixture (40 ml) contained buffer B (30 mM HEPES-KOH, pH 7.6, 10mM MgOAc, 2mM DTT, 150 mM $CaCl_2$, 10 mg/ml calmodulin, 0.1 mM ATP, 2 pmol [g-^{32}P] ATP(37 PBq/mol; Isotope), as well as preparation of substrate proteins and EF-2 kinase as indicated in the legends to figures. After 20 min incubation at 30° C samples were analysed by electrophoresis in SDS-PAAG according to Laemmli (1970) or 2D electrophoresis according to O'Farrell (1975). (B3:219)

A noticeable feature is the constant reference to rules formulated by earlier researchers, although these techniques are identified by attribution rather

than explicitly and many actions are not described at all. Laboratory experimentation is a practical activity involving craft skills, intuitions, social experience and technical equipment which are not often not only indescribable, but also all highly variable. The possibility of replication from such conventionalised descriptions is thus thought to be remote (Gilbert & Mulkay, 1984) and it is difficult to argue with Swales' assessment that:

> Methods sections ... are enigmatic, swift, presumptive of background knowledge, not designed for easy replication, and with little statement of rationale or discussion of the choices made. (Swales, 1990:170)

The decline of explicit reporting of methods is also reflected in the fact that this is now often physically downplayed in leading journals either by printing in small type (*FEBS Letters),* or by locating it at the end of articles (*The Plant Cell).* Moreover, the section may be meager indeed. This text constitutes the entire Methods section of a molecular biology article in my corpus:

> (5) Streptomycin-resistant Nicotiana lines 92/str^{R}6 and 92/str^{R}7, and line 92/wild type, were described by Fluhr et al (1985). Chloroplasts were isolated from percoll gradients as described by Fromm et al (1987). Chloroplast DNA were isolated (Fluer et al, 1981) and 16S rRNA genes cloned and sequenced (Fromm et al, 1987) as described. (F1:500)

Together with a highly routinised and formulaic description of procedures, and an apparently cavalier attitude to replicability, this physical deemphasis on Methods relates to the growing importance of new findings to busy readers. Scientists appear to be increasingly prepared to rely on editors and reviewers to validate the methods used (Berkenkotter & Huckin, 1995).

What function does Methods serve? For Gilbert and Mulkay this section offers the most obvious source of 'social accounting' in that it depicts scientists' actions in terms of invariant, objective terms for rhetorical purposes (1984: 51). A central aspect of justifying a scientific claim is to demonstrate that standardised and widely accepted procedures have been employed in ascertaining the 'truth'. Like citing accepted sources, the requirement to spell out procedures as highly routinised, with little room for variability or human initiative, acts as a compelling persuasive device by demonstrating a credible professional ethos. Adherence to sanctioned procedures is therefore a means of establishing the acceptance of the discourse community. The descriptions of

scientific methods are important in that they provide the warrants upon which the scientist's claim rests:

> Typically, procedures scientists choose and describe in research reports are those the authors think will meet the audience's approval. Procedures reported are likely to be standardised ways of getting at the facts or of calculating data. If the audience understands or endorses the procedures, they are more likely to judge the authors and their claims favourably... (Prelli, 1989: 156)

The writer's purpose is to establish a claim as a contribution to knowledge and aligning one's research procedures with accepted scientific methods makes a powerful appeal to have that claim acknowledged. The Methods section is therefore not a means of allowing other scientists to repeat one's experiments, but serves instead to build a scientific ethos, providing rhetorical support for the claim by demonstrating that appropriate procedures were employed.

Results Sections

Rhetorically, Results constitutes the core of the RA by conveying new knowledge through the presentation, explanation and interpretation of data, it thus represents a carefully constructed discourse to persuade readers of the validity of the scientific facts which underlie a particular knowledge claim. Style manuals state that the purpose of Results sections is to "communicate new scientific findings" by presenting the observations and experimental data obtained by the procedures already discussed (Day, 1988:2). Quite how Results differ from other sections however is unclear, as they frequently contain a discussion of experimental techniques which one might expect to appear in Methods:

(6) The most widely used experimental procedure to dissociate the 17 and 23 kDa polypeptides is an exposure of PSII membranes to 1 to 2 M NaCl at a pH around 6.3 (Ghanotakis & Yokum, 1981). (A4:31)

(7) To determine the ability of wheat germ EF-2 to serve as substrate for phosphorylation the purified EF-2 was incubated with EF-2 kinase from rabbit reticulocyte in the presence of [g-^{32} P] ATP and then analysed by 2D electrophoresis (Fig 1B and C). (B3:220)

(8) The soluble protein was purified from large nodules by a procedure involving anion-exchange chromatography on a DEAE-Sephadex column

followed by preparative SDS-PAGE. The purified protein was digested with trypsin and the derived peptides were separated. (D2:261)

The high frequency of methodological justifications is also noted by Thompson (1993), who found such rationales in 95% of her sample of 20 biochemical articles. These statements explain the reason for adopting a particular approach rather than detail the actual method used, and the purpose of this is clearly rhetorical. Thompson (1993:111) argues that "justifying experimental method is critical in demonstrating the validity of experimental data", as results depend on methodological choices which have to be defended in the discourse. Such methodological justifications anticipate reader objections, and pre-empt challenges to their knowledge claims, particularly when the methods chosen depart from established techniques (Bernhardt, 1985):

(9) In spite of its shortcomings, this method has been widely employed to evidence this type of modification in a number of genomes including plastid and nuclear ones (22-24) (F3:16)

In addition to justifications of procedures which look back to the Methods section, Results also look forward to the Discussion by offering interpretations of the data (Brett, 1994). Writers frequently intrude into the text to draw conclusions, often in probabilistic and subjective terms:

(10) On the basis of our earlier studies and the results described above, we suspected that the remarkable action of betaine is not specific for this solute. (A4:31)

(11) As these fragments result from limited plasmic digestion, one cannot discard the possibility that changes in the conformation of fragment E may occur. (C2:270)

(12) Most T-GUS nodules showed relatively uniform expression throughout the core (Fig. 2C), whereas P-GUS expression seemed to be more intense at the periphery of the central core, probably just internal to the vascular tissue. (D1:635)

Interestingly, unequivocal assertions about the results are rare and interpretations are often hedged, but the objective is not to establish absolute truth.

Instead the goal is to " bring the reader into agreement with the author on what the experimental results mean" (Thompson, 1993: 118). Hedges serve the persuasive function of proposing a plausible interpretation based on experimental data in order to begin the process which transforms subjective interpretation into scientific fact.

Brett and Thompson also observed the use of other rhetorical moves in Results. These were generally explicit evaluative comments assessing the accuracy of the data and supporting previous studies, reflecting writers' need to urge and persuade the reader. The rhetorical effect of evaluating material and citing supporting studies relies on explicit argumentation for its rhetorical effectiveness. On the other hand, openly confessing to experimental disparities or uncertain results seems to persuade by projecting a trustworthy and dependable personal character.

(13) What determines the constitutive expression in nightshade leaves, or lack of constitutive expression in tomato leaves is unknown and could be speculated to occur at several levels of regulation. (F2:335)

(14) The presence of a light activation site in the SCPEPCD1 protein is surprising, since all other features of this gene and its product seem to indicate that the SCPEPCD1 product is a non-photosynthetic PEPC isozyme, and thus would not be expected to be light activated. (F4:669)

Clearly the demonstration of integrity and a willing admission of research problems and unexpected results are central to the ideals of science, but it is no less important to anticipate possible critical responses to the research.

Despite their apparently objective role, Results sections involve complex and unacknowledged rhetorical steps. The presentation of findings is thus not only expository but also argumentative. That such an apparently rhetorically untainted section contains these moves suggests that the validity of the results themselves is considered arguable.

Discussions and Conclusions Sections

The final sections of RAs often coalesce in terms of discourse function, but here we find the most persuasive part of the RA which continues the interpre-

tation of data begun in Results. Here the writer seeks to highlight findings and situate them in the context of a wider body of knowledge by relating results to the work of others.

The best known work on Discussion sections extends Swales' approach and identifies a cyclical pattern of 11 moves (Dudley-Evans, 1986 & 1993; Hopkins and Dudley-Evans, 1988) which emphasises the explicitly rhetorical nature of Discussions. The basic pattern appears to be a statement of result, followed by a comment which evaluates the result, the main goal being "to make claims about the results very cautiously", providing what Toulmin refers to as 'warrants' for the claims through support from the literature" (Dudley-Evans, 1993:144). Further work is needed to verify this descriptive framework, refine the functional categories and linguistic exponents, and establish the disciplinary variation of the model. However, the existence of such a cyclic pattern appears to be a primary organising feature of Discussions. Swales (1990:172) notes for example that the first 'statement of results' move will carry the strongest claim, with weaker results in later cycles. This fragment illustrates such a decline in the significance of successive claims:

(15)
Cycle 1: These data demonstrate that it is possible to carry out optical spectroscopic studies at cryogenic temperatures on single crystals of photosynthetic pigment-protein complexes.

Cycle 2: The methods presented here establish the means of assaying the structural integrity of the crystallised complex both at room and low temperatures and confirm the feasibility of carrying out other techniques of structural analysis either requiring or benefiting from low temperatures.

Cycle 3: The present data show very small (<3nm) absolute wavelength differences in the BChl spectral lines compared to the values observed for the purified complex (Figs 2 and 3). (A2:97)

Support for the general steps of this model also comes from Berkenkotter and Huckin (1995: 41) who propose a rhetorical structure for Discussions which reverses that suggested for Introductions by Swales. Thus Discussions are said to open with the researchers' strongest claim to reassert a title to the 'niche' created in the Introduction and occupied in Methods and Results. The

next move conflates several of Hopkin's and Dudley-Evans' steps which show how the findings relate to the literature and to larger issues, re-establishing the claimed niche. Finally, the writers generally comment on implications, occasionally drawing on speculation and suggesting ways forward to establish additional territory. In this way the Discussion moves reverse the direction of the Introduction by moving from the study to the field as a whole, emphasising what is novel and interesting in the study before situating these in the accepted knowledge of the community.

The length and complexity of Discussions appears to correspond to the rhetorical effort involved in establishing the importance of claims and gaining their acceptance. Thus the current research is promoted heavily in both its location, and the way it is supported by rhetorical moves which set out its relationship to the previous knowledge of the discourse community. As we have seen, this is because the scientific audience shares an acceptance of the judgement criteria against which claims are measured. When such claims are inconsistent with current knowledge, the recycling of rhetorical moves is one means by which a writer is able to provide a 'context of justification' (Gilbert, 1976), accommodating the newly announced findings into the established literature. In this way the writer can demonstrate that arguments in the paper are compatible with those approved by colleagues.

It is worth observing here that the work on rhetorical structure draws on content-based categories to determine the boundaries of generic staging in texts. Where boundaries are not signaled by headings and paragraph indentations, it is "a *cognitive* rather than a *linguistic* sense that guides our perception of textual division" (Paltridge, 1994:295). That is, our understandings of discourse functions are largely distinguished by the way we perceive the world, rather than the rules or conventions of language. This is not to assert that scientific texts are without any distinctive linguistic features, but simply to suggest that not everything that can be said about a text relates to observable, textual phenomena. Once again, this points away from the physical aspects of language to how readers process and use it, directing our consideration to relevant aspects of external reality reflected in texts.

Disciplinary culture and rhetorical context

The purpose of this chapter has been to offer a broad characterisation of the context in which hedging occurs. A complex social activity, such as developing

community validated knowledge, relies on language for its achievement. I have therefore tried to show that regularities such as hedging are the result of particular social activities, and that writer purposes are influenced by the norms, values and epistemological assumptions of the community within which such forms have meaning. Hedging is a convention firmly embodied in a "rhetoric of science", a linguistic resource adopted by writers for particular rhetorical ends and which both encodes and reflects particular cultural practices.

The research discussed here has suggested that the construction of scientific discourse is part of the construction of scientific communities and the socially ratified roles, responsibilities and interactions of scientists. Scientific knowledge is created in a process of negotiation focusing on texts as much as test-tubes, and labs have been characterised as places primarily concerned with textual operations: citing, criticising and forming knowledge claims in competition with other labs (Latour & Woolgar, 1979). Like all skilled writing, scientific RAs display a concern for audience and an understanding of the social consequences of writing, helping to establish the scientist's credibility and persuade others of their findings. Moreover, we have seen that by contrasting laboratory activities with their conventionalised published representation, research discourse not only contributes to the construction of scientific knowledge, but actually helps create science itself, as Bazerman and Paradis (1991:3) observe:

> Writing structures our relations with others and organises our perceptions of the world. By studying texts within their contexts we study as well the dynamics of context building. In particular, by understanding texts within the professions, we understand how the professions constitute themselves and carry out their work through texts.

In short, because scientific writers must engage with their colleagues via texts and use recurring rhetorical practices to help establish the validity of their claims, meaning draws heavily on the contextual knowledge that writers have in common with readers, particularly the beliefs they hold as members of a scientific community. The shared reality of scientists thus provides a great deal of the context which enables discourse features to be correctly interpreted. Consequently, any justification that a particular meaning is present in a text cannot be confined to the relations between syntax and semantics. Bazerman (1988: Chapter 2), suggests that the contexts which most influence academic writers are the object under study, the literature and the anticipated audience. It is clear

however that the assumptions in the reader's "cognitive environment" (Sperber & Wilson, 1986: 39) that can affect text interpretation are diverse, and interpretation can always draw on a variety of sources.

Although the research reported here demonstrates the culturally organised and culturally dependent nature of hedging, it is impossible at the present time to provide a comprehensive model of its context; indeed, even a single precise definition of the term 'context' may not be feasible (Goodwin & Duranti, 1992). The phenomena which contribute to the shared assumptions and discoursal situation within which hedges are embedded are many, and more may be invoked by the use of hedging itself. Even by examining the activities that writers use to constitute their social worlds, we may not gain access to all the phenomena that participants utilize as an interpretive context. This study, for example, draws on texts and the understandings of experts who use them, but it cannot exactly recreate the authentic reading practices of working scientists selectively scanning articles in a rapid search for novelty and personal relevance.

Simply, the real and multiple constraints which act on the production and consumption of texts cannot be replicated by analysts, and we cannot be sure that all relevant aspects of context have been captured in our descriptions.

The difficulties of specifying relevant aspects of context are considerable and have been discussed at length in the literature (eg Fillmore, 1977; Levinson, 1983). Even in the study of such a well defined and established genre as the research article, it is difficult to identify what is germane in the myriad aspects of norms and expectations which will enable us to clearly account for pragmatic features such as hedges. It is clear however that we must go beyond linguistics to combine the strengths of work in generic structure, composition studies, and the sociology of knowledge in creating a descriptive framework. This reinforces the linguistic study of meaning with a sociological component, but it also increases the linguist's role by expanding the relevant contextual factors to include:

- elements of the co-text (such as agentivity, the type of claim and cooperation in the case of hedges),
- schematic constructs which incorporate factors of the social context (such as the goals of the participants, the purpose of the discourse and the institutional role of the genre)
- aspects of the participants knowledge of their field, role relationships and shared expectations concerning appropriacy and community membership

While such aspects are not fully understood, it is clear that analysis of language in social contexts must be based on the study of those contexts. This means understanding hedging as situated in particular discoursal events and examining them as contributions to texts, as aspects of the discursive practices of researchers, and as elements in the social practices of scientific discourse communities (Fairclough, 1995). Without such a multi-layered understanding of context it is easy to overlook or misunderstand the ways that writers use hedges to adopt positions, form alliances or express deference to their readers. Examination of scientific discourse must be linked with examination of scientific practices, because that is where communities are sustained and knowledge is created.

3. Perspective and hedging in discourse

The discussion of scientific research writing in the previous chapter has highlighted a number of features which help provide a linguistic and pragmatic context for academic hedging. In particular, it has demonstrated the rhetorical and dialogic nature of such discourse and emphasised the importance of relating structure and textual elements to social and cultural variables of language behavior. Meanings result from interactions among writers and readers, via texts, employed in particular settings and subject to conventions of genre, purpose and social relations. Persuasiveness means that a text's propositional structure and socio-rhetorical functions are coherent to a particular audience. In scientific RAs this concerns epistemic change, involving the readers' interpretation of both semantic acts of meaning and reference and the pragmatic understanding of knowledge claims.

Hedging is critical in scientific discourse because it helps gain communal acceptance for knowledge. Scientific "truth" is as much a social as an intellectual category, and the distinction writers make between their subject matter and how they want readers to understand their relationship to it is crucial to such a highly self-conscious form of discourse. Not only does it influence the effectiveness and credibility of argumentation, but helps define what it means to write science. Degrees of cognitive probability and generality often facilitate, or even determine, the comprehension of a message. The assessment of propositions therefore places epistemic modality, the handling of potential fact, in a critical role, because the selection of devices can influence the ratification of arguments.

This chapter focuses on this issue and unites earlier chapters by examining more closely the means by which arguments are negotiated in scientific communication. It first considers the concepts of modality and hedging as they are discussed in the speech act and linguistics literature. It then goes on to ex-

amine how hedging has been viewed in scientific writing, exploring the relationship between fact, opinion and audience in scientific prose.

Linguistic approaches to hedging

Speech acts and interpretive maxims

Hedging and indirectness have been of interest to speech act theorists as they involve the writer's beliefs concerning the truth, purpose or appropriateness of the proposition in assigning illocutionary force to speech acts. Thus while each of these sentences carries the force of an 'assertion' in that they seek to alter the beliefs of the reader, the differences in expression are not simply stylistic.

(1) Such a mechanism might serve to balance the synthesis of the products throughout this pathway in wild type Arabidopsis. (D3: 176)

(2) We speculate that the inhibitor interacts specifically with a protease, the origin of which is still to be determined (plant or bacteroids), that are important for the maintenance of symbiosis. (D2:267)

(3) The presence of a light activation site in the SCPEPCD 1 protein is surprising since all other features of this gene and its product seem to indicate... (F4:669)

Underlying the significance of illocutionary force modifications then, are issues concerning both writer-knowledge and writer-reader relationships, both of which raise problems of speech act identification and recognition.

A well known response to this issue abandons Austin's canonical expression of performatives to posit a class of 'hedged performatives' which can be used to modify the act specified (Fraser, 1974). Thus (4) and (5) are 'strongly performative', for while the sentence meanings are not identical to the corresponding performative sentences, "they are easily seen as counting as the act denoted in the performative verb in the sentence" (ibid:1). So *can* permits us to infer the writer's intent to perform the act of concluding, and *must* simply reports an obligation, rather than performing the illocutionary act itself.

(4) ...we can conclude that the decreased α and β areas are not due to... (A3:7)

(5) We must note that β-turn type I or III structures may absorb at this frequency. (C2:273)

Fraser accounts for the force of such utterances in terms of certain "conversational principles", and compares them to cases such as (6) and (7) which do not appear to have the force named by the performative verb, which he calls "weak performatives":

(6) Under these condition, we may predict that seedlings will respond... (D4:41)

(7) One might speculate that what is now understood as carotenoid functions are not to be understood as common capabilities of all carotenoids .. (A5:176)

For Fraser such examples are not acts of *Predicting* or *Speculating* because the modals show a reluctance in performing acts where reasons for such reluctance are not apparent, although Searle might more simply argue that their form makes them *Representatives*.

While Fraser's observations point to contextual and syntactic information in determining the impact of forms which affect the interpretation of performatives, the circumstances in which a particular modal has a strengthening or weakening effect is not always explicit. This is because his principles are not justified in terms of their epistemological or explanatory status. In addition, his use of invented examples and focus on the hedging of paradigmatic performatives fails to capture the range of ways by which writers express their uncertainty about the propositions they make. Bach and Harnish (1979: 214) point out that form-function principles have to be shared to provide inferential bases and argue that a "communicative presumption" is sufficient to explain hedged performatives. This simply requires the reader "to search for a sufficient reason for the speaker utterance", so the hedged force is recognised because the reader infers the reason for the utterance as the execution of an intention, shifting the explanation to a pragmatic plane.

Inference approaches rely to some extent on Gricean implicatures because breeches of maxims of quality, quantity, relation and manner generate inferential procedures based on shared cooperative purposes for communicating (Grice, 1975). In conveying meanings, writers generally depend more on context than on linguistic form, so implicatures are an important way of drawing on and exploiting knowledge shared with readers about the purpose, subject and assumptions of the discourse. Science RAs might be less ambiguous from an illocutionary perspective than other forms of discourse in this regard, as readers are likely to bring fairly assured expectations of exactitude, clarity and relevance to the text. Anticipating these needs, writers will try to fulfil them by supplying no more or less than the text type requires.

An example of this is the tolerance exhibited by the genre towards imprecision, shown by the high frequency of approximations. These hedges are typical[1]:

(8) The area of the a phase is almost 50% smaller after 30 min. (A3:7)
... decrease in the in the overall intensity by about 35% (B4:231)
... gave over 40% inhibition. (C1:193)
... photosynthesis has obtained some 50% of its .... (C1:195)
.... contains most of the central domain of the molecule .. (C2:271)
...we calculate that more than half is present in the .. (C2:274)
...the host plant expresses a certain number of proteins... (D2:259)

The choice of these 'vague expressions' can be explained by reference to the Maxim of Quantity ("Make your contribution as informative as is required" ibid: 45). Channell's economist informants referred to such uses in terms of "adjusting the amount of information to fit what they think their readers will need to know" in the context of the argument (Channell, 1990: 108).

In addition to giving the right amount of information however, writers can maintain a commitment to truthfulness while simultaneously hedging their confidence in the reliability of their assertions. Clearly, identifying the strength the writer is prepared to guarantee for the truth of statements is part of the process of interpreting a claim. Writers can, for example, manipulate the quantity maxim to express the tentativeness of their beliefs:

(9) These data suggest that the ACC synthase isoenzyme expressed in E. coli behaves more or less like the native ACC synthase enzyme(s) of ripe tomato fruit in its response to substrate inactivation. (B5:107)

(10) Chylla et al (1987) observed that as much as 35% of the total PSII complement of relatively highlight grown spinach thylakoids was inactive. (A1:270)

(11) These results may have relevance to leaf behaviour during the hours after sunrise. (A3: 8)

Thus an important part of the information conveyed in these examples is the recognition that the writers are withholding full commitment to the propositions expressed.

In addition, it is generally expected that scientific statements should conform to requirements of honesty and truthfulness. Consequently, observation of the Maxim of Quality, "Do not say what you believe to be false" and "Do not say that for which you lack sufficient evidence" (Grice, 1975: 46), might explain the use of some hedged expressions. Restricting claims to what the writer knows to be true accounts for the heavy mitigation in (12), although less observance is made of the brevity principle (Maxim of Manner):

(12) .. together these data suggest that the protein synthesis in plants is not regulated by phosphorylation of EF-2. Nevertheless, it cannot be excluded that this mechanism is realised only at certain phases of plant development (for instance during gametogenesis) and such a possibility requires further study. (B3:222)

So, underlying the Maxim of Quality as a motivation for the use of modality in scientific statements is a more fundamental purpose of keeping within the boundaries of what is known and what might be known in the near future. In other words, "it is a way of taking out insurance on the statements one makes, limiting the damage that might result from bald propositions" (Nash, 1990:23).

In some cases informativity seems to be manipulated in order to focus readers' attention on what is important (Dubois, 1987; Channell, 1990). In (13), for example, imprecision in the comparison of oxygen concentrations

between the chamber and atmosphere is used to rhetorically foreground the precise quantities remaining after the experiment.

(13) The concentration of CO_2 in the chamber was 710μL CO_2 L^{-1}, which was approximately two times the concentration of laboratory atmospheric CO_2. The amount of CO_2 remaining after the experiment averaged less than 2% of the initial value. (E2:728)

Thus rhetorically motivated maxim deviations are common in academic texts. In (14) for example, the writer violates the Maxim of Quality by conceding a lack of adequate evidence while (15) appears to breech Quality, Quantity and Manner Maxims.

(14) We suggest here that the endogenous ligand for the NPA receptor is a conjugate of IAA, specific to the transport of the hormone. This suggestion is purely speculative, and there is no evidence as to the nature of this conjugate, but the high activity of NAIV raises the possibility that the molecule of interest may be IAA-valine (Fig 1). (E1: 294)

(15) Thus, the evidence strongly favours the conclusion that phytochrome A is the primary, if not exclusive, FR-HIR photoreceptor, consistent with previous physiological studies (Beggs et al, 1980; Holmes & Schafer, 1981; Smith & Whitelam, 1990). On the other hand, the possible involvement of phytochromes C, D or E in an interactive or synergistic manner with phytochrome A cannot presently be ruled out. (D5:45)

A strong case, drawing on research evidence and previous studies, is made for a "primary, if not exclusive" causative agent in the first sentence, then this is downgraded and mitigated in the second, suggesting lack of evidence, informativity and directness.

Such strategies appear to flout Grice's maxims by providing information which, from a (decontextualised) view of rational, economical and efficient communication, might be described as over-informative, irrelevant, imprecise or vague. Obviously the importance of participant assumptions cannot be overstated in the pragmatic interpretation of speech acts, and Sperber and Wilson (1986) have emphasised this by positing the presumption of contextual relevance in communication. If participants assume optimal relevance of in-

formation and the need for minimal processing effort, then readers will be able to retrieve an adequate interpretation of an utterance. Here then, hedges can be seen as a means of making contextual assumptions immediately accessible, ensuring that the reader draws the correct inferences about either the likely truth of the utterance, or just how strongly the writer is committed to it (Blakemore, 1992: Ch 6).

Epistemic modality

The expression of tentativeness and personal attitudes, of commitment and detachment, traditionally falls within the semantic domain of modality. The philosophical and linguistic literature on this subject is enormous and spans the philosophy of logic (eg White, 1975; Haack, 1978), semantic linguistics (eg Lyons, 1977; Perkins, 1983; Leech, 1987; Wierzbicka, 1987; Palmer, 1986 & 1990), discourse analysis (He, 1993), corpus studies (eg Coates, 1983), applied linguistics (eg Holmes, 1983; Brown, 1992) and pragmatics (eg Klinge, 1993 & 1995; Stubbs, 1986; Turnbull & Saxton, 1997). While there is no single, clearly definable semantic category of modality, Lyons (1977:452) suggestion that it concerns the "opinion and attitude of the speaker" captures its basic nature. Essentially it is concerned with a writer's standpoint, judging the truth of statements in terms of possibility, probability or certainty. In essence modality qualifies the categorical, and the qualification is "effected by showing that the validity of what one is saying to be relative to some conditional framework" (Perkins, 1983:19). Modals refer not to events but to their *potential* for fulfillment (Klinge, 1993).

It is common to distinguish *epistemic* and *deontic* modality (eg Lyons, 1977; Palmer, 1990). Epistemic modality expresses the speaker's opinion or belief concerning the truth of what is said, while deontic modality indicates his or her observations about the necessity or obligation to perform particular actions. While these senses are often confused (eg Holmes, 1984: 348; Stubbs, 1986) epistemic modality refers to items used to display confidence, or lack of confidence, in the truth of propositional information. Hedging is one aspect of epistemic modality, concerned with personal judgements based on a lack of knowledge. It therefore possesses two important properties: 'non factivity' (Lyons: 1977: 795) and subjectivity (Perkins, 1983: 68).

A non-factive predicator, such as "believe" or "suggest", commits the speaker to neither the truth nor falsity of a proposition, while an unmodalised

declarative sentence (a categorical assertion) is normally assumed to carry the speaker's belief that what he or she is saying is true[2]. Subjectivity concerns the personal source of the proposition. The source of the epistemic judgement can be explicitly attributed to the writer (16), or be evidential (17), intertextual (18), or non-explicit (19):

(16) I believe that the major organisational principle of thylakoids is ... (A1:267)

(17) The present work indicates that the aromatic ring ..... (E1:293)
From these calculations it appears that most of the... (A3:4)

(18) Trifonov (38) has suggested that the 530 loop ... (F1:503)

(19) The inhibitor is thought to be one of the inducible .... (F2:349)
It appears that the transformation of active reaction ... (A3:8)

Such strategies are not strictly 'objective', but are often employed to disguise the source of the judgement and withdraw the writer's involvement. Factuality and subjectivity can thus be manipulated to convey an impression of "detachment", or display what is often referred to as a particular "attitude" of commitment, tentativeness or formality.

Despite the tendency of linguists to focus on modal verbs as exponents of epistemic modality (Coates, 1983; Palmer, 1990), the modals are part of a much wider system. The literature confirms the importance of syntactic hedging through clauses of condition (Quirk et al, 1972; Perkins, 1983) and concession (Quirk et al, 1972), passive voice (Lachowicz, 1981; Rounds, 1982; Halloran, 1984), source attributions (Prince et al, 1982), tense (Lyons, 1981), questions (Perkins, 1983) and tags[3] (Holmes, 1984, 1990 & 1995; Coates, 1987 & 1988). Some examples from the corpus include lexical hedges (20), IF-clauses (21), questions (22), passivisation (23), contrast markers (24), impersonal phrases (25) and time reference (26)[4]

(20) Possibly phosphorylation of ACC synthase could contribute to the ... (B4:107)

(21) ...if the hemoglobin in the root were mainly at the tip, it would have a .. (D1: 638)

(22) It can be questioned whether these properties and structural implications apply to.. (B5:6)

(23) The large cells containing fluorescent inclusions are suspected to be ... (D2:262)

(24) Nevertheless, it cannot be excluded that this mechanism is realised only at certain... (B2:222)

(25) It now seems possible that the oxygen carrier function may be feasible because.. (D1:638)

(26) Such mutants should also help to elucidate the UV-B signal transduction pathway. (D3:177)

There also appear to be a number of hedging strategies available to scientific writers which will be discussed in more detail in chapter 6. These include hedging quantities (27), admitting to a lack of knowledge (28), citing a source (29), or referring to limitations of a model (30), experimental conditions (31) or method (32).

(27) ... which accounted for about 4% of the total bases present. (F3:16)

(28) To date and as far as we know, aminoacylation experiments on tRNA have a .. (B5:7)

(29) Also, the fluorescence quenching we have detected is at least partly reversible which is not thought to be the case with photoinhibitory damage to isolated chloroplasts (Krause, 1988). (A3: 8)

(30) Based on and consistent with the model proposed above, ... (A1:273)

(31) The quantitative measurement of PEPc kinase activity in extracts was difficult, because of the presence of endogenous proteinase... (C1:192)

(32) The bare platinum employed to detect oxygen in the (...) electrode is not well suited to making absolute measurements since its sensitivity can be altered. This makes it difficult to compare signal sizes between different samples. (A3:6)

Epistemic modality clearly encompasses what has been termed 'evidentiality', concerned with the reliability of knowledge in terms of sources, mode of knowing and matching information against prior expectations (Chafe & Nichols, 1986). This might be referred to as an 'epistemic' sense. It also includes an affective dimension: the interpersonal and interactional factors which affect how knowledge and knowing are linguistically realised.

Halliday (1994) argues that modality falls within "the interpersonal semantic system of language" which mediates the role between writer and reader and thus represents the personal involvement of the writer in the text. Modality is therefore:

> a form of participation by the speaker in the speech event. Through modality, the speaker associates with the thesis an indication of its status and validity in his own judgment; he intrudes, and takes up a position .
> (Halliday, 1970:335)

Halliday also gives syntactic reasons for assigning modality to the interpersonal component, arguing it is not subject to distinctions of polarity and tense (Ibid: 245-50), and therefore does not form part of the ideational aspect of the clause. This claim for functional discreteness is rather unconvincing however (eg Butler, 1985:175), and I will later argue that modality performs both interpersonal and ideational functions.

Halliday's categorisation, however, draws attention to the role of modality in signaling the writers' involvement in a social context and suggesting how it actualises a persuasive purpose in a text. Thus Smith (1986) for example, shows how the use of features such as impersonalisation and modal adjuncts in the interpersonal component reflect both 'personal tenor', or degree of formality, and 'functional tenor', or language purpose, which can shift the interpersonal stance of a text through an essay. Similarly, in a study of counselor-student discourse He (1993) found modal choices functioned "as a resource for structuring the interpersonal relationship between the participants and for organising the institutional activity of academic advice giving" (Ibid: 527). He's study suggests an interactive basis for speakers' use of different modal values,

with tentativeness used to offer solutions, give advice, and state preferences. In these and other studies, then, modality is seen as a means of indicating both truth status of utterances and the discourse status of speakers. That is, modality expresses both epistemic and interactional meanings.

An important interpersonal aspect of modality discussed in the literature is that of 'mitigation' which has widely been seen as a strategy for softening or reducing the strength of a speech act whose effects may be unwelcome to the hearer.

Fraser (1978 & 1980; Fraser & Nolan, 1981) distinguishes mitigation from hedging in Lakoff's (1972) sense of making things fuzzier. There are considerable similarities between the two terms however, as Fraser recognises, for hedging items can have a mitigating effect on the force of an illocutionary act Fraser suggests a number of options for mitigation in spoken discourse which include strategies such as indirectness, providing justifications and impersonalising on one hand, and linguistic forms such as parenthetical verbs, disclaimers and tag questions on the other. Fraser observes that there is no way to insure that mitigation, the intended perlocutionary effect, will be successful as "it is ultimately, up to the hearer to determine whether or not an unwelcome effect has been softened" (ibid. p349). If mitigation strategies follow particular conventions of usage however, there is at least a possibility of the desired perlocutionary act occurring.

Holmes (1984, 1990 & 1995) widens the discussion of mitigation of speech acts to include both attenuation (weakening) and boosting (emphasising) and discusses a range of lexical, prosodic and syntactic exponents. While many of these realisations are not relevant to academic writing, Holmes observes that some speech act categories can be modified by reference to the sincerity conditions of the act. Searle's (1979) Assertives, for example, can be "boosted or attenuated by referring explicitly to the extent of the speaker's belief in or commitment to the proposition being asserted" (Holmes, 1984: 347). Thus propositional commitment is hedged in (33) and strengthened in (34):

(33) We speculate that the inhibitor interacts specifically with a protease, the origin of which is still to be determined (plant or bacteroids), that are important for the maintenance of .. (B2:267)

(34) We have demonstrated that the promoters of the haemoglobin genes of the two nonlegumes. (D1:635)

While Holmes does not develop this further, this kind of mitigation is clearly of major importance in academic discourse (Hyland & Milton, 1997), and indicates a central element of writer perspective.

The notion of mitigation has probably received the most systematic attention in the work of Brown and Levinson (1987) who suggest that a central goal of interaction is face, the maintenance or creation of desired images, of self, other and relationship (Scollon & Scollon, 1995). In this view everyone has both Negative Face, the desire to be unimpeded, and Positive Face, the desire for approval. Based on an assessment of social distance, relative power and the degree of imposition, people choose courses of action which maximise benefits and minimise face-loss. Mitigation is thus seen as a form of politeness conceptualised as strategic conflict avoidance. It reflects a relation between the writer and readers, not only the degree of probability of a statement. Hedging has been found to be a central component of the work routinely done by conversationalists to repair face damage by conveying reluctance and obligation when refusing requests (Turnbull and Saxton, 1997) and other face threatening acts (House & Kasper, 1981). Myers' (1989) strong claim that mitigation plays a similar interpersonal role in academic writing will be discussed later in this chapter.

Leech (1983) appears to be alone in explicitly associating mitigation with rhetoric. Leech bases his "interpersonal rhetoric" on a Cooperative Principle, following Grice's maxims, and a Politeness Principle, based on maxims of tact, generosity, approbation, modesty, agreement and sympathy. While there may be reservations about the theoretical justification for these maxims (Dillon et al, 1985), Leech draws on a rich theoretical framework to explain politeness. Maxims of Tact (minimising costs to other), Approbation (minimising dispraise of other) and Agreement (minimising disagreement between self and other) appear to be the most relevant to scientific discourse. Moreover, by situating mitigation within a rhetorical framework, Leech overcomes the weakness of Brown and Levinson's analysis which assumes that the use of mitigation can be predicted by the simple application of the variables of power, social distance and weight of imposition. Cherry's (1988) study of persuasive letters to a university president, for example, found variations in the use of mitigation devices unrelated to status or rank, junior staff exploiting the potential of

hedges for rhetorical effect. This illustrates the importance of a flexible view of mitigation and the need to regard pragmatic principles as probabilistic rather than invariant.

Metadiscourse

Another approach to hedging is to see it as a form of metadiscourse, because hedging devices contribute to the means for directing readers to how they should understand, evaluate and respond to propositional information. This is clear in Crismore's definition:

> metadiscourse allows writers to show readers how different parts of the text are related and how they should be interpreted. Metadiscourse also permits writers to express their attitudes toward the propositional content of the text and toward their readers. (Crismore et al, 1993:40).

A number of writers have thus taken a comprehensive view of metatext and include in it both text-organising and interactive elements such as the writer's attitudes and expressions of uncertainty (eg. Crismore & Farnsworth, 1990; Hyland, in press a & b; Vande Kopple, 1985). This approach clearly owes a great deal to the Functional Components Hypothesis of Systemic Grammar, and these categories roughly correspond to Halliday's textual and interpersonal macro-functions with the former referring to text connectives, code glosses and illocution markers and the latter to attitude markers (*surprisingly, fortunately*), directives (*you may wish to read the last section first*), source attributions (*according to X*), and epistemic validity markers *(probably, might)*.

While some analysts have preferred to focus on features of textual organisation (Mauranen, 1993; Valero-Garces, 1996) or explicit illocutionary predicates (Beauvais, 1989), such a terminological extension draws attention to an important dimension of texts, that is, the means by which writers reveal both themselves and their communicative intentions. Studies have therefore suggested the importance of metadiscourse in casual conversation (Schiffrin, 1980), school texts (Crismore, 1989), science popularisations (Crismore & Farnsworth, 1990), undergraduate textbooks (Hyland, in press b), postgraduate dissertations (Swales, 1990), Darwin's *Origin of the Species* (Crismore & Farnsworth, 1989) and company annual reports (Hyland, 1998). It has also been shown to be a critical feature of good ESL and native speaker student writing (Intraprawat & Steffensen, 1995; Cheng & Steffensen, 1996), and an

essential element of persuasive academic writing (Crismore & Farnsworth, 1990; Hyland, in press a).

In a recent study of 28 RAs in four academic disciplines, hedges were found to be the most frequent from of metadiscourse in all but the astrophysics articles (Hyland, in press a). While biology articles contained relatively few explicit markers of attitude or commentaries that directly appealed to readers, hedges and attributions to sources were an important rhetorical means of supporting the writer's position. Thus the biology texts displayed a marked reluctance of writers to project a prominent authorial presence compared to those in 'soft-applied' disciplines, but these writers did deploy hedges as rhetorical devices. From a metadiscourse perspective, hedges represent clear attempts to negotiate academic knowledge in ways that are meaningful and appropriate to particular disciplinary communities. They indicate the writer's acknowledgement of the discipline's interpersonal conventions and build writer-audience relationships by drawing the reader into the discourse as a participant (Crismore et al, 1993).

Appropriate definitions and analytical approaches to the study of metadiscourse however remain unresolved. Several possibilities exist for framing a systematic theory as metadiscourse can be considered as either a syntactic, semantic or pragmatic phenomena. Clear and consistent criteria are essential when formulating basic terms, particularly when delineating subcategories. Some of this work however has been characterised by an inappropriate and inconsistent use of form-based and function-based categories which leads to confusion in theoretical approaches (cf. Beauvais, 1989). The metadiscourse research however has confirmed the significance of hedging in a range of genres, domains and languages and has begun to describe how preferred uses of hedging may contribute to rhetorical patterns which reflect broad areas of intellectual inquiry and knowledge. Its importance lies in situating hedges in wider interpersonal and rhetorical practices and seeking to link these practices with the ways specific groups use language to achieve particular social goals.

Hedging in academic writing

Despite the attention given to modality by linguists and the fact that many statements in scientific discourse express opinions, hypotheses or lack of cer-

tainty, until recently there was surprisingly little empirical study dedicated to describing or explaining qualification in academic writing. This section briefly reviews the main quantitative studies in this area to demonstrate the importance and variation of hedging in academic prose.

Skelton's comparative study of 40 RAs in sciences and humanities focuses on the expression of 'commentative language', or "the way in which we can express the extent to which we commit ourselves to particular propositions" (Skelton, 1988b: 98). Skelton identifies four realisation categories of comment: copulas other than "be" (35), modal verbs (36), adjectivals and adverbials which are clause initial (37) or introduced by *There is*, *It is*, *This is* (38), and lexical verbs (39):

(35) The insertions appear to be concentrated in four major loop regions on the periphery of this core. (C3:285)

(36) Although it is premature to answer this question it might be suggested that synthetases present in nuclei [33] could be involved in the regulation of the processing of tRNA precursors. (B6:7)

(37) Possibly, phosphorylation of ACC synthase could contribute to the existence of its pI forms. (B5:107)

(38) It is also possible that phosphorylation of PEPc actually stimulates a proteolytic cleavage that removes the phosphorylation site. (C1:192)

(39) Thus we propose that this insert is the major site of interaction with the membrane. (C3:284)

Skelton's initial scheme was far from comprehensive, but more recently he has characterised commentative language as any statement by which we modulate our attitude to propositions, marking either "truth judgements" or "value judgements" (1997:124). Clearly such commentative language is far broader than hedging and also includes remarks which intensify commitment or comment on the emotional status of assertions, conveying surprise, importance, agreement and so on. However, Skelton's work indicates that hedging is firmly located within a repertoire which allows writers to represent the truth value of what they say in different ways and points to the fact that such com-

mentative language is functionally suasive. It also begins to suggest the extent to which academic writing is hedged: "Comments are common in academic writing, and about equally common in arts and sciences. (...) they certainly appear to occur, overall, in between one third and one half of all sentences" (1988b: 103).

This observation is supported by Adams Smith (1984) who found one comment every 3.7 lines rising to one every 2.2 lines in Discussion sections, and by Butler (1990) who observes that modal verbs account for about one word in every 100 in scientific articles. The studies also show that the occurrence of these features varies considerably according to the communicative purpose of different sections, being most strongly represented in the Discussion sections (Table 1).

Table 1: Distribution of Modality in scientific discourse.

Focus of study	Introduction	Methods	Results	Discussion
Modal themes (%) (Gosden)	6.8	1.0	6.1	13.6
Modal verbs (%) (Hanania & Akhtar)	11.1	11.8	6.9	15.1
Modal verbs (per 1000 words) (Butler)	9.74	0.65	3.14	12.51
Epistemic comment (per line) (Adam Smith)	High 1: 3	Low 1:21	Low n/a	High 1:2.2
Hedging comment (per 1000 words) (Skelton)	High 9.7	Low -> 4.39 <-	Low	Very High 19.33

Thus this research suggests that academic commentary is overwhelmingly epistemic. About 70% of the author's comments in Adams Smith's study of medical RAs related "to probability, the truth or definiteness of the thesis"

(ibid: 28) and were typically (54%) marked by modals. Dudley-Evans' (1993) analysis of moves in an economics debate found that 37% of the claims made were hedged. Butler found that modals are an important means of allowing authors to "make claims about what it is legitimate to conclude from the results, what may or must be the case, what phenomena are sometimes or generally observable and so on" (ibid: 166). Gosden's survey of thematic functions in science RAs showed that 7.6% of grammatical subjects expressed uncertainty, realised through markers of modality (Gosden, 1993:68), and modals accounted for 8.1% of all finite verbs in Hanania and Akhtar's (1985) corpus of MSc dissertations.

Another area of hedging to receive attention in the literature is numerical approximations. These have been discussed both in terms of their truth-conditional implications (eg Lakoff, 1972; Wachtel, 1980) and as options of purposive vagueness in pragmatics (Channell, 1980 & 1994; Powell, 1985). Only Dubois (1987) has examined imprecision as a rhetorical technique in a scientific genre. She found that hedges are used as a means of diminishing quantitative precision in biomedical slide talks and are realised by a limited range of adjectives and adverbs, most often *approximately* and *about.* While more frequent in spoken discourse, my RA corpus also reveals a considerable amount of such imprecision, principally when generalising from findings or considering tabular or graphic data in Results and Discussion sections:

(40) .. more than about five bilayers into a grana stack.	(A1:268)
...partial reaction rate was about 25% as a result of ..	(A3:7)
... the violaxanthin content is around 18% of ..	(A5:173)
Most of the PS II, about 85%, ...	(A5:175)
... decreases by approximately 60% at 44°C, ...	(B2:192)
...the number of such residues is about 20.	(B4:231)
...an increased followed, after some 90 min ..	(C1:192)
..sufficient to inhibit approx 90% of protein synthesis	(C4:165)

While imprecision can be seen as adherence to Gricean maxims, made necessary by preliminary results, uncertain predictions, small samples, or imperfect techniques, it may also have a rhetorical rationale. Thus while the modal + comparative in (41) may be motivated by the variability of the data, it nevertheless helps reinforce the point, and the effect of replacing the construc-

tion with a simple 'be' leaves the statement semantically unaltered but less convincing.

(41) The fact that grana stacks in some shade plants can be as much as 100 bilayers thick (Anderson et al, 1973) would therefore appear to be in significant conflict with the theory that suggests grana increase light absorption under low intensity conditions. (A1:269)

Hedges can also be used to rhetorically down-play or background results in order that findings considered more important can be emphasised. This may explain the movement from hedged to precise figures in (42), for example, and the way which the authors introduce their study in (43):

(42) Addition of 10ng of 5-methylcytosine to a crDNA hydrolysate gave a clearly distinct peak (Fig 3e) which accounted for about 4% of the total bases present. In contrast, leaf nuclear DNA showed a variety of modified bases (comprising about 10% of the total DNA, fig 3b). The G + C of tomato leaf nuclear DNA was 48% and that of tomato plastid DNA was 35% (Fig 3). (F3:16)

(43) Recently we found that fluorescence emission from isolated spinach chloroplasts which had been exposed to 650 nm light (6 $\mu Em^{-2}s^{-1}$, 30 min) and then inhibited with DCMU, was about 15% lower than the dark control (Sinclair & Spence, 1988). (.....)

In this paper, we present the results of a more detailed investigation of this fluorescence quenching along with observations on the changes in the electron transport properties of the chloroplasts which occur following exposure to low light intensities. (A3:1)

Dubois (1987:537) argues that numerical hedging can also cite the results of other writers one believes to be in error, highlighting the precision of the speaker's analysis against their approximations, but this function does not occur in my corpus. RAs generally appear to contain few cases of approximation in fact, but Dubois' work suggests that imprecise numerical expressions "which are used to express views of one's own findings, as well as those of others" should be included in the range of hedging markers (ibid: 359).

The work discussed here demonstrates the importance of hedging in academic writing, but equally illustrates the limited attention it has received. Applied linguistics has only just begun to scratch the surface of an enormous topic. These studies reveal that hedging is an important means by which writers express their judgements and press their claims, intruding into the text and initiating a dialogue with the reader. Its use greatly increases the interactive role of the writer, allowing a viewpoint to be expressed and objections to be anticipated. This central purpose of scientific hedging, as an aspect of a researcher's participation in the written text, is considered more fully below.

Knowledge claims and scientific truth

Skelton (1997) depicts three main kinds of scientific truth. These refer to the shared knowledge contained in the literature (contextual truth), areas deemed true because of the statistical or experimental results of a study (evidential truth), and non-statistical judgements of fact or value about findings (interpreted truth). Each of these types can be manipulated to persuade readers of the author's contribution to a debate, and hedges play a large part in accomplishing this. Writers may thus hedge their commitment to accepted knowledge, to new findings, or to what those findings mean. Essentially, the goal of scientific researchers is to establish their claims as facts, using the resources of the language to promote their work, and ensure that interpreted and evidential truths become part of the accepted truth of the discipline. At the heart of a scientific RA, then, is a statement of a research finding or knowledge claim, and the writer's purpose is to create a text where such claims seem unproblematically related to observed data.

However, the process by which a scientist's research findings gain the status of accredited knowledge is a long and complex one. The evaluation of academic claims by members of the scientific community begins during the course of the research itself, when immediate colleagues and an informal communication network operate (Bazerman, 1983). The claims are then subjected to a detailed review process by journal editors and referees which often involves negotiation and revision (Myers, 1990). Finally the published claims undergo evaluations of usefulness and importance by peers.

Detailed reading by the discourse community is motivated by the possibility of using a paper's results and each reader brings his or her own purposes and background knowledge to the paper. It is this relevance to the researcher's own activities and understanding of the field which determines the fate of claims. As Bazerman (1985:21) notes "the statements that will have a continuing life in the literature will be those that readers will consistently integrate into their work". Some claims will be ignored or rejected as invalid, others will be explicitly recognised through citation, providing background or support for additional future claims (Bloch & Chi, 1995). As the successful claim grows older, it will no longer be referred to but implicitly incorporated into arguments as a taken-for-granted assumption. Readers are therefore only fully persuaded when all sources of support have disappeared, "when readers are sufficiently convinced there is no debate about it and the processes of literary inscription are forgotten" (Latour & Woolgar, 1979: 76).

The process of fact accreditation can be seen most clearly as scientific claims pass from research articles into other rhetorical environments. A number of studies have described this transformation of information into popular accounts or textbooks (Winstanley, 1976; Latour & Woolgar, 1979; Fahnestock, 1986; Myers, 1991, 1992 & 1994).

Winstanley's (1976) early account traced the coverage given to the genetic function of DNA in a variety of biology/genetics genres for the 10 years following Crick and Watson's 1953 discovery. References to the description of DNA structure first appeared in the professional literature, later in university textbooks, then in school textbooks and finally in popularisations. The number of popular journals carrying science reports for a lay audience has dramatically increased in recent years however, and presumably the most newsworthy reports now occur much sooner in popularised form than in textbooks.

An important aspect of this process involves variations in how knowledge is presented in different genres, mainly by the removal of hedges to confer greater certainty on claims. Latour and Woolgar (1979) and Myers (1992) observe that textbooks contain a higher proportion of unmodified assertions than RAs because of a different attitude to facts. As Myers (1989:9) points out:

> Authors of textbooks try to arrange currently accepted knowledge into a coherent whole, whereas authors of journal articles try to make the strongest possible claim for which they think they can get agreement.

Part of gaining acceptance of claims lies in the use of hedges to evaluate the status and value of information (Hunston, 1993). In an earlier work, Myers

(1989) has argued that virtually all new claims are hedged while, in contrast, textbooks present claims as accredited facts that need no hedging. The examples in this thesis clearly demonstrate the extent and variability of hedging in this genre, with each RA in my corpus averaging 60 hedges.

While RAs show a reluctance to assert that claims represent an unequivocal truth, textbooks appear to exhibit no such scruples; they contain fewer hedges, and these generally indicate marginal matters about which some uncertainty remains. A similar picture emerges with the presentation of scientific results in popular magazines. These seek to glamorise material for a wider audience, upgrading the significance of claims in order to emphasise their uniqueness, rarity or originality. Fahnestock (1986) observes that this is accomplished by the removal of qualifications that a knowledgeable scientific audience would require. For the researcher such qualifications are a necessary defence against refutation from an expert readership, for the science accommodator they simply reduce the news-value of a statement by undermining its certainty.

Fahnestock shows how the qualified conclusions from an article in *Science,* reporting a longitudinal study of mathematical aptitude, were transformed by two popular magazines. The original research article looked like this:

> (44) We favour the hypothesis that sex differences in achievement in and attitude toward mathematics result from superior male mathematical ability, which may in turn be related to greater male mathematical ability in spacial tasks. This male superiority is probably an expression of a combination of both endogenous and exogenous variables. We recognise, however, that our data are consistent with numerous alternative hypotheses. Nonetheless, the hypothesis of differential course-taking was not supported. It also seems likely that putting one's faith in boy-versus-girl socialisation processes as the only permissible explanation of the sex difference in mathematics is premature. (Benowa and Stanley, 1980).

When the reports appeared in *Newsweek* (45) and *Time* (46) however, these qualifications and hedges had disappeared, increasing the certainty of the claims and the impact of the stories:

> (45) The authors' conclusions: "Sex differences in achievement in and attitude toward mathematics result from superior male mathematical ability".

(46) According to its authors, Doctoral Candidate Camilla Persson Benbow and Psychologist Julian C. Stanley of Johns Hopkins University, males inherently have more mathematical ability than females.

The reduction or elimination of hedges in textbooks and popularisations has different rhetorical purposes. While the textbook writer seeks to present what is taken-for-granted as fact, the populariser attempts to add to the significance of the subject. The importance of hedging in distinguishing degrees of certainty in knowledge claims also points to its major rhetorical function in scientific discourse. It additionally suggests the substantial pedagogical implication that the different purposes and audiences of these genres undermines the value of these sources as models for science writing courses.

Latour and Woolgar's (1979) analysis of statements in scientific discourse seeks to formalise these observations by providing a hierarchical taxonomy of claims. This work distinguishes five types of statement type according to the degree of certainty each conveys. Type 5 statements refer to taken-for-granted assumptions which are so self-evident to the discourse community that they are presupposed rather than stated. These "merged imperceptibly into a background of routine inquiry, skills and tacit knowledge" (ibid: 76). Type 4 statements concern uncontroversial matter that is explicitly asserted. These expressed certainties are the stock-in-trade of the science textbook and popular account but generally only seem to occur in RAs as contextualising material:

(47) Mitochondrial cholesterol side-chain cleavage enzyme (P-450$_{scc}$) catalyses the rate-limiting initial step in steroid hormone biosynthesis. (C3:281)

(48) Hemoglogin is present in the nitrogen-fixing nodules of both legumes and nonlegumes, where it transports oxygen to the nitrogen-fixing endophytes. (D1:633)

While type 4 statements are part of the scientist's knowledge-base, types 2 and 3 contain hedges or qualifications which indicate that the information conveyed is not regarded as indisputable by the writer. Type 3 statements express these uncertainties quite subtly by citing a source for the assertion or an expression concerning prior work. In Latour and Woolgar's system such features do not necessarily reinforce the certainty of a claim, but weaken its

strength by suggesting that human agency, rather than objectivity, supports the assertion. The statement is thus made less assured by indicating the need for backing.

(49) Photoinhibition is characterised by a decrease in PSII electron transport activity, chlorophyll fluorescence and quantum yield (Powles, 1984). (A1:273)

(50) Protein fluorescence, mainly that of tryptophan residues (Trp), is known to be a useful intrinsic probe of the local polarizability and confirmation changes in proteins [7,8]. (B4:229)

The removal of these sources is enough to transform type 3 into type 4 statements, but type 2 statements are altogether more tentative and contentious. These explicitly draw attention to the quality of evidence which underlies the claim, and involve a wider range of items. They frequently employ lexical hedges such as modals, adjectives or lexical verbs.

(51) One cannot exclude a possibility that the activity of EF-2 kinase in wheat germ is inhibited at a given stage of ontogenesis in some manner making its detection difficult. (B3:222)

(52) Although it is premature to answer this question it might be suggested that synthetases present in nuclei [33] could be involved in the regulation of the processing of tRNA precursors. (B6:7)

(53) Although a causal relationship between the latter processes remains to be verified, the correlation may not reflect pure coincidence. (B2:192)

Finally, type 1 statements are openly speculative and conjectural. They represent the tenuous assertions of lab conversations but may also occur in the conclusions of RAs.

(54) What determines the constitutive expression in nightshade leaves, or lack of constitutive expression in tomato leaves is unknown and could be speculated to occur at several levels of regulation. (F2:335)

(55) One might speculate that what is now understood as carotenoid functions are not to be understood as common capabilities of all carotenoids, but rather a set of quite compound specific, primary functions associated to various carotenoids. (A5:176)

(56) This suggestion is purely speculative, and there is no evidence as to the nature of this conjugate, but the high activity of NAIV raises the possibility that the molecule of interest may be IAA-valine (Fig 1). (E1:294)

The transformation of scientific knowledge from RA claim to textbook fact can therefore be represented in terms of Latour and Woolgar's taxonomy of modalities. But while the scale suggests a preliminary representation of how hedging acts on the facticity of statements, this discussion of the movement of a claim from artifact to fact has serious limitations. Principally, the methodology is descriptive not analytic. It is designed to provide sociological generalisations about the nature of scientific activities rather than a linguistic account of discourse features, and it therefore fails to offer a systematic framework for analysis. The units are not clearly defined, the means of identifying them is unspecified, and their validity is empirically unverified. The scheme appears to be largely impressionistic in that few authentic examples are given to illustrate the categories, and no procedure is suggested for categorically assigning statements based on linguistic features. The scale does not identify the forms used to signal each class of statement, nor are important semantic or pragmatic distinctions discussed. So, while the discussion warrants attention, it is difficult to employ in a practical way for the analysis of hedging in scientific writing.

In sum, a cautious tendency to remain close to experimental data is a distinguishing feature of RAs when compared to popular accounts or textbooks, demonstrating a central function of hedging to scientific communities. Hedges allow writers to weigh up the certainty of their claims and signal an attitude to them. Removing hedges changes the nature of the information, and while science is not unique in this respect, the constraints on presenting claims may be more rigorous than in other disciplines (eg Simpson, 1990). However, the studies I have discussed thus far only focus on an 'epistemic' function of hedges. The variation in their occurrence in different contexts shows that scientists not only evaluate the importance of their claims against the idea of an absolute and independent natural world, but also against the knowledge-base of their audiences. The modification of statements can therefore also reveal a

great deal about the negotiation of a relationship between writer and audience, and it is to this interpersonal dimension of hedges that I now turn.

Knowledge claims and audience design

If the problem for scientists is to persuade their readers to accept their statements as facts, and the presence of hedging reduces the factual status of statements, how can we account for the widespread use of modality in scientific discourse? The answer to this is that propositional qualification also serves a suasive function by meeting the affective expectations of a scientific audience. Hedges here function to "damp down (the) force or intensity or directness" of an utterance (Holmes, 1995:74). In this regard they have been referred to as 'softeners' (Crystal & Davy, 1975), 'downgraders' (House & Kasper, 1981), 'compromisers' (James, 1983) 'weakeners' (Brown & Levinson, 1987), and backgrounding terms (Low, 1996).

In science we have seen that grammatical modalities help express the rhetorical "weight" of a statement as part of the "organisation of persuasion through literary inscription" (Latour & Woolgar, 1979: 84). This viewpoint has been explored by Myers (1990), who has investigated the revision of claims through successive versions of articles and grant proposals in the biological sciences. Myers argues that in both genres claim definition involves a tension: claims must be significant to be published or funded, but must follow earlier work to be science. The importance accorded a claim by the scientific community requires that it be both original, and closely related to the concerns and methods of current research. Writers thus seek to gain acceptance for the most significant claims possible, while placing those claims appropriately within the 'evidential context' of the scientific literature. This tension is resolved by the addition of hedges to later drafts to modify claims.

Applications for research funding comprise the most obviously rhetorical scientific genre as they essentially involve both a presentation of a worth-while project and a self-projection as a competent researcher. However, because the standard proposal format and impersonal report style contradict such self-assertion, the most effective means of self-presentation is "understatement, toning down - not one's claims for one's research, but one's language" (Myers, 1990: 48). Writers have to be extremely cautious in how they define their rela-

tionship to the research community, and be particularly careful about claims of priority and untried theoretical approaches. New work has to be placed into an existing literature, and the increasing use of hedges in successive text revisions is an important means of accomplishing this, achieving a closer fit between a proposal and a disciplinary consensus. Myers cites the following changes between drafts as typical:

(57) courtship behaviour ... is dependent on androgens
courtship behaviour might depend on androgens

(58) the implications of this observation have not been appreciated
the implications of this observation have not been fully appreciated

(59) we proposed that .....
one interpretation would be that ....

The same considerations apply to research papers submitted for publication, for the status the scientific community assigns to a claim is a critical consideration in framing research findings. On one hand, writers try to make the highest level claims possible because they desire the recognition and credit for adding to the knowledge accepted within their field. On the other hand, such high level claims are more likely to contradict the relevant literature and challenge assumptions underlying colleagues' research programmes. The status of the claim is therefore negotiated through the process of referee review and subsequent author revisions. While the writer seeks credit for something new, reviewers try to ensure the claim is grounded in the body of scientific knowledge. As a result, earlier expressions of high-level claims are diluted in subsequent drafts (Myers, 1990, Ch 3).

Appropriate hedging therefore constitutes a central dimension in audience design and in negotiating the acceptance of claims between a writer and reader. It tempers claims by acknowledging the place of the work in the research literature, thereby strengthening the acceptability of arguments. In the same way, citations are employed to demonstrate an awareness of parallel research and indebtedness to other contributors for "claims of originality are risky and criticisms of opposing views can seldom be explicit" (Myers, 1990: 59). Claims must be both new and related to the discipline to be worth publishing or funding, achieving a balance between the profound, but hazardous,

and the correct, but trivial. This tension constitutes an essential dimension of the writer-audience relationship. Hedging in scientific discourse is not simply a means of distinguishing the factuality of claims, but is a rational interpersonal strategy, crucial to defining a relation with other researchers and their work.

Myers (1989) has discussed this interpersonal aspect of hedging through an innovative extension of Brown and Levinson's model of politeness to scientific texts. He argues that claims and denials of others' claims are potential Face Threatening Acts (FTAs) which challenge both a general scientific audience and a specialised readership of colleagues in the same field. According to this view, such impositions must be carefully modified, and many surface features often treated as stylistic conventions can be seen as politeness devices. Hedging is therefore a strategy for addressing social interactions in RAs; writers must present themselves as servants of the discipline while also asserting an individual contribution. Claims threaten the community's Positive Face as they solicit acceptance by a scientific audience while denying or superseding the claims of others. They also threaten the Negative Face of other researchers by placing restrictions on how their own research can proceed.

Opportunities for positive politeness, which demonstrates social closeness by reference to the reader's wants, are restricted in scientific writing. However, the strategic use of pronouns, modifiers assuming common ground and indications of shared knowledge all express membership of the discourse community, while citation choices, as 'gifts of credit', attend to the face needs of other researchers. Scientific hedging falls within the compass of Negative Politeness which is the more formal, socially distancing component of politeness, indicating respect and minimising the imposition that an FTA effects. So, for Myers:

> Hedging is a politeness strategy when it marks a claim, or any other statement, as provisional, pending acceptance in the literature, acceptance by the community - in other words, acceptance by the readers.
>
> (Myers, 1989: 12)

Myers argues that scientific discourse assumes a large social distance between participants, and that while relations between individual scientists require little deference, a researcher must always defer to the discourse community as a whole. Many conventional features of scientific discourse then, such as impersonal constructions, hedging and the assertion of general rules, can be interpreted as negative politeness devices.

It is certainly the case that modified claims are so common that an unqualified assertion is unlikely to be intended as an addition to scientific knowledge. Even Watson and Crick, who informally announced to anyone who would listen that they had discovered the secret of life (Watson, 1968), were coy in presenting their claim to their peers in their celebrated 1953 paper:

(60) We wish to suggest a structure for the salt of deoxyriose nucleic acid (DNA). This structure has novel features which are of considerable biological interest. (Watson & Crick, 1953: 737)

Clearly Watson and Crick were well aware that the world would acknowledge the enormous consequences of their breakthrough. But while this degree of understatement may be rhetorically unusual, writers can normally depend on the significance of their claims being largely understood. Bald assertions are rare, and examples throughout this thesis demonstrate the variety and extent of the mitigation that takes place.

One form of hedging relevant to interpersonal issues is the use of personal attribution in criticising other researchers' views. The possessive and personal pronouns in the examples below hedge criticisms by allowing the reader to choose the more persuasive explanation which follows. Example 63 goes further by explaining inconsistent findings:

(61) There has been some disagreement concerning whether illumination increases the V_{max} of PEPc (eg refs 1, 2, 4, 11). In our hands, there was no significant change in V_{max} on illumination. (C1:191)

(62) Quick et al (27) also noted that there was no consistent effect of reduced Rubisco content on the stomatal conductance of their antisense plants but observed that conductance was insensitive to CO_2 (30). Our results do not support the latter observation, because conductance decreased with increasing p_1 in both control and anti-SSu plants (fig 5). (E4:301)

(63) We do not know the reason for the discrepancy between our results and those of Ngernprasiritsiri et al [16, 23], but it might reflect genetic differences in the cultivars employed. (F3:17)

Public criticism of other work is rare however, and any controversy that does arise is generally conducted civilly to promote the free flow of information. Hagstrom (1965), for example, quotes one of his physical science informants as saying "Work rarely gets challenged: it is often superseded or passed over, but not challenged." This is confirmed by my data and by Barton (1995) and by Bloch & Chi, (1995: 257) who found very few critical citations in the physical sciences. Similarly, Pickard's (1993a) study shows refutation accounting for only 5% of 315 citations in an applied linguistics RA corpus, with all rebuttals hedged.

The survival of claims depends on citation, and a failure to be cited can have severe consequences. Rival claims are usually ignored, and 'Bald-on-record' criticism is so rare that I had to search outside my corpus to find an example:

> (64) We do therefore not agree with and our data do not support the contention that "photoinhibition is known to induce degradation and resynthesis of D1 protein" (Virgin et al, 1988). To our knowledge no data published so far have unambiguously demonstrated a direct correlation between D1 turnover and photoinhibition. (Cleland et al, 1990)

A more common way of dealing with such delicate interactions as denials of others' claims is to hedge by the use of impersonalisation. This may involve avoiding attributing a criticised view to any single researcher (65-6), or impersonalising one's own position, either by use of an impersonal subject (67) or by appealing to those with whom one agrees (68):

> (65) However, beyond a general realisation that these solutes act as stabilisers of PS II preparations and as cryoprotectants during their storage, other implications of their presence are rarely considered. (A4:29)

> (66) This procedure only identifies methylated nucleotides located within the recognition sequences of the sensitive enzymes. In spite of its shortcomings, the method has been widely employed to evidence this type of modification in a number of genomes including plastid and nuclear ones [22-24]. (F3:16)

(67) Matoo et al (10) have suggested that the rapid turnover may in some way be related to a control mechanism for PS II function. Alternatively, Arntzen et al (1) hypothesized that the unusually high rate of turnover of the Q_B protein in the light could be a natural consequence of its in vivo function as the secondary acceptor of PS II. (...) This manuscript will present data that are interpreted to support the latter idea. (Ohad, et al, 1984)

(68) The fact that ... would therefore appear to be in significant conflict with the theory that suggests grana increase light absorption under low intensity conditions. Additional support for the shielding function comes from in vitro experiments by Jennings & Zucchelli (1985) which show that stacked thylakoids absorb ... (A1:269)

Thus, while refutation is often thought to be central to argumentation, criticism is unusual.

While the devices discussed here have clear interpersonal significance, the applicability of Brown and Levinson's model to scientific discourse is less - certain. While there are many strengths to this approach, the theory neglects the fact that writers may attempt to enhance their face and denigrate that of the other (Wood & Kroger, 1994). More centrally, it assumes that principles of language use are governed by universal needs and, by attributing regularities of scientific style to the avoidance of imposition, Myers is assuming the universality of a scientific culture which is expressed in terms of conflict avoidance. Scientific discourse is thus seen as a fundamentally dangerous and antagonistic endeavour. As Schmidt observes, Brown and Levinson's theory:

> represents an overly pessimistic, rather paranoid view of human social interaction in language, viewing politeness as a response to threats of face rather than as an essentially positive phenomenon.
>
> (Schmidt, 1980: 104)

Moreover, there has been a growing reaction against this kind of universalism, and a number of recent studies have shown that the approach has a strong anglocentric bias, being inappropriate for speakers of Chinese (Gu, 1990), Polish (Wierzbicka, 1985), Israeli (Katriel, 1986) and Japanese (Matsumoto, 1988; Ide, 1989). (See Turner, 1996, for a recent review of Brown and Levinson).

Myers' belief that hedging in science simply reflects the politeness principles of an (apparently antagonistic) context of everyday conversation ignores crucial differences in the social norms operating in the two forms of discourse.

It fails to give sufficient weight to institutional factors which influence scientific writing and does not take full account of the constitutive nature of consensus in confirming knowledge claims. As Becher argues:

> Communication is the force that binds together the sociological and the epistemological, giving shape and substance to the links between knowledge forms and knowledge communities. (Becher,1989: 76)

Communication is critical not only to the promotion of knowledge, but also to the establishment of reputation in science. It is the key cognitive and social impetus in the scientific world and as such reflects the central practices of the discipline. The integral interdependence between the communication system, the reward system and the validation of knowledge means that the function of mitigation in speech act modification is likely to play a more rhetorical role than assumed in Myers' theory.

Treating RA hedging as politeness over-emphasises the instrumental aspects of language use at the expense of the normative, under-estimating the importance of the scientific peer group in maintaining standards, judging merit and evaluating reputations. Simply, the theory neglects the exercise of power and conformity in the discourse community:

> Power, in the form of control over tenure, research funds, recommendations for jobs, research leave and so on, tends to cluster at the top. Those at the top are if not the architects, then the hard-working builders of the dominant paradigm. An open challenge on strictly scientific grounds to the theory on which the life work of one's teacher/ chairman/ patron is based is viewed not only as unscientific. It can also be seen as disloyal, reckless, and probably dangerous to career prospects.
>
> (Boissevian, 1974)

Rarely is power exercised so bluntly however, and I am not suggesting that resources are used to protect vested theoretical interests. However, underlying controls operate to encourage conformity and regulate the linguistic behaviour of publishing academics. Both the generation of knowledge and the professional recognition given to individual scientists rest on publication. Thus the ways writers express their views is critical to their success.

It is probable that hedging plays a more rhetorical and normative role in the accomplishment of consensus in scientific persuasion than it does in the achievement of understanding in social encounters. So while the choice of lin-

guistic form in science is undoubtedly partly determined by the writer's appreciation of a responsibility to the reader, a failure to observe 'politeness' will not merely prevent individuals securing goals, but will incur social sanctions. This is a position which a face-caring model overlooks. We therefore have to reject the politeness view as an adequate explanation for the use of hedging in science and conclude that discourse community norms are likely to play a larger part than credited by the Myers/ Brown and Levinson model. To be 'polite' is to abide by the rules of a relationship established by the scientific discourse community. It involves adherence to an empirical viewpoint and acting as if one trusted all other scientists to do likewise; without such trust, the edifice of scientific knowledge production would collapse.

Contributing to a scientific debate involves the writer entering into an interactional contract with specific rights and obligations, among which are limits on self-assurance and norms concerning the deference due to the views of other researchers. Adherence to such an interactional contract may exert a stronger influence on scientists than considerations of face, and may prove to be a more insightful means of analysing the interpersonal use of hedges in scientific prose. This issue is pursued further in chapter 6.

Knowledge claims and writer commitment

A further motivation for hedging is suggested in the work of Prince, Frader and Bosk (1982) and Rounds (1982) who both include the notion of writer commitment in their frameworks. As briefly noted above when discussing the maxim of quality, hedging here is seen as a withdrawal, a prudent insurance against overstating an assertion, limiting the damage that might result from the disproving of bald propositions. Not only is this a useful strategy in economic forecasting (Pindi & Bloor, 1987), but is crucial in academic research writing. Bloor and Bloor (1993) suggest that knowledge claims in economics articles vary in the extent to which they are "field central", involving modeling and interpretations crucial to the discipline. It is these claims, rather than more substantive observations, that attract the greatest hedging, as it is upon these that the authors' reputations rest. Hedging here then represents a desire of writers to shield themselves from challenges, indicating once again the influence of the discourse community on language use.

Prince et al's analysis of spoken physician-physician discourse led them to distinguish two different senses of Lakoff's definition of a hedge as "making things fuzzy":

> One class of hedges introduces, or is responsible for, fuzziness within the propositional content proper, while the other class of hedges correlates with fuzziness in the relationship between the propositional content and the speaker, that is, in the speaker's commitment to the truth of the proposition conveyed. (Prince, Frader & Bosk, 1982: 85)

The first type they call 'approximators' and the second 'shields'. These types relate to hedges within the proposition, and between the proposition and the speaker respectively. Broadly, the former concern exactitude, and the latter speaker commitment.

Rounds (1982: 17) argues that the use of approximators serve "to weaken the relationship between the subject and object or complement", but such hedges rather seem to modify the extent of the process or state denoted by the verb or noun, rather than loosen propositional ties:

(69) This appearance of kinase activity correlates quite well with the time-course of the changes in the kinetic properties of PEPc that is observed on ... (C1:193)

(70) ... in the presence of high concentrations of betane, the dissociating effect of NaCl could be almost entirely suppressed, especially with respect to... (A4:31)

Rounds also appears mistaken in claiming that approximators are most often realised by non-existential verbs, modals, adverbs and adjectives of quantity, as these seem to behave in very different ways to those in (69 & 70) above:

(71) From these calculations it appears that most of the decrease ... (A3:4)

(72) ...the phosphorylation of EF-2 may play a role in the regulation of protein biosynthesis in animal cells. (B3:219)

(73) No Pre-PS has been detected, presumably because it has not yet been sought. (A1:270)

All these devices have the entire proposition as their scope, rather than the relationship between elements and appear to qualify the writer's full commitment to the whole claim.

In other words, these hedges specify the extent to which assertions may be said to be true by restricting the range of the claim as regards time (74), quality (75) or generalisability (76-7):

(74) ...and this often coincides with the development of senescent cells in the nodule. (D3:173)

(75) ..it can be seen that the violaxanthin content is around 18% of total carotenoid ... (A5:173)

(76) To date and as far as we know, aminoacyl-tRNA synthetase with precursor tRNA.. (B5:7)

(77) Viewed in this way the concept of lateral heterogeneity becomes obsolete ... (A1:268)

In a Speech Act framework, which makes a sharp distinction between truth-conditional semantics and non truth-conditional pragmatics (eg Gazdar, 1979; Karttunen & Peters, 1979), approximators fit into a semantic perspective. This is because they affect the truth conditions of associated propositions and, following Lakoff (1972) indicate a markedness, or non-prototypicalness, with respect to class membership. Consider, for example, the following statements based on Lakoff:

(78) a. A robin is a bird (TRUE)
b. A chicken is a bird (TRUE?)
c. A bat is a bird (FALSE)
d. A cow is a bird (FALSE)

(79) a. A robin is virtually a bird (FALSE)
b. In a manner of speaking a chicken is a bird (FALSE?)
c. A bat is essentially a bird (TRUE?)
d. A cow is essentially a bird (FALSE)

(80) a. A robin is a bird par excellence
b. Strictly speaking a bat is not a bird
c. Loosely speaking a bat is a bird
d. A bat is a bird in that it has wings and can fly

The addition of an approximator to a proposition may result in a statement which differs from the original in terms of its truth-conditions. Those in (78) are straightforward categorical assertions which can be confirmed in the real world or by reference to linguistic categories. Those in (79) include hedges from Lakoff's list and show that while the propositions remain unchanged, hedges can reveal degrees of category membership. Here values have been reversed, *virtually* and *in a manner of speaking* entail the presupposition that chickens are not really birds, which is false, and *essentially* picks out secondary criteria (wings, ability to fly, etc) for category membership. The examples in (80) demonstrate that hedges can help distinguish between central and peripheral category members (a), or between degrees of non-membership (b). In addition, they can also expose the flexibility of boundaries (c) and redefine categories by re-weighting given attributes (d).

Approximators can therefore help to show how observations vary from the absolute, or prototypical, categories which natural languages impose on the world, allowing greater precision in describing reality. So while writers and readers may have to rely on a relatively fixed lexicon of relevant terms with which to classify and describe real-world objects and relationships, hedges can serve to match perceived reality with an appropriate term more precisely. I will return to this issue in chapter 6.

While approximators refer to descriptive adequacy and the precision of terms, the second type of hedge in this framework, shields, imply that "the speaker is less than fully committed, or committed in some marked way to the truth of the proposition" (R. Lakoff, 1972: 85). Unlike Approximators, the use of Shields leaves the truth conditions of propositions unchanged and simply reduces the assurance with which one can accept the writer's certainty. In (81) for example, the author merely indicates that she believes there is a causal connection between the items and in (82) the claim is made dependent on data which is open to several interpretations:

(81)... I believe that the overall orientation of grana stacks and stroma lamellae is dependent on the direction of the light. (A1:271)

(82) Taken together, these data suggest that the protein synthesis in plants is not regulated by phosphorylation of EF-2. Nevertheless, it cannot be excluded that this mechanism is realised only at certain phases of plant development (for instance during gametogenesis) and such a possibility requires further study. (B3:222)

Prince et al identify two types of Shield which they call plausibility shields and attribution shields. The first simply indicates a lack of speaker certainty, employing forms which indicate a degree of tentativeness and serve to distance the writer from the proposition:

(83) It appears that the transformation of active reaction centres into quenching centres is partly reversible if..... (A3:9)

(84) This would probably suggest that the C-terminal processing is not achieved in the nodules, but this needs to be demonstrated. (D2:265)

(85) Presumably some 'signal' generated in the choloroplasts initiates a sequence of events that lead to phosphorylation or dephosphorylation of PEPc in the cytosol. (C1:190)

(86) We speculate that the inhibtor interacts specifically with a protease, the origin of which is still to be determined... (D2:267)

There are cases where this lack of commitment seems to virtually remove the writer from the responsibility for the assertion altogether so that it may become difficult to precisely identify the actual work for which responsibility is being claimed:

(87) We suggest here that the endogenous ligand for the NPA receptor is a conjugate of IAA, specific to the transport of the hormone. This suggestion is purely speculative, and there is no evidence as to the nature of this conjugate, but the high activity of NAIV raises the possibility that the molecule of interest may be IAA-valine. (E1:294)

Assertions marked by plausibility shields imply that the writer is stating a belief based on plausible reasoning. Without such hedges the reader will as-

sume that the writer has knowledge, either based on logical deduction or empirical data, that the proposition is true. Typically, plausibility shields are realised by lexical verbs and phrases which indicate uncertainty in some way. Prince et al mention, but do not pursue, the possibility that plausibility shields may convey different levels of uncertainty. This observation is also made by Lyons (1977:800) but there appear to have been no studies which demonstrate that readers distinguish significantly between different degrees of plausibility reasoning.

Attribution shields have a different function, implying that the "belief in the truth of the proposition conveyed is to be attributed to another" (ibid: 91). In scientific RAs these most often occur in Introduction and Method sections where the writer reports background studies which have not been personally witnessed. Hoey (1983: 128-9; 175) suggests that a non factive presentation of another writer's viewpoint is a "hypothetical" term which strongly predicts a contrastive "real" statement by the author. This is not always the case however, and Prince's position is that such citations are appeals to another authority, and while this may imply a reservation, it does not anticipate a refutation.

The attributee may not always be specified, and as layers of interpretation are added (eg Pinch, 1985), it may change to an appeal to a consensus, seeking to merge the hedge with a logical deduction. Such "summary citations" (Jacoby, 1987: 55) are rare in science:

> (88) It is also clearly recognised that the structure and function of the thylakoid membrane system must undergo dynamic changes to accommodate varying environmental conditions and maintain high rates of photosynthesis and quantum yields. (A1:266)

These cases clearly invoke agreement and are therefore available for rhetorical manipulation, but any form of citation can provide persuasive support by placing current research in a wider academic arena and demonstrating its force as 'valid science'. More commonly the source of the attribution is explicitly mentioned, as academic authors are required to specify previous relevant work and cite those who published it. My corpus of 26 RAs, for example, has an average of 30 attributions to one or more sources per paper, a figure identical to that found in a study of 11 applied linguistics articles (Pickard, 1993b). Non-integral citation forms (Swales, 1990: 148) predominate in my

corpus with the cited author in parentheses, although numerical footnoting is used in four of the six journals:

(89) ... presumably the result of differences in the position of integration of the chimeric genes (Dean et al, 1988). (D1:634)

(90) The nitrogen-fixing nonlegume Parasponia andersonii has a single haemoglobin gene (Landsmann et al, 1986) that is ... (D1:633)

(91) ...phosphorylation of ACC synthase has recently been observed (Boller, T. personal communication). (B5:107)

While attribution shields "implicate that the speaker is speaking from knowledge or beliefs acquired via hearsay" (ibid: 91), the writer's own level of commitment can often be inferred from the context, particularly by appeal to Gricean Maxims. Put simply, an unhedged statement implies the writer has adequate evidence for believing an assertion to be true (as required by the Maxim of Quality), whereas the introduction of another person's commitment to the proposition is not strictly relevant. By citing an attributee, the Maxim of Quantity is violated by providing more information than is required for the understanding of the message. The additional linguistic material therefore generates the implicature that the cited source is able to provide adequate evidence for the proposition, whereas the writer cannot. Thus while the writer might demonstrate sufficient commitment to meet Grice's admonishment to make one's contribution as informative as required, he or she can indicate that this is below an appropriate level by invoking external support.

The situation is complicated however by the choice of reporting verb which allows a stance to be adopted, demonstrating exactly how strongly the writer wishes to be aligned with the cited work. Prince et al do not discuss this issue, but while some attributions require a wider discourse context to disambiguate the writer's stance towards cited information (92), it is clear that (93) indicates a more positive commitment than (94):

(92) In a theoretical study, Sorokin (1985) investigated the effects ... (A3:7)

(93) Jofuku and Goldberg (1989) showed the expression of several Kunitz protease inhibitors in soybean tissue. (D2:266)

(94) Henninger et al (3) speculated that tuber size is correlated with extent of necrosis in field-grown potato. (E3:199)

Reporting (or lexical) verbs are therefore an important means of expressing modality in scientific discourse because they enable fine distinctions of precision to be made in expressing commitment to the truth of the statements they report. This issue is discussed further in chapter 5.

Rounds (1982) adopts Prince et al's schema in her unpublished study of abstracts from the Behavioural and Brain Sciences Journal and suggests a third type of shield, in addition to plausibility and attribution shields, related to the impersonalisation of propositional content. While Rounds does not always adequately distinguish hedges acting on content from those indicating the writer's relationship to the proposition, she accepts the view that shields serve to distance the writer from the proposition. So commitment can be hedged by either by indicating tentativeness (95), stating the source or status of information (96 & 97) or impersonalising (98 & 99).

(95) One might speculate that what is now understood as carotenoid functions are not to be understood as common capabilities of all cartenoids,.. (A5:176)

(96) According to these authors' concept, the determining factor for the response of the protein was whether the thermodynamics of the ... (A4:34)

(97) I believe that the major organisational principle of the thylakoids is that of continuous unstacking and restacking of sections of ... (A1:267)

(98) The model implies that the function of grana is to shield .... (A1:268)

(99) The inhibitor is thought to be one of the inducible defensive proteins... (F2:349)

A similar classification has been proposed by Hunston (1994a) who distinguishes modal meanings according to the source of the judgements. In science RAs this source may be explicitly the author (95, 97), implicitly the author (99), an intertextual source (96), or the evidence itself (98).

The limited occurrence of personal pronoun agents in Rounds corpus is confirmed by my data and a computer frequency analysis of 817,000 words of academic discourse. There is little doubt that impersonal phrases are typical of academic writing:

(100) The present work has revealed ... (A2:99)
...together these data suggest that ..... (B3:222)
Table II shows that..... (C1:194)
The model implies that the function... (A1:268)
This study has shown the existence of .. (A5:177)
These results demonstrate that .... (D3:176)
The proposed model reconciles a number... (A1:274)

The use of the passive voice without an agent is a further impersonalising technique, allowing writers to withhold full commitment to their propositions:

(101) No detectable signals could be observed .... (F3:17)
On the basis of these studies a model has been proposed.. (C2:268)
It should be mentioned in this respect, that ... (B4:231)
It might be expected, however, that the inhibitory... (D5:46)

The use of the passive voice to display objectivity in scientific discourse is a textbook staple, but the claim here is that it is a means of avoiding commitment. The passive enables the writer to achieve some distance from a proposition, thus effecting what Rounds (1982: 21) refers to as "minimising the dimension of personal agitation". Interestingly, Lachowicz (1981) has argued the opposite, that the authoritative influence of the passive allows the scientific writer to take greater responsibility for statements. Moreover, it could also be claimed that such forms are less to protect the writer from criticism, than to disguise the weaknesses of inductive procedures, concealing interpretive practices behind an empiricist style which appears to report the workings of nature directly. Clearly the role of the passive is a complex one and claims for particular rhetorical functions must be supported with contextual evidence in specific cases.

The need to avoid full commitment is clearly important in a discourse where new information is regularly being added to knowledge of a topic. But

while the distinction between Approximators and Shields seems to be a useful one, the value of "diffusers" as a third category of hedging is less certain. Rounds argues that diffusers are "tactics designed to short-cut arguments or to sidestep an issue", but this seems to be less a separate class than a more specific aspect of the other categories; the removal of commitment itself being a form of diffusion. Nor are the realisation strategies she discusses convincing, as neither avoidance, the use of jargon nor footnoting seem rhetorically effective means of weakening opposition among a scientific audience. This may explain her failure to cite examples from her own data and their absence from mine.

Rounds argues that the absence of what the audience expects can be seen as a hedge, as it deliberately flouts the Maxim of Quantity by failing to make a contribution as informative as required. The implicatures generated by such a violation in an RA, however, are likely to reflect negatively on the writer, while the specialist knowledge required to identify such a hedge, places such an omission beyond applied linguistics. Similarly, the use of jargon as a hedging strategy seems irrelevant to an understanding of writing in the sciences. There seems little obvious benefit in resorting to obfuscation and confusion when trying to persuade a specialist readership and, indeed, it is difficult to imagine either a circumstance or terminology which might accomplish it. The role of footnotes focuses on more central issues, but the inclusion of this feature as a hedging *device* confuses function with location, as the kinds of diffusion they may contain can also occur in the text proper.

The inclusion of footnotes as a class of hedge is therefore curious and the notion of 'Diffusers' itself open to question. Not only is it unjustified as a distinctive category, but it creates ambiguity in the classificatory system itself. Unlike Shields and Approximators, which are defined in terms of their grammatical and lexical realisations, Diffusers have no linguistic expression and involve either obfuscation, non-occurrence or discourse location. The generalisability of Rounds analysis is also limited by a corpus of 14 abstracts, a discourse usually characterised as a distinctive genre (Swales, 1990). In particular, its language form eliminates redundancy (eg Graetz, 1985: 125) and places constraints on space. These aspects are likely to restrict the kind and extent of hedging used in the abstract and might therefore distinguish it from the RA in how hedging is used. The study is therefore unable to provide a comprehensive categorisation of hedges as they occur in journal articles.

Overall however, Price et al and Rounds make some important distinctions concerning academic hedging and their division between hedges which affect propositional relationships and those which influence writer commitment appears to be a particularly fruitful one. How writers mark their commitment to their propositions and indicate who is responsible for what claims is at the heart of skilful scientific writing. The means by which this is achieved however remain largely unexplored, and there has been little attempt to follow up these initial studies by examining particular genres.

Conclusion

The use of hedges to express the writer's perspective towards his or her propositions and audience is critical in science. They represent an important means by which authors appear in their texts to adjust claims and anticipate audience response, allowing writers to make distinctions concerning the certainty they give to their propositions, such that a statement without qualification is probably not a statement of new knowledge. In essence, hedges are rhetorical means for projecting due caution, modesty and humility when making statements, and their removal is a major linguistic means of conferring greater certainty on propositions. These strategies are therefore central to the whole enterprise of science, a fact recognised when Boyle took care to separate the interpretation of facts from the facts themselves at the dawn of scientific writing 300 years ago.

Given this significance, the absence of research interest in the distribution and meaning of hedging in particular genres is surprising. No further studies have sought to provide an account of the contextual and interactional forces which shape a writer's stance or the factors which cause the variation in the use of hedges. Writers approaching the subject from different directions however have recognised at least two factors which motivate the modification of speech act expression. Roughly, these concern 'epistemic' meaning, which is proposition or speaker oriented, and 'affective' meaning, which is addressee oriented. In other words, hedging is both an expression of the writer's assessment of referential information and a form of social interplay with readers. While these categories may not be exhaustive, any adequate description must consider this functional duality.

In sum, an explanatory model of hedging has to describe not only linguistic exponents, but also the epistemic and interpersonal principles behind

them. It therefore needs to include consideration of both the writer's perception of the truth value of statements and the socially and contextually defined discourse relationship of the participants. Such an examination of the resources available to scientific writers to convey their stance to both their statements and addressees will be presented in the following chapters.

4. Theoretical and methodological considerations

The approaches and findings discussed in the previous chapters have determined, in general terms, the significance and purpose of hedging in academic genres. In this chapter I will build on this work and provide a bridge to the analyses which follow, establishing some preliminaries to the surface-level description and the functional-rhetorical analysis in the following chapters. Here the research design, theoretical framework and methodology of the study will be considered. I will begin by explicitly restating my assumptions regarding the constructedness of scientific knowledge and the importance of a shared disciplinary context to my analytical framework. Next I will explore the role of hedging in science writing and outline the functional categories of the model. The last two sections will give a description of the corpus and the research methodology employed. Short text samples will be employed here for exposition, but longer extracts may be consulted in chapter 7.

Scientific claims in the construction of knowledge

A fundamental assumption of my understanding of hedging is that academic communication is a social activity which functions in disciplinary cultures to facilitate the production of knowledge. We have seen that the epistemological foundation of scientific knowledge has been much debated, and this section examines in more detail the central concepts of a constructionist view.

In broad strokes, on one hand there is the traditional positivist conception that the work of science is the discovery of truth about the natural world. This position assumes that the material world exists independently of the knowing subject, and that truth is found through formulation of theories tested by experimental method. Claims are either true or false and what makes them so is their

them. It therefore needs to include consideration of both the writer's perception of the truth value of statements and the socially and contextually defined discourse relationship of the participants. Such an examination of the resources available to scientific writers to convey their stance to both their statements and addressees will be presented in the following chapters.

4. Theoretical and methodological considerations

The approaches and findings discussed in the previous chapters have determined, in general terms, the significance and purpose of hedging in academic genres. In this chapter I will build on this work and provide a bridge to the analyses which follow, establishing some preliminaries to the surface-level description and the functional-rhetorical analysis in the following chapters. Here the research design, theoretical framework and methodology of the study will be considered. I will begin by explicitly restating my assumptions regarding the constructedness of scientific knowledge and the importance of a shared disciplinary context to my analytical framework. Next I will explore the role of hedging in science writing and outline the functional categories of the model. The last two sections will give a description of the corpus and the research methodology employed. Short text samples will be employed here for exposition, but longer extracts may be consulted in chapter 7.

Scientific claims in the construction of knowledge

A fundamental assumption of my understanding of hedging is that academic communication is a social activity which functions in disciplinary cultures to facilitate the production of knowledge. We have seen that the epistemological foundation of scientific knowledge has been much debated, and this section examines in more detail the central concepts of a constructionist view.

In broad strokes, on one hand there is the traditional positivist conception that the work of science is the discovery of truth about the natural world. This position assumes that the material world exists independently of the knowing subject, and that truth is found through formulation of theories tested by experimental method. Claims are either true or false and what makes them so is their

approximation to features of external reality. Theories can be subjected to the proof of falsification which means that science is beyond the influence of contingent factors, such as sense impressions or language, and transcends the opinions of individuals. 'Normal science' is "predicated on the assumption that the scientific community knows what the world is like" (Kuhn, 1970: 5); it is firmly based on past achievements which provide a foundation for the determination of further problems and their solutions. In sum, knowledge is believed to be built on an objective, universal foundation which assures its certainty. Scientists aim to provide a literally true description of the world, and acceptance of a theory involves the belief that it is *true*, or at least not yet falsified.

On the other hand, a social-constructionist position is insistent on the relevance and importance of *context* in the creation of knowledge, and argues that scientific results are an achievement of interpretation rather than a description with a universal foundation. This perspective assumes that reality, knowledge and facts are language constructs generated for the time being by scientific communities and used to maintain community coherence. We make judgements about the world in terms of our power to represent it, and this 'representation' actually helps construct it. Truth belongs to this construction. This position originates with Kuhn, and behind him Wittgenstein and Dewey (Bruffee, 1986), and essentially sees scientific knowledge as "intrinsically the common property of the group or else nothing at all" (Kuhn, 1970, 210).

Constructionists argue that there can be no context-independent criteria of rational justification that establish some hypotheses as more credible than others. Science has not succeeded in defining universal grounds for the validation of theories, and the fact there are any number of possible interpretations of a set of observations means it cannot therefore be a coherent body of objective truth about the world. As a result, science can only provide theories which are empirically adequate, and acceptance of a theory merely involves the belief that it is empirically adequate.

This outline polarises the two positions to emphasise important distinctions between foundational and non-foundational assumptions of cognition and knowledge, but clearly belief in a material universe does not preclude theorising and speculation. Moreover, the dispute might be considered peripheral to my topic. After all, scientists are virtually obliged to believe in the existence of an external world and to the discovery of order in its nature. Its importance, however, lies in the fact that a constructionist perspective locates language at the centre of scientific knowledge. Rather than seeing language as a mere con-

duit by which ideas and theories are channeled from one individual mind to another, it links language, knowledge, and a community of peers into an inseparable matrix.

But constructionism is not relativism. The acceptance of theories is located in the discourse community and the constraints on justifiable belief are socially constructed among individuals. The model of science is of "independent creativity disciplined by accountability to shared experience" (Richards, 1987: 200). 'Objectivity' thus becomes "consensual intersubjectivity" (Ziman, 1984:107), as scientific methods and findings are coordinated and approved through public appraisal and peer review. The communication system is therefore the fundamental structural component of the scientific community, and an understanding of knowledge involves an understanding of how it is employed in the social justification of belief .

While I do not wish to identify knowledge *entirely* with the language in which it is formulated (eg Rorty, 1979), the critical role of the group in socially authorising knowledge is persuasive. It is the collective agreement of scientists which establishes that a statement has been adequately tested, and it is the verdict of a specially trained audience which is authorised to establish a claim as knowledge (Overington, 1977; Longino, 1990). Research is therefore less a search for truth than a quest for agreement (eg Polyanyi, 1964: 13), as claims must be critically reassessed by other scientists before they can be regarded as "well established". In other words, because scientific writing seeks to produce specific responses in an active audience, its effectiveness depends on the success of writers in analysing and accommodating their readers' needs and what they are likely to find interesting, intelligible and plausible. As Overington (1977: 144) observes:

> science as a knowledge-producing activity may be treated rhetorically <u>because</u> the construction of scientific knowledge involves argumentation before an audience.

Such a concept of audience thus focuses on the external circumstances which define the rhetorical context and require the text to have certain characteristics in response. Audience refers to a writer's sense of awareness of appropriate discourse strategies which shape a text as "a complex set of conventions, estimations, implied responses and attitudes" (Park, 1982: 251). Discipline, audience and context therefore significantly influence the language required for argument. This is not however only because scientific discourse imposes con-

straints on linguistic format, schematic structure and style, but because linguistic choices simply assist readers to produce meaning:

> language is only one of the ingredients involved in the production of meaning. The contribution of the knowledge base of the addressee (and the speaker) is crucial. (Lee, 1992: 191)

While the probability of shared universes of knowledge diminishes when one writes for an audience, the notion of discourse community provides a means of accounting for the possibility of shared presuppositions. It also allows us to account for the rhetorical strategies which can be employed to invoke them.

Scientific reality and knowledge are embodied in the texts produced for a particular discourse community and can best be revealed by analysis of those communities and of the genres they create:

> social construction assumes that the matrix of thought is not the individual self but some community of knowledgeable peers and the vernacular knowledge of that community. That is, social construction understands knowledge and the authority of knowledge as community generated, community maintaining symbolic artifacts.
>
> (Bruffee, 1986: 777)

Writers must organise data and observations into meaningful patterns for readers, so part of an academic's competence involves familiarity with the conventional discursive practices of a particular disciplinary community. The notion of *discourse community* has been much debated in recent years (Harris, 1989; Lyon, 1992; Swales, 1990, 1993 & 1996), and there is now a tendency to see discourse communities as rhetorical constructs which "persist by instantiation and engagement, rather than existing through membership and collectivity" (Swales, 1993:696). However, their influence on the rhetorical practices of writers is no less important than if they are viewed as having more material substance.

Moreover, the fact of scientific specialisation and the participation of individuals in multiple communities also makes it necessary to distinguish the relevance of particular communities. Varied interests, allegiances and relationships mean that writers may be influenced by a *significant* audience, smaller than the disciplinary community, from whom recognition is sought and from where reputations are gained. This may be an invisible college of perhaps a few hundred researchers engaged in the same specialist field. Thus Myers (1989: 3) makes a useful distinction between the esoteric audience involved in the ongo-

ing research problem, and the exoteric audience of a wider nature. The former group shares certain practices, assumptions and conventions of language use and interpretation which are, at least partly, constitutive of the group's knowledge.

The role of human judgement in the interpretation of data, widely accepted in the philosophy of science (eg Faust, 1984; Van Frasen, 1980; Kuhn, 1970 & 1977; Overington, 1977; Kourany, 1987), implies that the study of scientific statements is central to understanding the processes involved in the creation of knowledge. Knowledge is influenced by the basic elements of the communication process: writer, audience and language (eg Berlin, 1982; Johns, 1990), and in expressing an interpretation a writer must employ linguistic and rhetorical means to persuade an audience to accept it. This need for consensus provides the foundation for the expression of commitment and hedging in scientific writing, because statements must reflect the certainty the writer wishes to invest in them given the conventional means of establishing the truth of claims. In presenting claims, writers recognise that some ideas have been previously confirmed by the discourse community as constituting truth about the world. They can therefore assume that these statements form part of the set of background beliefs for the interpretation of evidence shared with the reader, and that they can be relied on when presenting new hypotheses. Statements which employ such evidential support of hypotheses will therefore have a *factive* character. Every other statement, by which the writer asserts the propositional content to be true as far as he or she knows, and for which responsibility is assumed, is a hedged or *non-factive* statement. This division into factive and non-factive assertions can be presented in terms of types of knowledge in scientific writing. Here I propose a provisional 5 scale division of knowledge:

1. What is known, or accepted, to be true
2. What is known, or accepted, to be possibly true
3. What is known, or accepted, to be not necessarily true
4. What is unknown
5. What is known to be untrue

Groups 1 and 5 contain understandings to which the scientific community is committed in its practices, and accepts as "facts". They also include statements which might be called *quasi-factive* and which Lyons refers to as categorical assertions (1977: 797). Such assertions, of course, represent the attitude

of the writer to the proposition, and can certainly be rejected by the reader, but they also carry a matter-of-fact presupposition of certainty concerning their validity. The writer seems to take the truth of a proposition for granted and therefore conveys the impression that he or she wants to communicate that he or she believes the proposition. The act of relying on the beliefs assumed to be shared with one's interlocutors has been called a pragmatic presupposition (Stalnaker, 1991; van der Auwera, 1979).

In these examples the writer appears sure of his or her claims and does not expect to be challenged:

> (1) In higher plants, the first events of photosynthesis, the light dependent reactions, take place in the thylakoid membrane. (A5:171)

> (2) Nitrogen fixation in legumes is performed by endosymbiotic bacteria of the genus Rhizobium or Bradyrhizobrium. (D2:259)

This relative certainty stems from research occurring within the intellectual limits established by the discourse community and is characterised by a study of issues about which there is the possibility of widespread agreement (i.e. 'normal science'). The recognition of knowledge, of a relationship between a state of affairs said to be evidence in support of a particular hypothesis and the hypothesis itself, is not self-evident, but depends on agreement for whatever epistemic status it achieves. So in expressing a categorical assertion, the writer draws on previously agreed understandings confirmed by the research community. Such assurance is rare when making new claims however, and most examples occur where authors are providing background in Introduction sections.

In expressing a proposition categorically, the writer may seek to draw on two sources of agreement. Firstly, reference may be made to previously published arguments which have been transformed by consensus into knowledge:

> (3) Light-harvesting in photosynthetic systems is carried out by well-defined antenna pigment-protein complexes which absorb light and transfer the energy with high efficiency to the reaction center (Codgell and Thornber, 1980; Thornber et al, 1983; Codgell, 1986). (A2:93)

> (4) Some time ago, Biswas and Gorini [4] showed that streptomycin specifically binds to 16S rRNA in vitro. (F1:499)

(5) Inhibitor I (Monomer M_r 8300) is one of two serine proteinase inhibitors that accumulate in leaves of tomato and potato plants in response to wounding by predator attacks or by other mechanical damage [16]. (F2:349)

Authoritative judgements bestowed by colleagues on a claim, generally after some time, have the status of knowledge and can be referred to in quasi-factive, or unhedged terms.

Alternatively, a writer may choose to present an unhedged claim by establishing a connection to claims previously accepted as true. The forms of argument may differ in establishing the strength of such connections. One kind of connection involves more or less complete deductive arguments, whereby the truth of the previously recognised premises logically guarantees the conclusion, or at least purports to do so. This may also apply to the validation of methods (8).

(6) We have demonstrated that the promoters of the haemoglobin genes of the two nonlegumes *Parasponia* and *Trema* function at elevated levels in the transgenic nodules of the legume *Lotus*.. (D1:635)

(7) The results of global analysis of the set of spectrally and time-resolved data are given in Fig. 2 in the form of emission decay-associated spectra (DAS). It follows that each of the four components is characterised by its own emission spectrum, whose FWHM is within 45-48 nm .... (B4:230)

(8) The methods presented here establish the means of assaying the structural integrity of the crystallised complex both at room and low temperatures and confirm the feasibility of carrying out other techniques of structural analysis either requiring or benefiting from low temperatures. (A2:97)

The connection of a claim to accepted knowledge may also be presented in causal terms, i.e. how the phenomenon conforms to accepted universal rules of behaviour or statistical regularities (9, 10), or in terms of how a particular instance represents an example of a phenomenon widely accepted as an element of the external world (11):

(9) Since the fungus does not invade the leaf blade, this study demonstrated unequivocally that it can significantly affect N metabolism in parts of the grass which remain entirely free of fungal mycelia as well as in parts that of the plant which are actually colonised. (E2:731)

(10) The reduction in Rubisco content by antisense RNA had no effect on stomatal conductance despite the reduced rate of CO_2 assimilation. (E4:301)

(11) Here we report the successful transformation of *Solanum nigrum* (black nightshade) with a tomato Inhibitor I gene containing 725 bp of the 5' region. We present data showing that the tomato Inhibitor I protein is expressed in a wound inducible, tissue specific manner in leaves and that the inhibitor is correctly processed to the same mature form found in wounded tomato leaves. (F2:350)

These forms of argument enable writers to present their claims without mitigation, and any modality is implicit or unmarked. Thus Type 1 statements relate directly to what is known, either because they report conclusive results which can be anticipated from what is known or present arguments or logically incontestable claims which are assimilable to what is known. Group 5 statements, concerning what is known to be false, can also be expressed with similar certainty.

Groups 2, 3 and 4 comprise the majority of work in normal science, contributing to the cumulative extension of the scope and precision of scientific description. Group 2 statements, i.e. what is possibly true, include the vast majority of claims made in scientific discourse (12 & 13), while what is believed to be not necessarily true (14 & 15), and statements concerning what is unknown (16) are less frequent:

(12) Possibly, phosphorylation of ACC synthase could contribute to the existence of its different pl forms. Alternatively, phosphorylation-dephosphorylation may be involved in the inactivation / activation of the enzyme in vivo. (B5:107)

(13) I suggest therefore that D1 degradability must be causally linked to Q_B site occupation which in turn determines PEST region accessibility... (A1: 266)

(14) Such effects presumably can also be rationalised within the framework of the concepts of Timasheff's group since cryoprotection is critically dependent on a preferential hydration of the protinaccous object under study (Crowe et al, 1990) (A4:35)

(15) .. this protein was suggested to be EF-2. However, there was not any direct evidence for this suggestion. (B3:221)

(16) Although these mediators induce synthesis of SAP, the precise mechanism involved is not clear. (C4:162)

Once again, there is a reliance on presuppositions concerning what is accepted, but the evidential status of the assumptions that mediate inferences between data and hypotheses are less assured. Claims thus exhibit more or less confidence, and are seen to require varying degrees of hedging. These are more typical of the data as few knowledge claims are presented in unmitigated form; induction and inference rather than deduction and direct causality constitute most arguments in scientific discourse. Writers may have strong grounds for trusting the relations between accepted premises and new results, but uncertainty, caution and interpersonal factors will prevent the categorical assertion of such claims. In these groups then, probability is explicit and negatability anticipated. Writers must acknowledge gaps and limitations in order to present their conclusions in the light of accepted knowledge. This provides a provisional characterisation of scientific statements, illustrated in Fig 1 below.

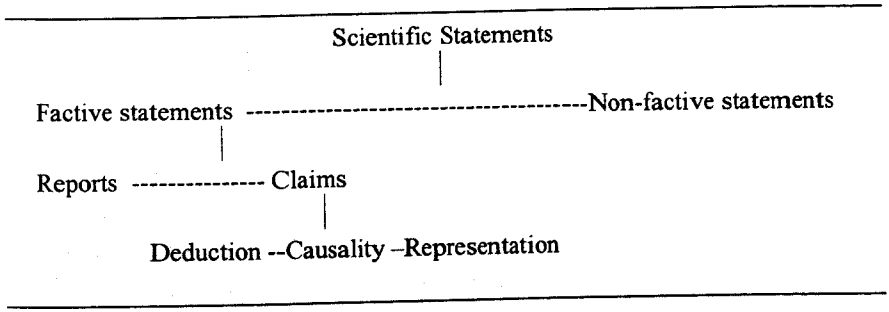


Figure 1: Types of statement in science.

The next section turns to a discussion of the right hand branch of the diagram dealing with group 2, 3 and 4 types of knowledge and the expression of non-factive statements.

A characterisation of hedging in scientific RAs

In general terms the chief function, or extra-linguistic purpose, of scientific endeavour is to extend understanding of those phenomena and theories that the paradigm displays as worthy of study. It is therefore axiomatic that the writer should always make the strongest claim for which there is epistemic authority (Lyons, 1977: 808). This is also how individual research scientists gain academic credibility, by achieving readers' ratification of the most significant claims their findings will support. Gaining this acceptance constitutes the highest level purpose in academic writing. At a lower level this involves both informational and persuasive functions and these in turn presuppose the reader's comprehension of the message (its denotational meaning), and acceptance of the propositions it contains.

In speech act terms, gaining acceptance for claims involves relating illocutionary acts to perlocutionary effects. A writer wants a message to be understood (an illocutionary effect or uptake), and also to be accepted (a hoped for perlocutionary effect). Relating these two factors is by no means straightforward, because however clearly, convincingly, and appropriately reader-oriented information may be articulated, the writer's ability to influence the reader's response is severely restricted. Basically, there is no independent scientific framework for distinguishing the observational from the theoretical, and this means that the classification of each will change depending on what can be contested and what taken for granted (Longino, 1990). So, if the attainment of "truth" does not lie exclusively in the external world, and there is always more than one plausible interpretation of a given piece of data, the reader may be persuaded to judge a claim as acceptable, or may decide to reject it.

This perspective draws on Perfetti & McCutchen's (1987) influential model of reading competence which distinguishes comprehension from interpretation. They argue that meaning is driven by the text in accordance with the constraints of lexical and syntactic forms, whereas interpretation is unconstrained and subject to knowledge effects beyond the writer's control. While one may hesitate to demarcate too rigidly between the two, and thereby tie the

reader's knowledge-base exclusively to acceptance rather than understanding, the distinction is useful in broad terms. Readers largely employ lower-level linguistic processing skills for understanding but rely on higher-level reasoning skills and world knowledge for acceptance. They therefore always retain the option of rejecting the writer's message. In other words, the scientific writer can only guide the reader to a particular interpretation through the use of formal meanings; factors beyond the text, particularly the reader's prior knowledge and beliefs, affect interpretation.

The right to refute a claim demonstrates the active role readers play in the communication process, as it is readers who control the desired perlocutionary effect. The fact that all statements require ratification reveals their negatability, and because readers are guarantors of the negatability of claims, they have an active and constitutive role in how writers construct them. This in turn is why mitigation is central to academic writing, as hedged propositions signal the writer's anticipation of the negatability of his/her claim.

The fact the reader's standpoint is not predetermined means there can be opposition to any sentence through negation of its propositional content. Such opposition can be divided into two types. First, the claims made within propositions clearly have to meet standards of correspondence with what is known (or believed) to be true in the world. This relationship between the proposition and an extra-linguistic representation of reality will be referred to as adequacy conditions. In addition, a proposition which from a purely objective perspective could be presented as a categorical assertion, may be explicitly hedged because of reader considerations. Claims must also incorporate an awareness of interpersonal factors and this aspect of claim adjustment will be called acceptability conditions.

For example, in (17) there is an acknowledged possibility that the claim expressing a causal relationship might have an alternative explanation. Presumably the evidence is insufficiently conclusive to allow a categorical commitment to the matter and the writer is aware that the statement may be negated for failing to meet adequacy conditions:

> (17) Since the rate of oxygen evolution with the far-red light was much lower than with the 650nm light, it appears possible that the mechanism causing the light-activated fluorescence quenching may be triggered by either photosystem. (A3:7)

In (18), on the other hand, it appears that the writer is not hedging adequacy conditions but attending to interpersonal aspects of negatability.

> (18) These results may have relevance to leaf behaviour during the hours after sunrise. (A3:9)

Presumably the writer has some confidence in the consequence of the conclusions and can assume that readers are able to judge their significance for themselves. The hedge is therefore a reader-oriented strategy which accents the consequences of the work without appearing to be too directive.

In sum, the need to have the claim ratified by the reader means the writer must make assumptions both about the nature of reality and about the acceptability of the hypothesis to an audience. The question of adequacy corresponds to the objective negatability of an utterance and acceptability to its subjective negatability; the former require content-oriented hedges and the latter reader-oriented hedges. The model of scientific statements now conforms to Fig 2.

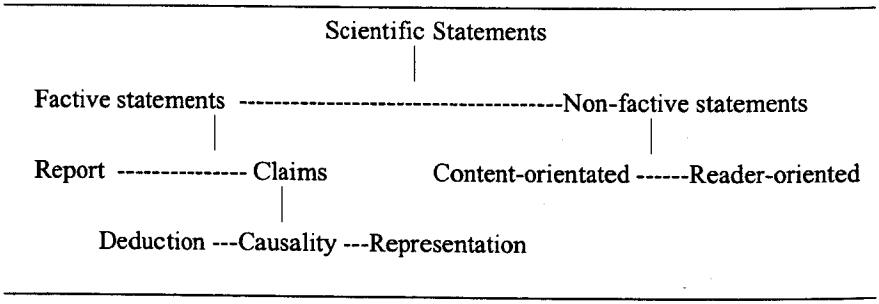


Figure 2: Types of statement in science.

This view corresponds with Toulmin's (1958) well known discussion of practical reasoning structures which assumes sentences have an argumentation basis which can be explicitly justified. Figure 3 shows that the choice of raw data results in findings which can be converted to a modally qualified claim (Q = modal qualification, C = Claim) based on certain justifications (W = Warrants). The findings are subject to possible rejections (R = Rebuttals).

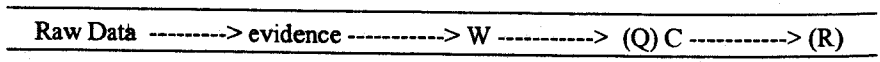


Figure 3: Toulmin's system of logic.

This formula can be read in terms of content-oriented and of reader-oriented rebuttals. Both interpersonal and propositional considerations operate on sentence construction and may entail changes in what the statement literally conveys. The repertoire of devices used to anticipate possible Rebuttals in English however does not always allow readers to differentiate between these. So a complex overlap of usage suggests that the precise motivation for employing a mitigating device may not always be clear. The resulting ambiguity can become a strategic option for writers when presenting claims.

Finally, content-orientation may be expressed at two further levels of delicacy. First there seem to be two fundamentally different reasons for seeking to modify propositional correspondence with perceived extralinguistic reality. The first originates in the obligation to meet academic requirements of presenting claims as accurately as possible; the second concerns the need to anticipate what may be harmful or advantageous to the writer. The two are likely to overlap in actual statements but forms will be referred to respectively as *accuracy-oriented* and *writer-oriented.* Accuracy-oriented hedges can be further distinguished according to whether they concern the qualification of predicate intensity (*attribute hedges*), or writer confidence (*reliability hedges*). The first reflect the difficulties of using a limited language to describe natural phenomena, while the second implicate the writer's confidence in the certainty of his/her knowledge. This final version of the model is shown in Fig 4.

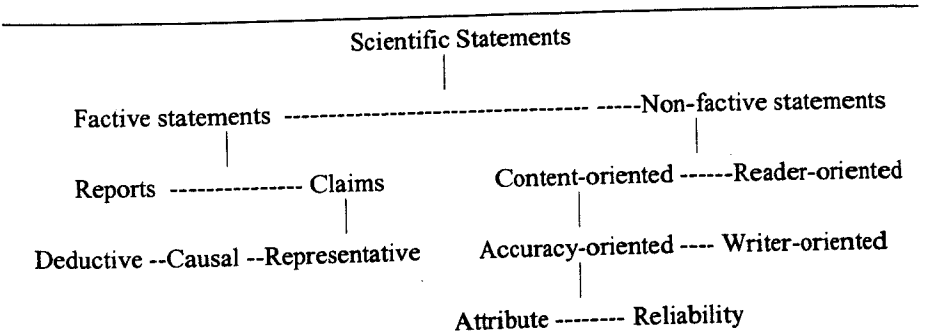


Figure 4: Types of statement in science.

To summarise the motivation for hedging in this outline, my explanatory system seeks to account for the form of sentences by taking possible objections (R) into consideration. Scientists gain and keep reputations by making the

highest level claims they can, demonstrating that they deserve credit for something new. As one informant put it:

> When you are writing scientifically you want to cover all the bets, but you want to be acknowledged as the first person to say something. So if someone follows up an idea in your paper they will cite you, but they won't criticise you for being rash.

So in presenting such claims, scientists must meet both adequacy and acceptability conditions. They therefore use hedges to reduce the risk of negation on objective grounds, i.e. in terms of the match between propositional content and what external reality is believed to be like, and on subjective grounds, relating to acceptable levels of self assertion, deference and willingness to debate. Both are epistemically modalised forms of speech, whereby what is said is related to what is meant by a regard for reader perspective. These forms of hedging will be discussed in greater detail in chapter 6, and applied to text extracts in chapter 7.

Data selection and description of the corpora

A further, and largely implicit, assumption adopted in this thesis is that an analysis of a representative corpus can identify the most common features of the linguistic system we use, frequency being seen as one measure of significance. Data was thus obtained from two sources: a large computer database of published academic writing and research articles selected from a number of leading scientific journals.

The computer corpora

The data for a preliminary frequency survey was provided by three well-known computer corpora of academic English: the JDEST science corpus and the academic (J) sections of the Brown University and Lancaster/Oslo-Bergen (LOB) corpora. The main features of the corpora are shown in Table 1.

Table 1: Computer corpora analysed in the study

Corpus	Field	Words analysed
Brown University	US Academic	175, 000
Lancaster/Oslo-Bergen	UK Academic	175, 000
JDEST Corpus & Technology	Various Science	435, 850
Total words:		785, 850

The JDEST Corpus was compiled by Jiao Dong University in Shanghai in 1985. It consists of English texts from ten subjects in science and technology, i.e. physics, nuclear energy, metallurgy, computer science, aeronautics, mechanics, electrical engineering, chemical engineering, architectural engineering and shipbuilding. The texts were originally randomly selected from theses, textbooks, academic works, popular science and digests published in the UK, US and elsewhere. The corpus does not include running discourse but consists of 2, 000 units of about 500 words each, making a total size of 1, 000, 000 words.

The Brown and LOB corpora contain almost equivalent numbers of texts from the same varieties, the LOB corpus being compiled from British sources and organised to facilitate comparison with the American Brown database. They each consist of some 1 million words in 500 text extracts of around 2000 words (each text ending at the first sentence break after the 2, 000th word). The corpora consist of 15 categories which cover a widely representative range of both fiction and non-fiction text-types. The data considered here are from category J ("learned") which is typical of written academic English, while representing no particular subject area. Each corpus consists of 80 extracts covering natural sciences, mathematics, law, social and behavioral sciences, politics, medicine, education, humanities, technology and engineering. Together the two academic sections represent over 160 authors, and the approximately 350, 000 words are equivalent in length to some 1, 000 pages of text (Kennedy, 1987: 78).

This database was used to provide an overview of lexical markers of mitigation in an extensive set of written academic texts, as it is by no means clear in the literature which of the various ways of expressing epistemic modality are typical in academic discourse. Coates (1983) used the entire Lancaster corpus in her comprehensive corpus survey, but this focused on the modal auxiliaries and was not specific to academic genres. Palmer's (1990) study

made use of data from the Survey of English Usage for exemplification only and, once again, was restricted to modals. Holmes (1988b) drew on a variety of empirical sources but was concerned with epistemic devices to express certainty as well as to mitigate. Her use of the academic sections of Brown and LOB, however, is useful for comparative purposes in some cases in the present study. The computer corpora were analysed by the Longman Mini Concordancer and Wordsmith Tools (OUP) software on a PC.

The journal corpus

The main study is based on a corpus of 26 research articles in the field of cell and molecular biology[1] consisting of 75, 000 words. The RAs appeared in 6 mainstream, professional, refereed international journals published in the Netherlands (4), USA (1) and France (1) between 1988 and 1993. The articles were written in English by scientists affiliated to institutions in 11 countries, principally the US and UK, and the publication frequencies of the journals range from 6 to 63 issues per year. The corpus is comprised entirely of experimental research articles, which "report experimental or *ex-post facto* research designed to test a hypothesis or theory" (Hill et al, 1982: 344).

The rationale for the selection involved two principal considerations. The first was a desire to relate the findings of the investigation to the needs of second language students, therefore an attempt was made to identify key disciplines in which NNSs were enrolled. Informal inquiries to a number of universities revealed a relatively high proportion of L2 students in applied cell and molecular biology. This field has grown out of molecular biology, a discipline which dates from the 1930s when investigations first began to be made into the ultimate units of the cell (Judson, 1995). The objective of the science is to understand cellular, genetic and molecular aspects of plant biology, particularly the structure and function of photo-synthetic apparatus and the biological basis for plant molecular improvement. This applied dimension has obvious attractions to students from countries where agricultural development plays an economically significant role. As a result, this field of biology is now a rapidly growing science which attracts substantial numbers of international students, particularly at post graduate level, to Western universities.

The second selection criterion involved the need to base the study on a range of texts drawn from this discipline which were representative of its research articles. The RA is a recognisable genre and has emerged as a result of a well-defined and mutually understood communicative purpose, reporting new

experiments intended to add to the existing knowledge of the discipline. The RA is therefore distinguished from other types of scientific texts such as theoretical papers (Morris, 1966: 204), review articles and short papers. The abstract is also excluded from analysis as it is not considered to be an integral part of RA discourse, but a separate distilled version of it (eg Rounds, 1982; Gosden, 1993; Bhatia, 1993)[1].

The choice of journals was made from specialist informants' recommendations together with reference to the 1992 Journal Citation Reports (SCI, 1993). Valid sampling procedures are crucial to text analyses to avoid highly valued and possibly idiosyncratic text examples (Crookes, 1986). Consequently both avenues were explored to find "prototypical exemplars" of the genre (Swales, 1990: 58). Consultation with experienced biologists at three universities produced a list of widely read international English language journals in plant molecular biology which were also common destinations for the researchers' own submissions. Reference to the Journal Citation Reports (JCR) was used to minimise possible subjective affects of informants' recommendations on the sample. The JCR ranks each journal according to the number of times it is cited in a particular year, allowing the importance of a journal to be determined in terms of the likelihood of it being read by anyone engaged in a certain field. This index was also employed by Crookes (1986) and Paltridge (1993) in their studies of scientific text structure and involves working down ranked listings to build a picture of the most popular journals.

The JCR provides information which allows identification of the small set of publications that typically account for the majority of articles in particular specialties, and of the number of times they were formally cited in the current year. It also reports the average number of current citations to articles a journal published in the previous two years to give an 'impact factor'. The impact factor thus indicates the relative frequency with which the journal's 'average' article has been cited in 1992 and is useful in gauging the significance of absolute citation frequencies, as it discounts the advantage of size and frequency of issue. A further useful indicator of use is the 'cited half-life' integer which equals the number of whole years it took to accumulate 50% of the current citations. Together these figures reveal a journal's impact on its field.

This information revealed the particular importance to the research community of 6 journals from those recommended by the expert informants. Two of the journals, *Biochimica et Biophysica Acta* and *FEBS Letters*, are among the top 25 largest scientific journals in terms of source items and total citations,

and all six rank highly in their specialist fields. Having chosen the journals, articles were selected by assigning numbers to papers on the basis of their location within volumes covering the five year time span and consulting a list of random numbers. Letters, notes and reviews were ignored, as were articles written by an author already selected. Informants confirmed that the selected texts fulfilled the typical expectations of the parent community in terms of purpose, structure, style, content and intended audience. The journal corpus can therefore be seen as representative of the highly conventionalised, socially embedded models of refereed, published articles in this field.

Table 2 shows the titles of the journals chosen, the number of times they were formally cited in 1992, their impact factor and their cited half-lives and the number of texts taken from each journal for analysis in this study.

Table 2. Corpus journals, citations and impact rankings.

Journal	SCI category	Rank	Total Citations 1992	Impact Factor	Cited Half-life	Number of texts
FEBS Letters	Biophysics	8	36, 202	3.505	4.9	5
Biochimica et Biophysica Acta	Biophysics	9	67, 903	2.610	8.2	4
Plant Cell	Plant Sciences	2	3, 123	6.342	2.5	4
Plant Molecular Biology	Plant Sciences	7	4, 673	3.361	3.6	4
Photosynthesis Research	Plant Sciences	9	1, 453	3.017	4.1	5
Plant Physiology	Plant Sciences	10	23, 219	2.925	6.5	4

Method of analysis

A comprehensive understanding of hedging in scientific research writing involves at least three levels of linguistic description and inquiry: a surface-level

analysis of hedges employed in the particular genre, a pragmatic analysis of their functions, and oral interviews with expert informants concerning text extracts. Each approach provides crucial data and offers corroboration or corrective to the other sources. The first type of analysis provides empirical validity for an explanatory framework to replace impressionistic ideas about the incidence of particular forms likely to be encountered in this variety. The second seeks to account for pragmatic choices and demonstrate how linguistic features take on restricted values in particular contexts. The third attempts to validate the interpretations and determine the actual discoursal practices of insiders and their interpretations of the features under study. Together these promise a viable analytical approach to hedging and an understanding of the rationale underlying the selection and distribution of surface features.

The purpose of the quantitative analysis was to characterise the extent of hedging and its principal forms in a representative sample of scientific RAs. The absence of an existing corpus meeting the above criteria meant that one had to be compiled from published texts as described, then analysed manually. A detailed investigation of the RAs revealed both the principle lexical forms of hedging and their distribution, and also highlighted the significance of three strategies. By referring to experimental weaknesses, limitations of the model, theory or method used, or to inadequate knowledge, writers qualify commitment in ways previously overlooked in the literature but which should be considered among the hedging devices available to scientific writers. A parallel analysis of the computer corpus of heterogeneous academic material was also made to determine the range and numerical significance of lexical hedges in more general contexts.

The study not only provided a description of the major hedging expressions in this genre, but also suggested aspects of their use. In examining and categorising surface features, a number of regularities became clear which provided generalisations concerning their functions. These were subsequently examined at the next stage of analysis. This involved searching for further exemplars of different functions, seeking to identify the purposes served by items in particular cases, and continuously refining a series of categorisations until all cases could be plausibly accounted for.

The procedure was as follows. First, a list of likely high frequency items used to express mitigation was compiled by reference to published studies of epistemic or related notions in computer corpora, principally Holmes (1988b), and Kennedy (1987). Both of these writers used the J Sections of the Brown

and LOB corpora, Holmes to determine the relative frequency of items expressing epistemic modality, and Kennedy to determine the devices used to refer to temporal frequency. The latter research appeared useful as many adverbials of indefinite frequency such as *often*, *usually* and *generally* are also used to hedge and express probability (Halliday, 1994). Holmes' work indicated the likelihood of encountering epistemic verbs such as *appear*, *assume* and *suggest* in academic discourse, for example, while Kennedy shows that tentative expressions such as *seldom* occur only rarely in the corpora, and *infrequently*, *hardly ever*, and *on occasion* do not occur at all. These forms were supplemented with items from grammars, linguistics texts and introspection.

The next stage in the study was to determine how often each of the words or phrases identified in the lists appeared in the general academic corpora. These were computer scanned to produce frequency counts of the target items. In order to ensure that each use expressed mitigation, a key word concordance, which places the item in a context of 15 to 20 words, was performed on a large sample of the modal auxiliaries and all other items where there was a likelihood of semantic indeterminacy. Such studies must be treated with some care however as generalised corpus-based frequency counts, even within the various genre which comprise academic English, can provide misleading information when compared with a specific professional sub-field (cf. Kennedy, 1995).

Following this, a detailed study was made of the journal corpus using the information gained from these investigations. Every word or phrase expressing a hedging function in the RAs was noted. Once again, the judgements rested on close contextual scrutiny and estimations concerning whether the devices performed hedging functions. Devices meeting this criteria were coded to reflect provisional lexical and syntactic categories such as those suggested by Lyons (1977), Perkins (1983) and Brown (1992). Further study of these devices revealed particular functions which were eventually refined into three strategic categories. Interviews with disciplinary informants were central to both the generation of categories and the validation of individual cases.

The oral interview is a common data collection instrument in social science and composition research, but has only recently begun to be been used in the analysis of written science texts (eg Berkenkotter & Huckin, 1995; Myers, 1990). In this study it has three main purposes: to provide access to insider interpretations, to test and refine the categorisation scheme, and to validate particular interpretations. The interviews used here were semi-structured (Cohen & Manion, 1994) and involved open-ended prompts related to research objectives,

follow-up probes to interesting observations and think-aloud responses to texts (Smagorinsky, 1994; Witte & Cherry, 1994). These protocols simply required participants to read text extracts or complete RAs and respond to textual features. In addition, focus groups of two or three members were also used to help analyse the extracts discussed in chapter 7.

All interviews were taped with participants' knowledge using an unobtrusive Walkman recorder and were written up as a summary highlighting key issues immediately after the session (Oppenheim, 1992). The advantage of taping was that they could be returned to later and the significance of some observations only became clear further into the analysis. In addition, the actual language used and the degree of certainty expressed through intonation was preserved. However, while these ethnographic methods offer a means of bridging the gap between the understandings of the outsider and the interpretive practices of working scientists, they are by no means an unproblematic means of gaining direct access to those practices. Chin (1994) observes that interviewing is a "meaning constructive activity" and responses do not transparently reflect a stable set of views or understandings. The value in obtaining insider perceptions however lies in obtaining a fuller representation of what is going on, supplementing and correcting linguistic text analyses and researcher interpretations with a third dimension which gets the native perspective. Only in this way can we begin to describe the culture in terms of how its members understand it and participate in it.

As I observed earlier, there are various ways of classifying hedges and the pragmatic functions of these devices to mitigate propositional connections or build relationships with colleagues will be discussed in chapters 6 and 7. I begin, however, with a detailed analysis of the formal characteristics of hedging. The following chapter therefore seeks to describe how epistemic modality is accomplished in scientific writing, by focusing on the results and importance of frequency information.

5. Surface Features of Hedging

The literature on hedging has largely focused on the semantic categorisation of modal verbs or on the expression of modality at general levels, comparing spoken and written language for example, rather than how hedging is organised and employed in particular contexts. As a result, we have little information on either the particular purposes or frequencies of hedging in different varieties. This chapter identifies the most numerically significant lexico-grammatical features used to hedge in science articles. It shows that specialist writers in biology appear to be fairly consistent in the way they mitigate statements, and reveals their preferred forms of expression.

In addition to a formal description of hedging in the science corpus, the chapter offers some broad comparisons with general academic writing for illustration, although the nature of the data prevents a detailed statistical analysis or a demonstration of significance. It should be remembered that the comparative data sets do not include running discourse but are composed of text extracts of either 500 words (JDEST), or 2,000 words (Brown/LOB). While statistical analysis of these corpora may well reveal differences with my more homogeneous sample, such differences are likely to be due to factors related to incompatible corpora type rather than the categories under study. However, while these do not allow accurate comparisons with the articles, they provide a general reference against which the frequency of target forms in the articles can be judged, and suggest the extent to which particular forms can be said to characterise these text types. The description of hedging offered here then, fills a gap in our knowledge of the subject and is preliminary to the development of a viable explanation of underlying purposes which follows.

Lexical and strategic hedges

In this discussion I will refer to lexical and strategic hedges to distinguish lexical signals from regularly used grammatical patterns and other means of expressing reservation in the articles. This division is not meant to imply that lexical devices are not routinely employed to realise such strategies, but simply to draw attention to the overwhelming preponderance of lexical mitigation in the corpus and to conveniently categorise a heterogeneous diversity of expressions that might otherwise fail to be included. These strategies I have labeled according to their principal functions, which are to (i) refer to experimental conditions, (ii) question the model or method employed, and (iii) to acknowledge inadequate knowledge. The means by which they are expressed vary, but include questions, conditional clauses, and contrast markers, as well as a limited range of formulaic phrases.

Lexical hedges represent the most common means of realising epistemic modality in English and the language offers at least 350 devices for this purpose (Holmes, 1988b: 27). The significance of individual words, over phrasal and syntactic realisations, is inevitably over-emphasised by a computational approach however, but the method is useful in distinguishing the items most frequently used to express a concept in a particular register. Unfortunately linguists have tended to exaggerate the significance of modal verbs (eg Coates, 1983; Palmer, 1990, Butler, 1990), but Holmes' (1983 & 1988b) demonstrates the importance of other forms of hedges to native speakers.

The use of particular devices also appears to differ across domains. This is shown in the comparison with Holmes' (1988b) data for the academic sections of the Brown and LOB corpora and informal and semi-formal sections of the Lund corpus of spoken English in Table 1. These corpora differ in a number of ways, but most striking is the apparently greater use of epistemic adjectives and adverbs in the journals compared with academic writing more generally, which may be related to the high lexical density of scientific writing. In addition, writing seems to contain more epistemic nouns and speech more modal auxiliaries, with biologists appearing to make considerably less use of the latter than other language users. The problem here, of course, is the absence of formal tests of significance to demonstrate meaningful differences, which may help explain why such distributions have gone unnoticed elsewhere in the literature. However this overview clearly illustrates the importance of not restricting an analysis of hedging to modal verbs.

Table 1. Relative frequency (%) of grammatical categories used to express epistemic modality

Class	Journal corpus	Holmes' Corpus		
		Writing	Speech	Total
Lexical verbs	27.4	35.9	31.5	33.3
Adverbs	24.7	12.8	21.5	18.1
Adjectives	22.1	6.6	2.3	4.0
Modal verbs	19.4	36.8	42.4	40.2
Nouns	6.4	7.7	2.3	4.5

Table 2 shows the frequency of hedging in the corpus by grammatical and strategic categories. As can be seen, scientific hedging is principally a lexical phenomenon, with 85% of cases realised by lexical verbs, adjectives, adverbs, nouns and modals. In addition to the grammatical classes, however, non-lexical exponents, which I have categorised as hedging strategies, constitute 15% of hedging in the corpus. These strategic expressions will be discussed further below. First I will consider the findings related to the different lexical expressions, organised by grammatical classes.

Table 2. Relative frequency of various categories used to express hedging in the journal corpus

Category		Items per 1,000 words	percent	Raw number
Lexical	lexical verbs	4.9	23.3	366
	adverbs	4.4	21.0	329
	adjectives	3.9	18.8	294
	modal verbs	3.5	16.6	259
	modal nouns	1.1	5.4	85
	Total	17.8	85.1	1333
Strategic	reference to limiting conditions	1.3	6.1	97
	reference to a model, theory, method	1.1	5.3	83
	admission to a lack of knowledge	0.7	3.5	55
	Total	3.1	14.9	235
Overall Total		20.9	100	1568

Modal auxiliaries

Previous corpus studies demonstrate that the distribution of epistemic modal auxiliaries varies with context, occurring much more often in informal speech and in personal writing than in published texts (eg. Coates, 1983: 22; Holmes, 1988b; Collins, 1991). In scientific discourse, Flowerdew (1993b) found modals to be far more frequent in a biology lecture course than in the textbook it was based on. This variation is best illustrated by Coates' (1983: 24) analysis of the Survey of English Usage and LOB corpora which shows that *will* and *can* both occur about twice as often in speech than writing. *May* and *must* are the only modals that figure more often in writing.

But while the modal verbs can be readily identified and extensive information exists on their frequencies, the actual meanings expressed by modals in different registers is less clear. The modals cover a wide semantic range and each one represents an overlap of meanings and stylistic differences. Establishing these meanings involves analysing every instance, a time-consuming and painstaking task, as Holmes points out:

> Since a million word corpus produces between 3, 000 and 4, 000 tokens of modal forms such as 'would', each requiring detailed contextual analysis, the dearth of frequency information is perhaps not surprising.
>
> (Holmes, 1988b: 28)

The polysemeous nature of the modal verbs is acknowledged by Huddleston (1971), for example, who proposes that there are six distinct meanings of *may*, and by Coates (1983), who finds that *could* has seven meanings and *should* five. These variations in meaning have received different treatments and there is considerable confusion and terminological proliferation in the description of modal verbs (eg Greenbaum & Quirk, 1990; Halliday, 1970; Leech, 1987; Young, 1980). The concept of modality is generally categorised according to the nature of the possible worlds to which the validity of an utterance is relativised (Lyons, 1977; Perkins, 1983). Often however these are reduced to the modality of social and natural laws such as volition, ability and obligation (deontic and causal modality), generally referred to as 'root' meanings, and the modality of reasoning and belief (epistemic modality). Hedging only concerns the latter and, as we shall see, it is important to keep these distinctions in mind.

Coates is the primary source of information on modal semantics, and the following discussion relies heavily on her insightful and coherent treatment.

Like Lyons (1977:791), Perkins (1983), Klinge (1993 & 1995) and others, Coates views the polysemy of the modals as a function of the contexts in which they occur, that is, meanings do not reside in the modals themselves, but are assigned to utterances which contain them. An adequate descriptive framework for modality therefore requires an explication of the link between linguistic semantics, or invariant traits of items, and pragmatic contexts, their situated meaning in science articles. Table 3 summarises her major findings and categories.

Table 3. The meanings on modal verbs and the proportion expressing epistemic modality in Coates (1983) LOB samples

Modal	No. in sample	Epistemic Function Meaning	%	Paraphrase	Primary meaning
Must	236	confident inference based on deduction	31.3	I am sure	obligation/necessity
Should	229	tentative assumption based on inference	12.2	I assume / probably	weak obligation
Ought	104	tentative assumption based on inference	12.5	I assume / probably	weak obligation
Can	231	none			Root possibility (possible for..)
Could	210	tentative possibility	7.1	I believe / perhaps	Root possibility
May	236	epistemic possibility	60.1	I believe/ perhaps	epistemic possibility
Might	191[1]	epistemic possibility	78.5	I believe/ perhaps	epistemic possibility
Will	203	prediction about present based on repeated experience	18.2	I confidently expect	epistemic prediction (future reference)
Shall	225	prediction about present based on repeated experience	35.1	I confidently expect	obligation/intention
Would	219	past prediction	25.1	I confidently expected	hypothetical
		hypothetical prediction	58.0	I expect given unlikely conditions	(unreal condition)

Essentially Coates identifies four distinct "semantic clusters" of primary meanings based on a sample of around 200 cases of each modal taken from a

written corpus of various genres. Epistemic uses are related by avoiding commitment to categorical assertion and the primary epistemic forms can be placed on two parallel scales moving from confidence to doubt and from inference to non-inference. The modals relating to assumption are *must*, *should* and *ought*, and those relating to possibilities are *will*, *may*, *might* and *could*. *Shall* and *would* represent hypothetical epistemic uses. Only affirmative *can* and *need* have no epistemic senses.

Using Coates' semantic definitions, a random section of 25, 000 words from the JDEST[2] corpus was computer scanned, producing 330 instances of modal verbs which when examined in contexts of 25 words revealed 130 (39 %) epistemic cases. The journals yielded 395 instances of modal verbs, of which 257 (65 %) were used epistemically, the most frequent being *would, may* and *could* which accounted for 76.6% of the total. Table 4 presents the forms and frequencies of epistemic modal verbs in the JDEST and journal corpora together with Butler's (1990) findings for 12 physical science journals and textbooks and data supplied in Holmes (1988b), for her written corpus and an extract from the Brown corpus analysed by Hermeren. In all cases I have extracted the figures for hedging functions and converted them for a standardised corpus of 10,000 words.

Table 4. Frequencies of modal verbs used to express hedging in various corpora (per 10, 000 words)

Modal verb	Frequencies per 10, 000 words				
	Journals 75, 000 words	Butler 32,000 words	JDEST 25, 000 words	Holmes 25, 000 words	Hermeren 70, 000 words
would	10.4	6.6	16.8	18.8	16.0
may	9.2	11.6	8.0	8.8	4.4
could	6.4	4.7	3.2	4.8	0.4
might	3.6	2.5	2.4	4.4	4.0
should	2.4	1.2	0.8	0.8	0.4
cannot	0.8	0.3	0.0	2.0	0.4
will	0.8	0.3	18.0	24.4	5.6
must	0.8	1.6	2.8	2.4	2.4
shall	0.0	0.0	0.0	0.0	0.0
ought to	0.0	0.0	0.0	0.0	no data
Totals	34.4	29.4	52.0	66.4	33.6

Despite similarities, the general academic and the science corpora exhibit some clear differences. The highest overall densities are found in Holmes' sample of the academic sections of Brown/LOB and my sample of the JDEST corpus. This may be because both are relatively small samples of diverse subject matter and genres; the JDEST material containing a variety of technology and applied science theses, manuals and other sources, and the J sections heterogeneous academic texts. It may, however, indicate a greater use of hedging in more general academic writing. Holmes does not discuss the disparity between her results and those of Hermeren for the same J section, although Hermeren's sample was far larger and was exclusively composed of the American Brown corpus. Thus the considerable differences that exist in the distribution of particular modals and their meanings in different fields could reflect sampling variations (eg. Butler, 1990: 144).

The frequency of *will* is the main variation between the general academic corpora and the science material. One reason for this may be that its hedging sense, to make predictions based on anticipated outcomes, is perhaps inappropriate in such prudent domains as the hard sciences. Butler's science material and the RA corpus show similar overall densities of epistemic modals and broadly similar rankings for individual items with *would, could, may* and *might* figuring significantly. The clearest differences occur between the use of *would* and *may*, which alter their rankings. The distribution of both modals in the science data shows considerable similarity compared with the other corpora however, with *would* ,*will* and *must* occurring less often, and *should* more often. Overall, scientific writing seems to contain a lower density of epistemic modals and employs a more restricted range than academic writing generally, with *would, may* and *could* accounting for 75% of all cases. Neither epistemic 'shall' (expressing expectation), nor 'ought' (tentative assumption) occur. I will now discuss individual modals in turn.

Must / Need. Must is the predominant modal of inferential certainty, ranging from strong to weak, in spoken discourse (eg. Collins, 1991) although it occurs infrequently in all these written samples, particularly the science data, and only six times in the RA corpus. Need, as the equivalent of must in certain negative environments, does not occur as an epistemic modal in my data at all and only once in Butler's sample. The relative infrequency of must in scientific research discourse, where there is often a need to make deductions from known

facts, may suggest writers are reluctant to express even weak convictions concerning the truth of their propositions. Indeed, must is frequently found with explicit hedges, underlying its essentially subjective and less than categorical nature:

> (1) I suggest therefore that D1 degradability must be causally linked to Q_B site occupation which in turn determines PEST region accessibility to protease through allosteric effects. (A1:266)

It is noticeable that *be* frequently co-occurs with *must* in the corpus, in both its epistemic and root meaning.

Can / Could: While can occurs with epistemic meaning only in interrogative or negative environments, epistemic could is similar in meaning to may and might in expressing tentative possibility (eg Perkins, 1983: 37; Butler, 1990: 154). Its frequency in the journal corpora is undoubtedly affected by the speculative and tentative nature of scientific research compared with the digest and textbook material found in the JDEST corpus. The following examples are typical:

> (2) Another conceivable reason for this decrease could be a change in TRP τo value under the influence of the electric field induced by P^+ and Q^- charges. (B4: 231)

> (3) What determines the constitutive expression in nightshade leaves, or lack of constitutive expression in tomato leaves, is unknown and could be speculated to occur at several levels of regulation. (F2:353)

> (4) Such an alteration in the spectrum of synthented proteins could be due to the influence of the heat shock on gene expression processing and/or translation of RNAs [4]. (B2: 189)

The similarity between epistemic uses of *could* and *may* is illustrated by their combination in this extract together with the co-occurrence of *possibly*:

> (5) Possibly, phosphorylation of ACC synthase could contribute to the existence of its different pI forms. Alternatively, phosphorylation-

> dephosphorylation may be involved in the inactivation/activation of the enzyme in vivo. (B5:107)

Coates recognises two other principal uses of *could* which involve possibility: as a past tense form of 'root possibility' *can* (6) and a hypothetical form to express root possibility in unreal conditions (7):

> (6) Long ago it was shown that the components of the photosystems could be partially separated by detergents.... (A5:171)

= it was possible to separate the components by the use of detergents.

> (7) From these findings and from electron microscopy studies, a model implying almost complete lateral heterogeneity in the thylakoid membrane could be suggested .... (A5:172)

= the findings and other studies enabled a model to be suggested.

This acknowledges a crucial distinction between cases where external circumstances, rather than an authority source, affect the outcome of the proposition (Coates, 1983: 113). Thus whereas epistemic possibility refers to the writer's assessment of the likelihood of the truth of *p*, root meanings concern the role of enabling conditions and external constraints on its occurrence. The determining characteristic is therefore the objectivity of the event, and the clearest cases are those where such objective enabling conditions are made explicit (7).

The difficulties in always distinguishing between the conceptually possible and the experientially possible in the use of *could* is admitted by Palmer (1990: 185). He recognises a similar distinction to Coates, which he terms epistemic and dynamic, the latter "concerned with the ability or volition of the subject of the sentence, rather than the opinions" of the writer (1990: 36). These examples illustrate the problem:

> (8) Transformation of a flavonoid-deficient cultivar with a recombinant gene that directs tissue-specific expression of the wild-type enzyme could restore UV-B tolerance without inducing undesirable characteristics. (D3:178)

= a) it is possible that this explains what happens OR
b) it is possible for this transformation to occur when these conditions prevail

(9) This could be sufficient for hemoglobin to function in the facilitation of diffusion of O_2 to the rapidly respiring cells in the vicinity of the root tip and in ..... (D1:638)

= a) I believe it is possible that this causes the effect OR
b) In some circumstances it is possible for the effect to occur

(10) An increase in NO_3^- assimilation in the roots could function to provide additional N to the sheath where fungal growth is most extensive. (E2:731)

= a) it is possible that this phenomenon has this effect, OR
b) this would probably result if we could create these conditions.

In each case the context allows for both an epistemic or a root reading, referring to either the writer's assessment of possibility, or whether the situation would occur given appropriate circumstances. The semantic contrast is therefore *realis* vs. *irrealis:* either 'this is one of the results of an increase in NO_3' vs. 'I believe this might explain what happens'. If a statement represents that an outcome depends on external enabling or disabling circumstances, it carries a root meaning, if not, then it is epistemic. Indeed, this kind of ambiguity may provide additional security for writers by enabling them to establish greater distance from their propositions.

Would. Coates (1983), Perkins (1983: 50) and Leech (1987) all observe that hypotheticality receives its principal grammatical expression through the modals would, could, should and might. Coates recognises would as the main hypothetical modal with epistemic meaning, root forms occurring infrequently. As in Coates' corpus, a common use of epistemic would in scientific writing is as the hypothetical variant of will (= marker of prediction):

(11) It is not very difficult to imagine that in the long term this 3D structure would change with changing light intensity, as crudely re-

flected in the ratio of appressed to non-appressed membranes, and that granna would be generally less extensive the higher the light intensity. (A1:271)

(12) ..if the haemoglobin in the root were mainly at the tip, it would have a local concentration 100-fold or greater than originally deduced. (D1:638)

(13) An increased demand for N in the sheath would decrease the N available to the blade, resulting in the necessity for the more efficient N utilisation that would be provided by an increase in NH_4^+ reassimilation in the blade. (E2:731)

Such hypotheses are often based on prior theoretical or experimental premises and, as these examples illustrate, the conditions required to fulfil the hypothesis are commonly expressed, although not usually in an overt conditional clause. Syntactically, the majority of examples of *would* are active, and one of the most frequent co-occurring main verbs is *be,* followed by *appear* and *suggest.*

Leech (1987) suggests that hypothetical auxiliaries with no expressed conditions can substitute for 'primary modals' to add greater diffidence and tact. Such an interpretation is suggestive in the case of *would,* but some examples, where there seems little justification for diffidence and where *will* cannot be easily substituted, appear to be only hypothetical:

(14) If our prediction were valid, then the *hy* 8 locus would either represent the *phy* A gene itself or another gene whose product is closely linked to the synthesis or action of phytochrome A. (D5:41)

However, many examples of hypothetical *would* do suggest a pragmatic interpretation:

(15) From the results of Fig 1 one would predict that in the absence of sucrose, Cl depletion by Na_2SO_4 should occur ... (A4:31)

(16) This result would favour the hypothesis that the plant protein is targeted to the bacteroids. (D2:260)

(17) Increased N economy would be consistent with observations that the endophyte decreases the effects of stress on growth and survival, ... (E2: 732)

(18) From this discussion, then, it would appear that some of the changes in amino acid concentrations .... (E2: 732)

In these cases the modal attenuates the proposition unnecessarily and expresses a tentative expression rather than a genuine hypothesis. It implies no lack of commitment to the propositions, as the substitution of *might* indicates, but softens a categorical assertion and avoids forcing the reader to accept a forthright insistence on the recognition of a claim. There appears to be an attempt to create a relationship with the reader through shared understandings as fellow scientists.

Coates also refers to epistemic *would* as marking the past tense of *will* (= 'predicability') which "make a confident assertion or 'prediction' about some action or state in the past" (1983: 208). She is referring to here to assertions based on actions or states which become predictable through repetition, in science this most often involves "backshifting" the reference to a past orientation point, although it occurs infrequently in the RA data:

(19) Under these conditions, we predicted that seedlings would respond through the action of phytochrome B, and thus allow .... (D5:40)

(20) Our screening strategy was designed to target the action of the FR-HIR photoreceptor, which, based on indirect evidence, seemed most likely to be phytochrome A. If our predictions were valid, then the hy8 locus would either represent the phyA gene itself or another gene whose product .... (D5:41)

Shall / Should. Shall is principally used in the corpus with first person plural subjects as an alternative to will (eg Huddleston, 1971: 309-11), often tinged with volition in informing the reader of future sections of the text. None of the written data contained examples of confident expectation. Epistemic should typically refers to the future and consequently has a more tentative meaning than would, expressing a less confident assessment of probability based on

facts known to the writer. It occurs far less often than would in my data, although my results concur with Butler in showing a higher frequency than in the other corpora.

> (21) Indeed, it should be possible to test predictions about the carbon gain efficiency of sunflecks depending on their duration and intensity ... (A1:272)

> (22) Such mutants should also help to elucidate the UV-B signal transduction pathway. (D3:177)

> (23) However, this information is forthcoming and should lead to a more thorough discussion of the data, including an explicit calculation of ... (A2:99)

Epistemic *should* thus links subjectivity (i.e. the writer's attitude to p , what he or she believes is probable) and logical assumption (what is inferred from known facts). This more objective dimension often appears to be paramount in the RA examples.

There is also a hypothetical meaning of epistemic *should* expressing unreal conditions with a negative implication. This accounts for only two (11%) of the RA cases:

> (24) Those solutes should have been equally effective if Ca^{2+} release were a function merely of salt concentration .... (A4: 33)

> = the solutes were not equally effective

> (25) However, if polyethyleneglycol had no stabilising action on the binding of the 23 kDa polypeptide, it should not have acted like the other solutes in opposing the loss of functional Cl from the water oxydising complexes. (A4:32)

> = it did act like other solutes and therefore did have a stabilising action

Will. There is a large discrepancy in the use of epistemic will compared with its use in the other domains, with both Butler and myself finding only a handful of instances. Everyday epistemic uses involve claims about the present, while

in scientific writing the reference is usually to general truths with only a peripheral element of futurity:

(26) If significant energy is lost by the plasma particles in this process, the plasma will be cooled. (JDEST)

(27) However, yielding at the ends simply means that further increase in the load will result in an increase of positive moment without any great increase in the rate of inflection. (JDEST)

These examples refer to predicted logical outcomes and indicate a general truth, i.e. whenever X occurs, it is predictable that Y happens. The justification for the assurance is based on repeated experience rather than logical inference. Leech (1987: 85) argues that this sense is "naturally suited to scientific or quasi-scientific statements" and, in corroboration of this view, Coates' found that 38% of examples in her Lancaster sample occurred in the J (learned/ scientific) section.

Will (= prediction) is like the logical necessity meaning of *must*, and has a similar distribution in the molecular biology corpus. One might speculate that this is for similar reasons and may be due to the fact that the genre is a forum for reporting conclusions and results which are not indisputable. *Will* might well be considered an inappropriate means of presenting propositions in such an environment. *Could* and, particularly, *would* are the predominant means of expressing present time probabilities. These verbs can be used epistemically to express predictions based on tentative deductions from experiments. Their prevalence in the data may therefore be the result of writers substituting more conditional and tentative forms for the confidently predictive *will*. Examples from the corpus demonstrate the difference such choices make in expressing a proposition (also 18-21):

(28) The faint 21-kD band observed in the PBM lane (Fig 2) would reflect the transient passage of this protein across the PBM from the plant cell cytoplasm to the bacteroids. (D2:260)

(29) The prediction of this model is that aminoxyisovalerates with aromatic rings characteristic of potent synthetic auxins would be good candidates for active molecules. (E1:295)

(30) .... it might be suggested that synthetases present in nuclei [33] could be involved in the regulation of the processing of tRNA precursors. (B6:7)

It is extremely difficult to distinguish an epistemic from a future interpretation where *will* refers to a future action, as reference to the future inevitably involves some uncertainty or doubt. When events are so predetermined as to be habitual, shown in sentences such as *The sun rises in the east* or *The south-bound train leaves at 4.30 daily*, to talk of predictability introduces an inappropriate degree of doubt, for, as Leech observes (ibid: 85), it is a poor train service if departure times are only typical. Recent literature tends to the view that the future time reference of *will* is secondary to a more modal function (e.g. Lyons, 1977: 816; Perkins, 1983: 42; Palmer, 1990: 57; Quirk et al, 1972: 87), and Coates recognises a gradience from prediction about the present (predictability) to prediction about the future (prediction) with overlap between them. In many cases *will* (= prediction) is simply a neutral marker of future tense, but there is often a tinge of uncertainty which gives it its epistemic sense.

(31) Further purification of PEPc kinase will be required to resolve this issue. (C1:195)

(32) The relationship between the root-specific and nodule-specific DNA signals in the two non-legume promoters will eventually be determined by a functional analysis of the promoters in transgenic lotus and tobacco. (D1:638)

May / Might. Figures for might and may, often considered to be the prototypical hedges, are relatively consistent across these corpora, although may is slightly more frequent in Butler's scientific corpus than my own[3]. While both modals can be used interchangeably to indicate a 50-50 assessment of possibilities, epistemic may appears to occur over twice as often in published writing. It is the only modal which figures significantly more often in academic than other genres. Moreover, its frequency in formal contexts contrasts with spoken discourse, where it is beginning to yield to might as an exponent of epistemic possibility (Coates, 1983: 153). A broad generalisation is that while might appears to occur slightly less frequently in scientific writing, may occurs more often.

While *may* has a "permission" sense, this is relatively uncommon and the basic distinction is once again represented by root and epistemic meanings, relativising either the occurrence of the event described, or its truth. This division is often neutralised with *may* in formal texts however, and the RA corpus reveals many examples of an overlap between enabling circumstances which permit *X* and the writer's lack of confidence in the possibility that *X*:

(33) Regulation of D1 protein degradation may be controlled by occupation of the Q_B binding site. (A1:267)

= either a) it is possible that this explains the event.
OR b) it is possible for degradation to be controlled given certain conditions.

(34) ... the Tyr-96 region and / or the β3 segments in the helix-poor domain may undergo topological adjustments. (C3:285)

= a) I believe it is possible that this can occur OR
b) particular circumstances allow these adjustments to occur

In both examples the root and epistemic meanings are equally possible and do not conflict, allowing substitution of the paraphrases 'it is possible for x' and 'it is possible that x' ('perhaps'). My data concurs with other studies which show the active voice to be far more frequent than passive for epistemic uses (Butler, 1990:151; Huddleston, 1971:304).

Although both auxiliaries focus on the current verifiability of propositional truth, both Perkins (1983: 50) and Palmer (1990:58) have argued that *might* is the more remote form of *may* and that it expresses a higher degree of 'conditionality' or tentativeness. This is given some credence in my data where it appears that *might* is the preferred choice when the context permits ambiguity between the two 'possibility' interpretations.

(35) Such a mechanism might serve to balance the synthesis of the products throughout this pathway in wild type Arabidopsis. (D3: 176)

(36) This implies that the extent to which the endophyte might affect N metabolism under field conditions could also depend considerably upon other interacting factors. (E2: 731)

The possibility that *might* indicates less certainty than *may* is strengthened by its greater co-occurrence with more tentative epistemic items in 'harmonic combinations' (Lyons, 1977: 807). Coates found the occurrence of such mutually reinforcing forms was very similar for both *may* and *might* in her spoken data, but in the journal material they are overwhelmingly associated with *might*:

(37) One might speculate that what is now understood as carotenoid functions are not to be understood as common capabilities of all carotenoids. (A5:176)

(38) Although it is premature to answer this question, it might be suggested that synthetases present in nuclei could be involved in the regulation of the processing of tRNA precursors. (B6: 7)

(39) It might be speculated that the lack of crDNA methylation in cv. Platenese could result in enhanced amounts of mRNA in chloroplasts. (F3:17)

(40) This insertion, which we suspect is the membrane anchor, could associate peripherally with the membrane or might span half the bilayer in a manner similar to model 1 depicted for microsmal enzymes by Black [22]. (C3: 283)

There may be insufficient instances to make a strong claim in this regard, but *might* appears to express a more distant possibility than *may*, perhaps because the condition of realisation is more remote. It therefore allows scientific writers to represent their statements in a more tentative way.

Two more uses, both relatively infrequent in the journal data, deserve comment. First, the collocations *may well* and *might well* are both epistemic and appear to reduce the degree of qualification and to express greater confidence in the proposition, an observation reinforced by the modal adjective *probable* in (42) (rather than the harmonic *possible*):

(41) This increase may well play a significant role in regulating the phosphorylation state of PEPc. (C1:195)

(42) The moderately hydrophobic leader-sequence of these mitochondrial enzymes, which are cleared off after import, might well be the evolutionary precursor of a probable permanent N-terminal transmembrane anchor of the microsomal enzymes. (C3:285)

The second observation concerns cases where the writer seeks to orientate the statement to the reader's perspective to influence his or her opinion:

(43) Furthermore, the synthesis of these compounds is induced by UV-B in a variety of plant species, as might be expected for an adaptive response. (D3:171)

(44) It might be expected, however, that the inhibitory effect on elongation growth would be transient..... (D4:46)

By ascribing the writer's point of view to the addressee, such uses encourage the reader to adopt a common logic shared with the writer and understand the statement as a colleague engaged in the same professional quest for truth.

Overall, the modal auxiliaries are used less frequently to express epistemic modality in scientific research articles, and their distribution shows a preference for more tentative forms than typically occur in other published writings, including other science genres. RAs in molecular biology do not employ all epistemic modal verbs, and *will*, *would* and *must* are under-represented. The auxiliaries *may* and *might* occur in similar proportions across all domains and only *should* and *could* appear more often in the science articles.

Epistemic lexical verbs

Epistemic lexical verbs such as *suggest*, *indicate*, and *predict* are the most common means of expressing mitigation in the science corpus with 366 cases. Epistemic verbs represent the most transparent means of coding the subjectivity of the epistemic source and are generally used to hedge either commitment or assertiveness. Their numerical significance thus reflects their rhetorical versa-

tility in contexts where categorical assertions rarely represent the most effective means of expression. By indicating the writer's confidence in a statement they contribute to the evidential reasoning between grounds and claims (Toulmin, 1958), and allow an evaluation of the intended degree of certainty.

There are at least four ways a writer may express the non-factual status of a proposition which Palmer (1986: 51) calls speculative, deductive, quotative and sensory; that is, writers can mitigate claims by indicating that they are presenting information as subjective opinion, deductive conclusion, hearsay, or as based on the evidence of their senses. Speculative and deductive categories involve epistemic judgements by the writer, allowing uncertainty and the tentative assertion of hypotheses. They can be paraphrased as 'I believe that' and 'I conclude that'. The quotative and sensory categories concern the nature of the evidence the writer employs to support a claim. They therefore imply a certain amount of doubt as evidentiary justification is required for their support. They can be paraphrased as 'X said that' and 'It seems that'. Epistemic verbs therefore mark both the mode of knowing and its source (belief, deduction, report, perception), and thereby carry implications about the reliability of the knowledge itself. The following discussion also introduces a third category of evidentials concerned with hedging the outcome of the research activity.

Epistemic Judgement verbs

These reflect appraisals by the speaker of the factive status of events and include speculation and deduction. They are distinguished by the fact that the degree of commitment to the truth of a proposition is predicated on a reference to the uncertainty of human evaluation, thereby differing from other verbs of knowing and saying by their epistemic orientation to the proposition. Within the judgement subset, speculative verbs indicate there is some conjecture about the truth of a proposition. It comprises mainly conventional "performative verbs" (cf. Perkins, 1983: 94; Brown, 1992) which perform, rather than describe, the acts they label:

> (45) Thus, we propose that this insert is the major site of interaction with the membrane. (A1:267)

(46) I suggest here that the photosynthetic apparatus in higher plants not only has to be dynamic, but that it is in fact totally dynamic in the sense ... (C3:283)

The tentative nature of these verbs is clearly illustrated by the following example, where the writer explicitly emphasises the distinction between the speculative and the categorical:

(47) Although suggested, it has not been demonstrated that the isolated French bean proteases are expressed by the plant and not by the bacteroids. (D2:266)

The speculative category however also includes verbs which involve unobservable cognitive states or processes, and which do not obviously "perform" tangible acts in the same way as the core examples of speech act verbs:

(48) I believe that the major organisational principle of thylakoids is that of continuous unstacking and restacking of sections of the membranes ... (A1:267)

(49) We speculate that the inhibitor interacts specifically with a protease, the origin of which is still to be determined .... (D2:267)

(50) This insertion, which we suspect is the membrane anchor, could associate peripherally with the membrane or might span half the bilayer ... (C3:283)

These verbs appear to give a more conjectural than assertive meaning to the propositions which follow, hypothesising a world in which *p* might be true. They can thus be seen as a more tentative means of expressing a claim than the 'performative' speculative verbs.

The second type of judgements derive more obviously from inferential reasoning or theoretical calculation than from speculation, and are presented as deductions or conclusions:

(51) Of this percentage, we calculate that more than half is present in the coiled-coil portion and the rest in the gobular domain. (C2: 274)

(52) We infer that the rate becomes limited by the by the rate of regeneration of RuBP because both the calculated rate of electron transport and ... (E4:301)

(53) We conclude that the fluorescence quenching seen here is not due ... (A3:8)

(54) This information is used to study and deduce the secondary structure of the globular domains of fibrogen. (C2:269)

One observation to be made here is that the mitigating force of these devices is not affected by conventional performative formats[4], and many examples employ a variety of devices used to suppress human agency, commonly through the use of passive voice:

(55) It has been calculated that the concentration of these metals in soybean bacteroids is 14mM ..... (D2:262)

(56) What determines the constitutive expression ... is unknown and could be speculated to occur at several levels of regulation. (F2:353)

The use of the passive often allows a writer to hedge a responsibility to statements by appearing more objective and distanced. A number of indirect measures of passive voice, such as terminal -ed and -en morpheme verbs and the auxiliaries *be, have* and *has*, indicate that the academic sections of the Brown and LOB data contain some 5,000 passive forms. Flowerdew (1993b: 84) confirms the importance of the passive in a written biology corpus, where it appeared almost twice as often compared with the same material presented in lecture form. However, while a manual analysis of the RAs shows that the passive voice constitutes 46.9% of main sentence verbs in the corpus, it suggests that the use of the passive in science is correlated principally with non-epistemic verbs.

Table 5 shows that a much higher proportion of epistemic lexical verbs occur as active forms, with less than a third of the total occurring as passives. The distribution shows that passive forms of epistemic verbs overwhelmingly occur in Discussion and Result sections, while almost half of the passive forms

of non-epistemic verbs are to be found in Methods, the only section where the passive predominates.

Table 5. Distribution of active and passive forms in RA corpus (%)

RA section	Epistemic lexical verbs			Other main verbs		
	active	passive	total	active	passive	total
Introduction	13.7	14.0	13.8	15.8	9.4	12.7
Methods	2.8	3.9	3.2	11.7	45.7	28.0
Results	29.7	32.1	30.4	29.7	21.1	25.6
Discussion	34.1	33.6	34.0	25.6	10.4	18.3
R&D	19.7	16.4	18.6	17.2	13.4	15.4
Overall	69.3	31.7	100	52.1	47.9	100

In addition to the use of passives, writers can distance themselves from their propositions through the manufacture of "abstract rhetors" (Halloran, 1984) which imply that rhetorical acts can be accomplished without human volition:

(57) Taxonomic evidence suggests, therefore, that a species ....(D1:638)

(58) The model implies that the function of grana is to *shield* varying amounts of PSII from light and to provide a PSII pool ..... (A1:268)

(59) Third, the present work indicates that the aromatic ring to which the carboxyl group is bound is not necessary, provided that ... (E1:293)

It is worth noting that this convention of scientific discourse allows the writer to employ verbs such as *indicate* and *imply* which contribute to the impersonalisation of the discourse by appearing to make the text or the data the source of epistemic judgements. These represent a linguistic sleight of hand by implying that any reasonable and informed reader would draw the same conclusions. Such forms obscure the functioning of verbal or mental processes by

nominalising a personal projection, thereby encouraging an interpretation close to 'makes us think that X', or 'leads us to the conclusion that X', rather than 'my interpretation is that X' (eg Halliday, 1988: 174).

Epistemic Evidential verbs

The second category of lexical verbs refers to evidentiary justification, either based on the reports of others, the evidence of the writer's senses, or the feasibility of matching evidence to goals.

First, scientific writers rely heavily on 'hearsay' evidence from the research literature, and the choice of reporting verb can indicate their commitment to what is reported. Verbs used for quotative evidence both specify and acknowledge previous findings, and also take a stance towards those findings by referring to either speculative or deductive judgements. These examples reveal different justifications and levels of commitment:

(60) Jofuku and Goldberg (1989) showed the expression of several Kunitz protease inhibitors in soybean tissue. (D2:266)

(61) Jofuku et al [1989] deduced that an N-terminal signal sequence.. (D2:264)

(62) Sandal et al (1987) and Metz et al (1988) have suggested that ... (D1:637)

(63) .. Henninger et al [3] speculated that tuber size is correlated with extent of necrosis in field-grown potato. (E3:119)

(64) Sorokin's study predicted that the creation of quenching centres .. (A3:7)

The literature on "reporting verbs" shows a range of available semantic options (eg Thompson & Ye, 1991; Pickard, 1993b; Thomas & Hawes, 1994). It is clear that by varying the support offered to cited statements writers can effectively hedge their attendant propositions. Quotative evidential verbs of mitigation all demonstrate a reserved position to the material introduced. There is no consensus concerning the number of reporting verbs in the literature, with

Swales (1990: 151) estimating about 50, and Thompson and Ye (1991) putting the figure at over 400. The range of forms in the corpus however is extremely limited, with *suggest* and *propose* being the most common epistemic items, but the preponderance of non-integral citations in RAs obviously restricts this use of lexical verbs.

The second group of evidential lexical verbs refer to perception or apprehending. In science writing items distinguishing sensory evidence from categorical assertion are often visual:

> (65) These changes appeared to involve both assimilatory and basic N metabolism, because the concentrations of NO_3^-, NH_4^+, and amino acids, ... (E2:731)

> (66) This hypothesis seems plausible because UV-B-absorbing flavonoids accumulate in leaf epidermal cells, where they may protect the .... (D3:171)

> (67) The presence of a light activation site in the SCPEPCD 1 protein is surprising since all other features of this gene and its product seem to indicate... (F4:669)

Academic discourse also appears to include a third evidential subcategory. Items here hedge the strength by which the goals of the research or the possibility of acquiring appropriate evidence are expressed. I refer to these as "rationalising narrators", as they appear to contrast the goal of the study with the results achieved, thereby foregrounding the purpose of the research and downplaying the extent to which adequate supporting evidence was acquired. The hedge functions to create a context which removes responsibility from the researcher for the degree of success achieved in meeting these objectives. It allows the writer to express modesty in undertaking the study and caution in anticipating its degree of success. These metadiscoursal devices thus contribute to the construction of an identity and a narrative, while relating to evidence by hinting at the fallibility of knowing.

> (68) We sought to investigate this by studying the regulatory properties of PEPc kinase. (C1:191)

(69) In these FTIR studies we attempt to gain insight into the secondary structure of fibrinogen and its domains. (C2:272)

(70) Because the use of low temperatures to study photosynthetic pigment-protein complexes usually ...we were prompted to attempt optical absorption and linear dichroism spectroscopy on single crystals of the B800-850 complex ... (A2:94)

(71) To evaluate the merit of using the bacterial system for ... we sought to see if other factors in the bacterial protein extract affected its inactivation. (B5:106)

The distribution of the most common hedging verbs across these epistemic categories in the RAs is shown in Table 6.

Table 6. Most frequent hedging lexical verbs in RAs (per 10,000 words)

Judgemental verbs			**Evidential verbs**		
Category	Item		Category	Item	
Speculative	indicate	10.8	**Quotative**$^{+}$	report	1.7
	suggest	9.1		note	0.8
	propose	2.8	**Sensory**	appear	4.0
	predict	1.5		seem	2.7
	assume	1.1	**Narrators**	attempt	0.5
	speculate	0.8		seek	0.3
	suspect	0.7			
	believe	0.5			
	imply	1.1			
Deductive	estimate	1.2			
	calculate	0.4			
Total Judgemental		29.9			
Total Evidential		10.0			
Overall Totals		39.9			

Note: $^{+}$ Quotative evidence utilises a range of writer-oriented judgement verbs.

Clearly, scientific writers make substantially more use of devices in the speculative judgement category which seems to support Palmer's (1986:53) suggestion that the English grammatical system permits only one type of epis-

temic modality. However, not only are evidential hedges well represented in the sensory and quotative categories, but the categorisation is misleading in that verb choice does not simply reflect claims based on either conjecture or analysis. All research statements, however hedged, result from careful theorising or rigorous laboratory procedures and writers are conscious of experimental limitations, possible exceptions, alternative explanations and entrenched opinion. They recognise their results can never include every variable, nor account for all aberrations, and are invariably the product of incomplete information.

The choice of device is therefore essentially strategic, as writers use an appropriate hedge to adjust the strength of their claims and modify their confidence in statements; they do not just distinguish judgement from evidence. Epistemic modality involves both a) the 'confidence' or degree of commitment by the writer to the proposition and b) the indication that an inference is made from available information or "known facts". These categories therefore represent the different degrees of certitude that writers confer on their statements rather than the procedures employed. The fact that RAs demonstrate a clear preference for speculative over deductive judgements, and of judgemental over evidential hedges suggests the adoption of pragmatically cautious positions, rather than any emphasis on the evidence used to support them.

Turning to the forms themselves, lexical verbs comprise the largest group and the greatest range of items with 38 different forms represented. The RA data confirm unmistakable distinctions of channel, with the absence of common conversational hedges such as *guess, reckon, feel* and *think* (which occurs over 100 times more often in speech than writing) and the high frequency of *appear* and *assume* which occur overwhelmingly more often in writing (Holmes, 1988b:31-2). Almost all the forms listed in Table 6 are more likely to occur in the academic sections than the other categories of the Brown/LOB corpora, including 'factual' categories such as newspaper reportage. A frequency search across different domains reveals that *calculate* (77% of the Brown/LOB total in J), *speculate* (50%), *assume* (43%) and *indicate* (41%) are particularly highly represented in academic discourse. Only *claim, believe* and *propose* are comparatively under-represented in the academic texts. Published scientific work is more likely to contain instances of *predict, propose, suggest* and *indicate* with the varied JDEST data displaying more cases of *assume, believe* and *claim.* Overall, the general academic texts demonstrate the greatest range of items, with higher frequencies for most devices. Table 7 compares the most frequently occurring items in the three corpora.

The RA data employs the most restricted range of items but demonstrates the clearest preference for speculative judgements. *Indicate, suggest, appear* and *propose* constitute 56% of all cases in my corpus. A comparison with the other academic corpora shows that all four verbs, and particularly *indicate* and *suggest*, are more prominent in scientific writing, while *seem* and *assume* occur far less often. There were only seven negative forms of lexical verbs, a pattern similar to that of modal auxiliary verbs. Negation in scientific RAs shows a preference for negative quantifiers (*few, little*) and lexical negation (*rarely, overlook).*

Table 7. Most commonly occurring lexical verbs expressing mitigation

Item	Research articles	JDEST	Brown/LOB (J)
indicate	10.8	3.2	4.3
suggest	9.1	3.7	3.9
appear	4.0	2.7	3.7
propose	2.8	1.6	0.9
seem	2.3	4.0	7.7
report	1.7	2.5	2.0
predict	1.5	1.2	0.4
assume	1.1	2.3	2.9

Figures adjusted to show devices per 10, 000 words of text

The syntactic forms of the lexical verbs demonstrate a striking uniformity across the domains, as shown in Table 8.

Table 8. Grammatical forms of major mitigating lexical verbs

Verb form	Research articles		Brown/LOB (J)	
	No.	%	No.	%
Simple Present	**178**	**48.7**	**668**	**49.7**
base form	109	29.8	391	29.1
-s form	69	18.9	277	20.6
***-ed* Forms**	**169**	**46.1**	**596**	**44.3**
Past form	53	14.4	no data	
Past participle	116	31.7	no data	
***-ing* participle**	**19**	**5.2**	**81**	**6.0**
Totals:	**366**	**100**	**1345**	**100**

Generally, over 80% of all verb forms in spoken or written English are in the simple present or past (e.g., Ota, 1963; Kennedy, 1995). Simple present forms, finite and nonfinite uses, predominate in both the RA and general academic data, with the base form accounting for some 30% of the totals. Barber (1962) also found a preponderance of this form in an early statistical analysis of scientific writing, although his figure of 64% refers to all verbs in his corpus. As noted above, the passive accounts for almost 50% of main verbs in the corpus.

Attempts to describe tense usage in scientific discourse (eg Lackstrom et al, 1973; Trimble, 1985; Ard, 1985; Een, 1982; Malcolm, 1987) have not yet produced an adequate theory, although it is clear that strategic considerations play an important role in tense choice. Tense reflects rhetorical as well as syntactic judgements, and decisions reveal the degree of generality and proximity in presenting information. The reasons why the active present simple tense should be used in over half the cases of epistemic lexical verbs are no doubt complex, but choices may reflect a concern with making claims immediate and relevant to present concerns, or with the degree of the writer's assurance in their truth. The use of the present form to hedge might be seen as a present estimation of the validity of a statement, since the use of non-present forms allows a greater distance to be created between the statement and the writer. According to this view, past or perfective forms may serve to increase the strength of the hedge by removing the statement from the present, making the claim more remote, and thus de-emphasising its current importance.

In sum, epistemic lexical verbs are the most frequent means of expressing mitigation in the RA corpus, as they provide writers with a number of ways to signify the non-factual status of a proposition. These verbs indicate both precision concerning the degree of commitment to what is said and whether the warrant for it is judgemental or evidential. Judgements appear to predominate over evidentials in the data, but this may simply reflect the use of caution in making claims about findings or interpretations. *Indicate, suggest* and *appear* are the most commonly employed items, active voice predominates over passive, and the present simple tense represents almost half the forms in the sample.

Epistemic adjectives, adverbs and nouns

Adjectives, adverbs and nouns constitute 53% of the major grammatical classes expressing hedging in the journal corpus (Table 1). It is therefore surprising that they have received little attention beyond a recognition of their epistemic role (eg Lyons, 1977; Perkins, 1983; Holmes, 1988b; Brown, 1992). Both epistemic adjectives and nouns are more frequent in writing than speech with adverbs and adjectives particularly prevalent in the RA data. Table 9 shows the principal forms in the corpora.

Table 9. Most frequent nouns, adjectives and adverbs expressing hedging

Item	RAs	JDEST	Brown/ LOB	Item	RAs	JDEST	Brown/ LOB
about	30	20	39	approximate (ly)	16	13	14
(un) likely	23	19	20	consistent (with)	15	5	3
possible (ly)	30	14	15	partially	15	8	4
apparent (ly)	21	15	21	most	17	15	11
probable (ly)	27	25	34	slightly	14	10	9
essentially	18	9	8	presumably	11	3	7
relatively	17	24	17	somewhat	9	3	11
generally	16	25	17	possibility	7	12	17

Note: All frequencies adjusted to 75, 000 words.

Table 9 reveals that *possible, essentially,* and *presumably* are major forms of hedging in the RAs but occur less frequently in the other data. On the other hand, a word count showed that *almost, estimate* (noun), *virtually* and *usual (ly)* are rare in the journal corpus, but are quite common in academic data (Kennedy, 1987; Allison, 1994). I will now discuss epistemic adjectives and adverbs in turn. The use of these items to express numerical qualification treated separately.

Epistemic adjectives

20% of the lexical hedges in the RAs were adjectives with the most frequent items being *(un) likely* and *possible* (each 3.1 per 10,000 words), *most* (2.3), and *consistent with* (2.0). All these items serve to reduce the writer's categori-

cal commitment, the inclusion of *consistent with* as a hedge being confirmed by a contrast with *prove* (73):

> (72) For wheat germ the existence of such a mechanism is unlikely since though the division of cells proceeds synchronously for at least 100h, ... (B3:222)

> (73) Together these results are consistent with the possibility, although do not prove, that the capacity to mediate the FR-HIR may be an intrinsic ... (D5:40)

> (74) ..they indicate that the photoreceptor involved is somehow related to the photosynthetic apparatus itself. (C1:191)

Two frequent adjectives used to express a degree of variation to an expected norm have not been listed above. *Similar* occurred 79 times in the data, but represents an objective assessment of resemblance rather than the epistemic evaluation of the writer (75). Use of the term *significant* in scientific writing has taken on a technical meaning, losing its everyday hedging sense to become an objective and measurable appraisal (76-7):

> (75) Examination of the data indicate that the induction phase of SAP mRNA in-vitro is similar to that observed in in-vivo experiments. (C4:167)

> (76) Separations of the different treatment means were not conducted, since ANOVA indicated, in most cases, that significant interactions occurred among the.. (E2:272)

> (77) Although the degree of activity of DAIV in the NPA binding assay was sometimes low and varied from experiment to experiment, it was always statistically significant. (E1:292)

A more familiar hedge is *possible*, although only a handful of the 523 occurrences in the computer corpora express epistemic modality. *Possible* has both root and epistemic meanings, roughly parallel to the modal verbs *can* and *may*, or more accurately, to the notions of dispositions or abilities and to writer beliefs, paraphrased as 'possible for' and 'possible that'. The majority of mo-

dal uses in my corpora involve root meanings (i.e. circumstances allow or do not preclude *p*), although the distribution is more balanced in the journal sample, indicating a higher relative frequency of epistemic cases in the RAs. Typical epistemic examples are:

> (78) .... it appears possible that the mechanism causing the light activated fluorescence quenching may be triggered by either photosystem. (A3:7)

> (79) It is also possible that phosphorylation of PEPc actually stimulates a proteolytic cleavage that removes the phosphorylation site. (C1:192)

Perkins (1983: 67) argues that the use of the verb *be* in such modal expressions asserts the predicated state of affairs categorically. Thus in a modal expression it categorically asserts the modality expressed and indicates that the possibility is based on evidence external to the writer, i.e. that it expresses objective modality. In the examples above however, there is an explicit qualification of commitment to the truth of the proposition, although impersonal expressions have been used to obscure the source of the judgements. Consider also the above examples with sentences containing nominal and adverbial forms:

> (80) One cannot exclude a possibility that the activity of EF-2 Kinase in wheat germ is inhibited at a given stage of ontogenesis in some manner ... (B3:222)

> (81) Possibly, phosphorylation of ACC synthase could contribute to .. (E4:301)

While such constructions are common in scientific discourse, they are not true examples of objective modality, but devices to disguise the hedge, attributing the possibility to vague external circumstances. Thus the use of nominalisation (80) helps transform an opinion into an abstract quality in expository prose (eg Martin, 1987:32). Processes and participants can then appear as remote and autonomous which helps to conceal the source of the modality and enables writers to make statements more distanced and objective. Epistemic possibility, however, always involves the writer's confidence in the truth rather

than an assessment of enabling conditions, even if the writer seeks to hide this judgement.

Often it is difficult to be definitive about when hedging is occurring. In both these cases of theoretical model building (82-3) it is unclear whether the writer is referring to an assessment of what is feasible or to what is conceivable, a difference which hinges on the knowledge of the field possessed by the writer (and readers).

> (82) Two types of conveyor belt model are possible in the photosynthetic apparatus. (A1: 267)

> (83) Further refinements of the proposed structure are possible in these regions. (C3:286)

Clearly an epistemic meaning would attach to an interpretation where the outcomes are currently only imaginable rather than likely. The distinction is nevertheless a useful one and captures a critical semantic difference in most uses.

In addition to taking complementation with *be*, many modal adjectives are used attributively with an apparently restricted, but unspecified, range of nouns. About 40% of the cases of *possible* in the RAs occur in this syntactic environment.

> (84) However, the existence of such a possible mechanism of translation regulation in plant cells was not investigated. (B3:219)

> (85) ..the possible involvement of phytochrome C, D or E in an interactive or synergistic manner with phytochrome A cannot presently be ruled out. (D5:45)

This use restricts the reference of the noun while still qualifying the writer's position. The use of *possible* in these contexts is epistemic as, like other uses, it does not refer to physical objects in the real world, but to what Lyons (1977: 442ff) calls second or third order entities, i.e. states of affairs or abstract propositions which may or may not exist.

Attributive uses also illustrate the prevalence of a feature of writing in the physical sciences known as noun phrase stacking (eg Dubois, 1982). This involves packaging a complicated phenomenon into a single, syntactically com-

plex, semantic entity by making it one element of clause structure. Epistemic adjectives are common in such environments:

(86) The proposed substrate binding site appears just below.... (C3:284)

(87) There are other potential hydrogen blood donors nearby which could also participate in heme binding. (C3:285)

(88) ... might well be the evolutionary precursor of a probable permanent N-terminal transmembrane anchor of the microsomal enzymes. (C3:285)

Thus concepts and experiential content can be organised into impersonal constructions, the contribution of modal adjectives allowing the writer to hedge the process described in the nominal group. In Halliday's view (1988:170) such stacking provides a means of making the rhetorical function of the phenomenon explicit in terms of theme/rheme and given/new information. More simply, these NP constructions demonstrate an identifiable and conventional shorthand means of referring to objects and relations in an environment where succinctness is highly valued. They also point to a discernible feature of the RA as a peer-group object which draws on shared specialist knowledge.

Epistemic adverbs

Adverbials are the second most frequent means of hedging in the journal corpus after lexical verbs. They comprise a greater range of items than either adjectives or nouns and while most forms are closely related to equivalent adjectival forms, it is the adverbs which predominate in the corpus. Over 36 different forms occur in the RAs, the most frequent being *probably, apparently* (each 2.8 per 10,000 words), *relatively* and *essentially* (2.4 each), and *generally* (2.1).

Modal adverbs differ from semantically equivalent nominal and adjectival forms in that they are "peripheral in clause structure" (Quirk et al, 1972:421). They are not syntactically integrated as an element of the clause but can appear in a number of positions without affecting the meaning relation between clause and adverb. This distributional mobility allows the modality of an utterance to be either 'thematised' (89 & 90), or inserted intrasententially (91 & 92), the scope of the modality (S or VP) being the deciding factor:

(89) Presumably, some 'signal' generated in the chloroplasts initiates a sequence of events that lead to phosphorylation... (C1:190)

(90) Possibly, phosphorylation of ACC synthase could contribute ... (B5:107)

(91) Viewed in this way the concept of lateral heterogeneity becomes obsolete because the distinction between granal and stromal lamellae for any given membrane is relatively short-lived. (A1:268)

(92) No Pre-PS II has been detected, presumably because it has not yet been sought. (A1:270)

While the concept of theme is controversial, the choice of initial position can nevertheless serve to accent the hedge and provide the reader with an alternative interpretation for the entire sentence, marking what follows as hypothetical and subjective.

Downtoners: Semantically, epistemic adverbs function principally as adjuncts and disjuncts. Quirk et al (ibid: 452ff) refer to a category of intensifying adjuncts they call ' downtoners' which have a "lowering effect on the force of the modified verb". A number of hedging and frequency adverbs can be identified as downtoners and the categories represent scalable distinctions from 'compromisers' (93), through 'diminishers' (94), and 'minimisers' (95) to 'Approximators' (96).

(93) This appearance of kinase activity correlates quite well with ... (C1:193)

(94) This shift could be partially caused by solvent-exposed α-helical segments of its coiled-coil portion. (C2:273)

(95)... that light will rarely penetrate further than about... (A1:268)

(96) Eventually the PS II pool would become virtually depleted ... (A1:273)

In realising the greatest degree of intensity, approximators almost deny the truth denoted by the verb, negating the full implication of the predicate:

> (97) ...the dissociating effect of NaCl could be almost entirely suppressed, especially with respect to the 23 kDa polypeptide. (A4:31)

The other categories all provide a revision that expresses a more precise version of the proposition, an option also identified by Flowerdew (1993b) in his textbook corpus.

Disjuncts: Disjuncts, or what Halliday (1994) calls probability adjuncts, are detached from the rest of the sentence and can be divided into two broad types: style and content disjuncts. The former convey:

> the speaker's comment on the style and form of what he is being said and defining in some way the conditions under which 'authority' is being assumed for the statement. (Greenbaum and Quirk, 1990:181)

Some items in this relatively small class of adverbs therefore indicate that a generalisation is being made (Quirk et al, 1972: 509) and therefore hedge the accompanying statement:

> (98) Staining was generally confined to the vascular tissues ... (D1:635)

> (99) The results are broadly similar to those obtained by ... (C1:192)

Content disjuncts, on the other hand, are more numerous and include a wide range of hedges relating to contingency and degrees of certainty. Disjuncts concerned with certainty comment on the truth-value of what is said, "firmly endorsing it, expressing doubt, or posing contingencies such as conditions or reasons" (ibid: 183). Quirk et al (1972: 511-513) distinguish three further groups in this category which correspond to Palmer's distinctions used above to categorise lexical verbs. Once again, writers can qualify the status of their knowledge by expressing propositions as resting on speculative, deductive, quotative and sensory judgements.

The largest group simply expresses doubt without carrying implications about the truth of the statement, or the sense in which it is seen to be true or false:

(100) No Pre-PS II has been detected, presumably because it has not yet been sought. (A1:270)

(101) These EGTA clots are transparent and possibly comprised of.... (C2:273)

(102) There was a low level of staining that was difficult to localise but was probably associated with the vascular tissues. (D1:635)

The second group is similar to the perceptual group of evidential lexical verbs and conveys how the truth of the proposition can be mentally perceived. Most adverbs in this class communicate conviction, however, and items expressing doubt are rare in the corpus:

(103) ... were observed in the cytoplasm, which is apparently in contradiction to results obtained by subcellular fractionation experiments. (D2:262)

(104) Evidently, far-reaching structure-organising effects appear to emanate from the binding sites of Cl and Mn or, in a more general sense, from the ... (A4:33)

(105) This can be rationalized intuitively with the favorable effect of ..... (A4:34)

Interestingly, adverbials with *-ed* participle stems such as *allegedly, reportedly, reputedly* and *supposedly*, which comment on quotative evidence, are rare in the corpus. This is probably because the frankly skeptical view with which these items treat the reliability of reported information conflicts with expectations regarding the mitigation of criticism. The range of reporting verbs provides more resources for greater delicacy in commenting on the work of others.

Items in the third group are judgemental and express the sense in which the writer believes what is said to be true or false. Quirk et al observe that these adverbial disjuncts typically carry a reference to the reality of what is said, so epistemic items express either a contrast with reality (106), or signify that what is said is true only in principle (107).

(106) This suggests that though the activity of EF-2 from wheat germ can be potentially regulated, by phosphorylation, this mechanism ... (B3:221)

(107) Our samples gave essentially the same patterns irrespective of .. (F3:16)

It is worth noting that the mobility of adverbs means that they can be selected to operate at either clause level or within a group or phrase, which means that both downtoners and disjuncts may be employed in restricted roles. This can obviously considerably limit the scope of what is hedged:

(108) In view of the seemingly wide distribution of DNA methylation... (F3:18)

(109)... are strikingly homologous to the bovine enzyme and have a signal sequence of approximately the same length. (C3:281)

An approximate comparison of the relative occurrence of adverb items in different classes is summarised in Table 10 below.

While employing the classifications of adverbial adjuncts recognised by Quirk et al, I have interpreted them more broadly and have also included a range of indefinite time-frequency adjuncts which are used to express probability. In general, hedging adverbs are accuracy-oriented, and items which mitigate the force of the verb occur slightly more often than those that comment on what is said. Most downtoners have a limited modifying effect on the predicate, while in the disjunct categories, judgements predominate over a concern with how truth is perceived. Overall, neutral items are the most frequent, expressing doubt without implying any sense in which this is judged or perceived.

Table 10. Approximate comparison of hedging adverbials in different classes.

Category	Function	Principal Examples	%
Adjunct			
Downtoner	lowering effect on force		
Compromiser	slight effect	quite; usually; normally	13
Diminisher	increased effect	partially; slightly	26
Minimiser	considerable effect	rarely, occasionally	2
Approximator	approximation to force	almost; virtually; relatively	12
		Sub-total	53
Disjunct			
Style	conveys the sense in which statement is true	approximately; generally	4
Content	conveys comment on the truth-value of proposition		
Certainty	certainty and doubt	likely; probably; presumably	26
Truth	mental perception	apparently; evidently	5
Sense	judgement of truth	essentially; potentially	12
		Sub-total	47

Hedging numerical data

Variability is a crucial factor in scientific data, and a great deal of hedging in science involves the manipulation of precision in quantification, with 91 cases in the corpus[5]. Numerical expressions and quantities are hedged by disjuncts, approximators and adjectives and principally by a limited range of style disjuncts. The most frequently employed expressions are *about* (30), *approximately* (16) *some* (6) and *around* (5). These figures exactly mirror Dubois' rankings for oral presentations in the biomedical field (1987:531).

> (110) The synthesis of the 94, 85 and 74 kDa HSPs decreases by approximately 60% at 44°C while that of the 64-60 kDa class is more thermostable. (B2:192)

Reference to limited knowledge

In judging claims readers want to know how much confidence the writer invests in them. One of the clearest means of distinguishing between conditionally true statements and speculative possibilities is to comment on the state of existing knowledge. This use is distinct from that discussed by Swales (1990) for establishing a research niche, and while it appears to have gone unnoticed elsewhere, it is an important means by which writers are able to fix their work in an evidential context of uncertainty.

Many examples are closely followed by a presentation of the writer's thesis and appear to help 'frame' the related claim in an appropriate environment of doubt and probabilities. Examples (117-8) assist in adjusting the reader's expectations of exactitude as a context for the writer's view on the matter, while (119-20) follow the presentation of results and serve to hedge an interpretation.

> (117) Nothing is known about the chemical constitution of the fluorescent material. This could be due to protein-metal complexes such as .... (D2:262)

> (118) We have no knowledge whatsoever about very rapid and short term changes in the molecular arrangements of the thylakoids. (...). I suggest here that the photosynthetic apparatus in higher plants not only has to be ... (A1:267)

> (119) We do not know whether the increase in intensity of illumination from 250 to 1000 μ E/m^2 per s causes induction of one specific..... (C1:192)

> (120) It is not known whether such a weak temperature response .. (B2:190)

Other cases simply signal possible alternative explanations, as in these earlier examples:

> (81) One cannot exclude a possibility that the activity of EF-2 Kinase in wheat germ is inhibited at a given stage of ontogenesis in some manner ... (B3:222)

(85) ..the possible involvement of phytochrome C, D or E in an interactive or synergistic manner with phytochrome A cannot presently be ruled out. (D5:45)

As can be seen in these examples, the form know appears commonly in this strategy, as do structures and phrases which suggest options, such as and/or, alternatively and on the other hand. Questions also occur in the expression of this strategy. Questions are a relatively uncommon means of highlighting knowledge limitations, but they can be used to hedge the truth of a proposition by making it relative to a writer's state of knowledge. While they generally seek to engage and elicit a response from the reader, questions typically draw attention to the existing state of knowledge in order to signal an important unresolved issue (121), or to withhold commitment from a claim (122-3).

(121) Is the sole function of phytochrome A to mediate the FR-HIR and, if so, under what circumstances in the natural spectral environment would this be important? Insufficient data are currently available to definitively assess these questions. (D5:46)

(122) By what signal is this expression regulated? Jofuku and Goldberg (1989) showed the expression of several Kunitz protease inhibitors in ... (D2:266)

(123) Could such a putative interaction of an aminoacyl-tRDA synthesase with precursor tRNA have a physiological significance? Although it is premature to answer this question, it might be suggested synthesase present in ... (B6:7)

Reference to limitations of model, theory or method

Writers also hedge their full commitment by referring to deficiencies in the research model (124), theory (125) or method (126) which may compromise the accuracy of their results. This allows the prudent researcher to present results having anticipated criticism of the premises or methods by which they were achieved.

(124) We are aware of the concerns expressed in the literature [26] concerning the application of homology based modelling to sequences at this level of similarity. Our initial attempt in modelling has revealed the conservation of .... (C3:286)

(125) Viewed in this way the concept of lateral heterogeneity becomes obsolete because the distinction between granal and stromal lamellae for ... (A1:268)

(126) The procedure only identifies methylated nucleotides located within the recognition sequences of the sensitive enzymes. In spite of its shortcomings, the method has been widely employed to evidence this type of ... (F3:16)

Where theoretical models are mentioned, they are rarely used to categorically assert their explanatory efficacy. Either direct reference is made to the limitations of the model (127), or results are presented in juxtaposition to it, establishing the reliability of the context in which the results were obtained and thus hedging their acceptability (128-9).

(127) Despite our careful alignment and model building procedures we do not claim that our model structure is comparable to a crystal structure. (C3:285)

(128) In the context of the proposed model the observations that ... (A1:272)

(129) Based on and consistent with the model proposed above I suggest the following mechanism for photoinhibition ... (A1:273)

Comments on the limitations of the method used vary considerably, but divide into those seeking to justify the technique employed (130), and those which hedge the replicability of the technique or its generalisability (131-32):

(130) Internal necrosis of harvested tubers was evaluated visually, which is the standard procedure used to characterise this disorder in potatoes. (E3:118)

(131) In our hands there was no significant change in V_{max} on illumination. (C1:191)

(132) ...approx 70% according to our method and some β-turn .. (C1:191)

Similarly, writers occasionally refer to the theoretical context of their work as a means of hedging the reliability of results (133). Again, this is a way of limiting the damage that might result from a bald proposition; the hedge functioning to contextualise the claims made by suggesting parameters within which they might be accepted as true. Example 134 shows the cumulative effect of placing such a context-referenced hedge among other devices to create a sense of proceeding speculatively beyond the data.

(133) Our interpretation of these results is that the total level UV-B ... (D3:175)

(134) Previously reported differences in UV-B susceptibility of varieties of plants within a species might be caused by variations in levels of these UV-B-absorptive phenolic compounds. In this scenario, breeding plants for agronomically useful traits, such as reduced colored flavonoid pigment synthesis or increased palatability due to loss of phenolic compounds, might inadvertently cause enhanced UV-B sensitivity. (D3:176)

Specification of the formal realisations of these strategies can only be expressed in terms of probabilities. Personal pronouns, often accompanied by verbs of sense or cognition, are common, as are the lexical items which label this strategy. In addition, conditionals are a common means of qualifying commitment to the methodology, theory or model as they make one circumstance dependent on another and thereby hedge the certainty of outcomes. The corpus included 29 conditional clauses, and these were principally used to hedge propositions by removing them to a possible alternative world where the expressed facts are not epistemically accessible, but relate to a speculative state of affairs:

(135) If regular D1 protein degradation and shedding of LHCII drive this "conveyor belt" system (....), then this model also provides an explanation as to how very rapid adjustments can be made in this system. (A1:271)

(136) If this scheme is correct, then the orientation of the heme plane will almost be parallel to the membrane plane as determined by EPR experimental studies. (C3:284)

The connections established here are between propositions whose truth is contingent upon the situation in the conditional clause, so that the *if-* clause specifies the conditions under which the *then-* clause is satisfied. Both propose the advantages of an explanatory model while withholding full commitment to them: the fulfillment of the condition is left unresolved and the asserted claims remain dependent on the validity of the propositions in the *if-* clauses.

The contrast between claims and real world possibilities is expressed far more explicitly using hypothetical conditionals. In the following examples, the modal operator in the *if-* clause conveys the writer's assessment that necessary conditions are unlikely to be fulfilled, and therefore that the hedged consequences will not occur. The implication is that the hypotheses are unlikely to be true:

(137) These results suggest that if a flavonoid mutant with unaltered sinapate accumulation were available, it would be more sensitive to UV-B than is *tt* 4. (D3:175)
= but it isn't available

(138) Those solutes should have been equally effective if Ca^{2+} release were a function merely of salt concentration, of a competition between Na^{+} and Ca^{2+} for the binding site, and of the .... (A4:33)
= but it is not a function only of salt concentration

Most examples in the data, however, leave the condition open and the truth of the proposition unresolved, thereby hedging the accuracy of the theoretical or descriptive claims:

(139) If correct, this prediction might explain why previous exhaustive screen s have not detected mutants in phytochromes other than phytochrome B. (D5:40)

(140) If we assume that the antisense gene is also active in guard cells, it seems that the stomata do not use the Calvin cycle (either in the guard cells or neighbouring mesophyll cells) to monitor p_i to set the appropriate conductance. (E4:301)

Not all conditionals hedge the efficacy of models or theories however; some express real world contingencies (141), or hedge the precision of results (142):

(141) Initially, the gradient of the variable fluorescence curve will increase if there is a sigmoidal component present but will decrease if the curve is .. (A3:2)

(142) ... indicating that only very small amounts, if any, of additional carotenoids like antheraxanthin could be present. (A5:173)

Reference to experimental limitations

The most common strategy in the corpus was to refer to doubts surrounding the experimental conditions. While this form of hedging reduces the writer's conviction, it nevertheless suggests greater confidence than admitting to insufficient knowledge, because questioning the effectiveness of an experiment can also indicate the conditions under which results might be regarded as valid. Writers can comment on the uncertainties of experimental conditions to suggest different levels of confidence. On one hand, some statements fail to guarantee the precision of experimental results with any assurance:

(143) We have not been able to determine precisely whether GUS expression and Lotus leghemoglobin synthesis are initiated simultaneously, but .. (D1:635)

(144) Under these conditions phosphorylation of PEPc by... (C1:192)

(145) We did not succeed in obtaining the complete transcript. (D2:262)

At the other extreme, statements can withhold endorsement of the validity of results, indicating that claims are neither decisive or utterly convincing:

(146) The bare platinum employed to detect oxygen in the Joliot type oxygen electrode is not well suited to making absolute measurements since its sensitivity can be altered, eg by material adhering to platinum. This makes it difficult to compare signal sizes between different chloroplast samples. (A3:6)

(147) Therefore, results under these conditions were not considered conclusive. (E2:730)

(148) It should be noted also that in these experiments very high concentrations of fusidic acid was used (up to 3 mM) resulting in non-specific decrease of phosphorylation of many other proteins. So it is difficult to conclude whether the 100 kDa protein mentioned above is actually the EF-2. (B3:221)

The realisation of this strategy typically involved negative determiners and items which have negative connotations such as *difficulty, problem* and *fail*. Like the other strategies however, the recognition of this means of hedging depends principally on its content rather than its form. It is not therefore possible to offer a more precise formal characterisation, as there is no established and widely accepted knowledge about their realisations in the literature, and the corpus yields insufficient examples to draw broad conclusions. Nevertheless, they are important means of expressing caution in RAs and should be considered among the hedging devices available to scientific writers.

Distributional information

Overall frequencies show the particular importance of lexical verbs, modal verbs, modal adverbs and adjectives in expressing hedging in the RAs, constituting 79% of all devices. Table 12 compares the figures with other written corpora. It shows that the items *indicate, suggest, could, propose* and *should* are far more frequent in the RAs. It is unclear why these forms are so significant for writers, but their choice may result from conventionalisation within the discourse community as a result of readers' constant exposure to them. An informant with long experience as a reviewer and journal editor commented on the repetition of some items in research articles:

These are familiar expressions in just about every paper I receive. They are kind of formulas for reporting research, particularly in the discussion section where you have the greatest freedom, but are still limited by the boundaries of your data.

Table 12. Rank order of most frequently occurring forms used to hedge in the corpora (per 10, 000 words)

Item	Journal RAs (75, 000)	JDEST (435, 850)	Brown/LOB (J) (350, 000)
indicate	10.8	3.2	4.3
would (not)	10.4	16.8	16.0
may (not)	9.2	8.0	4.4
suggest	9.1	3.7	3.9
could	6.4	3.2	0.4
about	4.0	*	*
appear	4.0	2.7	3.7
might (not)	3.6	2.4	4.0
likely	2.8	2.5	2.7
propose	2.8	1.6	0.9
probably	2.7	1.6	2.8
apparently	2.7	*	2.8
should	2.4	0.8	0.4
seem	2.3	4.0	7.7
possible	2.3	1.5	1.3

The significance of these relative frequencies can be illustrated by the fact that some 100 high frequency words account for half the words in any text (eg Sinclair, 1991; Kennedy, 1993 & 1995; Murison-Bowie, 1993), and the remainder is made up of words that occur only once (Sinclair, 1991). The dominance of a few high frequency grammar words is confirmed by the Bank of English corpus of 200 million words which shows that 700 words make-up 60% of all English texts (Cobuild, 1995). *The* occurs 40 times in every 1000 words and *of* 20 times, for example, and the very highest frequency content words figure less than 1.5 times per 1,000. Clearly then, many of these epistemic items have a very high currency in scientific writing and will appear among the highest frequency words.

Harmonic combinations

An interesting aspect of the distribution of hedges is their tendency to cluster together in "harmonic combinations" (Halliday, 1970: 331). Table 13 summarises the distribution of these combinations.

Table 13. Combinations of hedges in the same sentence

First item	Subsequent items								
	Lexical verbs	Adjectives	Adverbs	Modals	Nouns	Limited condtns	Model / method	Limit knowledge	Total
Lexical verbs	12	43	31	34	9	3	4	8	142
Adjectives	11	10	14	11	3	0	0	2	51
Adverbs	6	10	9	8	2	0	2	2	39
Modal Verbs	11	10	16	12	10	3	3	4	69
Modal nouns	0	2	1	2	0	0	1	0	6
Limited conditions	2	0	2	2	0	1	0	0	7
Model, theory method	3	2	0	4	2	1	1	0	13
Limited Knowledge	1	0	0	5	0	0	1	0	7
Total	46	77	73	78	26	8	12	14	334

Of 1568 instances of hedges in the corpus, 668 (42.6%) occur in the same sentence as at least one other device. Lexical verbs (188) and modal verbs (147) are most frequently found in combination with other hedges, representing 51% and 57% of all epistemic and modal verbs in the corpus respectively. As can be seen, the most commonly occurring collocations involve lexical verbs with adjectives, modal verbs with adverbs, and adjectives with adverbs. Lexical verbs appear in 28% of all combinations, while nouns and hedging strategies are found least frequently in combinations. Some 75% of lexical verbs occupy initial position in a cluster, but adverbs are far less likely to be found in this position.

The hedges which occur most frequently in clusters are *suggest, would, possible* and *indicate,* with the most commonly occurring pairs being *may* with

suggest (8 times), *possible(ly)* (7 times), and *indicate* (5 times), *suggest* with *might* (5 times), and *could* with *possible (ly)* (5 times). The following examples show typical patterns:

(149) It has been suggested that the affects of light on PEPc may be mediated by a Calvin cycle product originating in the bundle sheath tissue. (C1:195)

(150) It now seems possible that the oxygen carrier function may be feasible because if the hemoglobin in the root were mainly in the tip, it ... (D1:638)

(151) Although it is premature to answer this question, it might be suggested that synthetases present in nuclei [33] could be involved in the regulation .. (B6:7)

(152)..this modification could possibly play a role in substrate binding (C3:284)

(153) These properties indicate that SAP may play an important role in the mediation of an inflammatory response; however, the principal ... (C4:162)

Clusters containing three or more items occur only 43 times. The most frequent patterns involve a lexical verb and modal verb together with either an adjective (12 times), a strategy (7 times), or an adverb (6 times).

(154) If we assume that the apparent molecular weight obtained by SDS PAGE is correct, this suggests that only a few amino acids are missing from the N-terminal end and that a leader sequence may be encoded. (D2:262)

(155) ... by aminoacyl-tRNA synthases of precursor tRNAs containing introns appears unlikely, it may be a likely possibility for other precursors.... (B6:7)

(156) Thus, if this scenario is correct, phytochromes A and B would appear to have transiently antagonistic roles in regulating early seeding photomorphogenesis. (D5: 47)

(157) This would probably suggest that the C-terminal processing is not achieved in the nodules, but this needs to be demonstrated. (D2:265)

Table 14 shows the most frequently occurring combinations in the corpus.

Table 14. Most commonly co-occurring hedging items

	might	possible	probable/ probably	would	appear/ apparent	could	some (how)	may	likely	predict
indicate	4	1	4	1	4	3	3	5	-	-
suggest	5	2	3	2	-	2	-	8	3	-
may	-	7	3	-	-	3	2	1	-	2
would	-	2	1	5	4	3			4	4
could	-	5	-	3	1	-	-	3	-	1
seem	2	1	4	2	-	-	-	-	2	1
appear	1	2	1	4	-	1	1	2	2	1
speculate	3	-	-	1	-	2	-	1	-	-
cannot	-	4	-	-	-	-	-	-	1	-
consistent with	-	3	1	2	1	-	2	-	-	-

Lyons (1977: 807) employs the term "modally harmonic" to refer to contexts where two or more forms express the same degree of modality, such that "there is a kind of concord running through the clause, which results in the double realisation of a single modality" (ibid: 808). That is, a single cluster tends to express either certainty, probability or possibility, with few semantically variant items. Thus the forms *may* and *possible* act in concord in a clause to strengthen the hedge, while *may* and *certain* do not. Clusters in the corpus overwhelmingly function to reinforce the epistemic strength of items, what Banks (1994) refers to as "fertilising hedges". However, while non-harmonic combinations are relatively rare in this corpus, forms expressing a varying de-

grees of hedging eg. *may* and *probably* (which occurs 3 times), are perfectly interpretable, and readers generally accept that the higher value was intended.

Because of this tendency of writers to combine items of similar value, this type of analysis suggests the possibility of eventually constructing scales of certainty and tentativeness. *Could*, for example, occurs with *possible* on 5 occasions but never with *probable*, indicating a greater harmony with the less assured form. Credence is also given to the view that *might* expresses a more remote form of *may*, as the former principally co-occurs with more speculative hedges. *Might* is generally found with *suggest, predict, suspect, speculate, seem, imply* and *presume*, while *may* occurs with less tentative forms like *some, could, probably* and *most*, as well as *possible*. *May* also seems to co-occur less frequently overall with epistemic verb forms. The scientific data therefore imply that these forms express different degrees of modality, with *might* being the preferred choice to indicate a more remote possibility than *may*.

Rhetorical distribution

The distribution of hedges in the macrostructure of the RAs reflects their essentially rhetorical purpose with the majority of items occurring in the more discursive sections (Table 15).

Table 15. Overall distribution of mitigating devices in various sections

	Totals	RA Section[1]				
		Intro	Methods	Results	Discussion	R &D[2]
Total words	71,635	8,989	17,904	16,791	15,770	12,181
Total Devices	1,474	180	61	336	574	323
Percentages	100	12.2	4.1	22.8	38.9	22.0
Devices per 1000 words	20.6	20.0	3.4	20.0	36.4	26.5

1 Figures excluded for one RA which does not conform to the IMRD structure.

2 Includes 10 RAs with combined Results and Discussion sections.

The labeled macrostructures of the texts in the corpus differed slightly, with 10 articles having a combined Results and Discussion section, and one article containing headings which could not readily be classified into an IMRD structure. The Discussion sections had both the highest total frequencies of hedges and the highest density per thousand words. 80% of all items occurred in the Research and Discussion sections with only 4% in Methods. In RAs

with an IMRD organisation, Discussion sections averaged 36.4 devices per thousand words, with Results and Introductions yielding 20.0 each.

Table 16 shows that while the distribution of items differed across rhetorical sections, densities were always highest in Discussions. Modal verbs (7.8 per 1,000), adverbs (7.7), lexical verbs (7.1), and adjectives (6.6) were particularly prominent here. In the four part texts, 60% of epistemic modal verbs, 52% of epistemic nouns, and 46% of adverbs were found in Discussions. While the density of items in Results sections is about a quarter of that found in Discussions, 42% of lexical verbs and 52% of references to limiting experimental conditions are found there.

Table 16. Distribution of various devices

Epistemic Category	RA Section[1] (devices per 1,000 words)					Average per RA[2]	Raw totals
	Intro	Methods	Results	Discussion	R &D		
lexical verbs	4.9	0.6	5.7	7.1	6.2	14.1	337
adverbs	4.0	0.6	4.0	7.7	6.1	12.7	313
adjectives	3.7	1.2	4.3	6.6	4.3	11.3	285
modal verbs	2.8	0.0	2.4	7.8	3.9	9.7	237
limiting conditions	0.6	0.3	1.4	2.1	2.1	3.7	91
modal nouns	1.7	0.2	0.6	2.2	1.5	3.3	82
model, method, etc	1.0	0.4	1.2	1.8	1.1	3.2	79
lack of knowledge	1.3	0.1	0.4	1.1	1.3	2.1	53
Totals	20.0	3.4	20.0	36.4	26.5	60.1	1477

[1] Figures have been excluded for one RA not conforming to the IMRD structure.
[2] Includes all 26 articles in corpus.

The distribution is clearly related to the rhetorical functions of textual macrostructure. It is in Discussions that authors make their claims, consider the relevance of results and speculate about what they might mean, going beyond their data to offer the more general interpretations by which they gain their academic credibility. The level of generality, and therefore the density of hedges, is much higher here, as writers explore the ramifications of their results. In Introductions, hedging is largely concerned with mitigating reviews of prior research, speculating about the importance of the research and tentatively announcing findings. The Results section is where new knowledge is presented and relevant features of experimental method discussed, requiring the writer to justify the techniques used and qualify the findings produced. Hedges therefore

anticipate reader objections and pre-empt challenges to statements, largely by hedging the claims made for methods and results. Finally the Methods section, as the least obviously rhetorical part of the text, is also the least hedged with only one tenth of the items found in Discussions.

Conclusion

This chapter has provided a quantitative account of the ways hedging is linguistically realised in biology research articles. Results have been compared with those found in larger computer samples from general academic writing and with the findings of other researchers, principally those of Coates, Holmes and Butler. The findings demonstrate the numerical significance of hedging, help reveal the meanings conveyed, and offer an empirical basis for the explanatory categories discussed in the next chapter. I also hope to have shown that the interpretation of hedges in RAs cannot be captured by assigning particular semantic meanings to decontextualised forms: particular devices can only be interpreted as conveying hedging meanings in the utterance of actual sentences.

The discussion has again underlined the importance of hedging in structuring scientific communication and emphasised the variability of the means used to express it. Epistemic modality is used to hypothesise and generalise about what is possible in the behaviour of phenomena and the data show that very few original claims are presented categorically. An important finding has been that while scientific articles in the plant sciences employ a restricted number of epistemic items, they nevertheless draw on a range of grammatical and strategic categories, suggesting that the linguistic preoccupation with modal verbs is unjustified. Modal auxiliaries are by no means the most numerous items in the corpus and must be seen within a general framework in which they are one important way of indicating qualification of factuality and providing writer perspective.

Having identified how hedging is organised in the RAs and established that qualification is a central feature of scientific publication, I will now suggest a motivated account of why this should be so.

6. A pragmatic analysis of hedging

This chapter turns from a semantic to a pragmatic perspective, building on the forms identified in chapter 5 to propose a polypragmatic explanation of hedging in scientific writing. To do this I will examine text products and draw on the interpretations and understandings of biologists. I will first discuss the model introduced above (Fig 1) and then provide a detailed outline of the framework, examining each function in turn and focusing on the representative examples of each category.

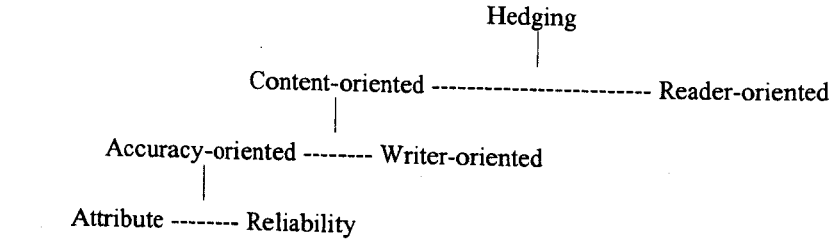


Figure 1: A model of scientific hedging

A polypragmatic model

Any analysis of epistemic uses of language must confront the fact that hedging devices are both polysemeous and polypragmatic. That is, not only do they often have different semantic interpretations, (for example between root and epistemic domains), but they may also convey a range of meanings for particular users in particular contexts. Because discourse is language used as a form of social practice, it is both facilitated and constrained by the social insti-

tutions within which it operates (Fairclough, 1992). Linguistic features therefore cannot be seen in isolation from particular socio-institutional activities and broader cultural understandings. This means that it is likely that members of different discourse communities use the resources of the linguistic system to express meanings which are particular to their own social structures and professional objectives.

I noted in the previous chapter that hedges are polysemeous in that they exhibit a number of related meanings, some more representative of a given category than others. Thus many of the cases discussed in this book reveal the following features:

1) an absence of discrete, invariant and qualitatively distinct semantic categories
2) an absence of invariant syntactic boundaries between categories
3) a qualitative similarity of characteristics which allow intermediate gradations.

These observations can also be extended to the pragmatic description of hedges. Analysis of text features cannot be isolated from the institutional and discoursal practices in which they are embedded, so in science modal expressions encode both writer-oriented and reader-oriented meanings. Moreover, not only is it impossible to relate particular forms to specific functions in any context-invariant way, but meanings are frequently expressed simultaneously, which introduces the problem of indeterminacy in specifying cases. So while hedging devices express writers' attitudes to both propositions and addressees, we cannot categorically state that a given device will always function to hedge propositional content, for example, nor that evidence for a particular function will always be found in the linguistic environment. Because indeterminacy appears to be an inherent feature of the epistemic use of language, an adequate account of hedging in scientific discourse must look beyond a mono-meaning model. In other words, hedges seem to require a 'more-or-less' rather than an 'all-or-nothing' account.

The polypragmatic nature of hedges prevents the possibility of always assigning specific meanings exclusively to particular forms[1]. This was clearly demonstrated in this study by the fact that expert informants occasionally differed in their interpretations of the same utterances, the disagreements focusing particularly on disambiguating writer and propositional hedging. The biologists

admitted that they regularly used hedges to mitigate their propositions, but they could not always identify the precise motivation for a given hedge. This therefore suggests that while the functions of hedges are important, they are not always apparent to insiders at a conscious level of awareness. The view taken here therefore predicts that there will invariably be some indeterminacy between hedging functions, so cases assigned to one category will include meanings associated with another.

In semantics, two main approaches have tried to incorporate indeterminacy in their explanations. These are gradience models, as suggested by Leech & Coates (1983), and by Palmer (1990) for modal verbs following Quirk's (1965) syntactic gradients theory, and a theory of fuzzy sets, as proposed by Zadeh (1972) and adopted by Coates (1983) and Hudson (1984). A description using a gradience model presupposes two extreme categories with distinct and clearly defined attributes to which a majority of cases can be assigned. Examples thought to be indeterminate will possess varying properties from each category and fall somewhere on a continuum between the two. Figure 2 represents this analysis.

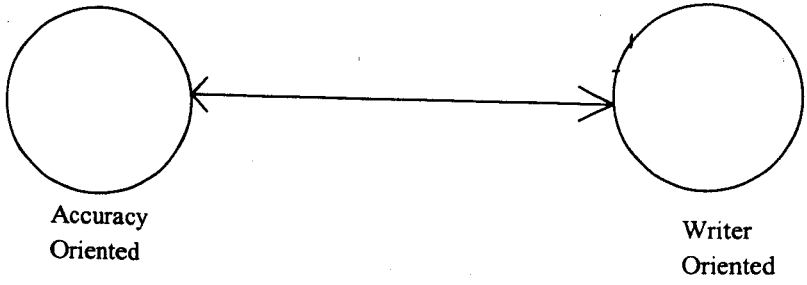


Figure 2: A gradience model of Content-oriented hedges

Such a model applied to my pragmatic account would therefore situate cases of content-oriented hedges on a cline between two poles defined by the criterial properties of "commitment" and "confidence". Most hedges would exhibit an exclusive concern with one or the other, with the remainder spread along the space between them.

While a gradience approach was initially adopted in this analysis, it quickly became apparent that such a model did not reflect the imprecision of my data. Hedging categories are virtually always indeterminate, each contain-

ing traces of meaning conveyed by the other. Again, the distinction between illocutionary and intended perlocutionary force is difficult to make as the writer's purpose is often opaque and not always accessible from surface forms, but most epistemic examples fell between the two extremes. A scale with precisely defined end points containing most examples therefore could not be sustained, and appeared inappropriate as an explanation of scientific hedging.

As a result, I followed Zadeh (1972) in using a fuzzy set model of graded membership: "A fuzzy-set is a class with unsharp boundaries, that is, a class in which the transition from membership to non-membership is gradual rather than abrupt (Zadeh, 1972: 4). Cases here are not restricted to binary endpoints but can range throughout the interval, with some examples denoting a greater degree of membership of one category than another. It is essentially a hedged theory of hedging, a perspective which makes use of the imprecise notion of category membership proposed by Rosch (1975) and developed in prototype theory. It is worth saying a little about prototypes here as they are central to the way categories are seen in this analysis.

Modern prototype theory is a central element of cognitive linguistics (Geeraerts, 1993; Rudzka-Ostyn, 1988; Taylor, 1989; Ungerer & Schmid, 1996). This perspective denies that categories exist independently of language users, and provides theoretical grounds for basing our understanding of language on the ways we conceptualise our experience of the world. In contrast to 'classical' or 'logical' approaches, cognitive linguists argue that categories are not homogeneous units with firm boundaries and clear criteria of membership. Instead, they propose a prototype model of categories which suggests that members do not necessarily share the same discrete attributes but can be linked by family resemblance. Categories are not homogeneous but have a prototype, good and bad members and fuzzy borders:

> Categories typically have fuzzy edges and might even merge into each other; some attributes might be shared by only a few members of a category; there might even be categories with no attributes shared by all their members. (Taylor, 1989:51)

The central members of a prototype category share similar meanings, and thus approach the ideal of a 'classical' category, while also admitting peripheral cases which exhibit fewer similarities. Numerous studies in experimental psychology have demonstrated the psychological reality of this structure, showing that we use prototypical colours, shapes, animals, birds, objects and so

on as points of orientation (Lakoff, 1987). Prototypical members of cognitive categories have the largest number of attributes in common with other members, and are maximally distinct from prototypical members of other categories. Marginal members share only a small number of attributes, and have some characteristics which belong to other categories as well. Thus we still recognise the ostrich as a 'bird', despite attributes which are either deviant or missing (an ability to run fast and inability to sing or fly), as a result of a family resemblance of other typical attributes (such as possessing wings, beak, feathers, laying eggs, etc.). Membership in a prototype category is thus a matter of degree: it is 'fuzzy'.

Apart from such categorisation experiments, a prototype can be seen as a schema or cognitive reference point which depends on our social and cultural knowledge. Language users' beliefs, based on the ways they routinely perceive the world and interact with their environment, are involved in identifying an item as a member of a certain category, and fuzziness represents the degree of deviation from that prototype. Part of my argument is, then, that the categories developed here to describe scientific hedging are grounded in experiential understandings and represent the ways scientists use language to get things done. The categories are culturally salient to writers because they are embedded in a social context shared by a particular disciplinary community and their fuzziness is a feature of their use. But while the motivation for hedging lies in the need to specify observations accurately and cautiously to acknowledge the views of colleagues, frequently there is a desire to accomplish more than one purpose at the same time.

In the terms of prototype theory, 'hedging' might be seen as a 'basic-level' category offering the largest number of correlated attributes, such as *weakens force of statements, contains modal expressions, expresses deference, signals uncertainty*, and so on. While forms are likely to overlap with other categories to express functions such as politeness or usuality (Halliday, 1994), the term allows a fairly reliable gestalt perception as it includes the greatest number of attributes not shared with members of other categories. We can, generally, recognise a hedge. This basic category also allows the identification of prototypical examples such as *possibility, might* and *suppose*. Member categories at the next level, content-oriented and reader-oriented hedges, contain greater overlaps so that the prototypical functions of propositional and interpersonal mitigation may be blurred. Similarly, at the next level, core examples of accuracy-oriented hedges will express a concern with precision of expression,

while we are likely to find that peripheral cases give more emphasis to the writer's self-protection.

Zadeh observes that fuzziness plays an essential role in human cognition and believes that most classes in the real world are fuzzy. He admits however that the context-dependency of most hedges prevents precision and that pragmatic interpretations require a degree of subjectivity. Often it is only possible to say that *X* is a core or peripheral example of *Y*, and I will therefore employ general descriptive terms such as "strong" and "weak" when discussing cases. The salient points of a fuzzy-set approach are thus preserved, while avoiding the need for precise quantification. Figure 3 is based on Coates (1983) and illustrates the essential feature of fuzzy-sets: categories such as "core", "skirt" and "periphery" have "unsharp boundaries" which allow a gradual transition to membership. To summarise, expressions at the core will most closely approximate to the meaning of that particular category, while examples at the skirt will exhibit less precise meaning.

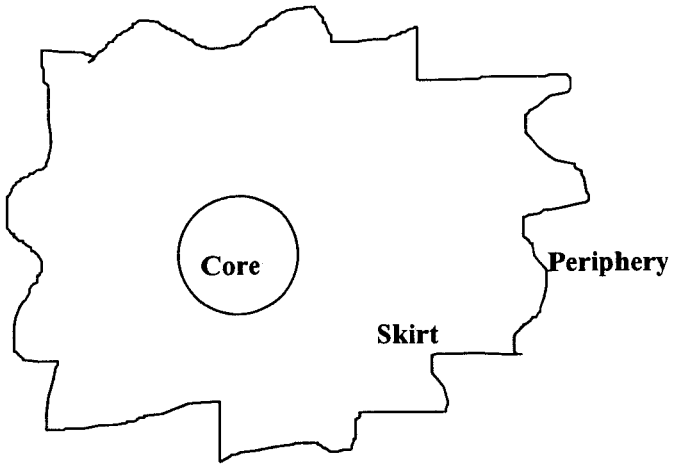


Figure 3: Representation of a fuzzy-set (after Coates, 1983)

The next two sections focus on content- and reader-oriented hedges, describing each type of function in detail, and examining their principal forms of realisation. The difficulties of precisely coding all individual examples were referred to above, but it should be noted that each of the examples in this

chapter was considered by three native speaker coders, all working independently and all familiar with the categorisation scheme employed. This procedure yielded a Kappa interrater reliability coefficient of .79, indicating relatively high agreement. The framework and each example utterance was then discussed with at least two scientist informants and their views elicited on the functions performed. While complete consensus was impossible, their views were used to fine-tune the framework and revise coding decisions for ambiguous cases.

Content-oriented hedges

Content-oriented hedges serve to mitigate the relationship between propositional content and a non-linguistic mental representation of reality; they hedge the correspondence between what the writer says about the world and what the world is thought to be like. The motivation for content hedges falls into two overlapping categories, concerning the writer's interest in i) stating propositional accord with reality, or ii) seeking self protection from the negative consequences of poor judgment. There is however, often an element of each purpose in any particular example, and while contextual information assists in distinguishing the dominant reasoning, it does not always determine precisely what it is. I have referred to these two forms of motivation as Accuracy and Writer-oriented hedges, and I will now discuss each in turn.

a) Accuracy-oriented hedges

These refer to the writer's desire to express propositions with greater precision in areas often characterised by revision and reinterpretation. Hedging here is an important means of accurately stating variable results or uncertain claims with appropriate indications of reliability (Rounds, 1982, Skelton, 1988b). It seeks to meet adequacy conditions by reducing the risk of negation on objective grounds. Almost all academic discourse is a balance of fact and evaluation, as writers try to present information as fully, accurately and objectively as possible. These Gricean constraints are evident in research articles because science is concerned with varieties of cognition, and cognition is often hedged with the writer's interpretive statements. To quote one biologist:

> With the biological system you are often not quite sure, while you can say that findings point to a particular conclusion, there are always some doubts about interpretations.

While accuracy-oriented hedges may serve certain rhetorical purposes[2], their principal function is to achieve precision; either by a) marking a departure from an ideal or b) indicating that a proposition is based on plausible reasoning or logical deduction in the absence of full knowledge. They request that the reader interpret what is stated is true as far as can be determined.

Accuracy-oriented hedges specifically address the writer's concern with the relationship between propositions, or propositional elements, and reality. They are the archetypal statements of academic writing since they help determine exactly what the state of knowledge is, promoting an objectivity which enables the reader to distinguish the actual from the simply potential or inferential. Accuracy-oriented hedges are a running contribution to the evaluation of the weight of statements in truth-seeking talk. They indicate that the writer has less than full warrant for unmodalised declarative assertions while commitment to the statement is either not involved or is secondary to this function. The two types of accuracy-oriented hedges, attribute hedges and reliability hedges, each has a different motivation, function and forms of realisation, and will be discussed separately.

Attribute hedges: Scientific schemata assume that there is a world independent of language which can be described, more or less faithfully, by linguistic expressions. The ability of words to represent an objective domain of non-linguistic facts, however, is mediated and organised by cognitive processing and crucially dependent on relevant background knowledge. Word meanings are characterised by shared, conventionalised and, to a large extent, idealised knowledge embedded in cultural beliefs and practices (eg Taylor, 1989). How writers construe a situation helps determine the meaning of the sentences chosen to describe it. But natural phenomena do not always dovetail with scientists' idealised cognitive models of what the world is like (cf. Lakoff, 1982), and experimental results frequently vary from how scientists imagine and structure them. Variations from an idealised conception of a particular relationship, behaviour, procedure, or appearance are common in science. In order to accurately describe such vagaries of experimental conduct, attributes are frequently hedged.

The use of attribute hedges allow deviations between idealised models of nature and instances of actual behaviour to be accurately expressed. They enable writers to restructure categories, define entities and processes exactly, and to distinguish how far results approximate to an idealised state. They specify more precisely the attributes of the phenomena described.

This is clear in the following examples, where hedges are used to indicate variability with respect to certain descriptive terms. In (1) the degree to which the detected response varies from what is considered "normal" for the behaviour of those proteins is hedged, while in (2) it is the comparison between two variables in terms of a particular feature. In (3) "essentially" denotes a deviation from an idealised conception of a particular process.

> (1) The response of the assembly of PSII proteins to the solute environment is unique in some ways, but quite normal and predictable in others. (A4:33)

> (2) Although variable, the isoelectric point of kunitz seed inhibitor is generally lower .... (D2:264)

> (3) The partially purified PEPc kinase phosphorylates PEPc with a stoichiometry approaching 1 and causes essentially the same reduction in sensitivity.. (C1:194)

Clearly researchers seek to classify and comment on findings as accurately as possible using a lexicon of scientific terms, but often there is a discrepancy between actual results and either an expected state, or the concept routinely available to explain it. A hedge therefore indicates that results vary from an assumed ideal of how nature behaves and allows a better match with familiar descriptive terms. Attribute hedges generally cluster around this pragmatic core, and involve the use of a finite set of items which Ernst (1984) labels "degree of precision adverbs". These specify the extent terms used to describe events and relationships vary from an endpoint on a cognitive scale and convey a range of items when used with numerical data. Such devices have received some attention in the literature (eg Powell, 1985; Channell, 1990 & 1994), and are similar to Prince et al's (1982) "rounders" which describe devices used to indicate degrees of category membership.

As discussed in the previous chapter, a number of these act as "downtoners" to weaken the force of an attribute and can be graded as to their strength. The examples below show items hedging deviations from "ideal" correlations, causes, behaviour, and so on. Forms which realise the greatest effect almost negate the force of the term modified (6 & 7).

(4) This appearance of kinase activity correlates quite well ... (C1:193)

(5) ..behaves more or less like the native ACC synthase enzyme.. (B5:106)

(6) The proteolysis of PEPc was almost completely prevented by .. (C1: 192)

(7) ... pTACC-C7 ACC synthase was barely detectable. (B5:107)

The class of adverbs discussed earlier as style disjuncts, together with some content disjuncts, also act as attribute hedges. These indicate the degree of precision intended and convey the sense in which an idea may be held to be true.

(8) .. at an acidity that generally guarantees a quite stable assembly of the PS II polypeptides. (A4:30)

(9) ..decreases by approximately 60% at 44°c, ... (B2:192)

The attribute can also be qualified by indicating the precise standpoint from which the reader might evaluate the truth of the claim, loosening a strict interpretation of how the descriptive terms used should be understood in characterising real-world entities:

(10) In a strict sense, these experiments measure inhibition of [^{3}H] NPA binding rather than competition ... (E1:291)

(11) Finally, from a practical point of view, the $tRNA^{asp}$ extended transcript could serve to produce large quantities of wild type $tRNA^{asp}$ transcripts.... (B6:7)

In each case the hedge acts to qualify the predicate intensity or the validity of the state of affairs expressed in the proposition. In sum, writers use attribute hedges to seek precision in expression, and core examples encode variability, rather than writer perspective.

Reliability hedges: These hedges acknowledge the writer's uncertain knowledge and indicate the confidence he or she is willing to invest in the validity of a claim: they say, 'I do not speak from secure knowledge'. Reliability hedges thus express simple subjective uncertainty in a proposition and are motivated by the writer's desire to explicitly convey the extent to which, and in which, it corresponds to his or her understanding of 'truth'. Informants emphasised that this is an inherent feature of biology:

> You're dealing with living organisms and, at whatever level you study, its very hard to say definitively 'this is how things are'. Actual truths are not easily dug out.

Reliability hedges deal with the epistemically possible and contingent but can reach uncomfortably close to objective possibilities and the logician's category of "alethic modality". However, the analyst's ability to establish the evidence for subjective inference in a text, as opposed to objectively verifiable likelihood, is often complicated by the inaccessibility of participant understandings.

For scientists, scientific truth must stand independently of its enunciation and enunciator. Reliability hedges therefore play a crucial institutional role in indicating an assessment of the reliability of truth, keeping interpretations close to findings, where claims may be less tenuous:

> (12) However, the opposite is also possible, and it cannot be ruled out that the 21-kD polypeptides seen in the bacteroid lane and in the soluble proteins (D2:260)

> (13) It is therefore possible that cv. Firstmore might be different not only in its chromosomal arrangement but also in displaying a particular mechanism .. (F3:17)

> (14) Together these results are consistent with the possibility, although do not prove, that the capacity to .... (D5:40)

In these examples the writers make clear that they have reservations concerning whether the stated situations actually obtain. The principle motivation is a desire to clarify the state of knowledge, a hedge against complete accuracy, rather than a wish to seek protection against overstatement. In these core cases acknowledgement of factual uncertainties predominate over attempts to disguise the author's opinion.

The most common means of expressing reliability accuracy-oriented content hedges is by conventional epistemic forms in the main grammatical classes, principally modal verbs, modal adjectives and nouns, and content disjunct adverbs expressing propositional uncertainty. Respondents agreed that the following examples appeared to be concerned with precision.

(15) This insertion, which we suspect is the membrane anchor, could associate peripherally with the membrane or might span half the bilayer .. (C3:283)

(16) .. but the high activity of NAIV raises the possibility that the molecule of interest may be IAA-valine. (E1:294)

(17) ...it appears possible that the mechanism causing the light-activated fluorescence quenching may be triggered by either photosystem. (A3:7)

(18) Oscillations in fluorescence and O_2 evolution activity are probably an expression of this short-term adjustment capacity. (A1:272)

Content disjuncts are a particularly common option available to writers wishing to mark reliability. These comment on the probability of the content of a proposition being true and include both adverbs of certainty, which simply convey doubt on the information (19-21), or of mental perception, which show how results are understood (22 & 23):

(19) These EGTA clots are transparent and possibly comprised of thinner fibres than the opaque Ca^{2+} clots. (C2:273)

(20) These screens were performed in white light, presumably precluding detection of mutants in the FR-HIR. (D5:40)

(21) .. is likely to be due to the modification of these factors. (C4:167)

(22) This can be rationalised intuitively with the favourable effect of ... (A4:34)

(23) Inactivation of the plastid genes is apparently not due to major recombinational events at the level of ptDNA, since restriction enzyme patterns ... (F3:12)

As reliability hedges specify the actual state of knowledge, there may be some hesitation about the strength of the connection between observed and assumed events:

(24) Slow PS I turnover may favour Q_B binding to PS II or at least not prevent it. (A1:268)

(25) This implies that the extent to which the endophyte might effect N metabolism under field conditions could also depend considerably upon other interacting factors. (E2:731)

In (24) the writer proposes two alternative formulations of the claim, while in (25) a note of caution is introduced concerning the possibility of other variables influencing the connections discussed. In both cases the writer avoids stating conclusions too confidently.

A characteristic reluctance to stray far from the data can create confusion with writer-oriented hedges, although these include an additional reluctance to make a commitment to the suggested explanations. In the following examples the choice of an epistemic device in initial position accents the hedge and provides the reader with an alternative interpretive framework, thus presenting the statement as a hypothesis and hedging the writers' commitment by distancing them from their propositions. In (26) then, the adjective *conceivable* suggests that while this explanation can be imagined, it is not necessarily the one subscribed to by the writers. In the case of (27) there appears to be a firmer commitment to the second option which hedges only confidence.

(26) Another conceivable reason for this decrease could be a change in Trp τ_0 value under the influence of the electric field induced by P^+ and Q^- charges. (B4:231)

(27) Possibly, phosphorylation of ACC synthase could contribute to the existence of its different pI forms. Alternatively, phosphorylation - dephosphorylation may be involved in the inactivation / activation of the enzyme in vivo. (B5:107)

Thus the major characteristic of accuracy-oriented hedges is that claims about relations are made contingent due to admitted limitations of knowledge. This is occasionally expressed by explicit reference to the existing state of knowledge:

(28) Almost nothing is known about the chemical constitution of the fluorescent material. This could be due to protein-metal complexes such as .... (D2:262)

(29) The mechanisms involved in the control of conductance through the stomata are poorly understood. The guard cells could respond to messages from the leaf mesophyll, or have their own CO_2 sensing mechanisms because they possess chloroplasts. (E4:301)

So while some examples are difficult to distinguish from defensive lack of commitment, all cases of reliability hedges express a conviction about propositional truth as warranted by deductions from available facts, relying on inference, deduction or repeated experience. They refer to present states and are usually in the active voice without writer agentivity.

To summarise, the defining characteristic of accuracy-oriented hedges is their contribution to the precision with which a claim is made. Their major function is to provide a specification of the state of knowledge rather than hedge the writer's commitment to the claim. Both forms are principally concerned with the interpretation of the world via laws of reason, and seek to increase the exactness with which a proposition is expressed, either by indicating how far entities approach an ideal endpoint on a cognitive scale, or by stating more precisely the writer's assessment of the certainty of a statement. In either case, core examples exhibit a concern with propositional precision. The writer's desire for protection from the consequences of the statement being wrong is either not implied, as in attribute hedges, or is of secondary importance.

The line between propositional and writer oriented motivation is a difficult one to draw, however, and hedging one's confidence in the accuracy of

stated knowledge and hedging one's personal commitment to those statements are often accomplished by the same linguistic means. Thus a precise functional assignment of cases is not always feasible. But while it was not always possible to interpret a particular utterance as a definitive example of a given class, in most cases analysis of the co-text and an awareness of the overall purpose of hedging in the genre allowed raters to arrive at corresponding interpretations.

b) Writer-oriented hedges

Scientific statements publicly link scientists with their knowledge claims and therefore represent careful decisions concerning the degree of commitment the writer wishes to invest in them. Writer-oriented hedges are often associated with higher level claims than accuracy-oriented ones. Usually the most significant claims are those which have the widest relevance and therefore assert the greatest generality. Increasing the generality of claims, however, necessarily weakens the certainty that can be invested in them, so writers must also protect themselves from the possible consequences of error by limiting their personal commitment (Hunston, 1994b). Thus where the writer is seeking to place results in a wider context and demonstrate a contribution to scientific knowledge rather than simply interpret findings, claims carry a considerable risk and an element of self-protection may be necessary. One of my informants expressed it like this:

> Scientists are fallible and so are their methods. Most of the time you could be right but there is often a chance that it might be something different and you'd better make sure you let people know that before they let you know.

Such hedges therefore *diminish* the author's presence in the text rather than *increase* the precision of claims, toning down the language they use to express their commitment to their research claims.

Writer-oriented hedges therefore create a clear pragmatic contrast with other content hedges: Accuracy-oriented hedges are proposition-focused and seek to increase precision by referring to the exact state of knowledge or to how the proposition is to be understood; writer-oriented hedges are writer-focused and aim to shield the writer from the possible consequences of negatability by limiting personal commitment. As suggested by the fuzzy-set model, this is necessarily, and almost invariably, a matter of degree. Prince et al (1982), Rounds (1982), Powell (1985) and Nash (1990) have all suggested in various ways that hedges serve as an insurance in helping writers protect their

reputations and limit any damage incurred from categorical commitments. Such devices are what Prince et al (1982) call *Shields,* and their main role is not to affect the propositional content of an utterance, but to imply that the writer is not fully committed to its truth.

To some extent this usage follows Lakoff (1972) in associating hedges with "fuzziness" but I am not suggesting that scientists seek to gain acceptance for claims through purposive vagueness or fudging. Instead, it is the strength of the relationship between the writer and the proposition that is made fuzzy. Hedges here represent strategies of saying less than one means by blurring the precision with which one associates oneself with one's claims. These hedges help minimise writers' personal involvement and allow them to maintain a distance from a proposition. This reduces the probability of refutation and allows scientists to "anticipate and discountenance negative reactions to the knowledge claims being advanced" (Swales, 1990: 175). Distinguishing speculation from certainty in this way is a hedge against being caught in error. Oswald Avery, the eminent British microbiologist, alluded to this function when he wrote: "It's lots of fun to blow bubbles, - but it is wiser to prick them yourself before someone else tries to" (quoted in Judson, 1995:39).

This is most obviously apparent in cases where writers seek to present claims for which they lack adequate grounds. The cluster of hedges in these examples, in conjunction with use of the passive, attests to the desire of the writer to gain some distance from the propositions rather than simply specify the accuracy of the claim. While a reference to inadequate knowledge (30) often suggests a propositional hedge, its use as a springboard into speculation here suggests a more writer-focused motivation. My informants concurred that such conjecture allows writers to stake a claim for priority without the attendant risks of rejection:

> (30) Although it is premature to answer this question it might be suggested that synthetases present in nuclei [33] could be involved in the regulation of the processing of tRNA precursors. (B6:7)

> (31)The present work indicates that the aromatic ring to which the carboxyl group is bound is not necessary, provided that a bulky substituent is ..... (E1:293)

In core writer-oriented examples care is taken to avoid assuming explicit responsibility for an assertion, while seeking to secure "uptake" by moving the reader to the writer's standpoint. There are a number of ways this may be achieved, and it was observed above that placing an epistemic element in initial position can serve this function. The most distinctive characteristic of writer-motivated hedges, however, is the absence of writer agentivity. Assertions can be coded to reduce the author's responsibility for performing the act by the use of passive constructions (32), or making a clause the subject (33):

(32) The BS fraction is assumed to originate from the center of the .... (A5:176)

(33) It was assumed that the phosphorylation of EF-2 may play a.... (B3:219)

A common strategy is the construction of "abstract rhetors" which, by nominalising a personal projection, suggests that the situation described is independent of human agency:

(34) These data indicate that phytochrome A possesses the... (D5:46)

(35) Taxonomic evidence suggests, therefore, that a species ... (D1:638)

(36) The model implies that the function of granna is to ... (A1:268)

Writers make creative use of limited conceptualising devices to describe experiences and in so doing apply semantic structures, such as agentivity, to non-paradigm situations, such as data or results, which lack animateness and intentionality. Cognitive linguists argue that "perspective choice determines the way a given situation is linguistically organised... representing different ways of structuring (and understanding) that situation" (Nishimura, 1993:501). A decision to bring "The data" in (34), for example, into perspective is a marked choice which presents a view where data, vested with agentivity, acts of its own accord with the primary responsibility for an interpretation attributed to it. The syntactic template also encourages us to credit that an inanimate or abstract entity may act as a cause of an event, while leaving open the question of predicate truth.

The role of epistemic lexical verbs, and particularly, judgmental lexical verbs is central in qualifying speculative claims, and is a principal means by which writer-oriented hedging is accomplished:

(37) The relative size of SCPEPCD1 transcript pools suggested by PCR assays is consistent with northern blot evidence for greening leaves and stems. (F4:671)

(38) Circumstantial evidence indicates specific shifts in the mRNA levels for different genes. (B2:189)

(39) The theoretical line assumes that the partial pressure of CO_2 is the same in the intercellular spaces and the chloroplast stroma ... (E4:299)

These verbs with non-writer agents indicate that the writer is not prepared to personally guarantee the proposition, the tentativeness relating principally to the commitment the author wishes to bestow on the statement rather than a strict concern with the truth of its propositional relationships. This form of hedging mainly involves the use of verbs of speculation (*assume, propose, predict,* etc) as the possibility of deductive verbs (*calculate, estimate, infer*, etc) occurring is obviously restricted in impersonal contexts.

A further means of implying that the writer does not wish to be thought fully and personally committed to a belief in the proposed state of affairs is the use of evidential lexical verbs, often in combination, and again in phrasings which avoid attributing beliefs to the writer:

(40) From this discussion, then, it would appear that some of the changes in the amino acid concentrations ... (E2:732)

(41) Thus, both molecular-genetic studies and chemical studies point towards an involvement of the 530 loop with streptomycin resistance. (F1:502)

(42) It seems that the stomata do not use the Calvin cycle .... (E4:301)

Rounds attributes such verbs to an accuracy-oriented function (Rounds, 1982:16). But the trouble taken by the writer to remove him or herself from the statement appears to signal a clear reduction of speaker commitment rather than

"acting to make the relationship between the subject and the predicate fuzzy". The avoidance of a strong alignment with a claim is the defining feature of writer-oriented hedges. Another strategy to achieve this is to attribute the source of a claim, or its underpinnings, elsewhere. Thus writers may refer to wider bodies of knowledge in the literature when moving away from what can be confidently implied by their results or methods.

> (43) This hypothesis seems plausible because UV-B-absorbing flavonoids accumulate in leaf epidermal cells, where they may protect the inner cell layers from UV-B damage (Caldwell, et al, 1983; Beggs et al, 1986). (D3:171)

As one informant observed here, "they know that the flavonoids are UV-B-absorbing and that the inner cell layers can be damaged by UV-B, but they haven't been able to prove that is what happens. They hedge because they want to claim it, but can't prove it". Such considerations are at the heart of writer-oriented uses.

It is interesting to observe that writer-oriented hedges, principally through the use of lexical verbs, foreground the procedures and methods of science, a feature less salient in other forms of hedging. Clearly, such strategies also imply a lack of confidence in propositional validity and can strongly suggest an accuracy-oriented interpretation, however, my scientist informants unanimously interpreted these forms as protective strategies. References to methods, the models employed or the conditions under which results were obtained, thus appear to be an important means of hedging personal commitment.

> (44) Therefore, results under these conditions were not considered conclusive. (E2:730)

> (45) There was a low level of staining that was difficult to localize but was probably associated with the vascular tissues. (D1:635)

> (46) Viewed in this way the concept of lateral heterogeneity becomes obsolete because the distinction between granal and stromal lamellae for ... (A1:268)

Frequently statements combine the two strategies, referring to both the warrant for the claim and citing supporting testimony. This is the case in (47), where the general tentativeness of the claim is reinforced not only by a reference to evidential and literature sources, but also by the use of two riders which hedge the writer's full commitment.

> (47) Thus the evidence strongly favours the conclusion that phytochrome A is the primary, if not exclusive, FR-HIR photoreceptor, consistent with previous physiological studies (Beggs et al, 1980; Holmes & Schaffer, 1981; Smith & Whitelam, 1990). On the other hand, the possible involvement of phytochromes C, D, or E in an interactive or synergistic manner with phytochrome A cannot presently be ruled out. (D5:45)

The pressures of modern careerism and the need to participate in the exchange of findings to achieve professional recognition in rapidly changing scientific fields encourage publication. Both cooperative and competitive forces, i.e. both discovery and recognition, therefore provide a context in which productivity is a measurement of worth, and create a situation which promotes early and frequent publication. James Watson, Nobel laureate for the discovery of the structure of DNA, argues this vigorously:

> It starts at the beginning. If you publish first, you become a professor first; your future depends on some indication that you can do something by yourself. It's that simple. Competitiveness is very dominant. The chief emotion in the field. (Quoted in Judson, 1995: 194)

At a more mundane level, several of my informants mentioned the effect of being required to publish at least two papers a year by university systems:

> As long as this attitude lasts, people are going to publish without being as sure as they'd like to be. They are therefore going to cover themselves.

Work in progress and preliminary conclusions therefore characterise much of the scientific literature, which accordingly has to be appropriately hedged against later falsification. The writer may be attracted to a position which lacks sufficiently convincing evidence, or may wish to present findings where hypothesis testing is extremely difficult.

In sum, writer-oriented accuracy hedges perform the pragmatic function of hedging the writer's commitment to the content of the proposition while leaving its truth value open. They diminish personal responsibility for a variety of reasons, and are often employed as a result of preliminary findings, small samples, doubtful evidence, uncertain predictions, limits of experimental conditions or an acknowledgment of imperfect measuring techniques (eg Latour & Woolgar, 1979: 75-87; Fahnestock, 1986).

Most importantly however, they result from writers' attempts to enhance their academic credibility by gaining acceptance for the highest level claims they believe they can make for their results. Writer-oriented hedges assist this while affording protection from the full effects of the eventual overthrow of the claim by colleagues. Indeed, most scientific assertions are eventually replaced, but a hedge can so alter the context of the statement in which it occurs that the resultant claim is very hard to falsify. A writer-oriented hedge serves to steer a reader's attitude away from a stipulatively literal interpretation which would be the case with an accuracy motivated usage. Attributional vagueness thus constitutes a viable communicative option which functions to protect the writer from the possible consequences of negatability.

To summarise this section, it is clear that these content-oriented hedges are a significant communicative resource. They enable scientific writers to negotiate the precision of a claim and convey an attitude to that claim, offering the reader a means of assessing the certainty of its truth and degree of personal commitment to it. On one hand the writer seeks to present statements with appropriate accuracy, on the other, to make the strongest claim possible while limiting the damage of being wrong. The achievement of adequacy conditions therefore involves mitigating the propositional content of an utterance in order to obtain ratification of that content. This is principally achieved by a repertoire of lexical markers and by various hedging strategies.

The difficulties of precisely categorising polypragmatic forms has been mentioned on several occasions, and while the interpretations of expert informants responding to particular examples was often helpful in resolving ambiguity, it also demonstrated the impossibility of reaching complete agreement in all cases. The fact that hedging devices can perform several functions simultaneously means that neither a purely formal treatment nor a detailed contextual analysis will invariably determine an unequivocal function, even for insiders[3]. The same forms, for instance, can convey elements of both accuracy-oriented and writer-oriented meanings as the distinction between the qualification of

writer confidence and of writer commitment is a thin one. In terms of the fuzzy-set model, core examples of accuracy-oriented hedges will carry proposition-focused meanings, but the majority of cases are likely to occur in the skirt and periphery, carrying increasing shades of writer-focused meanings, particularly as riskier and more significant claims are made. Pragmatic variables for locating particular examples at the core or periphery will be discussed further below, but now I turn to the second major dimension of hedges.

Reader-oriented hedges

While the characteristic features of epistemic mitigation are closely linked with an objective dimension, they are also deployed for a more interpersonal purpose. Securing ratification of scientific claims also involves reducing the risk of negation on subjective grounds, requiring writers to attend to the interactional effects of their statements. Core examples of reader-oriented hedges thus address the various dimensions of the social relationship between writer and reader in this genre.

Mitigating the illocutionary force of speech acts is a common conversational function of hedges (Kennedy, 1979; Holmes, 1982 & 1988a, 1995; Stubbe & Holmes, 1995). This has been attributed to the operation of rhetorical politeness (Fraser, 1990; Leech, 1983), the expression of deference (Fraser, 1978 & 1980; Fraser & Nolan, 1981), or the employment of strategic politeness in minimising threats to face (Brown & Levinson, 1978). Huebler (1983: 23) places such hedges in the sphere of ethics, as they "help to reinforce the moral maxims by which one's behaviour, verbal or otherwise, shall not harm the person addressed". Scientists themselves however are not concerned with why the conventions of hedging have emerged. While my informants clearly saw uncertainty and protection underlying hedges, they had difficulty in recognising that they might act to mitigate impositions on colleagues and were not conscious of courtesy or deference in their own writing.

Myers (1989; 1990) has suggested that avoidance of face-threats to the reader is the main motivation for hedges in scientific discourse, as writers must defer both to a general scientific audience and to a group of colleagues engaged in the same specialised field to gain acceptance for their work. But any similarity in the use of mitigation in these two domains is largely superficial, as a straight-forward application of imposition, distance and social power does not

explain scientific hedging. A simpler explanation is that such collegiate deference occurs because making apodictic statements about the world is inherently face threatening to one's peers. Essentially, in presenting a claim, a writer also projects a particular *persona* which subtly informs the reader of the writer's beliefs about the scientific endeavour, the communication process and conceptions of honesty and reasonableness (Campbell, 1975). This created personality is crucial to achieving rhetorical objectives as it also conveys an attitude about the reader and his or her role in the negotiation of claims.

Presenting claims as ex-cathedra assertions displays an unacceptable deviant persona since it ignores any involvement by the reader in the ratification of knowledge. Categorical assertions leave no room for negotiation: they imply a certainty in arguments that require no feedback, and this relegates readers to a passive role. Hedged statements, on the other hand, mark claims as provisional, they invite readers to orientate themselves to the discourse and engage in a dialogue. The effects produced by a scientific text depend as much on the attitude of the reader as on the text, and in this function hedges request the audience to respond in some way, they solicit collusion by addressing the reader as someone possessing good sense and an ability to participate in the discourse with an open mind. Good arguments are only "good" from a particular perspective and hedges work to create this perspective. Once this is achieved, arguments can be based on other criteria.

Together with this interpersonal dimension, there is a normative aspect of reader-oriented hedges. The reader-oriented use of hedges is not just an observation of ethical precepts, but represents conformity to research community expectations concerning deference due to colleagues in presenting information. One of my informants acknowledged this point in this way:

> We don't encourage people to go out and nail their colours to the mast, as this is not how you get your ideas heard. You must relate what you have to say to your colleagues

Underlying scientific practices is a dynamic social control which produces conformity to its norms and values. Such norms are not formally codified or legally binding, and would require extensive study to identify, but acting as a scientist involves implicit 'rules' of interpersonal behaviour (eg Becher, 1989). These are internalised and, consciously or unconsciously, reflected in scientists' communication styles. Convention, rather than calculation of personal cost, may drive this system but it is guided by relationships of power and underpinned by the possibility of sanctions which operate through the peer

validation of claims and which can influence one's career. Simply, the fact that knowledge accreditation is a social process means that the public achievements and reputation of every scientist depend on the opinion of colleagues within the same narrow specialism.

So while socialisation and interpersonal factors no doubt play an important part in acquiescence, conformity to normative controls is principally due to "institutional recognition" from colleagues (Hagstrom, 1965). Having research contributions accepted in reputable journals, being cited by others, acquiring research grants, receiving professional honours, etc. are badges of success which depend on adherence to scientific norms. The simple act of participating in the formal communication system emphasises to the individual the extent to which his or her investigations and discoveries are embedded in, and dependent on, the work and acceptance of others.

A paper must therefore carry conviction, but must also appeal to the reader as a thinking, knowledgeable scientist. Publishing a paper involves entering into an interactional contract with one's readers, endorsed by the research community and concerning the conduct of scientific debate. The elaborate communication system which manages publication and validates knowledge thus also regulates authors and referees, and helps enforce strict conventions on style and presentation. This framework of institutional rights and obligations governs interaction between scientists and enables participants to get on with the task at hand. The writer's adherence to this contract confirms a responsibility to the reader and helps determine the choice of linguistic form employed.

Once again, the data does not allow such "altruistic acts" to be unequivocally distinguished from "self serving acts" (Fraser, 1980), as many of the same devices are used to convey both types of hedge. It is worth emphasising, however, that from a formal point of view hedges constitute an extremely heterogeneous group, and while content-oriented hedges are normally expressed by conventional formats, principally epistemic lexical items, many realisations associated with writer-oriented hedges include expressions which would not usually be recognised by linguists as hedges. Semantically, however, all hedges are linguistic resources for writers to comment on the language they are using. In some circumstances this means they can only be identified by their function.

This is most apparent in the disguise of criticism or disagreement. Scientists tend to over-emphasise the degree of consensus in science, but disputes obviously occur and the use of hedges enables them to be acceptably managed

without open conflict. There is a lack of consensus regarding the strength of the many devices used to soften academic criticisms (Pickard, 1993a), but the framework proposed here suggests two major ways that disagreement can be hedged. First, the use of impersonal expressions to comment on others' work indicates a writer-oriented hedge. In such cases writers are likely to be mainly concerned with protecting their reputations and so criticisms are often diffuse, with a tendency not to attribute what is criticised to any particular source:

> (48) Thus, the existence of multiple gene families and expression of multiple ACC synthase isoforms question the validity of earlier results obtained with either crude plant extracts or partially purified enzyme preparations. (B5:103)

> (49) This procedure only identifies methylated nucleotides located within the recognition sequences of the sensitive enzymes. In spite of its shortcomings, the method has been widely employed to evidence this type of modification in a number of genomes including plastid and nuclear ones [22-24]. (F3:16)

This contrasts markedly with the use of reader-oriented hedges to damp-down critical comment of others' work. None of the following examples contains a locution approaching outright criticism, and it is the common choice of personal subjects which is critical:

> (50) We do not know the reason for the discrepancy between our results and those of Ngernprairitsiri et al [16, 23], but it might reflect genetic differences in the cultivars employed. (F3:17)

> (51) From our investigations we conclude that the data of Wydrzynski et al can be seen in a different light when the adverse effects of CI deprivation are seen in conjunction with the consequences of an omission of polyolic solute from the incubation medium. (A4:34)

> (52) Quick et al (27) also noted that there was no consistent effect of reduced Rubisco content on the stomatal conductance of their antisense plants but observed that conductance was insensitive to CO_2 (30). Our

results do not support the latter observation, because conductance decreased with increasing p1 in both control and anti-SSu plants (fig 5). (E4:301)

(53) There has been some disagreement concerning whether illumination increases the V_{max} of PEPc (eg refs 1, 2, 4, 11). In our hands, there was no significant change in V_{max} on illumination. (C1:191)

The frequency of personal subjects in science is confirmed by Banks (1994) who found that first person pronouns occurred four times more often with lexical hedging verbs than in his corpus as a whole. Banks however appears to be mistaken in concluding that "the combination has the effect of reducing the force of (of trimming) the hedge" (ibid: 590). My interpretation is that an overt acceptance of personal responsibility mitigates the expression of a proposition and signifies a reader-oriented hedge.

In the terms discussed earlier, reference to personal involvement moves the discourse from an empiricist to a contingent frame, alluding to an interpretive context in which facts do not speak for themselves but require human intervention to decode. One informant in this study suggested that writers often make use of the personal form to limit the acceptance of the view proposed to that writer only. Myers makes a similar point on this issue, observing that expressions of personal belief weaken claims because they are inconsistent with the supposed universality of scientific knowledge (Myers, 1989:14). Rhetorically then, this strategy functions to represent both the source and the status of the claim as individual rather than collective, making the claim dependent on activities which might be regarded as essentially fallible or limited in their applicability.

As we saw in chapter 2, the discursive routines developed for the presentation and accreditation of scientific knowledge seek to conceal evidence for the constructedness of that knowledge. The possibility of knowing the external world is limited to formal methodological procedures approved by the discipline and conveyed in a form which emphasises impersonality, cumulativeness and empiricism. To depart from this is to draw attention to a social form of accounting which requires a different response from readers, appealing to them as colleagues with similar insider knowledge and capable of drawing the same conclusions. In my data a reference to the writer as one who holds a different position therefore seems to explicitly mark it as alternative view rather than as

a definitive statement of the truth. The hedge signals that this is a personal opinion, allowing the reader to choose the more persuasive explanation.

Personal attribution is therefore also used to express reader-oriented hedges when introducing claims. Like writer-based hedges, these make frequent use of epistemic lexical verbs; verbs of judgment (54-55) and deduction (56-57) are common, while evidential verbs are restricted to personal subjects and therefore only occur in writer-oriented hedges:

(54) Thus we propose that this insert is the major site of interaction with the membrane. (C3:283)

(55) I believe that the major organisational principle of thylakoids is that of continuous unstacking and restacking of sections of the membrane .... (A1:267)

(56) Of this percentage, we calculate that more than half is present in the coiled-coil portion and the rest in the globular domain. (C2:274)

(57) We infer that the rate becomes limited by the rate of regeneration of RuBP-saturated Rubisco kinetics, ... (E4:301)

Thus explicit personal alignment with findings, models and analyses generally conveys a reader-based hedge[4]. By specifying a personal source however, the writer shifts the interpretive frame, drawing attention to the relation of the work to the investigator, and signaling that the claim is left open to the reader's judgment :

(58) Our analysis of the fluorescence results suggest that the largest effect of light exposure is manifested by the α phase where about 50% of the reaction centres are changed to quenching centres. (A3:8)

(59) My analogy is with a conveyor belt. (A1:267)

(60) Our interpretation of these results is that the total level UV-B ... (D3:175)

A more subtle indication of deferring to the reader in this way is to offer a claim as one possibility among many, and English offers a variety of means of suggesting alternative interpretations. These include hypothetical condition-

als (61) and, at the risk of seeming to mine the seam of hedging too deeply, even indefinite noun phrases (62):

(61) ...if the hemoglobin in the root were mainly at the tip, it would have a local concentration 100-fold or more greater than originally deduced. (D1: 638)

(62) From these findings and from electron microscopy studies, a model implying almost complete lateral heterogeneity in the thylakoid membrane could be suggested, ... (A5:172)

In addition to the use of personal subjects with lexical verbs, personal attribution, and indications of alternatives, a relationship with the reader may be invoked by the use of questions. Questions in RAs signal an important unresolved issue or the tentativeness of a solution, but they also genuinely seek a response. In so doing they involve the reader more closely in the research and convey the communality of the scientific quest:

(63) Could such a putative interaction of an aminoacyl-tRNA synthetase with precursor tRNA have a physiological significance? (B6:7)

(64) How is it, then, that this gene from a species that never forms nodules has the capacity to be expressed at elevated levels in nodules? (D2:637)

The writer thereby hedges the claim to be made by explicitly drawing the reader into the deductive process, rhetorically treating the audience as capable of making the same logical inferences. Readers may also be addressed directly to frame a theoretical possibility:

(65) One might speculate that what is now understood as carotenoid functions are not to be understood as common capabilities of all carotenoids, ... (A5:176)

In either case, an appeal is being made to the reader to consider associated claims on the basis of recognition as a fellow scientist and a respect for his/her judgements and views.

Deference to readers' opinions is similarly evident in the use of invitations. Frequently the Popperian dictum of falsifiability is invoked, and readers are invited to contribute to the solution of an ongoing problem. Reference to the testability of the findings or hypotheses or to the need for further research both acknowledge the role of readers in ratifying claims and allude to the community of scientists engaged in a joint quest for knowledge:

> (66) ...the key features of this working hypothesis relating to PSII are experimentally testable in detail and in a variety of ways in vivo as well as in vitro. (A1:274)

> (67) .. alerts to the intriguing and testable possibility that HSPs may participate in the stabilisation of plastid mRNA under heat-shock conditions. (B2:192)

> (68) Nevertheless, it cannot be excluded that this mechanism is realised only at certain phases of plant development (for instance, during gametogenesis) and such a possibility requires further study. (B3:222)

Such suggestions defer to the community by appealing to the reader as a scientist; emphasising a fellowship of interests and the pursuit of common goals.

In summary, all hedges are 'writer-oriented' in the sense that they function to reduce the risk of claim negatability, but reader-oriented hedges anticipate this possibility by addressing interpersonal rather than strictly epistemic issues. Reader acceptance is a powerful factor in communal accreditation, and hedging here results from the fact that claims may often present a challenge the research programmes, scientific beliefs or disciplinary values of readers. Writers must not only consider the plausibility of a proposition, but also its likely effect on colleagues' attitudes as both interested researchers and as disciplinary members. Power and conformity are influential in scientific cultures, and writers must attend to the needs of an audience which anticipates involvement in the negotiation of claims and recognises limits on self assurance. This is achieved is through expression of reader-oriented hedges, and is principally realised through personal attribution, particularly in conjunction with judgemental and deductive lexical verbs, the suggestion of alternatives, appeals to testability and an invocation of shared understandings.

Hedging in scientific writing

In science RAs hedging looks to the author, the reader and the proposition in seeking agreement for knowledge claims, the form of hedge representing how the writer has chosen to address the interactions inherent in negotiating such claims. Figure 4 summarises these functions.

Type	Function	Main hedging effect		
		Content	commit-ment	assertive-ness
Cat. Assertion.	full commitment to p and truth value	χ	χ	χ
Reader-oriented	hedges expression to gain reader acceptance of p	χ	χ	√
Content-oriented	hedges the extent to which p accounts for phenomenon.	√	√	χ
Accuracy-oriented	hedges correspondence of p to reality	√	χ	χ
Writer-oriented	hedges writer commitment to p	χ	√	χ

Figure 4: Principal functions of statements in scientific writing

Scientific writers always seek agreement for the strongest claims their evidence will support while also acknowledging the possibility of opposition. Thus hedges allow writers to upgrade or downplay their statements and thereby strengthen arguments by admitting limitations, uncertainties and the right of peers to participate in the ratification of knowledge.

The means by which these functions are often expressed in scientific discourse have also been tentatively identified and these are listed in Figure 5.

Since every hedging meaning is complex (i.e. contains a number of distinct components), most meanings share some components with most of the others. The different "uses" of a hedging device cannot be regarded as simply contextual variants of one meaning, however, as such a "common meaning" would be too general to be descriptively or predictively useful. Therefore a number of different, but related, meanings have been proposed. Writers may select a hedge to present a claim with greater accuracy, or to indicate an assessment of propositional reliability. They may also opt to use a hedge in order to gain some protection from the professional damage which might result from the claim's overthrow, or as a device which offers due recognition and deference to readers' views. Hedges which focus on the writer and the proposition

relate to strictly epistemic functions and express appropriate doubt or confidence in statements. Those which look to the reader more explicitly concern interpersonal issues in facilitating ratification.

Content Oriented		**Reader-Oriented**
Accuracy-Oriented	**Writer-Oriented**	
Hedges propositional content	Hedges writer commitment	Hedges assertiveness
Attribute type	Epistemic Lexical Verbs:	Epistemic Lexical Verbs:
Precision Adverbs:	judgmental	judgmental
content disjuncts	evidential	deductive
style disjuncts	Impersonal expressions:	Personal attribution
downtoners	passive voice	Personal reference to
Reliability type	abstract rhetors	methods
Epistemic Lexical Verbs	"empty" subjects	model
Modal verbs	Modal verbs	Assume shared goals
Epistemic Adjectives	Thematic epistemic device	Hypotheticals
Epistemic nouns	Attribution to literature	conditionals
Content disjunct adverbs	Impersonal reference to	*would*
Limited Knowledge	method	Involve Reader
	model	direct questions
	experimental conditions	refer to testability

Figure 5: Summary of Hedging functions and the devices often used to express them

To reiterate then, no device contains a single unequivocal meaning nor always permits a definite pragmatic interpretation. Writers are often seeking to achieve a number of concurrent goals, and the maximally efficient exchange of information in scientific writing frequently means that particular forms convey more than one function. To avoid distorting the reality of actual usage by assigning particular expressions exclusively to discrete categories, I have employed a fuzzy-set model which allows salient features of membership to be identified, while acknowledging that particular cases are likely to carry several meanings. Such a model captures the vagaries of my corpus and recognises the flexibility between language and context which serves a range of concurrent goals. It also means however that the categorisation of functions is rarely definitive.

Some instantiations, of course, are more representative or prototypical examples than others. In determining core cases four contextual variables were found useful: specification, verification, agentivity and combination. Each allows readers to make certain assumptions about the writer's attitude to a proposition or its context. In determining the particular function of a hedge, then, we can ask the following questions:

1. Is the content of *p* specified as far as possible?
Specification refers to propositional precision and concerns how far the terms used accurately describe the events or states referred to. A high degree of specification of propositional elements is most clearly seen by how a hedge acts to place restrictions on the scope of a statement. This is most closely associated with content-oriented strategies and particularly with attribute hedges. The writer places limits on certainty by restricting the time, quality or generalisability of the proposition. It is typically realised by downtoners and style disjuncts and indicates that something is true or accurate within certain limits. This type of hedge most closely resembles the 'institutional' tentativeness of science and is the function most clearly signaled by surface form.

2. Is the content of *p* verified as far as possible given the writer's knowledge?
Verification refers to the linguistic acknowledgment of uncertainties in the truth of a proposition. It clarifies the assurance that can be invested in the statement and indicates reservation in propositional accuracy. Verification is associated with reliability type accuracy hedges and is typically marked by the use of a range of epistemic lexical items and references to limited knowledge. The assessment of the certainty of the truth of the information expressed is often difficult to distinguish from self-protection, as it is often conveyed by the same forms. Informants acknowledged that these functions were distinct, but often referred to them as being closely related, hence they have been presented in my taxonomy as two sides of 'content-oriented' motivation.

3. Is the possibility of *p* being true directly attributed to the writer?
Agentivity entails the extent which the writer is explicitly linked with the proposition and is often critical in distinguishing between reader- and accuracy-oriented hedges, with non-writer agent contexts associated with content-oriented hedges, and writer-agency related to reader-oriented meanings. Thus

abstract rhetors, passivisation and impersonal reference to research processes, particularly in cases involving 'higher level' claims which proceed beyond the data into realms of conjecture, frequently indicate a reluctance by the writer to associate too closely with those claims. "Strong" agentivity, on the other hand, normally indicates an attempt to limit the generalisability of the proposition for interpersonal reasons, and marks the hedge as a reader-based strategy.

4. Has the writer constructed *p* juxtaposing a combination of hedges ?
Finally, compound hedges in the discourse, often suggest a writer-oriented motivation for hedging. If the linguistic context conceals the writer's viewpoint or surrounds the proposition with modal verbs, epistemic verbs, adverbs and other shields, then it is frequently the case that the writer is seeking to avoid personal responsibility for the validity of the proposition. The statement is therefore likely to be providing an example of writer-oriented hedging.

These general criteria are perhaps best clarified by applying them to actual data and therefore the next chapter explicates the model further by examining longer stretches of discourse.

7. Examining hedges in scientific discourse

In this chapter the major communicative uses of hedges are analysed in greater detail by studying three short text extracts and a full research article. Clearly, situating hedges in their appropriate linguistic and rhetorical context gives a better picture of the ways hedges interact with other rhetorical features and contribute to the persuasiveness of a research paper. In addition, we can also see more clearly how writers seek to vary their discussion of possibilities and manipulate their attitudes to truth of statements through the combination and concentration of devices. The analyses of longer stretches of discourse can thus more explicitly reveal how the distribution and use of hedges reflects the different rhetorical purposes of the RA stages.

The four texts were examined by myself and four native-speaker biologists, all experienced researchers and writers in botany, microbiology or cell biology. The analyses were carried out in one of two ways. The informants either individually read the texts and spoke aloud their reactions to underlined features, or they participated in small focus groups of two or three and discussed the texts together. The method did not appear to greatly affect either participation or the interpretations which were elicited. All the sessions were recorded and respondent sheets were collected and analysed. The interrater reliability for these individual raters (and myself) identifying epistemic functions was .77 (Kappa), which shows a relatively high level of agreement for this kind of analysis. All disagreement was subsequently reconciled by discussion. In addition, Extract three also involved several interviews with the author of the paper examined.

The chapter examines four extracts in turn: the first from an Introduction, the second from a Results section, the third from a controversial Discussion section, and the fourth from a full research paper.

Extract 1: Hedging in an introduction

The first extract is from *Plant Cell* and illustrates the use of hedging in an Introduction section. The extract is perhaps less typical than others in containing more mitigation, but demonstrates the use of hedging in Introductions to establish the significance and novelty of the research and its place in the area of shared knowledge to which the article contributes. The hedges here are predominantly content-oriented, not unexpectedly in a rhetorical component concerned with establishing a research space in which to locate the study. The writers are thus largely concerned to specify the reliability of the research findings and their commitment to its truth. There is some evaluation of prior research, but when the tone becomes more speculative and exploratory we see more use of writer-oriented hedges, allowing the authors to make their claims while defending themselves from the dangers of overconfident assertion. The hedges do not therefore compromise the writers' attempts to express the centrality of their work, but assist them in establishing that the field is lively and consequential, and that the claims to be made are significant and far-reaching.

Transgenic Arabidopdis seedlings over expressing phytochrome B exhibit enhanced sensitivity to Rc but wild-type responsiveness to FRc (Wagner et al, 1991; McCormac et al, 1993). This result is consistent with the behaviour of endogenous phytochrome B deduced from the *hy* 3 mutant studies and underscores the lack of involvement of this photoreceptor species in the FR-HIR. By contrast, transgenic Arabidopdis over expressing phytochrome A exhibits enhanced sensitivity to FRc (Whitelam et al, 1992; McCormac et al, 1993). Together these results are consistent with the possibility, although do not prove, that the capacity to mediate the FR-HIR may be an intrinsic property of phytochrome A.

Accumulated biochemical and physiological data also provide circumstantial evidence that phytochrome A may be responsible for the FR-HIR. Action spectoscopy has shown that the responsiveness of etiolated seedlings to FRc declines rapidly upon greening, indicating that this response is under the control of a physiologically defined photolabile phytochrome pool (Beggs et al, 1980; Holmes and Schäfer, 1981). Because the time course of

this decline was found to be similar to the decline in levels of the
22 abundant, spectrophotometrically detectable type 1 phytochrome,
the data are consistent with the possibility that this photolabile
24 phytochrome pool may be responsible for the FR-HIR.

26 Regardless of the validity of this specific proposal, however, be-
cause phytochrome B does not appear to be involved in the FR-
28 HIR, we reasoned that mutants defective in the activity of the
phytochrome mediating this response would retain phytochrome
30 B, and, therefore, retain responsiveness to Rc. If correct, this pre-
diction might explain why previous exhaustive screens have not
32 detected mutants in phytochromes other than phytochrome B.
These screens were performed in white light, presumably pre-
34 cluding detection of mutants in the FR-HIR. Based on this pre-
diction, we have devised a photobiological screen involving se-
36 quential screening of mutagenized populations of Arabidopsis
seedlings in FRc and Rc. We provide evidence that mutants se-
38 lected for lack of responsiveness to FRc, but which retain wild-
type responsiveness to Rc, are selectively deficient in functional
40 phytochrome A.

(D4:40)

The first paragraph addresses the literature in order to assert those matters that are taken to be true for the purpose of the paper. Because this background is assumed to be shared knowledge, the information is cited categorically in order to build a foundation of substantiated facts upon which the argument to follow can be constructed. The conclusions drawn from these premises are tentative, however, and there is an evident desire to represent the writers' beliefs more cautiously. This is achieved by the use of reliability accuracy hedges realised through epistemic adjectives, nouns and verbs. The precise degree of certainty is expressed by the multiple hedge in lines 10-11. The writers' objective here seems to be to gradually move the reader from the known through the less certain, and finally to the more speculative discussion in the following paragraphs.

Paragraph two continues to discuss the literature, again with a positive stance, and to make cautious inferences from it. The decision to focus on the attributed "action spectroscopy" findings (17) from the generality of the

"accumulated biochemical and physiological data" seems highly selective and may serve to hedge a body of work which may be less collectively supportive than suggested. However, the phrase "circumstantial evidence" together with the modal hedge in line 16 again indicate the degree of reliability with which the reader may accept the inferences made. Once more these claims are presented as being compatible with the authors' view, but not as substantiating it. There seems to be little hedging of commitment here; instead the use of reliability accuracy hedges are employed to present a claim which might be firmly held, but which lacks supporting proof. The writers thus appear reluctant to overstate their case beyond the acceptable warrant for it.

The third paragraph is altogether more speculative and exploratory, moving from the cautious expression of claims hedged by reliability devices to more rhetorical writer and reader-oriented hedges. The opening phrase (27) signals a departure, moving beyond what has tentatively been proposed on the basis of findings in the literature, and serves to hedge what follows by making this contrast explicit. First the authors insure themselves against the possibility of being incorrect about the involvement of phytochrome B in the process by use of a writer-oriented hedge (*not appear*, line 28). The claim which follows regarding the behaviour of mutants, is hedged by presenting it as the outcome of a personal cognitive process of deduction. The expression "We reasoned" (line 29), contains both a conjectural nuance and the tentativeness of a personal attribution which subverts the generalisability and reliability of the proposition.

As pointed out above, while such hedges seem to be strongly writer focused, the act of stressing the authors' involvement in the interpretive procedures which produce a claim actually cuts across the impersonal conventions of scientific practice. Reference to the writers' direct involvement in the research draws attention to contingent factors, highlighting the activities of the investigator rather than the research. It therefore marks the claim as a reader-oriented strategy, a conscious tactic to subtly hedge the generalisability of a claim and mark a position as an *individual* interpretation. Scientist informants to whom I showed this extract concur that the writer seems to be recontextualising the claim and inviting the audience to judge it according to the reasonableness of the deduction. The personal attribution serves to announce the proposition as simply a personal belief, while addressing the reader directly as someone who might well make similar deductions.

This is followed (lines 31-32) by an impersonal writer-motivated strategy which hedges the correctness of the proposal, and what it "might explain". This

rhetorical effect is skillfully achieved by a hypothetical conditional theme, an epistemic verb and a modal acting in close a harmony which simultaneously looks backward to downgrade the "reasoning" of line 29 to a "prediction". The combined effect of the hedge thus not only serves to protect the writers from the negatability of their claim, but also acts to mitigate an implicit criticism of earlier researchers whose studies have failed to detect the relevant mutant.

"Presumably" (line 34) is a reliability hedge, as it proposes a possible explanation for this failure, stating the degree of reliability of the guess with no necessity for self-protection. At the beginning of the next sentence (line 35) the proposal is, once again, hedged against consequences to the writers, indicating a clear reluctance to express commitment to the idea. Following this however are two sentences which contain personal attribution hedges and allow the reader the freedom to either accept or reject the accompanying propositions. Here the writers are mitigating the assertiveness of the expressions they employ rather than the probability of the statements. Neither the photobiological screen, nor the evidence presented on mutants involves any withholding of commitment, but the possibility is left open that readers may wish to judge either as incorrect.

Clearly the rhetorical work involved in delineating an appropriate context for this study is substantial. The task for the writers is to justify the importance of the research by creating a research space within their discussion of the literature. They must state what has been accomplished in the study of the phytochrome group of photoreceptors and establish in what ways their research extends this work. In this text, then, hedges play two important roles. First they, assist the authors in positioning their research and establishing a coherent warrant for their study by tentatively drawing conclusions from significant prior work. Second, they enable them to speculatively discuss the claims they wish to make for their own research and thereby construct the novelty and importance of the research.

Extract 2: Hedging results

This short extract illustrates the role of hedges in a Results section, taken from *The Plant Cell*. The long Results sections of this journal are typically subdivided by topic and the extract appears under the heading "Subcellular location of the late nodule-specific protein". It was noted earlier that there is con-

siderable variation in the extent to which Results sections simply describe results or contain the writer's comments on them. The following extract is typical in its inclusion of the authors' interpretation and discussion of findings, using hedges to set forward interpretations and claims which arise from the study itself. Attempts to explain results involve evidential reasoning which can invite refutation of the claims and criticism of the writers themselves, necessitating the use of both propositional and writer-oriented hedges.

A nodule-specific protein can be expressed by the plant or by the microsymbiont genome. To determine whether the 21-kD protein was associated with bacteroids, subcellular fractionation experiments were conducted on large nodules. Figure 2 shows the different fractions analyzed by immunoblotting with nodule-specific antibodies. Under the conditions used, most antibodies of the antiserum appeared to be directed against the bacteroid proteins represented by bacteroid proteins themselves and by the proteins of the peribacteroid fluid (PBF), instead of the plant proteins represented by the soluble fraction. This may be due to the large amount of bacteroid proteins in the tissue (thousands of bacteroids relative to the number of infected cells) and/or to the greater antigenicity of bacteroid proteins. The presence of the 21-kD polypeptide in the lane corresponding to the pure bacteroid fraction would suggest that it is a bacteroid protein, but a large amount of this protein is present in the soluble fraction and a small amount is present in the PBF and membrane fractions. This result would favour the hypothesis that the plant protein is targeted to the bacteroids.

The faint 21-kD band observed in the PBM lane (Figure 2) would reflect the transient passage of this protein across the PBM from the plant cell cytoplasm to the bacteroids. However, the opposite is also possible, and it cannot be ruled out that the 21-kD polypeptides seen in the bacteroid lane and in the soluble proteins lane are totally different proteins with the same apparent molecular weight. (D2:260)

The opening few lines (1-5) of the section represent the cognitive structuring of the research paper itself, stating a general assumption followed by a statement of purpose, then method and result. The transition to a more interpretive stage, involving subjective comment on the results, is marked by the writers' attempts to signal the uncertainty of their claims and their commitment to them.

This introduction is followed by a writer-oriented strategy which hedges the effectiveness of the experimental conditions used (7-8) and, with the sensory evidential verb "appear", the possibility of interpretive error. "Most" (line 7) is simply an attribute hedge, specifying the accuracy of the observation or the irrelevance of complete precision, my informants agreed on the latter. The writers then suggest a reason for this result, hedging the reliability of the claim with a modal verb (line 11) and the use of "and/or" (13), which in marking an alternative explanation helps specify the actual state of knowledge.

Up until now, the writers have used hedges as a means of avoiding misrepresenting their findings, seeking to account for their results impartially and hesitating to ascribe causes too strongly. But while lines 11-14 offer arguments in support of two possible explanations, the final two sentences (14-20) show that the writers are seeking to persuade readers of the validity of only one of them. They are here treading less certain ground and consequently choose to employ writer-oriented hedges to protect their exposure; the credence given to the bacteroid protein option helping to hedge commitment to the preferred answer. The impersonal constructions of these sentences suggest a relatively transparent deduction from the results to the claim, but also hedge the writers' degree of responsibility. The modal verbs and restriction of the claim to a more provisional "hypothesis" (19) further shield the authors from the embarrassment of possible error.

The second short paragraph seeks support from the gel blot results for the claim previously made. The use of "would" (22), where a judgemental lexical verb such as "indicates" might be expected, is interesting here and the choice of the hypothetical marker suggests a device to involve the reader in the deduction rather than a writer-oriented hedge. Given the final sentence, where the expression of a real possibility of an alternative explanation indicates a further reduction of writer commitment to the claim, it is possible that the authors are hedging certainty rather than commitment in order to express their tentative support for the claim, despite the mention of the second option.

It is apparent that the writers in this extract are doing more than simply presenting experimental findings, and the concentration of hedges is one way that the rhetorical nature of the extract becomes apparent. The use of mitigation in this short extract is a means of gaining reader support for how the results may be interpreted and what their significance might be. The caution with which this is accomplished suggests that establishing what might be considered true as a result of the study is by no means a straightforward procedure.

Extract 3: Hedging a controversial claim

The next extract shows how hedges are used as a reader-oriented strategy in presenting an extremely controversial claim which challenges a widely accepted position. In this text, taken from *Photosynthesis Research*, the writer proposes a radical explanation for the process of photoinhibition, an issue at the heart of cell and molecular biology. In making this proposal, the writer implicitly attributes serious inadequacies to current theories in their interpretations of critical data. She therefore runs the very real risk of having the claim rejected by a community of peers who, she perceives, have a great deal invested in the existing view and who are likely to defend it without giving serious consideration to her work.

This writer, then, sees herself as championing a new approach, as a researcher in the vanguard of a challenge to orthodoxy, and this accounts for the large number of hedges in the extract and in the paper overall. The paragraph is crowded with modal verbs and adverbs, epistemic lexical verbs, indefinite quantifiers and admissions of limiting conditions, creating a rhetorical and interpersonal context which seeks to pre-empt the reader's rejection.

Photoinhibition is characterised by a decrease in PS II electron transport activity, chlorophyll fluorescence and quantum yield (Poweles, 1984). It occurs every time a fluctuation upwards in the light intensity exceeds that to which photosynthesis is adjusted and/or the rate of rapid adjustment that is possible. Based on and consistent with the model proposed above, I suggest the following mechanism for photoinhibition: It is well established that an early manifestation of photoinhibition is the inability of the PSII reaction to reduce Q_B (Cleland et al, 1987). Since PS I

will continue to work through cyclic electron transport at high speed at high light intensities, the Q_B binding site would remain mostly unoccupied in some PSII complexes (Fig 3). This would render the PEST region inaccessible to protease (see Fig 1) and stop D1 degradation - LHCII would not be shed wither (Fig 2A and 3). Since not all active PSII units would be affected immediately and because more PS II complexes further into the stacks would be exposed to light, they would become successively inactivated while still absorbing light. Eventually the PS II pool would become virtually depleted because there would be no significant increase in D1 synthesis since the appropriate signal, i.e. Pre-PS II complexes available in the stroma Lamellae, would not be formed. Stopping chloroplast protein synthesis altogether with chloramphenicol has been shown to hasten the light-dependent demise of PS II and photosynthetic activity (Greer et al 1986). DCMU appears to protect against photoinhibition in some experimental systems because it keeps the grana together and thereby prevents many PS II complexes from absorbing too much light. This model also explains the increased severity of photoinhibition at low temperatures since the proper function of the conveyor belt would be extremely temperature sensitive. On the basis of this model very specific questions can be asked and experimentally tested concerning the control of DI turnover by titrating PS II with DCMU or the correlation of the speed of the conveyor belt, ie. DI turnover, with grana size.

In summary then I propose that photoinhibition occurs because the DI protein does *not* become degraded and LHCII is *not* shed, leading to inactivation of α centres, reduction in electron transport activity, and, of course quantum yield. (A1:273)

The opening statement establishes common ground, simply defining some known characteristics of the process of photoinhibition. The passage continues by making a number of claims said to be consequent on an acceptance of a model outlined earlier in the paper. It goes on to draw attention to the strengths of the model as an explanation for photoinhibition. Finally the author summarises the main features of the proposal.

The statement referring to the proposed model and to a "well established", i.e. widely accepted, aspect of the phenomenon (6-9), indicates a somewhat guarded opening. These strategies seek to hedge writer commitment by representing the argument as inductively consequent upon both impersonal scientific premises and the beliefs attributed to another source. However, the personal attribution in line 6 is critical and clearly acknowledges the desire of the writer to defer to the audience in presenting the new and radical claims she hopes to gain acceptance, and credit, for proposing. Thus the reader-oriented hedge "I suggest" (6) represents the tension between a personal conviction and the need to relate the claims to the existing knowledge currently sanctioned by the discourse community. At this stage the writer is stating that *she* believes her model supports a reinterpretation of the mechanism, but the readers are allowed to judge for themselves.

Following this personal assertion is an assumption of common ground, the marker of habitual predictability *will* (line 10) situating the certainty of a premise on which more speculative aspects of the process will be developed. In the subsequent argument the writer builds on the acceptance of this premise with the modal verb *would* (10 times) and epistemic adjuncts. The "downtoners" *mostly, some* and *virtually* (lines 12 and 19) all serve to reduce the force of their predicates, specifying the precision of the terms used. These attribute accuracy hedges simply indicate deviations from end points on an "ideal" scale and in no way modify the writer's belief in their generalisability. The writer is not restricting the validity of the claim, but simply hedging the expressions she employs to describe elements of the process. Her full commitment to the claim remains unaffected.

The repeated use of *would* (lines 11-22) is also interesting in this regard. Use of this modal often signals core examples of content-oriented hedges, mitigating the reliability of the conclusions by making them dependent on the validity of the premises. It is however almost quasi pre-suppositional in that it constructs a hypothetical world for the reader in which given an acceptance of the premise *A*, then *X, Y* and *Z* are likely to follow. In such a hypothetical world we can have confidence in the connections posited. *Would* here, then, appears not to weaken the writer's confidence in the relationships expressed but simply hedges the assertiveness of their presentation. *Would* therefore contributes to the persuasive character of the argument by deferring to the reader in emphasising the *potential* correspondence of these results to an actual situation in the external world.

There is a great deal of rhetorical work occurring in this paragraph and the author is working hard to gain a hearing for her perspective. This is a high-level, provocative claim which contradicts much well-supported current work in botany and microbiology and some strongly entrenched views. The author needs to ensure that her claim is not dismissed out-of-hand by adherents of a more orthodox opinion by presenting it with due deference to that opinion and with a degree of humility appropriate for the claim being made. The text suggests that the use of *would* is carefully selected to transmit a firmly-held belief in a manner which reduces the risk of challenging readers' assumptions too directly and thereby provoking the possibility of immediate negation of the claim.

That is, the claim will not succeed if it is expressed too humbly or too strongly and we can see here the negotiation of a claim through a focus on the form of its presentation. The writer advocates a position about which she holds a conviction, but uses the modal to convey that the theory is simply, as yet, unverified. Scientific truths are those upon which informed observers can agree and the writer here is soliciting credence for a position in which she firmly believes.

The writer then returns to more solid ground (lines 22-25) by once again referring to a claim in the literature, but this is only a strategy to launch further conjectural claims about the efficacy of the model. Here there is a switch to more writer-oriented hedges, relying on an impersonal form with the sensory evidential verb "appears" (line 22), and "the model" in agentive position (28 and 31). While the writer seeks some protection from negatability here, and thus appears to have less confidence in the particular claims for the model being made, reference to the value of the model to the research community (lines 31-35), while hedged, is maintained. Finally the writer restates the claim as a reader-oriented hedge (lines 37ff). Both "I propose" and "of course" negotiate a position with the reader, the first by stating the claim as an opinion and leaving it open to ratification, the second by establishing common ground with fellow researchers on what can be assumed about the claim.

The variable of agentivity is very helpful in distinguishing the pragmatic function of hedges in this discourse. The writer is exploring uncharted territory here and the desire to safeguard herself against later disproof would be an understandable reason for hedging, indeed this is an option which is adopted towards the end of the first paragraph. But while both reader- and writer-orientation seek to alter the context of a discourse, and not its truth conditions,

the use of personal attribution shows that the writer is trying to control the *attitude* the interlocutor takes towards the claim, rather than control her *commitment* to it. While self-protection may be peripherally present in this strategy and be a consequence of it, the claim is made with little of the uncertainty often seen when writers seek to limit commitment to ideas. In other words, the hedges function not to diminish responsibility but to offer the reader the opportunity to act on the claim and thereby increase its chances of ratification.

The extent of reader-oriented mitigation in this example is not surprising given the writer's awareness of both the novelty of her proposal and its implied criticism of currently accepted interpretations. On the author's own admission (Critchley, pc), the hypothesis is extremely inflammatory:

> It was just not possible to be definitive at the time and this was a very critical phase where I was saying to everyone in the field 'I don't believe what you're saying, the data just don't show what you are saying' ... this is a very much contentious issue about how you interpret certain kinds of data. I needed to break through the consensus and get them to listen to me

There is an attempt here to code these considerations linguistically and thereby avoid violating the code of conduct required by the discourse community. She is, in Fraser's terms, "abiding by the rules of the relationship" (Fraser & Nolan, 1981:96).

Extract 4: Hedging a research article

The final text consists of an entire RA selected for its relative brevity (2,346 words) and apparent representative density of hedging. It is titled 'Aryl-Substituted α–Aminooxycarboxylic Acids: A new class of auxin transport inhibitors' and was written by Gary Gardner and James Sanborn of a private research lab in the US, published in issue 90 of *Plant Physiology* in 1989. The abstract, tables, figures and reference list have been removed for economy, and the text broken into chunks for discussion, but otherwise it is presented in its entirety. The research reports experiments on the compound DAIV 1 to determine its role in binding to NPA sites in corn and interference with the normal transport of plant hormones causing abnormalities in the directional growth of roots.

The Introduction section is brief and classically organised, creating a research space by claiming centrality for the topic and reviewing items of prior research. The authors turn to their own research (line 13) by explicitly intruding into the discourse with a personal pronoun and switching to the past tense, a marked contrast to the impersonally expressed present tense general truths of the first paragraph. The personal attribution and the labeling of the finding as an "observation" help to stake out the writer's territory, but do this in a way that is unlikely to offend other researchers in the area. The switch to a passivised modal in the next line indicates a writer-motivated hedge against possible alternative interpretations, also seeking security by aligning the view with support from the literature.

The following sentence is again on surer ground, but the use of the contrast marker "however" and the return to the past tense again limits the issue to be addressed, rhetorically preparing the ground for the research to be reported. The next three lines (22-24) contain two cautious references to what might reasonably be deduced from current knowledge, the writers here using an accuracy hedge to rhetorically seek reader support for their interpretation. The hedges in this short introduction thus function to help establish a niche for the work by insinuating shared assumptions with the audience and avoiding aggressive assertions which might alienate the support of other researchers.

Herbicides and plant growth regulators which inhibit transport of the plant hormone auxin (indoleacetic acid) have been of interest for many years (11). Commercial compounds which act via this mode of action include the herbicide NPA3(naptalam) and the morphactin plant growth regulator chlorflurenol (chlorflurecol). These compounds bind to the same site on plant plasma membranes (16), and a common symptom of their action is the inhibition of root geotropism. Many new classes of chemistry which inhibit auxin transport and bind to the NPA receptor have been identified as a consequence of the abnormal geotropic response that they produce (10).

We observed that certain substituted α–aminooxyisovaleric acids demonstrated herbicidal activity (data not shown), and this observation could be logically explained by the known interaction of aminooexyacetic acid with pyridoxal phosphate-dependent en-

zymes (7). This and other aminooxy compounds inhibit ethylene synthesis in plants by inhibiting the synthesis of the ethylene precursor 1-aminocyclopropanecarboxylic acid from S-adenosylmethionine (1). However, several analogs, notably DAIV, compound 1, caused abnormal root geotropism, a symptom that would not be expected as a result of interference with ethylene production. Since these compounds do not obviously fall into a known class of auxin transport inhibitors, it was of interest to determine if the abnormal geotropism reflected a new mode of action for this class or a secondary consequence of a primary effect on ethylene synthesis. Therefore, DAIV was examined for its ability to bind to the NPA site on corn microsomal membranes.

The methods section contains only four hedges in 700 words. It is characterised by a dense presentation of agentless procedures, complex noun phrases, references to others' experiments and a heavy reliance on shared knowledge in the absence of explicit cohesion. The hedges (lines 55-58) weaken propositional reliability, drawing attention to the limitations of the procedures while tentatively associating them with those currently employed for studying the structure activity of this subject. There seems to be a desire to acknowledge the inherent restrictions of the approach rather than defer to readers or seek self-protection from criticism here. However, as one of my informants observed, it is highly probable that these could be pointed out by any informed reader.

MATERIALS AND METHODS

In vitro activities of various aryl-substituted α– Aminooxycarboxylic acids were compared by measuring their ability to displace [^{3}H] naphthylphthalamic acid from its binding site on membranes from etiolated corn coleoptiles (*Zea mays* L., cv WF9 x Bear 38). Preparation of the membrane particles and details of the centrifugation binding assay were carried out as described previously (14), similar to those used by Hertel et al. (5). Microsomal membranes were resuspended in assay buffer so that I mL contained the membranes from 1.5 to 2 g fresh weight of tissue.

[^{3}H]NPA was added to portions of the membrane suspension to a final concentration of 10^{-9} M. Nonradioactive compound at 10^{-5} or 10^{-7} M, or 1% appropriate solvent, was added, and triplicate 1mL aliquots were centrifuged at 16,000 rpm for 60 min. The supernatants were decanted, and triplicate aliquots were taken for determination of radioactivity. The end of each cellulose acetate butyrate centrifuge tube containing the membrane pellet was excised and extracted in 10 mL Beckman Ready-Solv HP scintillation cocktail for determination of radio activity by liquid scintillation spectrometry. 'Specific' or saturable binding in these experiments is the binding of 10^{-9} M [3H] NPA that is abolished by 10^{-7} or 10^{-5} M nonradioactive NPA or competing substance, according to the definition of Ray et al (12). Values in the tables are expressed relative to the specific binding displaced by the same concentration of NPA. In a strict sense, these experiments measure inhibition of [^{3}H] NPA binding rather than competition, and they do not speak to the type of inhibition. However, these procedures are consistent with the large body of literature that describes structure-activity requirements for the NPA receptor. (See, for example, ref 10.)

Polar transport of [^{3}H] indoleacetic acid (IAA) was directly measured through sections of 5-d-old etiolated corn coleoptiles of the same variety, as described by Hertel and Flory (4). Primary leaf tissue was removed, the apical 3 mm end of each coleoptile was discarded, and ten 4 mm sections were placed on a receiver block of 1.5 % agar containing 0.02 M potassium phosphate buffer (pH 6.4) and the test compound at one of several concentrations as indicated. A donor block in the same buffer containing 10^{-7} M [^{3}H] IAA was placed on top of the sections, and basipetal transport (apex to base) was carried out for 90 min. After the transport period, each receiver block was extracted in 10 mL of Beckman Ready-Solv HP scintillation cocktail, and radioactivity was determined by liquid scintillation spectrometry. Nonspecific, primarily diffusional, acropetal transport (base to apex) was determined in a similar way by inverting the coleoptile sections. Each transport test with 10 sections was carried out in duplicate. Spe-

cific (net) polar transport was calculated by subtracting from the radioactivity in the basipetal receiver block the amount in the receiver block of the acropetal control.

Whole plant phytotoxicity was measured on six species of weeds after both preemergence and postemergence applications as previously described (2). Preemergence herbicidal activity was evaluated in barnyard grass (*Echinochloa crusgalli*), garden cress (*Lepidium sativum*), downy brome (*Bromus tectorum*), velvetleaf (*Abutilon theophrasti*), yellow foxtail (*Setaria lutescens*), and sicklepod (*Cassia obtusifolia*). Postemergence activity was evaluated on redroot pigweed (*Amaranthus retroflexus*), johnsongrass (*Sorghum halepense*), crabgrass (*Digitaria sanguinalis*), velvetleaf, yellow foxtail and sicklepod. Phytotoxicity was evaluated on a numeric scale from 0 to 9, where 0 indicates no visible effect and 9 indicates complete kill (no living tissue). As a relative measure of *in vivo* activity, the scores of the species were summed; therefore, a value of 54 indicated complete kill of all six species.

The compounds were prepared either by the reaction of the sodium salt of the appropriate oxime with α–bromo-iso–valeric acid, which gives the acid in one step, or reaction of the sodium salt of the oxime (prepared by reaction of NaOH and azeotroping water with toluene in the presence of a phase transfer catalyst) with ethyl α–bromo-iso-valerate followed by alkaline hydrolysis.

[2,3,4,5^{-3} H]NPA (16 Ci/mmol) was purchased from Research Products International Corp. (Elk Grove, IL and was used from a 10^{-6} M aqueous stock solution. [5-(n) $^{-3}$H]IAA was also purchased from Research Products International Corp., and, for these experiments, the original specific activity of 25 Ci/mmol was diluted to 250 mCi/mmol with equimolar nonradioactive IAA.

The combined results and discussion section constitutes 57% of the text and contains about 85% of the hedges, demonstrating how writers increase the speculativeness of their texts in those sections which most clearly promote the

value of their work. I have divided the section into three, corresponding to three broad rhetorical purposes: to present the research findings, to discuss their relevance within the context of the existing literature, and to argue for their significance. The writers thus gradually move away from what can be safely assumed from the results to make wider claims for the research. This means that different kinds, combinations and frequencies of hedges have to be employed at each step. While these purposes sometimes overlap in the article, they are mainly presented in the linear fashion suggested, and I will discuss each section in turn.

RESULTS AND DISCUSSION

The observation that DAIV caused disoriented roots in a whole plant phytotoxicity screen led to testing that compound for its ability to displace NPA from its binding site on corn coleoptile membranes. A preliminary experiment indicated that the compound was, in fact, active (results not shown), and therefore, this compound was compared with several analogs in the experiment shown in Table 1. These analogs were chosen on the basis of abnormal root geotropism observed in the whole plant screen. The data express the relative activity of 10^{-5} M compound as compared with 10^{-5} M NPA in displacing 10^{-9} M [^{3}H]NPA from corn microsomes. As is clearly seen, DAIV showed significant affinity for the NPA receptor, with about half the activity of NPA at 10^{-5} M. Total preemergence and postemergence values from the whole plant phytotoxicity screen are shown for comparison. It should be stressed that these phytotoxicity values probably result from both modes of action, as auxin transport inhibitors and as typical α–aminooxycarboxylic acids.

As would be expected from previous structure activity in formation (16), the ethyl ester (compound 2) was inactive. The 4 Cl compound (3) also showed greatly reduced activity compared to DAIV. Because of this low activity and the analogs available, it is not possible at this time to ascertain the importance of substitution or lack thereof at R2 or the scope of the alkyl substitution at R'. The influence on activity of several aryl ring substitutions is

shown in Table 11. The only compound in addition to DAIV which showed significant binding activity was the meta CF3 analog, compound 8.

Although the degree of activity of DAIV in the NPA binding assay was sometimes low and varied from experiment to experiment, it was always statistically significant. This activity suggested an analogy between DAIV and NPA itself (Fig 1). Therefore, it seemed reasonable that increased activity might be achieved by synthesis of the naphthyl analog of DAIV. This molecule was prepared (NAIV, compound 10), and it demonstrated increased foliar activity in the whole plant screen and characteristic root disorientation. The results of a detailed binding study comparing NAIV and DAIV with NPA are shown in Figure 2. The naphthyl analog, with an estimated affinity constant (KD) of 1.7×10^{-7} M, was about an order of magnitude more active in displacing NPA than the 2, 4-dichloro compound. NAIV was still an order of magnitude less active, though, than NPA itself.

It was expected that these compounds would inhibit auxin transport since inhibition of root geotropism and binding to the NPA receptor are both properties of inhibitors of auxin transport. Direct confirmation of this hypothesis is shown in Table 111. At 10^{-5} M, both DAIV and NAIV inhibit the movement of [^{3}H] IAA through corn coleoptile sections. In all three assays—collar phytotoxicity, auxin transport, and NPA binding—the order of activity is the same: NPA > NAIV >DAIV. A more detailed comparison auxin transport inhibition is shown in Figure 3. NAIV is highly active with an l_{50} (the concentration at which auxin transport is inhibited by 50%) of 0.28μ M compared to 0.16μ M for NPA.

This first section presents, interprets and comments on the results of the study, preparing the rhetorical groundwork for the claims which follow. It is distinguished by an absence of personal pronouns and an emphasis on interpretations which seem to have no originators or which arise directly from the experimental methods themselves. The hedges in this extract are therefore

largely propositional, either accuracy-oriented or writer oriented. They focus on the accuracy of the statements to either specify precision or to shield the writers from criticism.

The opening sentence revisits the Introduction, contextualising the results in consensual knowledge and re-establishing the significance of the research. The writers then set about constructing a coherent story, a convincing "narrative" for biologists (Myers, 1990), which simultaneously contributes to the authority of science. The initial findings are presented with what appears to be a simple reliability hedge, but the attribution of reasoning powers to the procedure (115) suggests that an element of self-protection may also be intended. In any event, these beginnings are downgraded to the status of prefatory activities by the designation "preliminary". The results of the experiment are asserted confidently however, with the attribute hedge in line 123 subordinating the importance of a more exact figure. The final sentence explicitly emphasises doubts about the reasons for the results, using a reliability hedge to state the degree of certainty that can be invested in them.

The next paragraph begins with an insinuation of shared knowledge and a reference to prior work which provides a background to contextualise the results which follow and demonstrate their departure from what is known (131). The limits of knowledge are quickly re-established however with a propositional hedge (135). This is followed by a presentation of the main findings of the study (142ff.). While the results were "always statistically significant" (144), the conclusions to be drawn are hedged by reference to the variable behaviour of DAIV in individual experiments. The explanation for this is extremely cautious, the writers choosing to heavily hedge their interpretation with a lexical verb-lexical verb-modal adjective-modal verb harmonic cluster (144-146) to indicate their uncertainty and protect themselves against the dangers of overstatement. Ascribing an explanation for the variability of results to the results themselves demonstrates a reluctance to take responsibility for it, and the dense combination of hedges which follow creates a further smokescreen of defensive shields.

The avoidance of assertion here, then, results from an attempt to account for fluctuations in the experiments in a way that will allow the writers to take credit for priority should it turn out to be correct while protecting them from the full effects of its possible falsification. An informant observed that this seemed to be an increasingly common strategy in biology. The paragraph concludes with the binding study results, which are mitigated with two accuracy-

oriented hedges, the first placing limits on reliability and the second, again, apparently downgrading the need for scrupulous definiteness. The vagueness of the attributer here does not seem to be covering for lack of precision, but reducing the importance of exactness to better highlight the effect itself. This promotion of strong results is continued in the following paragraph which also links the findings to the initial 'problem', and through that to the wider significance of the research. The next extract is continuous with the present passage, but moves more explicitly into that wider realm.

Structure activity principles for inhibition of root geotropism have been elucidated by Katekar and his coworkers (8, 9). These require that active compounds should possess a carboxylic acid function attached to an aromatic ring, which is connected at the
175 *ortho* position to a second aromatic ring. The aromatic rings may be separated by a conjugated or planar system of atoms, and high activity is achieved when the distance between the centers of the two extreme aromatic rings is at least 0.73 nm. Further refinements of these rules questioned the requirement for coplanarity of
180 the rings and suggested that the most favored conformation of the carboxyl is the one which is out of plane with the aryl ring to which it is attached (9).

A complete consideration of the data however, requires a drastic
185 modification of these rules. First, the morphactins bind to the same receptor site, inhibit auxin transport, cause abnormal root geotropism, and do not fit Katekar's structural criteria. Second, the carboxyl group can be replaced by an acidic proton. A series of flavonoids was tested because of a report on the effects of fla-
190 vonoids on auxin transport (13). The flavonoid morin (3,5,7,2',4'-pentahy-droxyflavone) competes for NPA binding in vitro though it lacks a free carboxyl group (Table IV). Third, the present work indicates that the aromatic ring to which the carboxyl group is bound is not necessary, provided that a bulky substituent (in this
195 case isopropyl) is present to position the carboxyl group appropriately. A revised stereochemical view of structure activity requirements is simply that an acidic function is oriented in a precise spatial relationship to an aromatic / hydrophobic ring system.

Numerous novel chemical approaches should be possible to achieve these features.

During the preparation of this manuscript, a report appeared by Jacobs and Rubery (6) demonstrating that certain flavonoids effectively compete for NPA binding sites in microsomal membranes from zucchini hypocotyls. While it is not clear how their data in zucchini relate to the data in Table IV in corn membranes with regard to specific flavonoid substitution patterns, their report is an independent confirmation that a free carboxylic acid group per se is not an absolute requirement for significant binding affinity to the NPA receptor.

These three paragraphs engage with relevant aspects of the literature on geotropic inhibitors, challenging the work of one theory and drawing on another for support. The hedges here are therefore a mixture of accuracy and reader-based types, as the authors simultaneously seek to establish their own position while mitigating the criticisms of the other research.

The first paragraph sets out a model proposed over a decade earlier and simply reports the orthodox view without commentary. The hedging verbs in the final sentence (178-182) sustain the ambiguity about how far the writers are committed to the truth of these rules by framing the earlier claims in terms of the researchers', possibly flawed, interpretations of their results. The extent to which there may be intention to mitigate criticism here remains an open question, but the next paragraph clearly confronts Katekar's criteria quite directly, although this is introduced by a reader-oriented hedge (184) which prefaces the disagreement with a reference to empirical rather than personal factors. The hedge here therefore situates the review which follows in a context which focus on an analysis of the data and not its interpretation, thereby weakening the potential threat of bald criticism.

The rhetorical importance of this reinterpretation however, appears to be greater than the need to damp critical comment down too far in response to face or deference considerations. The argument here is central to the authors' attempts to recontextualise their findings in the body of truth accepted by the research tradition. The paragraph therefore directly contrasts the two studies in order to better present the current work and gain community acceptance for it.

assertion of novelty: that the experiments have revealed the existence of a new group of compounds which have implications for our understanding of the NPA binding site in corn. The authors admit however that further work is required to elucidate this understanding and a number of issues are raised for further study. The explicit references to the existing state of knowledge in lines 216-220 mark simple subjective uncertainty regarding the precise nature of the phenomenon. One informant, a journal editor, remarked that this also tells reviewers that the writers have "done their homework". Claims about the importance and possible implications of further studies into these areas however are more speculative and the forms "seem" and "may" carry stronger writer-oriented meanings.

The second paragraph is far more conjectural and tentative and takes advantage of the opportunities provided by the discussion section for writers to respond to larger issues and explore the perspectives on their work which will secure them academic credit for their insights. The writers here expand the discussion to the possible implications of their findings for chemical synthesis more generally. Taking their speculations beyond their results, they suggest that the molecule responsible for binding NPA is a conjugate of IAA (lines 235-237) and that it may be IAA-valine (lines 239-240). They then give reasons why readers should accept this.

The novelty of this proposal is indicated by the caution taken in expressing it and there is a great deal of hedging in this paragraph. It commences by directly addressing the reader with a question which both helps to frame the claims which follow and to gain reader support for them. It simultaneously indicates the current state of knowledge and addresses an intelligent collegial audience capable of making the same speculative leaps as the authors. The limited state of current knowledge against which to measure the importance of the findings is again referred to with accuracy hedges in line 231-3, and a reference to a tentative proposal in the literature (233-5). This citation is presented without comment and not directly addressed in the next sentence where the authors' view is implicitly presented as both building on that of Jacobs and Rubery's "suggestion", and as contributing to the ongoing research into a live problem.

By marking the proposition as a personal belief with "we suggest" (235) the authors recognise the possible interpersonal effects of their claim and simply label it as an alternative position without weakening its probability in any way. The extent to which the claim can be regarded as an uncertain possibility

is shown in the following sentence (line 237). This provides a barricade of writer and propositional hedges to both defend the writers from criticism and contextualise their claims within the limitations of current knowledge. Here the claim is explicitly labeled as speculative and the authors' commitment to it is carefully circumscribed. One informant called this clear signaling "a form of research honesty". This is done by the absence of human agents in the clause constructions, impersonalisation of the epistemic verbs "suggest", "speculate" and "possibility" and the two references to the absence of supporting evidence. The repetition of impersonal subjects, passive constructions and two references to the "model" (lines 241 and 246) continue to ensure that the authors are able to keep a distance from their proposal.

The ability of writers to combine hedges and concentrate them at points where their views are most vulnerable, where claims are most speculative, is clearly shown in this paragraph. The cumulative effect of 15 hedges in this paragraph clearly makes a powerful impact on the reader. But it also allows the writers to make the novel, high level claims for which they are likely to gain most credit. Representing these claims as possibilities and surrounding them with writer-oriented and propositional hedges is simply a prudent insurance. One senior biologist observed that this kind of self-protection was virtually mandatory:

> making a definitive statement based on inconclusive evidence, something that can't be tied up with a bow, would be a pretty dangerous thing to do and may well seriously damage your reputation.

The article concludes with a brief paragraph which re-establishes the importance of the territory previously staked out and restates the paper's main claim. The authority of this assertion is mediated by an explicit intrusion by the authors and a judgement verb, both of which clearly draw attention to the limited status of the proposition and once again defer to the possibility of reader adherence to an alternative position (256). The final suggestion, a procedural recommendation, is hedged with a concern for propositional precision, indicating less assurance by the writers on the direction that future research should take.

This example demonstrates the complex interplay of content and reader-oriented hedges in allowing writers to discuss speculative possibilities and express claims with varying degrees of personal assurance and commitment. Across sections, it is not surprising to find that propositional hedges constitute

the most frequent type overall and predominate in all sections except the Introduction and the more speculative section of the Discussion (Table 1). The most common uncertainties therefore relate to the those associated with confirming epistemic truth. Writer-based hedges constitute the next most common and outnumber accuracy-oriented hedges in the final, most speculative section. Reader-oriented mitigation did not figure prominently in this text but, again, was most frequent in the final section of the Discussion where the writers were seeking to promote their more novel, higher-level claims by stretching their interpretations beyond the data.

Table 1: Distribution of Hedging in Text 4

Text Section	**Type of hedging**			**Total**	**Total**	**hedges per**
	Accuracy	**Writer**	**Reader**	**Hedges**	**words**	**1,000 words**
Introduction	1	1	1	3	241	12
Methods	3	0	0	3	705	4
Results	5	4	0	9	519	1.7
Discussion 1	5	3	1	9	366	25
Discussion 2	8	10	3	21	458	45
Totals	22	18	5	35	2289	15

The analysis of a longer extract demonstrates the cumulative effect of hedging through a paper and how writers use different types of hedging to achieve different purposes. By the judicious use of different hedging forms the writers here are able to achieve a balance of scientific accuracy, self-protection and deference to their audience; presenting their claims with the precise degree of certainty, commitment and responsibility they wish to acknowledge. The analysis also revealed the rhetorical impact of concentrated hedges. By examining an unfolding text in this way it becomes clear how writers can increase the speculativeness of their claims through increasing the frequency of hedges and using them to work together in concert.

Some comments on the interpretations

The interpretations of these extracts is often difficult to confirm with certainty given the high degree of pragmatic indeterminacy of the devices employed.

Core examples are exceptional and the majority of cases exhibit semantic fuzziness, making categorical conclusions hazardous. The collection of interpretive data from scientists is an attempt to determine the meanings forms have for insiders, but the inherent subjectivity of utterance interpretation remains. Indeed, the interviews and focus groups produced considerable debate about some of the meanings. In particular, as I noted earlier, scientists appear to find it difficult to recognise an interpersonal motivation for hedging, although my informants were happy to acknowledge that caution and defensiveness were common in claim construction.

Thus while writers routinely use hedges to tone-down their statements, the precise reason for this may not be made at a conscious level and may therefore not be accessible to either the analyst or the scientist reader. Moreover, think aloud studies are introspective and give an artificial emphasis to features which may often receive only cursory attention when writers selectively search articles for novel results and details relevant to their own research. But although such procedures are inauthentic and lack the objective measurement of experimental studies, they nevertheless allow us to shift the research emphasis from products to processes and to gain the insights of insiders crucial to the development and confirmation of categorisation schemes (Smagorinsky, 1994).

Many of the functions assigned here therefore depend on close analyses of target forms which have uncertain weight for writers. Think aloud data is notoriously subject to idiosyncratic interpretation (Nisbet & Wilson, 1977) and can only elicit incomplete information (Cooper & Holzman, 1985). Ultimately, the actual practices and motivations behind a particular form might only be available to the actual writer and only then at the time of writing. Despite the views of expert informants then, I am more confident in some interpretations than others, and in a few cases doubts may be raised about some assignments.

I do not believe however that this is damagingly critical of either the analysis or the framework. It is simply a recognition that the creation of text and its interpretation are not exact sciences but depend on interpretation and the reader's perceptions. There is a band of variation possible and interpretations may place different emphasis and reliance on particular contextual factors. Thus there are some assignments about which there may be reservations and with these it is fruitful to reverse the normal process of analysis. Instead of focusing on the form and its context, we can ask what function an inference allows us to make in this case.

I am aware of no standards for comparison of the levels of tolerance acceptable in cases of doubt. It is, however, important for researchers to understand and delineate their approach to the data and devise a coding scheme that describes the processes that their theory delineates. The best means of interpretation therefore requires having as clear criteria as possible, taking the opinions of insider informants into account and being precise about areas of uncertainty. I have tried to draw the relationship between my perspective on scientific discourse and the functions served by hedging as clear as possible, but the coding of actual items in some cases may be ambiguous.

It should also be noted that while employing my analytical framework to discuss the hedges in these extracts, the views of insiders have tended to focus on an identification of their perceptions of the *primary* meanings conveyed. I acknowledge that this approach gives less attention to the other meanings which are being simultaneously expressed, but it does reveal more clearly the most salient features of the devices as they are used in specific contexts and thus helps to account for the pragmatic meanings expressed in the passages more distinctly. It should be borne in mind therefore that it is the "strongest" meanings which have been highlighted in the discussion and that this in no way implies a mono-semantic approach to hedges. Indeed, this consideration of examples of hedges in use should help confirm the view that a strict categorical model of discrete meanings cannot be supported by the empirical evidence.

Finally, I would like to reiterate the importance of seeing research papers as the products of cognitive activities influenced by disciplinary social practices. As Berkenkotter and Huckin (1995: 17) observe:

> No act of communication springs out of nothing. In one way or another, all acts of communication build on prior texts and text elements, elements that exist on different levels, including words, phrases, discourse patterns, illustrations and so on.

Academic writing is a processes of recurring rhetorical responses to particular social demands and its study must therefore involve situating writers and writing in a social, psychological, historical and political context. The increasingly accurate specification of this context offers one of the best ways of not only understanding the features of texts, but also of unlocking the mechanisms that community members bring to the production and interpretation of those features. Interviews and think aloud data can thus be remarkably illuminating of insiders' interpretations when these are seen as embedded in members' participation in the communicative activities of disciplinary life.

8. Hedging and second language learners

The considerable significance of hedges to scientists and to science discussed in previous chapters can also be extended to second language students acquiring the rhetorical skills they need for successful academic writing. But while writers need to make assertions which academic readers judge to be warranted and which reflect appropriate social interactions, the ability to express degrees of certainty is a difficult task for language learners (Allison, 1995; Hyland & Milton, 1997). However, the English for Specific Purposes (ESP) literature has generally contained little to guide L2 students towards an understanding of these forms or how they might be correctly used. The idea that academic writing is simply objective, impersonal and informational is still widespread among teachers and students and it is rare for modality to be either explicitly taught or adequately covered in textbooks. Because of this neglect, both native and non-native speaking students find it difficult to orientate themselves to using hedges in their writing.

The problem is particularly acute as English is now well established as the medium of international communication in science (Maher, 1986; Swales, 1987) and the ability to write effectively in the language is a prerequisite for full membership and involvement in one's research community (Bartholomae, 1986). The numbers of second language speakers studying scientific subjects in English has therefore grown enormously in recent years and foreign nationals now outnumber American students in US science and engineering graduate schools (Jenkins et al, 1993). Given the large numbers of foreign language students using English for academic courses or for scientific research, empirical findings on hedging use may have direct and immediate relevance to teachers, materials writers and students. The purpose of this chapter is therefore to place scientific hedging within the context of the teaching of academic and scientific

English and discuss how it is treated by teachers and in the pedagogic literature. Some suggestions will also be offered for encouraging learners to be aware of hedging in their reading and teaching them to use it appropriately in their writing.

Hedging in L2 students' academic writing

Throughout this study epistemic aspects of language have been shown to be a principal means by which writers can use English flexibly to adopt a stance to both their propositions and their audience. But while these forms can have a considerable effect on a reader's assessment of both referential and affective aspects of texts, they are generally acknowledged to be difficult for learners to acquire. The achievement of good academic communication skills represents a heavy burden on non-native speakers of English (Connor & Lauer, 1988; McCann, 1989; Ventola, 1992) and may be responsible for the low level of NNS contributions to the English language scientific literature.

As we have seen, hedging devices are complex for novice writers for a number of reasons. The main difficulty is the fact they can simultaneously convey a range of different meanings, not only signaling the writer's confidence in the truth of referential information, but also contributing to a relationship with the reader. Students also experience difficulties because epistemic meanings can be signaled in many different ways, with a variety of devices and expressions used for this purpose. The obstacle of learning appropriate strategies is further compounded by the absence of adequate pedagogical materials (Holmes, 1988b; Hyland, 1994) and the fact that different languages use different linguistic means for expressing modality. Nor are NNS students assisted by the fact that their efforts to master these forms are often measured against an unrealistic "expert writer" standard in terms of the requirements of an academic community to which they may have little experience.

A number of writers have commented on these difficulties. Skelton (1988a) and Bloor and Bloor (1991), for example, observe that direct and unqualified writing is more typical of EFL students than native speakers, even of poor adult writers, and this view receives support in studies by Johns (1990) and Silva (1993). Hu et al (1982) found Chinese L2 writers to be more direct and authoritative in tone and to make more use of strong modals than NSs, while Allison (1995) also expresses concern that ESL writers in Hong Kong

frequently make unjustifiably strong assertions. Arab students also appear to have difficulties in this area (Scarcella & Brunak, 1981), and Robberecht and Peteghem (1982) refer to the problems of Dutch and French students in mastering modality. These problems persist for L2 writers at post graduate level, where PhD supervisors are often required to counsel the need for appropriate degrees of qualification and confidence in expressing claims (Dudley-Evans, 1991: 47).

Principles of language behaviour and preferred patterns of exposition and argumentation have been found to differ enormously across cultural groups and such differences may influence how students write in English. Until recently the field of Academic English largely ignored the issue of students' culture, partly because of a traditional view of science as culturally neutral (Barron, 1991). Widdowson (1979) was influential in establishing the attitude that there is an independent scientific culture that is expressed by a "universal rhetoric". The development of academic writing among NNS learners, however, raises the issue of whether the structures of scientific knowledge are invariant and, in particular, whether the discourse features of academic writing are universal or culture-specific.

Thomas (1983) refers to the problems of sociolinguistic miscommunication as "cross-cultural pragmatic failure" which can result from either violations of the social conditions on language use (Sociopragmatic failure), or to different interpretations of equivalent speech acts (Pragmalinguistic failure). Absolute distinctions are impossible, however, as language use involves both an assessment of relevant contextual factors and the selection of an appropriate linguistic strategy for its expression. Transfer from the L1 may occur, then, as a result of the fact that different languages use different linguistic means for expressing modality. NNS students may therefore have an erroneous understanding of appropriate formality, directness, intensity, deference, politeness and other pragmatic rules required by the discourse community as a result of the different practices which operate in their own cultures. Thus Bloor and Bloor (1991) consider the degree of indirectness and concession permitted in academic writing in different languages a clear source of cultural difference.

While little understood, cultural variations in the norms of academic writing are a central issue. Pragmalinguistic factors are important as they involve cultural differences in the resources used for conveying particular illocutions and degrees of courtesy. Lyons (1981:238) points out that different languages use different linguistic means for expressing particular kinds of modal-

ity. In making a knowledge claim or discussing a rival theory, for example, a writer may select a strategy which conveys the appropriate illocution but has an inappropriate relational meaning, or deference.

A number of studies have identified variations in the certainty with which arguments are expressed in different languages. Academic writing in German and Czech, for example, appears more direct than in English (Bloor and Bloor, 1991:9; Clyne, 1987), while Finns (Mauranen, 1993; Ventola, 1992), Japanese (Hinds, 1983; Harder, 1984), Malays (Ahmad, 1995), Koreans (Choi, 1988; Eggington, 1987), and Chinese (Bloch & Chi, 1995; Hinkel, 1994) seem to favour a more cautious and indirect style when expressing opinions. However, in some cases it is unclear whether this means any more than an apparent preference for inductive discourse organisation (cf. Mohan & Lo, 1985; Young, 1982). There is also considerable evidence from the IEA study of written composition in 14 countries that cultural variation characterises specific uses and conventions in school leaver essay writing tasks (Purves & Purves, 1986).

Hinds (1987) has argued that such differences are related to expectations concerning the extent of reader involvement. English language cultures charge the writer with clarity, while Japanese requires a more active reader role. Similarly, Clyne (1987) contrasts 'Saxonic' and 'Teutonic' intellectual styles in terms of the different expectations of the reader and argues that texts by German academics are not designed to be easy to read. Scollon and Scollon (1995), on the other hand, suggest that an Asian preference for rhetorical strategies of indirectness and induction are due to the different cultural structuring of situations and participant roles, ultimately related to questions of interpersonal face politeness.

There are too few studies to draw definite conclusions on the issue of hedging. Some research suggests transfer may occur. Thus Connor (1996), for example, found a Japanese businessman writing in English made greater use of hedges than is common in L1 writing, while Crismore et al (1993) observed that the Finns tended to hedge their claims and conclusions much more than Americans when writing in the L2. Hinkel (1997) found that the essays of American, Korean, Japanese, Chinese and Indonesian students differed in their use of some indirectness markers but not others, demonstrating that perceptions of Asian students' writing as vague and indirect is only partially justified. Hyland and Milton's (1997) analysis of a one million word corpus of exam scripts revealed that L1 British school leavers used about twice as many hedges as Cantonese learners writing in English in similar circumstances, despite the fact

Chinese are often considered 'indirect' in their L1 argumentation. Clearly numerous factors may influence writing, and proficiency must be considered among these.

The studies in contrastive rhetoric do not predict the marked disparities between L1 and L2 argument paradigms, but they do suggest that students from different cultures may have preconceptions about the formal features of culturally and rhetorically appropriate writing which may differ from those which operate in English academic settings. The findings raise complex issues which cannot be explored here, but there is reliable evidence that the rhetorical preferences of different languages and cultures tend to manifest themselves consistently in the L2 writing of NNS students (eg. Sötter, 1988; Connor, 1996). Such differences can make NNSs vulnerable to the risk of violating communicative norms. Their writing may appear as too direct and run the risk of being considered as either brusque or dogmatic, or it may be seen as too tentative, and therefore be judged as equivocal, diffident or naive.

In summary, the need to present claims that are neither overstated nor understated in relation to evidence or reasonable assumption is an area of academic writing which many students, both native and non-native speakers, find seriously challenging.

Advice from the style guides

Growing awareness of the problems faced by both native and nonnative speakers of English in mastering the skills of academic and research writing has resulted in the growth of a large number of writing manuals and style guides. Directed principally at L1 speakers, these aim to assist all writers with prose and formatting style (eg O'Connor & Woodford, 1976; Booth, 1985). These texts generally include information and advice on editorial style and format as well as guidelines on grammar and usage. Some are prepared as the official guides of professional academic bodies which inform authors how to prepare manuscripts for publication (eg APA, 1994; Achtert & Gibaldi, 1985). Others are 'handbooks for writers' which address a range of generic expression and presentation skills (eg McKernan, 1988; Howell & Memering, 1986; Bate & Sharpe, 1990; Leggett et al; 1991; Beazley & Marr, 1992). There are also a number of style guides written specifically for scientific writers (eg Lindsay,

1984; Day, 1992) and for those producing research papers (eg Day, 1988; Winkler & McCuen, 1989; Roth, 1989).

Casanave & Hubbard (1992) found that members of science faculties rarely recommended a style manual to their students and largely left them to their own devices, occasionally referring them to a leading journal or to Strunk and White's *The Elements of Style*. The style guide literature however is generally heavily prescriptive and largely restricted to advice on grammar, spelling and admonishments concerning clarity, precision and organisation. It largely focuses on word and sentence level issues and generally ignores prose style. As Selzer (1982: 76) pointed out over a decade ago, such manuals are often "long on exhortation (...) but short on real instructions. They often describe a good technical style but seldom do much to help students achieve it". Indeed, Gross (1984) concludes that traditional advice on style is often not sound editorial practice as "stylistic changes are tactical choices made within the context of strategic presuppositions about the impersonal and descriptive nature of scientific prose".

The practical orientation of these texts has generally been at the expense of a sound theoretical or empirical foundation and consequently a great deal that is rhetorically important is omitted or the advice flawed. This is the case in the treatment style guides give to hedging, for while many authors have nothing to say on the subject (eg Leggett et al; 1991; Beazley & Marr, 1992; Roth, 1989; Day, 1992), others contain conflicting or contradictory assertions. Much of this work advises writers to avoid all tentative assessments, arguing that hedges rob writing of its certainty and power (Zinsser, 1976: 96) or serve to avoid commitments and undermine judgements (Rawlins, 1980). Thus Lindsay advises that:

> If you have no conclusive evidence don't dither around with expressions such as 'it may be possible that..' of (worse) 'the possibility exists that ...', which immediately suggests that you do not believe your own data.
>
> (Lindsay, 1984: 21)

Ignoring the fact precision is only appropriate in certain situations, the dominant view in this literature is that writers should eliminate hedging as it adds little to propositional information: "If your assertions are not made with precision, the value of your paper will be inevitably lowered" (Winkler & McCuen, 1989: 97).

The term *hedging* itself, as Skelton notes, has "unfortunately pejorative connotations in ordinary language" (1988b:98) and this also tends to be the

view of style guide authors' who call them "unnecessary words" (Yarber, 1985: 188), "padded expressions" (Muller, 1985: 328), "wasteful signposting" (Smith, 1985: 92), or "clutter words" (Lannon, 1986: 135). The most eloquent position is taken by the now dated but still widely used, *The Elements of Style* which calls hedges "the leeches that infest the pond of prose, sucking the blood of words" (Strunk & White, 1959: 59). Some writers are more positive, advising students to hedge material that is debatable or inconclusive (Booth, 1985:12; de Beaugrande, 1985:17; Hairston, 1981) or which expresses "doubt, wishes, probability, conditions contrary to fact, or conditional statements" (Howell & Memering, 1986:107). Recognition of the value of hypothesis in academic writing is, however, rare in this work.

Generally speaking the information and guidance given to writers on this subject is poor and fails to reflect the view emerging in applied linguistics concerning the value of teaching appropriate hedging strategies (eg Stubbs, 1986; Crismore & Vande Kopple, 1988; Holmes, 1988b). This stresses the importance of mitigation as a resource in distinguishing a proposition and the writer's commitment to it. Skelton (1988b: 107), for example, argues that the ability to hedge represents an important means by which NNSs are able "to use language with subtlety, to mean precisely and with discrimination". Dillon (1981:91) observes that good academic discourse requires hedging and suggests that a great deal of student writing "stands in need of a delicate qualifier here or there rather than the removal of them". Other writers suggest that a degree of tentativeness can help convey a sense of debate, compromise and sensitivity to the views of the audience (Laib, 1985).

The widely differing advice in the style guide literature is therefore largely subjective. It suffers from both an inadequate theoretical standpoint and a lack of analysis of authentic uses of English. Consequently it fails to confirm the assumptions of its prescriptions.

Hedging in ESP textbooks

The abundance of ESP writing materials is partly a result of the specialised nature of student needs in this area and partly because of the widespread belief that teachers should develop their own materials. Swales (1980: 18) voiced the predominant view of "buy a course and be damned or write a course and be praised" over a decade ago. But in reality, published texts are an ubiquitous

aspect of the ESP scene, being a major source of ideas for course structure, practice activities, language models and materials, even when not used as set texts (Richards, 1993). This reliance on published texts makes it imperative that they reflect actual language use, but textbooks are often deficient in their coverage of major items. Insufficient recognition is given to the pedagogical implications of current linguistic research, and crucial features of scientific discourse are neglected. Sheldon (1987: 3), for example, observes that "too often published materials fail to rest upon sound theoretical bases". With regard to hedges, these materials appear to provide little more assistance than the style guide literature. This issue will now be discussed in detail.

This section evaluates the adequacy of a range of ESP and EAP textbooks in providing learners with information on hedging, focusing on lexical items as markers of tentativeness. While this procedure ignores structural and strategic hedges, lexical choices represent the most frequently used means of expressing evidentiality and doubt in academic discourse, representing 85% of all hedges in my corpus. They are also the simplest devices for L2 learners to acquire, and we might therefore reasonably expect them to occur in textbooks more frequently than other devices.

A corpus of 22 textbooks was chosen as representative of a range of writing materials intended for L2 students, from post-beginner to advanced level, engaged in or preparing for academic study or specialist science courses in English. The selection was devised to provide a cross-section of popular commercial texts produced over the last 20 years or so. Consequently the books contrast in terms of their styles, organisation, assumptions of proficiency, intended audiences and pedagogic approaches. Taken together they constitute many of the texts in use around the world today to teach academic writing.

Texts covering a wide time scale were selected to ensure that a variety of approaches were included in the survey. As mentioned previously, commercial texts constitute the major source of teaching ideas for many teachers (eg Swales, 1980; Richards, 1993), and while older books may no longer be used as core texts, many continue to figure in EAP courses as reference or resource materials. The constraints of inexperience, institutional pressures and inadequate preparation time mean that resource banks of published texts from different eras are frequently drawn on for practice activities, language information and teaching models, indeed, in many countries older textbooks are the only available resources. As a result, large numbers of students continue to be ex-

posed to the materials in this selection, despite the rather outmoded pedagogical assumptions some reflect.

The texts therefore mirror many of the changes in ESP teaching in the last two decades and characterise the array of methods currently employed in the field. Older texts are predominantly organised around grammatical structures and parts of speech (eg Ewer & Lattore, 1969; Chaplen, 1981; Oshima & Hogue, 1981) or scientific topic areas (Royds-Irmak, 1975; Hutchinson and Waters, 1984). Later materials follow a functional direction, focusing on 'notions' or common patterns such as "classification", "definition" or "cause and effect" (eg Moore, 1980; Donovan, 1978; Cooper, 1979; Zimmerman, 1989; Mann and Mann, 1989; Jordan, 1990) or macro structures like "narration", "argument" or "exposition" (eg Neufeld, 1987). Methodologically both process (eg Kennedy & Smith, 1986; Brown et al, 1991), and product-based approaches (Arnaudet & Barrett, 1984; Weissberg & Buker, 1990) are represented, with some texts combining the two (eg. Neufeld, 1987; Reid, 1988; Currie & Cray, 1987). Reid (1988), Kennedy and Smith (1986) and Arnaudet and Barrett (1984) contain sections on writing up scientific research, but only Weissberg & Buker (1990) is devoted exclusively to the needs of research students, despite Swales' (1987) emphasis on its importance to L2 students.

Tables 1 and 2 summarise the results in terms of the primary focus of the textbook and the adequacy of the coverage given to lexical expressions of epistemic modality.

The evaluation is based on the number of exercises and quality of information devoted to relevant concepts and linguistic items. The hedging devices are grouped into their grammatical classes and broadly classified under three headings. "No coverage" refers to cases where the topic is not addressed, "minimal coverage", where little information, few examples and less than three exercises are provided, and "fair to extensive coverage" designates cases which offer a more thorough coverage and three or more exercises. In general, while the EAP writing texts (Table 1) seem to deal with the issue more comprehensively than the ESP materials, it appears that the interest in modality in the research literature is not widely reflected in the pedagogic materials.

Modal verbs receive the best coverage, representing a well established body of linguistic knowledge about their forms and meanings (eg Lyons, 1977; Perkins, 1983; Coates, 1983; Palmer, 1986 & 1990). Perkins (1983: 104) observes that the modal auxiliaries "provide the least marked, and thus the most straightfor-

ward, means of expressing modality in English". *Would*, *could* and *may* account for about $^{3}/_{4}$ of all cases in my corpus.

Table 1: Occurrence of modal expressions in selected EAP textbooks

EAP Writing Textbooks	Modal verbs	Lexical verbs	Modal Adverbs	Modal Adjectives	Modal nouns
Academic Writing Course Jordan (1990)	✔	✔ ✔	✔ ✔	✔	✘
Assignment Writing: McEverdy & Wyatt (1990)	✔	✔	✔	✔	✔
Writing up Research Weissberg & Buker (1990)	✔ ✔	✔ ✔	✘	✔	✔
Approaches to Academic Reading and Writing Arnaudet & Barrett (1984)	✔	✔ ✔	✔	✘	✘
Think and Link Cooper (1979)	✔	✘	✔ ✔	✘	✘
Strictly Academic Currie & Cray (1987)	✔	✔ ✔	✘	✘	✘
Write Ideas: An intermediate course in writing skills Glendinning & Mantell (1983)	✔	✔	✘	✘	✘
Writing Academic English Oshima & Hogue (1981)	✔	✔	✘	✘	✘
Plan, Write, Rewrite McEverdy & Smith (1990)	✔	✔	✘	✘	✘
Introduction to Academic Writing Oshima & Hogue (1988)	✔	✘	✘	✘	✘
Academic Writing Kennedy & Smith (1986)	✘	✘	✘	✘	✘
Essay Writing Mann & Mann (1989)	✘	✘	✘	✘	✘
The Process of Composition Reid (1988)	✘	✘	✘	✘	✘
Challenges: A Process Approach to Academic English Brown, Cohen, O'Day (1991)	✘	✘	✘	✘	✘

Key: ✘ = no coverage ✔ = minimal coverage ✔ ✔ = fair to extensive coverage

Table 2: Occurrence of modal expressions in selected ESP textbooks

ESP Textbooks	Modal verbs	Lexical verbs	Modal Adverbs	Modal Adjectives	Modal nouns
English for Science Zimmerman (1989)	✔ ✔	✔	✔	✘	✘
Basic English for Science Donovan (1978)	✔	✘	✔	✔	✘
Reading & Thinking in English Books 1-4 Moore (1980)	✔	✔	✘	✘	✘
A course in Intermediate Scientific English Chaplen (1981)	✔	✘	✔	✘	✘
A course in basic scientific English Ewer & Lattore (1969)	✔	✘	✔	✘	✘
Beginning Scientific English Books 1 & 2 Royds-Irmak (1975)	✔	✘	✘	✘	✘
Interface Hutchinson & Waters (1984)	✘	✘	✘	✘	✘
A Handbook for Technical Communication Neufeld (1987)					

Key: ✘ = no coverage ✔ = minimal coverage ✔ ✔ = fair to extensive coverage

Despite the applied linguistic preoccupation with modal verbs, only *English for Science* and *Writing up Research* deal with their epistemic use in anything but the most cursory way. Both discuss modal verbs as expressing an attitude of uncertainty, the former devoting several pages to means of expressing unproven hypotheses (pp. 54-7), and the latter providing a scale of tentativeness (pp. 82-4). This ranges from *will* as the most certain, through *would, should* and *may* to *could* as the most tentative, with an explanation of each and an example of their use. In addition, the authors offer the following advice:

> the statement of value is usually written in a way that suggests an attitude of tentativeness or modesty on the part of the author. When report-

> ing your own study, you should not sound too sure of the benefits, either practical or theoretical, of your work. It is conventional to sound more cautious. This is accomplished (...) by using modal auxiliaries, principally may. (Weissberg & Buker, 1990: 82)

Unfortunately such an awareness of how modals are used is rare. For the most part modal expressions are simply introduced without system or comment and are summarily dealt with in a single exercise which fails to emphasise either their function or importance.

Generally the range of modal verbs addressed and the information provided on their use is inadequate, given that modals are the most easily identified means of hedging. Most texts limit their coverage to *would*, *should*, *can* or *may* and these often appear under an unrelated topic (*Interface*: 15), or are concealed in a grammar exercise such as a passive transformation task (*Intermediate Scientific English*: 61; *Beginning Scientific English 1*: 93). Other textbooks simply point to their existence without incorporating them into a writing activity (*Strictly Academic:* 128*; Basic Scientific English:* 53; *Approaches to Academic Writing*: 83). Importantly, many texts fail to either maintain or clarify the distinction between epistemic and root modality, thereby confusing judgements of belief with those of obligation or volition, and often neglecting the hedging function of modals altogether (eg *Intermediate Scientific English*: 60).

Lexical verbs are the most common means of expressing epistemic modality in written academic discourse (6.4) and have received some attention in the literature (eg Perkins, 1983: 94-9; Brown, 1992). Once again, however, the textbook coverage is patchy and only *Approaches*, and *Strictly Academic* expose learners to more than a limited range of items. *Strictly Academic* asks students to complete sentences giving an appropriate source from a reading passage, then to list all the "verbs of assertion" and consider which are "strong" and which "weak" (50-1). The best coverage is offered by *Approaches,* which introduces reporting verbs in a reading sample and then discusses examples according to whether they express neutrality, connotation, uncertainty or opinion. This is followed by a paraphrase exercise where students select a verb to convey the appropriate meaning (pp. 153-7). *Writing up Research* (p 56) instructs students to use tentative report verbs in the past tense when citing the work of others. It then asks students to determine whether example statements present information as facts or tentatively, and to rewrite them expressing the opposite

attitude. There are also reading tasks which require students to circle tentative verbs and modal auxiliaries in a passage (57, 84, and 193).

Again, a number of texts present lexical verbs without giving adequate recognition to their discourse function. *Plan, Write, Rewrite* informs students how to incorporate quotations into their writing (p 78), for example, but fails to mention the 'stance' a writer can take towards the cited information shown by using hedging items. *Writing Academic English* includes *believe, think and estimate* when introducing the rules governing the sequences of tenses in noun clauses (p 142), while *Write Ideas* merely provides three examples in a substitution table as "useful language for making recommendations" (p 119). Jordan's *Academic Writing Course,* however, suggests that lexical verbs may be used to introduce conclusions (p 67), or the writer's point of view (p 77). *Writing up Research* advises writers to use "tentative verbs" such as *appear, seem* and *suggest* instead of modals to generalise from results when presenting findings (pp. 56 and 149), and to "emphasise the speculative nature" of statements (p 171).

Adverbials, nouns and adjectives are used extensively to express modality in written texts (Adams Smith, 1984; Skelton, 1988b) and together constitute 53% of the lexical devices expressing hedging in the RA corpus. Like lexical verbs, adverbials offer a wide range of means for expressing degrees of certainty and this is acknowledged by a number of textbooks. *Probably, possibly, apparently* and *unlikely* occur most often in the materials, with *Academic Writing Course* providing the most extensive list, including *hardly, scarcely, practically, virtually, presumably* and *slightly* (pp. 54-5 & 66). Both *Assignment Writing* and *Approaches* distinguish adverbs of tentativeness from those expressing certainty, but only *Academic Writing Course* (p 66) and *Think and Link* (p 42) indicate the relationship between different items in terms of a scale of certainty.

None of the textbooks gives much attention to the epistemic uses of adjectives and nouns, despite their distribution in academic writing. The nouns *assumption, claim* and *evidence* are the only examples cited and the absence of high frequency items such as *possibility,* and *estimate* are difficult to explain. Similarly, only a handful of adjectives are illustrated, and only *Basic English for Science* (101-113), *Developing Communication Skills* (p 90) and *Academic Writing Course* (p 66) explicitly include adjectives as a strategy for expressing uncertainty.

Clearly there is a wide variation in the treatment of epistemic devices in ESP/EAP textbooks and in the assistance they give L2 writers in exploiting grammatical choices to hedge statements. While some texts provide an impressive coverage, others confuse the semantic distinctions between different expressions and a number, particularly those taking a process approach, fail to address the topic at all. Generally the presentation of hedges in published texts is poor, with information scattered, explanations inadequate, practice material limited, and alternatives to modal verbs omitted. This failure to adequately represent hedges therefore gives misleading information to students concerning both the importance of the concept and the frequency of different devices.

Overall, the most comprehensive coverage is provided by Jordan's *Academic Writing Course,* McEverdy and Wyatt's *Assignment Writing,* Arnaudet and Barret's *Approaches to Academic Reading and Writing* and Weisberg and Buker's *Writing up Research.* The first two are for upper intermediate, pretertiary level students and each devotes a section to demonstrating the functions of a range of epistemic devices. McEverdy and Wyatt explicitly inform students that the use of tentative statements is "one of the distinguishing features of assignment writing" and list "important phrases and vocabulary" for completion exercises (85-93). Jordan provides a "scale of qualification" of quantity, frequency and probability (66) and lists a range of hedging devices but provides no practice tasks. Unfortunately, there is little exploitation of extended writing activities in either text.

The other two textbooks assume a higher level of English proficiency and are intended for undergraduates and research students. Both stress the importance of tentativeness in academic writing and give a clear indication of how students can accomplish this, particularly in terms of modal auxiliaries and lexical verbs. In addition, both emphasise the need for writers to express their views appropriately:

> as a student writer, you too should make use of this kind of language in your writing. Otherwise the conclusions you draw or the support you offer will be open to criticism from your professors and classmates as not being intellectually honest in that you are not "leaving the door open" for other points of view. (Arnaudet & Barrett, 1984:83)

Hedging a proposition as provisional, "leaving the door open" to alternatives, is a rare reference to the interactions in academic writing and the affective function of hedges.

With some notable exceptions therefore, textbooks fail to meet one of the four major criteria proposed by Candlin and Breen (1979) for analysis of teaching materials, namely: "Is the language of materials derived from a sociolinguistic analysis of the target repertoire?". Virtually all frameworks for evaluating textbooks include an emphasis on authentic language use (eg Sheldon, 1988; Richards, 1993) but there are clearly competing criteria for including items, not least pedagogical considerations of appropriate level. Consequently, the absence of hedging information in texts such as *Interface* and *Beginning Scientific English* may not seem particularly serious, as hedging devices are likely to be introduced at a later stage of learning. However, the coherent treatments offered in sources such as *English for Science* and *Basic English for Science* show that such significant items can be matched to both needs and lower ability levels. Hedges are important enough to warrant their inclusion in even introductory level textbooks, and their acquisition is undoubtedly a process learners should be exposed to from the earliest stages.

Thus despite the pragmatic and distributional importance of hedges and the emphasis ESP has placed on descriptive bases for materials, research interest in such rhetorical aspects of writing currently receives little recognition in coursebooks. Holmes (1988b) also found that four popular ELT textbooks failed to reflect frequency data in providing learners with reliable information on modality. A particular difficulty is that ESP materials frequently draw discourse features from contexts which are inappropriate to research writing. There is a heavy reliance on extracts from scientific textbooks (eg Chaplen, 1981; Arnaudet & Barrett, 1984; Currie & Cray, 1987; McEverdy & Wyatt, 1990; Jordan, 1990), government scientific reports (eg Neufeld, 1987), popular accounts (Mann & Mann, 1990) or simplified texts (eg Royds-Irmak, 1975; Oshima & Hogue, 1981 & 1988). Most coursebooks thus tend to underrepresent the importance of hedges by drawing on sources which offer a more consensual view, presenting statements as accredited facts. As Myers (1992:11) observes:

> A student who knows only the way textbooks use hedges for uncertainty is unprepared for the ways articles use them in polite statements of claims.

Disturbingly, this neglect of modality in popular textbooks may be duplicated by teachers who rely on them as sources for their own courses. Swales' observation is relevant here:

> the locally produced materials which traditionally fill the gap left by the abandonment the ESP textbook usually show a striking resemblance to the published materials that have been rejected. Swales (1980:11)

Textbooks provide harassed teachers with both inspiration and a sense that the relevant and important areas are being covered. In fact, there is a common assumption that any item in a textbook must be an important learning item and, conversely, that anything not included can be safely omitted from a course. So, while "textbooks and other commercial materials in many situations represent the hidden curriculum of the ESL course" (Richards, 1993: 1), such sources are often flawed from a linguistic and descriptive point of view. The shortcomings of textbooks can therefore have serious effects on students' successful acquisition of essential communicative discourse features.

Escape from this closed circle clearly requires more careful consideration of the language itself, turning to analyses of authentic corpora as sources of insight and instances of use. This is necessary as the inadequacies which have emerged here seem largely a result of authors' poor selection procedures, perhaps due to the common dependence on target needs analyses to determine teaching items. But while providing some indication of the tasks students will have to perform, these often say little about the nature of those tasks. As Dudley-Evans (1988: 28)observes, "materials writers need detailed analyses of the rhetorical and linguistic organisation of the tasks if they are not to be over-reliant on their own intuition".

Clearly textbooks need to be based on more thorough research than at present. While no textbook can be globally comprehensive, the high proportion of hedged propositions in academic writing, and the clear difficulty students have in acquiring them, means that information on this topic should be a high priority for inclusion.

Teaching ESP students to hedge

The purpose of this section is to suggest some ways that teachers can help develop students' understandings of the principles and mechanics of hedging, thereby overcoming the deficiencies of commercial materials and bridging a major 'rhetorical gap' in the communicative competence of many learners. While comprehension and production are obviously closely related, I will discuss them separately for clarity. First I will look at how expert writers' hedging

strategies can be made salient, then go on to suggest how students might be encouraged to use appropriate forms in their own work.

A great deal of academic writing instruction is designed to raise learners' consciousness of "prototypical examples of relevant genres" (Swales, 1990: 215). Discourse-centred courses typically use collections of authentic samples of the standard genre and its variations. These are provided as models and studied for features of form and rhetorical organisation to help students discover the link between purpose, discourse patterns and language. They are then often manipulated in some way or used to produce parallel texts (Dudley-Evans, 1985; Hall et al, 1986; Ramani, 1988; Richards, 1988; Swales, 1987; Jacoby et al, 1995). The goal is summarised in Davies' (1988:133) description of a university reading and writing course based on the integration of specific relevant genres:

> This analytic reading may yield for students new knowledge within a subject area but this is not its principal purpose, which is to discover what is distinctive about the rhetorical organisation of the text and the way this is realised linguistically.

Recognition of the devices to be taught must be based on the target language repertoire, and therefore one approach is to ask students to analyse and emulate the models they find in complete journal articles. An easier way of highlighting particular features, however, is the use of a concordancer, a computer programme designed to search large amounts of computer-readable text for particular words and combinations. It therefore allows the analyst to draw conclusions from data which is isolated, sorted, counted and revealed in its typical contextual environments. Concordance software for PCs is both widely available and easy to use, and millions of words of machine-readable text is now commercially available on CD ROM or can be down-loaded from websites. With the assistance of such computerised text sorters students can become researchers and search for patterns which are not otherwise readily apparent. Wu (1992:32) summarises the advantages of a concordance approach in this way:

> Only when words are in their habitual environments, presented in their most frequent forms and their relational patterns and structures, can they be learnt effectively, interpreted properly and used appropriately.

Such 'data-driven learning' (Johns and King, 1991) is therefore an important means of stimulating inquiry and encouraging independent engagement with the language.

Concordanced output provides authentic data for materials so that worksheets can be devised to concentrate attention on hedging forms widely used in science RAs. These may, for example, encourage students to predict missing forms and think of alternatives, or use contextual clues to complete gapped concordance printouts. This approach is especially useful if resources allow students to also analyse and compare the use of hedging in their coursebooks, since they may already be familiar with the content, or concordanced examples of ESL writing from corpora such as the International Corpus of English (Granger, 1994). Alternatively, students can conduct projects comparing the use of hedges in other genres, drawing on either electronic or printed corpora.

Using concordanced printouts of authentic scientific texts, students can classify hedges according to their grammatical forms or epistemic classes, or into their major functional categories. Printouts of collocations allow comparisons to be made between the use of hedges in different text types or the 'rules' for the appropriate use of 'semi-synonyms' or confusable pairs such as *infer* and *imply* or *suggest* and *propose* to be worked out. Again, students can examine the voice/tense choices writers make with lexical verbs, or study the different meanings expressed by modal verbs.

Corpus materials can also stimulate group work. This can involve advanced learners conducting small projects which focus on the quantitative and qualitative aspects of hedging in their own field. Thus each student might analyse a sample of articles from a discipline-relevant international journal in terms of the language of hedging. The class could then pool its expertise and discuss similarities and differences, compiling the results to draft an essay or research paper on the subject (eg Swales, 1987). Alternatively, concordanced data allows a deductive approach, encouraging groups to form hypotheses or problems, then use data to examine them. Relating tasks to students in this way means learners can activate their acquired linguistic competence and discipline knowledge in a context of immediate relevance to their field.

Another way to encourage an appreciation of hedges among students is through the use of text fragments selected for in-class examination. Here students can be asked to perform the following tasks which help raise awareness of the effect of hedges on statements:

- identify all hedges in a text, circling the forms used, and assign a meaning to them;
- distinguish statements in a text which report facts and those which are unproven;
- identify all hedges in a text and state what is being hedged in each case;
- locate and remove all hedges and discuss the effect on the meaning of the text;
- identify hedges in a text and substitute a statement of certainty and precision;
- identify hedges and rank them by the amount of certainty they express (high–mid-low);
- use this scale to comment on the accuracy of existing lists, such as that proposed by Jordon (1990: 66);
- examine reformulations of a text which vary the levels of certainty they contain and determine whether they accurately report the statements made in the text;
- compare the use of different realisations of hedges in terms of meaning, syntactic pattern, lexical environment, pragmatic function, etc;
- determine whether the forms vary cross-culturally by translating them into the L1.

Clearly there is a range of activities which can help students achieve an awareness of hedges, but in addition to consciousness raising tasks, students have to see their importance in a larger context. Why do authors choose to hedge at particular points in their work? Why do they use particular forms? Students require an understanding of hedges not only as text-based items, but also as discourse-based strategies, showing how they relate to the writer's overall text plan. Explicit links between reading and writing have to be established, developing both an ability to identify hedging forms and the functions they perform in meeting reader expectations. Here complete texts are required, with students focusing on particular rhetorical sections at first and entire RAs only later. Only in this way can sufficient context be provided to reveal the particular reader assumptions the author intends to draw on in using a hedge.

To accomplish this understanding, contrasts can be established where choices are related to criteria of acceptability and informativity. These can employ the pragmatic framework suggested in this thesis to determine whether uses involve precision, caution or deference. One powerful means of illustrat-

ing the effect of hedges on a text is to examine the same content in different evidential contexts, focusing on the different meanings and degree of certainty expressed. A text with all hedges removed clearly becomes a different type of text, with conjectures and speculations accorded an uncontroversial fact-like status. Thus students can be asked to compare two versions of the same material. Comparison can be made of the levels of certainty and deference implied in extracts from different drafts submitted for publication (eg Myers, 1990), or in treatments of the same topic in a textbook or news magazine and an RA. This awareness provides students with a simple grammatical technique for distinguishing facts from claims and identifying substantive assumptions made about a topic.

Additional techniques might require students to locate places where hedges have been removed or extraneous ones inserted, to comment on the consequences of different reporting verbs in citing prior work, to compare the meanings of the same forms in different contexts, or to suggest alternative hedges and examine their functional effects. A task focusing on demystifying language use is suggested by Swales (1987) in connection with citation studies: here students interview professors in their own disciplines concerning their hedging behaviour in publications. In all these ways, the goal must be to encourage students to analyse representative textual products of the genre, respond to surface features of the texts and reflect on the strength of the knowledge claims made.

In addition to an ability to identify hedging items and appreciate the circumstances under which they can be used appropriately, students require specific guidance to hedge their own writing. This can be most effectively achieved in programmes which emphasise the discoursal and cognitive aspects of academic writing, geared to helping the writer solve to the problems of using language to both generate ideas and to shape those ideas into a form which meets both readers' needs and the demands of effective persuasion. Here hedges must be seen as part of a wider process of creating and crafting complete texts, developing both a sense of audience and a sense of purpose in using hedges correctly to produce a complete, fully contextualised piece of work.

Of major importance is a supportive environment for the development of the processes of planning, organising, composing and revising their work. Many ESP courses therefore involve procedural methodologies that guide students "through a process of thinking, selection of evidence, writing, and revision" in the preparation of a rigorously defined product (Canesco & Byrd,

1989: 311). This is evident in project-based writing courses (Barron, 1991; Bloor & St John, 1988; Hyland & Hyland, 1992), which stress the purpose of writing and the audience for it, ensuring that students read, evaluate and learn from activities in an authentic shared context for writing. In particular, collaborative discussions concerning the levels of claims in drafting sessions, for example, or the addition or refinement of hedges by peer reviewers at the revision stage, provide a helpful focus. First, however, students need opportunities to practise the various hedges common in the genres of their disciplines. The following suggestions address this issue and are appropriate for even elementary learners.

While there is a wide variety of means to convey tentativeness, there is some justification for concentrating initially on lexical devices when developing the hedging skills of L2 writers. These are both the most frequent markers of hedging in science and are also likely to be the easiest to acquire, thus offering the greatest benefit for the least learning effort. Such a strategy will quickly provide learners with a means of expressing themselves with subtlety in situations where they might lack the confidence to use an appropriate syntactic pattern. Knowledge of lexical items will also help assist familiarity with the concept of hedging and the circumstances under which it can be employed, allowing more complex means of expression to be developed later. Acquiring the vocabulary to express this aspect of meaning is made easier by the fact that a comparatively limited range of devices has an extremely high frequency in science.

L2 students are frequently encouraged to master the semantic complexities of the modal verb system, although this often proves difficult for them due to the range of meanings these verbs express. A more sensible and effective strategy is to focus on function and develop an understanding of the use of the high frequency modal hedges *would, may* and *could,* which account for 77% of all epistemic uses. Moreover, as we noted in chapter 5, alternative forms predominate in scientific writing and these tend to be easier for L2 students to control (Hyland & Milton, 1997). Lexical verbs and adverbs in particular have an extremely high frequency and present students with far less semantic confusion. They also have a high currency in casual conversation (cf. Holmes, 1988b) and thus provide high learning returns by allowing a transfer to other contexts. The fact that lexical verbs are often used in combination with other hedges also allows devices to be taught in patterned phrases and can therefore

assist students to develop confidence in using different devices and an awareness of the degrees of certainty expressed by them.

Initially therefore, I suggest teachers focus on high frequency lexical devices in order to develop familiarity with the concept of hedging and confidence in its use. Early stages of ESP courses might therefore concentrate on *indicate, suggest, appear* and *propose*, which represent 56% of all mitigating lexical verbs and constitute a spread of both evidential and judgemental items. Students should also be introduced to a range of common disjunct adverbs such as *probably, possibly, apparently* and *essentially*, and downtoners of varying force, such as *quite, slightly, rarely* and *relatively.* Instruction can then proceed to related adjectival and nominal forms with more detailed information on frequent hedging strategies and syntactic realisations being introduced at later stages. This procedure may help students acquire an understanding of the importance of hedging, the resources to express it, and the assurance to employ forms appropriately.

Tasks which require an explicit focus on the productive use of hedges can request students to:

- complete sentence frames, in an appropriate context, which employ common hedges such as *in spite of .., if ....then, although ..., under these conditions ...., according to our method ..., the model implies ....., viewed in this way ....*, etc;
- present uncertain information or disputable beliefs in a variety of ways, exchange these with peers and discuss their interpersonal effectiveness;
- link statements by including or omitting hedges to convey an accurate assessment of the truth relations between them;
- discuss and write up various aspects of a topic that students are likely to hold different degrees of certainty about, eg the differences between seasons in a particular country or the details of a widely covered current affairs item;
- paraphrase a description of an experiment using hedges to refer to uncertain claims;
- rework RA titles from journals in appropriate fields by the addition or subtraction of hedges so their implied certainty and 'research space' is affected;
- use reader-oriented interpersonal hedges to present an alternative argument to the beliefs held by a politician or newspaper columnist;

- undertake free writing activities to explore imaginable possibilities in their lives, such as alternative directions at certain points in the past or options for the future;
- select the most appropriate supports for a particular claim from a list and link them together with appropriate hedges or certainty markers.

Many of the activities suggested here can provide the basis of 'reformulation' activities (eg Allwright et al, 1988) whereby a native speaker, perhaps a subject specialist, rewrites a NNS's draft for class discussion. Such a discussion could focus on the level of claims made and the form and function of the hedges employed. This would serve to both encourage reflection on the effectiveness of the suggested alterations and assist learners in drafting their own claims by offering an expert reader's perspective.

Obviously reader response is critical, and in addition to making appropriate cognitive choices in hedging use, students also need to make decisions about correct register, style and the likely background knowledge and expectations of their audiences. A major factor when considering the development of hedging strategies, then, is the context of the writing task. A social constructionist view encourages us to see academic writing as an acquired response to the preferred modes of communication and discourse conventions of a particular academic community (Bizzell, 1982; Coe, 1987). But while a perception of audience is crucial to the development of appropriate argumentation strategies (Berkenkotter, 1981; Connor, 1987), university science courses appear to rarely offer authentic contexts where a clear writing purpose and audience is specified (Braine, 1989; Leki & Carson, 1997). Pedagogic notions of audience tend to be relatively unsystematic and largely fail to assist students to an understanding of accepted writer-reader roles and how to express appropriate knowledge claims (Herrington (1985). It is of little surprise therefore that most college-level students experience difficulty in adapting their prose for readers (Witte, 1987; Redd-Boyd & Slater, 1989).

Consideration of the projected audience, however, moves the writer into a wider pragmatic context, which has an important influence on the interpersonal and rhetorical strategies employed. This can lead to an understanding of both the purpose of writing and the diversity of audience expectations, which are central to the development of hedging strategies and appropriately circumscribed claims. Richards and Skelton (1991: 28) point out that "talking to your reader is a complex game", for a significant aspect of academic competence is the ability to "establish a relationship with the reader and ... with the authorities

in the field" (ibid: 34). The competence to modulate the strength of knowledge claims is obviously critical to such a relationship.

Clearly some kind of content-centred instruction is valuable in offering learners authentic learning experiences in the English class, and this view has led to content-based instruction (Flowerdew, 1993a) and to 'writing across the curriculum'. There is little evidence however that either writing or learning is improved by conducting writing instruction in content classes themselves (Ackerman, 1993). Another solution has been 'text-responsible' writing classes (Leki & Carson, 1997) where learners are required to engage with content through source texts, and "adjunct" courses, where language teachers are paired with content teachers and derive their materials from them. Teacher/specialist collaboration, combining expertise in rhetorical and discipline knowledge and integrating students' subject knowledge and curriculum material with English skills, appear to be fairly widespread (Dudley-Evans, 1983; Snow & Brinton, 1988). As Grabe and Kaplan (1996) have recently observed, both EAP and content teachers are generally well aware of the genre-specific demands of disciplines and many are working towards the ideal of students attending to the content of their subjects while writing.

The development of such an environment, which seeks to emulate the writing processes of expert writers, is beyond the scope of this section, but genuinely communicative activities involve creating hypothetical or authentic contexts and motivating students to write appropriately for real or imaginary readers. Learners often have little experience in such writing and both teachers and students need to be sensitive to genre-specific and community-specific issues like hedging. Writers must ask who their readers are, what they know and expect and what they will use the information for. This requires tasks which incorporate students' particular socio-rhetorical situations into the language teaching materials, encouraging awareness of the kind of knowledge claims it is appropriate to make, and the level of conviction to invest in them.

To develop a consideration of audience and purpose in writing, students can:

- reformulate a textbook passage from the perspective of a scientist presenting the ideas for the first time in a research report, making the claims more tentative;
- write a speculative paper explaining possible causes of, say, the reasons for inflation, drug abuse, PhD dropout rates, juvenile crime, etc;

- write a Discussion section of an RA which hypothesises about future possibilities by projecting information from photographs, graphs, statistical trends, etc;
- contribute an essay for a class journal to inform teacher and classmates about the state of knowledge on a topic or the implications of a new scientific development;
- describe an experiment in the form of a Results or Discussion section of a research article for a subject teacher, using hedges to convey the appropriate level of claim;
- conduct a group investigation of the use of hedging in students' various disciplines and write an RA or essay on their similarities and differences.

The literature suggests various ways to teach audience awareness (Kroll, 1990; Brandt, 1990), but the use of audience analysis heuristics and peer response feedback (Schriver, 1992) seems a useful means of helping writers' anticipate reader needs. The pragmatic description proposed here suggests a framework for such think-aloud reading protocols which can be given to experts for taped feedback on essays. Other approaches have attempted to specify a clear communicative context for writing, either through classroom roles for students or by writing for experts. Johns (1993), for example, used an authentic genre and expert informants to teach grant proposal writing to practice arguments about issues which directly affected students. Relating writing tasks to real purposes and readers in this way can help students learn to use hedges effectively. Once again, the descriptive framework can be used to help students convey appropriate meanings by matching particular intentions with correct contexts and expressions.

An interesting idea is suggested by Swales (1990: 224 ff.) for teaching 'suggestions for further research' which seems eminently adaptable to work on hedging. After some initial familiarisation tasks, students listen to a research seminar presentation by an "outside speaker" (or the instructor) on which they provide critical comments. Following a review of the presentation and these comments on tape, they then switch roles to write a conclusion to the related research paper using both sources of information. This activity sets up authentic situations for the use of hedges in terms of both the need to be tentative in the oral discussion and in cautiously making claims for the research findings in writing.

In sum, acquiring an academic communicative competence means the learner needs to become familiar with the relevant discourse conventions and learn to operate within them. The teacher's task is to assist students to develop such an awareness by making these conventions salient, often by breaking complex genres down into manageable rhetorical strategies, and by providing opportunities to practice their use in contexts which approach authenticity. The tasks outlined here move the learners from the role of discussant, to presenter, to interactant, and help them to achieve an understanding the appropriate use of academic hedging. In each of these roles, students are able to develop both their understanding of hedges and the part that hedges play in the achievement of rhetorical purpose in a given genre.

Conclusion

NNSs who wish to function in the academic research world must be able to both recognise and use hedges in written texts. However, there seem to be major difficulties for learners as a result of the complexity of epistemic forms and their meanings, the apparent variations in the socio-linguistic rules of language use, and serious shortcomings in the materials used to teach hedging. All these factors provide considerable pedagogical justification for acknowledging the importance of mitigation and assisting learners to develop an awareness of the principles and mechanics of its correct use. But while ESP teachers have long recognised that the characterisation of students' target language is central to their work, the profession is only now developing a strong applied linguistics research base (Swales, 1988: 14). Thus a linguistic emphasis on the contribution of rhetorical choices to communicative effectiveness has not been substantially translated into the L2 classroom. In sum, ESP pedagogy urgently stands in need of revision based on an analysis of authentic scientific texts.

The use of authentic examples clearly addresses a number of criticisms often leveled at published textbooks (Sheldon, 1987; Kuo, 1993; Richards, 1993), by meeting requirements of relevance, face validity, learner independence, flexibility of use and disciplinary appropriacy. The absence of adequate textbook coverage can thus provide an incentive to develop resource banks of more reliable and learner-focused materials which can relate language practice to a particular topic. The limitations of current published materials demonstrate that the features of writing in particular disciplines must be based on more

thorough research than at present. Thus the more we can learn about the use of hedges in different contexts, the more confident teachers can be in selecting practice materials for their study.

The tasks outlined here simply suggest ways in which learners might be led to appropriate means of rhetorical expression by drawing on their own linguistic resources, authentic models of language use, and the analytical framework proposed in this book. The approach encourages students to consider the relationship between writers, claims and readers, and to introspect on the reasons for employing hedging in scientific writing. In this process the teacher and learner can each draw on their respective specialist knowledge and contribute as equals.

Students learn most of their discourse knowledge when expressing themselves in writing, and this, once again, emphasises the importance of audience design and genre-relevant writing experiences for novice science writers. The engagement of minds in a rational process of argument is at the heart of academic discourse, and therefore approaches which draw on task- and context-specific issues are important for teaching NNS scientists to hedge effectively in their work. The methods sketched here, focusing on purposeful assignments involving the acquisition of formal schemata through analysis and manipulation of model texts, suggest ways in which students can be helped towards an understanding of the language of their disciplines. The framework outlined in previous chapters may therefore enable teachers to make better informed judgements about the communicative usefulness of particular features, assisting them in decisions about what items to teach and when to teach them.

9. Conclusions and Implications

In this concluding chapter I will consider some of the implications of the study, point to remaining questions and, in programmatic outline, suggest directions for future research. First, I will briefly discuss four major themes of this study: contextuality, functionality, variability and indeterminacy.

Some principal issues of hedging

First, the notion of context has been central. In the past, introspective semantic/conceptual treatments of modality have greatly enriched our understanding of modality, but intuitions only explain clear-cut, nuclear cases which give a misleadingly orderly impression of hedging as it exists in natural language. Introspection can only be profitable for explaining language behaviour when it is used to attribute meaning to authentic examples, as this forces us to confront indeterminate cases and consider the influence of context on interpretation.

This work has studied hedges in contexts where they serve real purposes for real audiences and has suggested that the type of social organisation in which language is produced has a crucial influence on discoursal practices. Readers deploy non-grammatical inferences in interpreting discourse features. However, a semantic analyses can only provide 'default values' for items rather than the particular interpretations they are given in actual contexts of interaction (eg Perfetti & McCutchen, 1987). Scientific hedging can only be understood by attending to the sociocultural, institutional and interpersonal aspects of scientific environments as well as local features of the co-text such as specification, agentivity, verification and combination.

The second theme of this study has been functionality, the fact that hedging serves clear and important extra-linguistic purposes. Effective communication in any domain involves expressing commitment and detachment, personal and interactional allegiance, and qualification and certainty. The ability to use language flexibly in this way is probably endemic in most forms of language use but in scientific writing hedges are an essential part of presenting new claims for ratification. Here hedges contribute to the rhetoric of persuasive discourse by assisting the cognitive procedures of deductive reasoning and enabling the dialectical processing of objective fact and subjective evaluation that is at the heart of all academic writing. They are among the main pragmatic features which shape the research article as the principal vehicle for new knowledge and which distinguish it from other forms of academic discourse.

The social constructionist perspective adopted here has demonstrated the dual significance of hedges in shaping this key genre. Attention has consistently been drawn to the institutional and epistemological importance of language in the scientific community and to the pivotal role of hedges in encoding perceptions of both natural-world and interpersonal relationships. On one hand hedges contribute to the interactional nature of RAs, being created by the writer in anticipation of the reader's interpretation of the text. On the other hand they reflect the organisational and socio-cultural constraints and expectations of the discourse community, particularly in terms of how information is structured, presented and ultimately confirmed as knowledge. The study therefore implies a complex overlap of motivations for hedging, which can only be understood through study of the institutional, professional and linguistic contexts in which it is employed. The analysis has assumed that texts are the result of informational, rhetorical and personal choices made in a particular social context. Put differently, research articles clearly reveal the relationship between a discourse community, standards of knowledge and textual representations, and it is these in combination which clarify the use of hedges.

Hedges contribute to scientific argumentation through the achievement of consensus. Sociologists have convincingly argued that scientific knowledge is socially constituted by the collective agreement of scientists, and I have suggested that hedges assist in this accreditation by reflecting both the accuracy of statements and the certainty invested in them. The process of achieving a communally negotiated consensus depends on writers meeting community

requirements regarding full, accurate and objective disclosure and avoiding the possible negative consequences of overstatement when moving beyond their data. Probability is made explicit to clearly specify the epistemic authority for a statement, indicating how it should be interpreted in order to gain acceptance for the strongest justifiable claim. In their rhetorical role, then, hedges temper claims with caution in order to meet adequacy conditions, mitigating the relationship between hypotheses about the nature of reality and what is believed to be true in the world. Hedges are persuasive because they relate claims to accepted knowledge and thus form an essential element of practical reasoning structures (eg Toulmin, 1958).

As we have seen, in addition to their epistemological significance, hedges perform critical interpersonal and social roles which help create and confirm discourse community coherence. The system which validates knowledge also manages publication and governs interactions among scientists through a process which bestows institutional recognition on individual researchers. In other words, Interpersonally, hedges are employed to acknowledge the reader's right to refute claims by marking statements as provisional until accepted by peers. Institutionally, they assist writers in creating an appropriate professional persona, demonstrating a commitment to the norms of a fraternity of scientists in which deference, debate, a shared approach to truth values, and respect for the views of others are valued communicative principles.

Thus hedges are not the negative indicators of imprecision and fuzziness portrayed by style guides. They are an essential means by which speakers and writers encode a point of view and add perspective to their propositions. Hedging expressions are a positive contribution to the individual writer's efforts to make meaning and to the discipline's procedures to make knowledge. They are, in other words, a significant resource rather than an irritating problem.

The third theme in this study has been the frequency and variability of hedging in science. The rich variety of hedging devices in English demonstrates the importance of this function and is perhaps related to the predominance of English as the language of published academic discourse. Biology research articles are massively hedged. Hedges represent about one word in every 50 in the corpus, increasing to one every 36 in discussions. This is a frequency much higher than many other linguistic features which receive more attention in applied linguistics and in ESP pedagogy and is a figure which suggests that hedging must be treated with due seriousness.

Generally speaking, the literature has both tended to overlook scientific hedging and overstate the significance of modal verbs as devices for expressing doubt and certainty. The range of modal devices and meanings appropriately conveyed in science writing however appears to be extremely restricted, with would, could and may representing 75% of all cases in my corpus. Moreover, this neglect of alternative expressions is unjustified for, as the study shows, an enormous choice of lexical, syntactic and strategic means of expressing uncertainty is available to scientific writers, and some of these are more common than modal verbs. While domain comparisons show that science RAs may contain a more restricted range of items than the general academic corpus, variability is a main feature of the use of mitigation in science. Our analyses must therefore extend beyond the modal auxiliaries.

The final theme is indeterminacy. Modal expressions are acknowledged to be notoriously complex semantically because devices do not correspond with one particular meaning and may actually express more than one simultaneously. In some instances it is difficult to determine if an item is being used in a root or epistemic sense, i.e. whether it is concerned with "the ability or volition of the subject of the sentence" (Palmer, 1990: 36) or with writers' judgements about their state of knowledge. These are not always clearly distinguished in the literature and there is some debate about how best to account for the fact modals can have two (or more) interpretations. Some analysts propose monosemic solutions, which posit one basic meaning for a lexical item. Others favour polysemic approaches, which specify both root and epistemic senses disambiguated by context. Thus the complexity of modal expressions have generated a variety of theoretical approaches ranging from gradience (Leech & Coates, 1983) through continuums and scales (Halliday, 1994; Hemeren, 1978; Palmer, 1986), to theoretical primitives such as likelihood and execution (Marino, 1973), potentiality and representation (Klinge, 1993), or possibility and necessity (Lyons, 1977; Leech, 1987).

Traditional semantic approaches however often furnish sets of modal meanings without explaining how they were actually arrived at. I have relied on Coates' data-driven indeterminacy analysis and have also sought to distinguish semantic meaning and situational meaning to reveal regularities which emerge from a study of actual language behaviour. From this perspective it is the utterance of a statement which contains a modal expression which allows us to interpret it as a hedge; that is, it is the situation-specific factors involved in employing a particular device for a given purpose, genre

and audience which gives it pragmatic meaning. So while this study has examined formal realisations and semantic meanings of epistemic items, I have also considered the same forms as constituents of knowledge claims and the interpretations which can thereby attach to them.

The root/epistemic distinction at the semantic level is primary, and was emphasised when discussing a variety of lexical forms in chapter 5. This opposition represents the basic difference between the modality of social or natural laws on one side, and that of knowledge and belief on the other. Essentially modality encodes likelihood resulting from necessity and from possibility, with hedging clearly dependent on the latter meanings. Some devices, for example can, ought and must not, express only root meanings, while other items such as could, should, may and possible can express either writer beliefs or necessary conditions. Distinguishing the intended sense can be problematic. In this work, the distinction has been made by paraphrasing root and epistemic possibility as 'possible for' and 'possible that' respectively. But this is neither an elegant nor infallible test, and it is particularly unreliable where the expression of possibility is based on participants' expert knowledge, where it may be largely inaccessible to the analyst. In the absence of any explicit surface signals however, this is the most reliable way we currently have to discern a hedge from a consideration of existent constraints.

The issue of participants' knowledge also raises another source of indeterminacy. Because hedging is concerned with the beliefs and uncertainties of writers, subjectivity is fundamental. Palmer (1986:16) asserts that subjectivity is an essential criterion for modality, as it "is the grammaticalisation of the speaker's (subjective) attitudes and opinions". Lyons' (1977: 793 ff) distinction between subjective and objective modality locates the uncertainty in the speaker's mind in the former, and to an unverifiable state of external affairs in the latter. Hedges are thus subjective by definition, as they reveal the writer's assessment of possibilities, conveying reservations about the truth of a statement or the wisdom of expressing it categorically. In essence, hedges represent an evasion and refer to beliefs, while objective uses refer to possibilities which exist independently of the writer and are outside the writer's control.

While a preoccupation of logicians, objective epistemic modality, concerned with necessary truths, is actually rare in natural language (Palmer, 1990: 16). Halliday (1970: 337), in fact, restricts his use of modality to the subjective type and splits off objective possibility into the propositional content.

The distinction however allows writers to manipulate their claims by exercising a strategic choice in expressing possibility, either to make a claim inherently objective (1), explicitly subjective (2), or to rhetorically obscure the source of the epistemic judgement by using an impersonal expression (3):

> (1) Obviously in these plants only a very limited number of PSII complexes can function because most have been rendered inactive by the herbicide. (A1:272)

> (2) I believe that the major organisational principle of thylakoids is that of continuous unstacking and restacking of sections of the membranes ... (A1:267)

> (3) What determines the constitutive expression ... is unknown and could be speculated to occur at several levels of regulation. (F2:353)

One way in which scientists appear to objectivise their claims is through use of the verb be which acts to assert a state of affairs categorically, shifting the source of possibility to an independent state of affairs existing externally to the writer (Perkins, 1983:67):

> (4) Indeed it should be possible to test predictions about the carbon gain efficiency ... (A1:272)

> (5) Structural modifications are possible in a few regions which are less homologous or in which substitutions involving ..... (C3:283)

> (6) Two types of conveyor belt models are possible in the photosynthetic apparatus. (A1:267)

However, the verb *be* can itself be modalised, thereby hedging the assertion, and it is difficult to see how the uses of be in (7) and (8), for instance, serve to "categorically assert the modality of the statement":

(7) It might therefore be possible that cv. Firstmore is different not only in its chromosomal arrangement but also in displaying a particular ... (F3:17)

(8) However the opposite is also possible and it cannot be ruled out .. (D2:260)

In fact, these writers seem to be speculating about the plausibility of certain outcomes rather than the possibility of real world consequences "based on unquestioned evidence" (ibid: 67). Forms which stress the objectivity of possibilities are frequently only pseudo-objective in scientific writing. They merely disguise the subjectivity of the epistemic source rather than objectify the accompanying statement. Apparently impersonalised forms such as modal expressions with be, sentential adverbs or passives thus cannot be seen as inherently objective, but often rhetorically conceal subjectivity.

Indeterminacy is nevertheless a defining feature of modal forms, partly as a result of shared knowledge between discourse participants. A clear dividing line between what is conceptually possible, or believed by the writer, and what is experientially possible, concerning an unverifiable state of external affairs, is not always easy to draw in the absence of a means of gaining access to these understandings. Often an interpretation based on cryptic contextual considerations is denied the outsider and may even elude scientist informants:

(9) Further refinements of the proposed structure are possible in these regions. (C3:286)

(10) A definitive test of this hypothesis could be made if a known null tt4 mutation, such as a deletion allele, becomes available. (D3:176)

(11) Consequently, the effectiveness of the protectants may vary with the target proteins. (A4:34)

In every case, the question of whether an item hedges the proposition hangs on the issue of writer belief. The most clear-cut cases are those which make missing knowledge explicit or deny external enabling conditions to make the outcome contingent on the writers' assessments of probability.

(12) Another conceivable reason for this decrease could be a change in TRPo value under the influence of the electric field induced by P^+ and Q^- charges. (B4: 231)

(13) These results suggest that if a flavonoid mutant with unaltered sinapate accumulation were available, it would be more sensitive to UV-B than is tt 4. (D3:175)

The study confirms that hedging devices are equally indeterminate in the pragmatic domain. Hedges are polypragmatic in that they convey a range of functions simultaneously, preventing the formation of discrete descriptive categories. Cases are virtually always indeterminate in that they contain a greater or lesser overlap of meanings, so while some examples might suggest an exclusive concern with "commitment" and others "confidence", most are "fuzzy" and include shades of both. The analysis therefore adopted Zadeh's (1972) fuzzy set model based on the idea of graded category membership. This model reflects the view of a number of writers (eg Lakoff, 1972; Labov, 1973; Rosch, 1975; Taylor, 1989) that our linguistic theories should accommodate the fact that our understanding of the world may necessarily be imprecise. A perspective which accepts imprecision helps explain important characteristics of hedging which would otherwise be analytically intractable. Why, for example, are so few examples prototypically distinct? Why do some cases allow more than one equally plausible explanation? Why do some cases yield two mutually compatible readings? The answers lie in the fact that language is inherently indeterminate.

My model of scientific hedging therefore attempts to incorporate these subjective features. Hedging in academic writing is designed to achieve the single primary objective of overcoming the inherent negatability of statements to achieve the reader's acceptance of a claim. Writers are aware that the standpoint of the reader is subject to knowledge effects beyond their control and that opposition can only be neutralised by meeting adequacy and acceptability conditions. Statements thus often contain forms which simultaneously perform both content-oriented and reader-oriented functions. The degree of subjectivity involved in pragmatic interpretation therefore means that precise categorisation of scientific hedges is impossible, and we have to accept that examples may involve shades of precision, commitment or deference. Interpretation is necessarily fuzzy and variables such as writer-role

(agentivity), precise specification, harmonisation of hedges (combination) and writer corroboration (verification) can only offer guides to the writer's attitude to the proposition or its context. Ultimately only participants' knowledge may determinine which function is paramount in any expression, but pragmatic considerations will generally reveal that several hedging meanings have been expressed.

Hedging in science: proposing some answers

We have now arrived at a point to propose some answers to the questions put at the beginning of this book. These were:

1. Under what circumstances do scientific writers use hedges?
2 What pragmatic functions are served by hedges in scientific writing?
3. Do particular hedging devices express particular functions?
4. Do the answers to these questions distinguish scientific articles from other kinds of writing?

First, the circumstances of appropriate hedging use depend on the writer's appraisal of certain non-linguistic factors. In essence, the pragmatic conditions for the use and interpretation of particular items are governed by the institutional expectations of research writing held by the scientific discourse community and the personal goal so the writer. Ideological and professional understandings are crucial to the choice of linguistic forms, and I have sought to incorporate some of these as features of the analytical framework. In particular, the relevance of ratification and reward act as pragmatic constraints on the writer in constructing scientific statements. These are critical aspects of the writer's interpersonal and communicative environment and relate to the notion of negatability, which in turn influences how information is presented to meet both adequacy conditions and acceptability conditions.

Gaining audience acceptance of an argument depends on presenting claims that the disciplinary community recognises as 'reasonable'. In essence, reasonable claims seek to:

- respond to an existing and finite set of exigencies recognised by the community
- maintain or expand the community's understanding of natural phenomena
- represent empirical adequacy and accuracy in terms of prescribed methods

- correspond to existing assumptions, theories and bodies of knowledge believed to accurately depict nature
- adopt the most certain and general position readers are likely to accept
- demonstrate a scientific ethos which involves
    i. recognising previous work and acknowledging priority
    ii. concealing a rhetorical identity behind a pose of objectivity
    iii. presenting a modest and collegial persona, demonstrating deference to, and willingness to negotiate with, one's peers

These informal precepts require the use of hedges to avoid violating either adequacy or acceptability conditions. All statements are simultaneously oriented to both the world and the reader and the decision to hedge a statement is motivated by the need to address these issues. The circumstances which determine whether a writer will choose to hedge or to present information categorically therefore depend on an assessment of the non-linguistic context. This assessment involves a judgement of the extent to which the proposition corresponds with what is believed to be true of the world, and its potential for eliciting a required reader response. In answer to the first question then, writers are permitted to use hedges in precisely those conditions where an assertive claim would otherwise breach these pragmatic conditions.

The second question, concerning the functions served by hedges, has also been answered. Analysis of the data reveals that scientific writers have two principal orientations, towards the content of their propositions, and to their readers. These orientations in turn provide three main reasons for hedging in molecular biology RAs, which can be further sub-categorised. Hedging in scientific writing enables a writer to:

- convey an assessment of the truth of a proposition in order to express claims with an appropriate degree of explicitness and precision. This can relate to:
    i. the degree to which the linguistic terms employed accurately describe attributes of real-world phenomena.
    ii. the degree of reliability the writer wishes to claim for the proposition
- encode reservations concerning the extent of personal commitment he or she wishes to invest in a proposition as a protective insurance against the possibility of eventually being proved wrong.

- express a particular attitude to the reader in the context of utterance in order to recruit his or her interest and assent to claims and meet expectations of deference, modesty and negotiation.

The first two functions are epistemic and relate to the content of claims. These enable writers to present information as objective statements of likelihood and propose claims for which adequate epistemic warrant is limited. The third function is interpersonal and expresses an attitude to the reader, addressing notions of collegiality and deference in facilitating claim endorsement. Together the interpersonal and institutional factors acknowledge the necessity for reader ratification of statements and confirm the importance of hedges in according freedom of choice to the addressee. Once again, the fuzziness of categories allows writers to exploit the polypragmatic nature of hedges to convey more than one meaning and thereby achieve a number of these goals simultaneously.

The third issue, concerning the functions and rules of use for particular devices, has only been partially clarified as it has proved impossible to unambiguously associate particular hedging devices with given meanings and I have repeatedly cautioned against imposing a rigid form-function equation. The devices most often associated with the realisation of core cases have been discussed in chapter 6, but these are simply guides to interpretation. Indeterminacy has been noted as a principal feature of hedges and the fact that forms often carry more than one purpose means it is not always possible in practice to distinguish these functions on purely linguistic or textual criteria. Any classification scheme inevitably distorts reality by imposing hard and fast categories on the pragmatic fluidity and indefiniteness of natural language use, where shades of intention are expressed simultaneously. The fuzzy-set model has been used to capture this feature by allowing for the fact that cases will convey elements of both core and other meanings.

Tentative steps have been taken, however, to identify general probabilities for devices used to express particular hedging functions. Here we must remind ourselves of the importance of context in elucidating the function of particular forms and the necessity of attending to both linguistic and pragmatic factors in determining the effect of any item. Understandably, this determination cannot rely exclusively on observable surface forms, although syntactic position, scope, harmonization of devices, personal subjects, abstract rhetors, use of approximators, and passivisation all provide some clues to the

intended meaning. Pragmatic factors embrace matters like the nature and importance of the proposition being modified and, in particular, the variables of specification, agentivity and verification which may assist in determining the meaning of core cases.

The final question concerns the issue of the functional distinctiveness of RAs in terms of hedging, and the answer to this must necessarily be more uncertain. An answer depends on extensive comparative studies of other genres and domains and is beyond the scope of this book. Recent studies however reveal that hedging is extensive in the research writing of several academic disciplines and may be even more prevalent in the 'soft applied' areas such as business studies and applied linguistics (Hyland, in press a). Some business genres, on the other hand, appear to contain substantially less hedging. An analysis of a 500, 000 word corpus of chairman's letters in company reports indicates only about a quarter of the items found in science RAs (Hyland, 1998). Detailed functional examination of such corpora however await further research.

No thorough pragmatic analyses of the functional purposes of hedging were undertaken in the JDEST and Brown/LOB corpora as the heterogeneity of these samples made the possibility of profitable findings unlikely. A different balance of functional strategies to those identified in the RAs would presumably emerge but prove little about genre differentiation. Moreover, such analyses may not provide an assured and comprehensive response to this question, as it is by no means certain that language conventions actually characterise particular varieties (eg Coffey, 1984; Paltridge, 1993 & 1995). But while the value of specifying distinctive functional properties across genres may be questionable, it is certainly possible to draw some tentative comparisons with those domains which have received attention in the literature and I shall briefly consider these here.

The concept of hedging has received the most attention in the area of casual conversation where it is possibly twice as frequent as in written discourse and represents a significant resource for speakers (eg Holmes, 1983 & 1995; Stubbs, 1986; Coates, 1987; Stubbe & Holmes, 1995). Like scientific writers, conversationalists employ hedges to convey both referential and affective meanings, but there appear to be important differences in the nature of these functions. Unlike scientific writers, for example, speakers are said to use hedges when they wish to create an informal or convivial atmosphere, facilitate

discussion, show courtesy or to cover for a deficit of knowledge or suitable vocabulary.

While it is apparently common in speech (eg Powell, 1984; Channell, 1994), scientific writing understandably exhibits little use of deliberate vagueness in the sense of withholding necessary information. Vagueness is simply not a viable communicative option in most scientific writing where propositional precision is central to persuasive argument, and the clear and full expression of information is the only control a writer has in influencing the reader's views. Nor do writers seek to create conviviality or acknowledge power differences, and while a major function of hedges in scientific writing involves sensitivity to addressees, this differs considerably from conversational goals of negative politeness or the use of hedges for phatic communion.

Differences also exist in the functions of hedges between science RAs and other spoken genres. Dubois' (1987) research on bio-medical conference presentations, for example, has found that hedges serve to deliberately diminish the precision of findings in order to fudge accuracy, foreground more exact figures and to cite theories believed to be in error. These functions either do not occur in molecular biology RAs or play a very reduced role. Once again, it is doubtful whether they would be either useful or effective rhetorical strategies in influencing readers in this genre. In addition, doctors appear to make greater use of hedges in spoken interaction to indicate that greater precision is unnecessary (Prince et al, 1982), again a function not observed in my data. On the other hand, hedging seems to appear as a means of addressing interpersonal issues even in spoken informational discourse (Flowerdew & Miller, 1997). In contrast to Lakoff's (1989) view of academic lectures as a typical instance of transactional discourse, for example, Flowerdew (1992) observes that interactional elements, such as mitigation, typically occur even in acts of scientific defining in lectures.

Functional differences have also been found in academic counseling encounters, where hedges help structure organisational roles and negotiate discourse statuses between interview participants (He, 1993). Writer-reader relationships in science RAs, however, are institutionally and normatively defined by a communicative contract based on conventions of interpersonal conduct and underlined by community sanctions. Hedges may therefore help to project a persona and demonstrate adherence to the 'rules' of communication, but this is simply to achieve higher level functions of gaining acceptance of knowledge claims, and not to structure interpersonal relationships per se.

A number of written genres also display substantial differences to biology RAs, in both the circumstances in which hedges are employed and in the functions they perform. Textbooks, for example, reserve hedges for the purpose of representing matters which lack a consensus (Myers, 1992), while a taxonomy proposed for hedges in both medical case reports and research papers (Salager-Meyer, 1994) admits a very reduced role for the expression of authors' doubt compared with the scheme described here. The critical role of hedges in conveying uncertainty has been confirmed however for research articles in a range of disciplines including medicine (Adams Smith, 1984; Skelton, 1997), economics (Bloor & Bloor, 1993), general science (Fahnestock, 1986), biochemistry (Thompson, 1993), marketing and applied linguistics (Hyland, in press a). In the Biology RAs examined here, the use of hedges to indicate doubt is a primary function as it allows readers to makes inferences about the claims made by judging their truth and significance.

Hedging also appears to be important in literary criticism, although Simpson (1990) has observed an additional rhetorical function of hedges unknown in science writing. In the work of F.R. Leavis, hedges are used to present relatively uncontroversial material in order to deflect critical attention from more potentially contentious views presented as categorically self-evident. While this function appears to be an idiosyncratic use, such rhetorical manipulation would seem inappropriate in scientific contexts where readers place interpretive reliance on the use of hedges to distinguish the credibility of claims (reliability hedges) and the extent of writer commitment to them (writer-oriented hedges). Similarly, the use of hedges to support claims by deliberately fudging accurate numerical information in economics RAs (Channell, 1990) is not a feature of my data. However, the need to balance persuasive writing and factual reporting also applies in science and it may be that it is simply the conventional expectations of exactitude that vary between disciplines. Biologists appear no different from other academics in presenting data as precisely as required for the practical purposes of substantiating a claim and convincing peers of its efficacy.

In summary, while hedging is found to be an important dimension of all the domains of communication discussed here, the circumstances recognised as appropriate for its use and the functions it is believed to fulfill appear to differ markedly in different contexts. Confirmation of these differences however awaits further research.

Some implications

There are a number of implications arising from this research which are worth signaling here. These address our understanding of language, cross-cultural pragmatics, rhetoric and language teaching.

A central implication is that hedges cannot be understood independently of inferences based on their sentential and social contexts of use. The characterisation of one such context here has shown that contexts are multi-layered and include syntactic and lexical information, the type of discourse, the functions it serves for interactants and its role in the discourse community. Hedging is a form of language which corresponds to the particular social practice being engaged in. Its study must therefore acknowledge a functional understanding of writing contexts, and the notion that language is socially situated and employed to achieve personal and institutional purposes. Most importantly, then, the study of mitigation must be informed by the idea that it is writers who hedge, not forms, and it is the environment in which the writing occurs which gives it meaning. So, while hedging symbols have (syntactic and semantic) *meanings* which are contextually invariant, their (pragmatic) *interpretation* in a particular context depends on this meaning plus knowledge of the writer's goals and the typical role these symbols play in RA discourse.

This view requires that not only must descriptions be based on naturally occurring data, but that an analysis of scientific discourse must focus on the interactional and social aspects of scientific communities. This reflects Fairclough's (1992:4) concept of discourse as simultaneously "a piece of text, an instance of discursive practice and an instance of social practice". 'Text' is the linguistic manifestation of discourse, 'discursive practices' represent the processes of text production and interpretation, and 'social practices' link the text to the institutional and social circumstances of the event. The goal of analysis is to explain how discourses are produced and the purposes they serve in particular communities. Hedges can only be adequately understood when directly linked with individual goals and institutional structures. In scientific discourse the writer seeks to *ratify* knowledge, not simply report it, and is engaged in the pursuit of personal ambitions through professionally recognised channels. These goals and practices emphasise, perhaps with greater clarity than in less formal domains, the importance of social, institutional and ideological factors in the interpretation of meaning as it is negotiated in scientific settings.

It is increasingly accepted that context and communicative goals shape the use of language in many subtle ways and analysts' impressions alone cannot accurately reflect how hedges are used, nor identify the constraints which link them to their surroundings. This study has sought to utilise these contextual features through text analyses, the views and interpretations of scientist informants, and findings from the sociology of science. Only through such diverse approaches can we determine the frequency of forms, their co-occurrences, the sense in which they are used and the meanings they express. Only with respect for real instances and the opinions of those who create and use particular genres can features such as hedges be described, rather than prescribed in advance.

The model and results reported here also have implications for the growing research on interlanguage pragmatics. Findings in this field (eg Kasper & Blum-Kulker, 1993; Blum-Kulka et al, 1989) have demonstrated the variability of linguistic choices in the use of indirectness for achieving communicative goals. However, their work has been almost exclusively confined to elicited spoken samples; appropriate and valid means for assessing written competence have yet to be developed. The modality of language use in speech acts such as requests, apologies, disagreement and refusals has been a primary concern of these studies but findings of pragmatic variability have been restricted by the task effects of the methods used (Rose, 1994). Data collection techniques have generally involved rating tasks, closed role plays or discourse completion questionnaires which constrain responses elicited to particular situations (Kasper & Dahl, 1991). Rarely is authentic discourse observed or analysed and the instruments invariably both confine and predetermine the type of information collected, often precluding access to the type of interpersonal phenomena sought.

The framework I suggest here, on the other hand, offers a model of native speaker pragmatic competence in the characteristic use of mitigation in a particular genre. Such a model can assist researchers when comparing the expression and comprehension of modal devices in academic writing among different cultural groups. It can also contribute to the design of techniques to measure differences in pragmatic knowledge and production for purposes of comparison between native and non-native learners. It could, for example, help to create instruments to examine the role of sociopragmatic variables such as social solidarity among different discourse communities and the degree of imposition created by academic claims and criticisms. The framework might

also serve as a basis for procedures to infer learners' pragmatic comprehension of hedges in English, and determine whether the processing modes of NNS exhibit the same context sensitivity and conventionality as do NS in this field.

The work also illuminates the role of language in the expression and ratification of scientific knowledge and therefore has implications for our understanding of the work of research scientists and the processes of evidential reasoning they employ. Much of the work in the area of scientific knowledge has been carried out by sociologists and has sought to reveal the centrality of communication as the cement of the academic enterprise. In particular, it has shown how the knowledge claim is at the locus of two interconnecting systems. One is concerned with the evolution and communication of knowledge, and the other with the assessment of merit and the evaluation of reputations. However, while important observations have been made, the recurrent discourse features which help create this social world of science have remained unclear. This study has demonstrated that hedging is one such feature, due to its role in gaining acceptance for claims by embracing both epistemological and social aspects of scientific communication.

Patterns of hedging help reflect the characteristics of a scientific field of inquiry and its associated discourse community, as it both mirrors the social context of a discipline and contributes to the achievement of socially situated rhetorical and institutional goals. The framework suggested here has important implications for the study of scientific knowledge and its practitioners. This is because it begins to reveal the connections between the generation and acceptance of knowledge and the organisation of the scientific community. It sets up analytic categories firmly located in textual evidence which illuminate important aspects of the relationships between scientists, between scientists and their findings, and between the norms and conventions which connect these to wider social processes. It therefore helps explain critical aspects of scientific knowledge and shows how these are realised through interactions among scientists.

Information about hedging can also contribute to our understanding of the practice of evidential reasoning and the structure of argumentation (eg Toulmin, 1958; Longino, 1990). Argument is a critical aspect of persuasive discourse, and hedges contribute to the effectiveness of the system as devices which anticipate possible rebuttals. Toulmin suggests that all arguments are explicitly justifiable in terms of a configuration of grounds, warrants, qualifiers, claims and rebuttals, but this is essentially a formal structure and its

components are not elaborated in linguistic terms. Thus while hedges play an important role in this system by qualifying the relationship between grounds (or data) and claims, the devices used to achieve this are not discussed. By revealing the formal realisation of hedges in scientific discourse and the reasons for their deployment, this study has implications for the elucidation of argumentation by showing how writers move between grounds and claims in gaining reader ratification for statements.

Finally, as was discussed extensively in chapter 8, the results of this study have important implications for second language teaching and can be applied to the development of a principled pedagogy which addresses the issue of hedging. The frequency and significance of hedging in science means that learning to use these devices appropriately is both a crucial aspect of effective argument and one of the preferred forms of discourse which serves as symbol of membership in this community (Hyland, 1997). The most common patterns of hedging in scientific writing and the circumstances in which writers may legitimately hedge have been identified here. This information can be used by teachers to grade materials for students studying science in English and to help them make explicit the techniques for gaining admission into this discourse community.

Conclusions and directions

It is clear that while this study has offered a descriptive and an empirical account of hedges in scientific research articles, it has also raised a number of questions which were unclear before and which remain to be resolved in the future. This is not surprising because the study has focused on an area which is only now beginning to attract research interest. Moreover, an empirical orientation not only reveals important aspects of language, it inevitably unearths issues which remain hidden when only invented and decontextualised examples are considered. Consequently, the continued study of hedging offers considerable potential for further interesting results in several areas and in this last section I briefly suggest some future directions in which a study of hedging might move.

Firstly, the results provide base-line data for comparisons with other academic fields and genres, and with other domains of language use outside academic contexts. If hedging is an inherent aspect of language use, as I have

suggested, then it would seem crucial to determine which, if any, particular forms and meanings characterise specific genres. This would involve identifying both the relative frequencies of the items employed, and the aspects of context which most affect the functions they convey. Thus the approach and analysis reported here may be of potential interest in revealing the distinguishing features and contextual rules which operate in other areas.

Secondly, the analytical framework and methodology employed here could be extended to investigate studies of hedging in languages other than English. While identifiable differences have been established in the extent of tentativeness and indirection in academic argumentation in various languages, more research is needed to specify how these rhetorical differences are actually manifested in surface phenomena. The study of the pragmatic interpretation and expression of mitigation by speakers from non-English speaking backgrounds has largely focused on spoken language and much work needs to be done on cultural variations in writing and in the acquisition of various forms by NNSs. We also know little about how such phenomena are connected to non native speakers' perceptions of appropriate social relationships and related behaviour in different contexts in their cultures, and how these might differ from those in parallel discourse communities in English. The approach here, which seeks to identify salient aspects of the specific cultural context and relate them to relevant discourse forms, may help to elucidate some of these issues.

Thirdly, in the area of lexical analysis, more work is needed to explicate the strategies used to hedge. A number of hedging expressions examined here involve phrases which I have simply discussed in terms of frequency and discourse effect. I have not explored the role that conventionalisation plays in the selection of forms, nor the possibility that particular multi-word expressions may take on an idiomatic quality for a specific discourse community. In addition, there is clearly scope for further research into the nature of harmonic clusters and their effects on mitigation in different genres. Additional work in lexical analysis must also involve exploration of polysemy and indeterminacy in order to ascertain the possibility of more precisely quantifying the pragmatic interpretation of particular examples in terms of grades of membership. Clearly, a more detailed understanding of the subtle gradations of meaning in hedging forms would be a great contribution to linguistic analysis.

Fourthly, future studies might seek to examine the role of various factors in reader responses to hedges in different texts, examining both native and non-native speakers. Research by Low (1996) has suggested that hedges are

"lexically invisible" to respondents when included in questionnaire items, for example, and it would be interesting to determine the extent to which readers respond to hedges in other genres. Potential areas include the psychological effects of hedges on text recall, comprehension and persuasiveness. These factors might be linguistic (such as the role of tense or voice choices or the selection of a particular hedging form) or non-linguistic (such as the role of previous teaching or exposure to hedges or the effects of various contextual variables). The systematic use of introspective reports and interviews is likely to enrich our understanding of these areas.

Finally, further research is clearly warranted in the theoretical approach to hedges where a number of critical issues remain unresolved. The most significant of these is to determine a means of more precisely distinguishing objective from subjective modality. While the distinctions drawn here are accurate as far as can be determined, a more principled means of differentiating examples and identifying hedging uses is needed. In addition, the use of adjectives and adverbs to manipulate the certainty and precision conveyed by numerical expressions has only been briefly touched on here and is another area likely to reward further investigation. Again, there are many aspects of language indirectness such as understatement, irony, metaphor and humour which have not been explored here. Presumably these principally relate to informal uses of hedging, but their study may throw light on its use in formal contexts. Finally, the fact that hedges are not amenable to traditional semantic categorisation creates considerable difficulties of interpretation and analysis. Clearly, the whole field of discourse pragmatics, and how readers understand polypragmatic forms, awaits more detailed explanation.

In conclusion, I hope to have shown the intrinsic importance of this particular aspect of language use to science, scientists and to the forms of argument they employ. The considerable scope for uncertainty in science and the critical rhetorical, cognitive and interpersonal roles it plays in this genre throws the role of hedges into sharp relief, but there are no grounds to assume that hedges are any less important in other areas of communication. Over a decade ago Stubbs (1986: 8) observed that "..to discover just how many utterances are qualified, and to what degree and in what manner in different contexts, is a matter of prolonged fieldwork". This study has begun to address these points in a particular genre in one academic discipline, but much more needs to be done to advance and extend the issues raised. Hedges constitute a rich and virtually unmined seam in linguistic analysis and further research of

their forms, their frequency and their functions in different contexts and cultures presents a fascinating and valuable enterprise.

Notes

1. On Hedges and Hedging

1. All examples from my corpus are code referenced according to source and page number. The sources are given in the appendix. I am aware that some readers would mark the hedging more extensively than I have in these examples, but I use underlining to refer to the principal signals of hedging rather than the parts of the proposition that are hedged. I hope the latter are generally clear.

2 Towards a context of scientific Hedging

1. "Assertions" are called "representatives" in Searle 1976.
2. In this example I have replaced the original bracketed footnote citation forms with author name attribution. I think this is clearer.

3. Perspective and Hedging in Discourse

1. Expressions of imprecision have also been discussed by Dubois (1987) in the genre of biomedical slide talks.
2. The philosophers distinction between knowledge and correct belief is irrelevant here as it is the linguistic expression of modality rather than ts conceptual status that interests us here. "Belief" and not "knowledge" is the issue. To exclude cases where the speaker may be mistaken r lying, many writers identify epistemic modality with the concept of "belief" (eg. White, 1975: 80).
3. Cameron, et al (1988) however argue that the majority of tags are facilitative rather than epistemic.

4. Lyons (1977: 819) and Perkins (1983: 106-110) regard time reference as a means for expressing modality n the sense of remoteness v. unremoteness (Lyons) or as the denotation of non-actuality or non-validity.

4. Theoretical and Methodlogical Considerations

1. The corpus is listed in the appendix.

5. Surface Features of Hedging

1. The LOB figures were not given, therefore the data for *might* refers to the Survey of English Usage. This is a corpus of 725, 000 words of spoken and unpublished written data.
2. The sample consisted of five text units from each of the ten science and technology disciplines represented.
3. It is possible that this result is an efect of Butler's significantly smaller sample. His corpus consisted of only two RA and two textbook texts from each of three science disciplines.
4. Conventional perfomatives are those expressed in the declarative form of the present simple tense with first person subject.
5. The use of imprecise numerical expressions was referred to in Chapter 3.

6. A Pragmatic Analysis of Hedging

1. An extreme example of this is the term quite which can either hedge or intensify a proposition (Hyland, 1996).
2. As I noted earlier, Channel (1990 & 1994) and Dubois (1987) observe that when used with numerical data, such hedges may serve the rhetorical objective of foregrounding more precise quantities. Their use may therefore draw on assumptions about what can be safely taken-for-granted as non-salient.
3. It is always possible however that a sufficiently large sample of disciplinary informants' readings of the texts might nevertheless reveal particular in-group codings that would distinguish specific meanings for them.
4. It is interesting in this regard that Crismore and Vande Kopple (1988) found that the addition of hedges in personal voice contributed to students remembering far more of the referential information in texts.

Appendix: Corpus of journal articles.

The articles are presented here according to the journals in which they appeared. The codes identify the examples in the text.

A: Photosynthesis Research

A1: Critchley, C. (1988). The chloroplast thylakoid membrane system is a molecular conveyor belt. *Photosynthesis Research, 19,* 265-276.

A2: Frank, H., Aldema, M., Parot, P. & Cogdell, R. (1990). Absorption and linear dichroism spectroscopy at room and low temperatures of single crystals of the B800-850 light-harvesting complex from Rhodopseudomanus acidophila strain 10050. *Photosynthesis Research, 26,* 93-99.

A3: Sinclair, J., Spence, S., & Langille, L. (1991). The appearance of quenching centres in photosystem II. *Photosynthesis Research, 29,* 1-9.

A4: Homann, P. (1992). Stabilization of the water oxidizing polypeptide assembly on Photosystem II membranes by osmolytes and other solutes. *Photosynthesis Research, 33,* 29-36.

A5: Juhler, R., Andreasson, E., Yu, S.-G., & Albertsson, P.-A. (1993). Composition of hotosynthetic pigments in thylakoid membrane vesicles from spinach. *Photosynthesis Research, 35,* 171-178.

B: FEBS Letters (Journal of The Federation of European Biochemical Societies)

B1: Kusnetsov, V., Mikulovich, T., Kukina, I., Cherepneva, G., Herrmann, R., et al. (1993). Changes in the level of chloroplast transcripts in pumpkin cotyledons during heat shock. *FEBS Letters, 321* (2, 3), 189-193.

B2: Smailov, S., Lee, A., & Iskakov, B. (1993). Study of phosphorylation of translation elongation factor 2 (EF-2) from wheat germ. *FEBS Letters, 321* (2, 3), 219-223.

B3: Godik, V., Blankenship, R., Causgrove, T. & Woodbury, N. (1993). Time-resolved tryptophan fluorescence in photosynthetic reaction centers from Rhodbacter sphaeroides. *FEBS Letters, 321*(2, 3), 229-232.

B4: Li, N.,Weisman, Z., Liu, D., & Mattoo, A. (1992). A functional tomato ACC synthase expressed in Escherichia coli demonstrates suicidal inactivation by its substrate S-adenosylmethionine. *FEBS Letters, 306* (2, 3), 103-107.

B5: Perret, V., Florentz, C., & Giege, R. (1990). Efficient aminoacylation of a yeast $tRNA^{Asp}$ transcript with a 5' extension. *FEBS Letters, 270* (1, 2), 4-8.

C: Biochimica et Biophysica Acta

C1: McNaughton, G., MacKintosh, C., Fewson, C., Wilkins, M., & Nimmo, H. (1991). Illumination increases the phosphorylation state of maize leaf phosphoroenol pyruvate carboxylase by causing an increase in the activity of a protein kinase. *Biochimica et Biophysica Acta* (1093), 189-195.

C2: Azpiazu, I. & Chapman, D. (1992). Spectroscopic studies of fibrogen and its plasmin-derived fragments. *Biochimica et Biophysica Acta* (1119), 268-274.

C3: Vijayakumar, S., & Salerno, J. (1992). Molecular modeling of the 3-D structure of cytochrome $p\text{-}450_{sec}$. *Biochimica et Biophysica Acta* (1160), 281-286.

C4: Zahedi, K., & Whitehead, A. (1993). Regulation of mouse serum amyloid P gene expression by cytokines in vitro. *Biochimica et Biophysica Acta* (1176), 162-168.

D: The Plant Cell (Journal of The American Society of Plant Physiologists)

D1: Bogusz, D., Llewellyn, D., Craig, S., Dennis, E., Appleby, C., et al. (1990). Nonlegume hemoglobin genes retain organ-specific expression in heterologous transgenic plants. *The Plant Cell, 2*, 633-641.

D2: Manen, J.-F., Simon, P., Van Slooten, J.-C., Osteras, M., Frutiger, S., et al. (1991). A nodulin specifically expressed in senescent nodules of winged bean is a protease inhibitor. *The Plant Cell, 3*, 259-270.

D3: Li, J., Ou-Lee, T.-M., Raba, R., Amundson, R., & Last, R. (1993). Arabidopsis flavonoid mutants are hypersensitive to UV-B irradiation. *The Plant Cell, 5*, 171-179.

D4: Parks, B., & Quail, P. (1993). hy8, a new class of arabidopsis long hypocotyl mutants deficient in functional phytochrome A. *The Plant Cell, 5*, 39-48.

E: Plant Physiology

E1: Gardner, G. & Sanborn, J. (1989). Aryl-substituted α–Aminooxycarboxylix acids: A new class of auxin transport inhibitors. *Plant Physiology*(90), 291-295.

E2: Lyons, P., Evans, J., & Bacon, C. (1990). Effects of fungal endophyte acremonium coenophialum on nitrogen accumulation and metabolism in tall fescue. *Plant Physiology* (92), 726-732.

E3: Win, K., Berkowitz, G., & Henninger, M. (1991). Antitranspirant-induced increases in leaf water potential increase tuber calcium and decrease tuber necrosis in water-stressed potato plants. *Plant Physiology* (96), 116-120.

E4: Hudson, G., Evans, J., von Caemmerer, S., Arvidsson, Y. & Andrews, T. (1992). Reduction of ribulose-1, 5-Bisphosphate carboxylase/oxygenase content by antisense RNA reduces photosynthesis in transgenic tobacco plants. *Plant Physiology* (98), 294-302.

F: Plant Molecular Biology

F1: Fromm, E., Gallun, E. & Edelman, M. (1989). A novel site for streptomycin resistance in the "530 loop" of chloroplast 16S ribosmal RNA. *Plant Molecular Biology, 14*, 499-505.

F2: Johnson, R., Jong, S. L., & Ryan, C. (1990). Regulation of expression of a wound-inducible tomato inhibitor I gene in transgenic nightshade plants. *Plant Molecular Biology, 14*, 349-356.

F3: Marano, M., & Carrillo, N. (1991). Chromoplast formation during tomato fruit ripening. No evidence for plastid DNA methylation. *Plant Molecular Biology, 16*, 11-19.

F4: Albert, H., Martin, T., & Sun, S. (1992). Structure and expression of a sugarcane gene encoding a housekeeping phosphoenolpyruvate carboxylase. *Plant Molecular Biology, 20*, 663-671.

References

Achtert, W., & Gibaldi, J.

1985 *The MLA Style Manual.* New York: Modern Language Association.

Ackerman, J.

1993 The promise of writing to learn. *Written Communication.* 10, 334-70

Adams Smith, D.

1984 Medical discourse: Aspects of author's comment. *English for Specific Purposes, 3,* 25-36.

Ahmad, U. K.

1995 *Academic language and culture: some observations on scientific Malay and scientific English.* Exploring language, culture and literature in language learning, RELC Conference: Singapore.

Allison, D.

1994 Why *often* isn't *always.* In D. Nunan, R. Berry & V. Berry (eds.) *Language awareness in language education.* (pp. 33-50) Hong Kong: University of Hong Kong Press.

Allison, D.

1995 Assertions and alternatives: helping ESL undergraduates extend their choices in academic writing. *Journal of Second Language writing, 4,* 1-15.

Allwright, R., Woodley, M.-P. & Allwright, J.

1988 Investigating reformulations as a practical strategy for the teaching of academic writing. *Applied Linguistics, 9* (3), 236-256.

APA

1994 *Publication Manual of the American Psychological Association. (4th ed.)* Washington: American Psychological Association.

Arnaudet, M., & Barrett, M.

1984 *Approaches to Academic Reading and Writing*. Englewood Cliffs, NJ: Prentice Hall.

Atkinson, D.

1996 The Philosophical Transactions of the Royal society of London, 1675-1975: a sociohistorical discourse analysis. *Language in Society*. 25, 333-371.

Austin, J.

1962 *How to do things with words*. London: OUP.

Bach, K. & Harnish, R.

1979 *Linguistic communication and speech acts*. Cambridge, Mass: MIT.

Bacon, F.

1960 *The new organon and related writings*. New York: The Liberal Arts Press.

Banks, D.

1994 Hedges and how to trim them. In O. A. M. Brekke, T. Dahl & J. Myking (eds.), *9th European symposium of LSP*, (pp. 589-92) University of Bergen.

Barber, C. L.

1962 Some measurable characteristics of modern scientific prose. In *Contributions to English Syntax and Phonology*. (pp. 1-23) Stockholm: Almquist & Wiksell.

Barnes, B., & Edge, D. (eds.)

1982 *Science in Context*. Cambridge, MA: MIT Press.

Barron, C.

1991 Material thoughts: ESP and culture. *English for Specific Purposes, 10*, 173-187.

Bartholomae, D.

1986 Inventing the university. *Journal of Basic Writing, 5* , 4-23.

Barton, E.L.

1995 Contrastive and non-contrastive connectives. *Written Communication*. 12, 219-239.

Bate, D. & Sharpe, P.

1990 *Student writer's handbook*. Sydney: Harcourt Brace Jovanovich.

Bazerman, C.

1983 Scientific writing as a social act: A review of the literature of the sociology of science. In P.V. Anderson, R.J. Brockman & C.R. Miller

(eds.), *New Essays in technical and scientific communication: Research, theory, practice* (pp. 156-184) Farmingdale, NY: Baywood.

Bazerman, C.
1985 Physicists reading physics: Schema-laden purposes and purpose-laden schema. *Written Communication, 2* 1), 3-23.

Bazerman, C.
1988 *Shaping written knowledge.* University of Wisconsin Press.

Bazerman, C. & Paradis, J. (eds.)
1991 *Textual Dynamics of the Professions.* Madison: University of Wisconsin Press.

Beaugrande, de R.
1985 *Writing step by step: Easy strategies for writing and revising.* Chicago: Harcourt Brace Jovanovich.

Beaugrande de, R.
1989 Special purpose language as a complex system: the case for linguistics. In C. Lauren & M. Nordman (eds.), *Special language: from human thinking to thinking machines* (pp. 3-29) Clevedon: Multilingual Matters.

Beaugrande, de R. & Dressler, W.
1981 *An introduction to text linguistics.* London: Longman.

Beauvais, P.
1989 A speech-act theory of metadiscourse. *Written Communication, 6* (1), 11-30.

Beazley, M. & Marr, G.
1992 *The Writer's Handbook.* Victoria: Phoenix.

Becher, T.
1989 *Academic tribes and territories: Intellectual inquiry and the cultures of disciplines.* Milton Keynes: SRHE/OUP.

Berkenkotter, C.
1981 Understanding a writer's awareness of audience. *College Composition and Communication, 32,* 388-399.

Berkenkotter, C. & Huckin, T.
1995 *Genre Knowledge in Disciplinary Communication.* Hillsdale, New Jersey: Lawrence Erlbaum.

Berlin, J. A.
1982 Contemporary composition: the major pedagogical theories. *College English,* 44, 765-777.

Bernhardt, S.

1985 The writer, the reader and the scientific text. *Journal of Technical Writing and Communication, 15* (2), 163-174.

Bhatia, V.

1993 *Analysing Genre: Language use in professional settings.* London: Longman.

Bizzell, P.

1982 Cognition, convention and certainty: what we need to know about writing. *PRE/TEXT, 3* (213-241).

Blakemore, D.

1992 *Understanding utterances: an introduction to pragmatics.* Oxford: Blackwell.

Bloch, J. & Chi L.

1995 A comparison of the use of citations in Chinese and English academic discourse. In D. Belcher & G. Braine (eds.) *Academic writing in a second language: essays on research and pedagogy.* Ablex: New Jersey.

Bloor, M. & Bloor, T.

1991 Cultural expectations and socio-pragmatic failure in academic writing. In P. Adams, B. Heaton & P. Howarth (eds.), *Socio-cultural issues in English for Academic Purposes.* (pp. 1-12). Basingstoke: Modern English Publications/British Council.

Bloor, M. & Bloor, T.

1993 How economists modify propositions. In W. Henderson, T. Dudley-Evans, & R. Backhouse (eds.) *Economics and Language.* (pp. 153-169) London: Routledge.

Bloor, M. & St John, M.

1988 Project writing: the marriage of process and product. In P. Robinson (ed.), *Academic writing: process and product.* Basingstoke: Modern English Publications.

Blum-Kulker. S., House, J. & Kasper, G. (eds.)

1989 Cross-cultural pragmatics: requests and apologies. Norwood, NJ: Ablex.

Boissevain, J.

1974 Towards a sociology of social anthropology. *Theory and Society, 1,* 211-30.

Booth, V.

1985 *Communicating in science writing and speaking.* Cambridge: CUP.

Botany Dept, University of Queensland
1993 *Plant and fungal molecular biology: Laboratory manual* . University of Queensland.

Braine, G.
1989 Writing in science and technology: An analysis of assignments from ten undergraduate courses. *English for Specific Purposes, 8* , 3-15.

Brandt, D.
1990 *Literacy as involvement: the acts of writers, readers and texts.* Carbondale, Ill.: Southern Illinois University Press.

Brett, P.
1994 A genre analysis of the results section of sociology articles. *English for Specific Purposes, 13* (1), 47-59.

Brown, H., Cohen, D., & O' Day, J.
1991 *Challenges: A process approach to academic English..* Englewood Cliffs, NJ: Prentice Hall.

Brown, K.
1992 Describing modality in English. In R. Bowers & C. Brumfit (eds.), *Applied linguistics and English Language Teaching* Review of ELT, 2.

Brown, P. & Levinson, S.
1987 *Politeness: Some universals in language usage.* Cambridge: CUP.

Bruffee, K.
1986 Social construction: Language and the authority of knowledge. A bibliographical essay. *College English, 48* , 773-779.

Butler, C. S.
1985 *Systemic linguistics: theory and applications.* London: Batsford.

Butler, C. S.
1990 Qualifications in science: Modal meanings in scientific texts. In W. Nash (ed.), *The writing scholar: Studies in academic discourse.* (pp. 137-170). Newbury Park, Ca: Sage.

Campbell, P.
1975 The *Personae* of scientific discourse. *Quarterly Journal of Speech., 61*, 391-405.

Candlin, C. & Breen, M.
1979 Evaluating and designing language teaching materials. In *Practical Papers in English Language Education II* (pp. 175-188). Lancaster: University of Lancaster.

Canesco, G. & Byrd, P.

1989 Writing required in graduate courses in business administration. *TESOL Quarterly, 23* (2), 305-316.

Casanave, C. & Hubbard, P.

1992 The writing assignments and writing problems of doctoral students: faculty perceptions, pedagogical issues and needed research. *English for Specific Purposes, 11*, 33-49.

Chafe, W. & Nichols, J. (eds.).

1986 *Evidentiality: the linguistic coding of epistemology*. Norwood, NJ: Ablex.

Channell, J.

1980 More on approximations: A reply to Wachtell. *Journal of Pragmatics, 4*, 461-476.

Channell, J.

1990 Precise and vague expressions in writing on economics. In W. Nash (ed.), *The writing scholar: Studies in academic discourse* (pp. 95-117). Newbury Park, Ca: Sage.

Channell, J.

1994 *Vague Language*. Oxford: OUP.

Chaplen, F.

1981 *A Course in Intermediate Scientific English*. Walton-on-Thames: Nelson.

Cheng, X. & Steffensen, M.

1996 Metadiscourse: A technique for improving student writing. *Research in the teaching of English*, 30, 2, 149-181.

Cherry, R.

1988 Politeness in written persuasion. *Journal of Pragmatics, 12* , 63-81.

Chin, E.

1994 Ethnographic interviews and writing research: a critical examination of the methodology. In Smagorinsky, P. (ed.) *Speaking about writing: reflections on research methodology*. (pp. 247-272). Thousand Oaks, Ca: Sage.

Choi, Y.

1988 Text structure of Korean speakers' argumentative essays in English. *World Englishes, 7* (2), 129-142.

Clyne, M.
1987 Cultural differences in the organisation of academic texts. *Journal of Pragmatics, 11* , 211-247.

Coates, J.
1983 *The semantics of the modal auxiliaries*. Beckenham: Croom Helm.

Coates, J.
1987 Epistemic modality and spoken discourse. *Transactions of the Philological society, 85* , 100-131.

Coates, J.
1988 Gossip revisited: language in all-female groups. In J. Coates &. D. Cameron (eds.) *Women in their speech communities.* (pp. 94-122). Harlow: Longman.

Coates, J. &. Cameron, D. (eds.)
1988 *Women in their speech communities.* Harlow: Longman.

Cobuild 1995 *Collins Cobuild English Dictionary.* London: Harper Collins.

Coe, R. 1987 An apology for form: or, who took the form out of process? *College English, 49* , 13-28.

Coffey, B.
1984 ESP: English for Specific Purposes. *Language Teaching, 17* (1), 2-16.

Cohen, M. & Manion, L.
1994 *Research methods in education. (Fourth Ed)* London: Croom Helm.

Collins, P.
1991 The modals of obligation and necessity in Australian English. In K. Aijmer & B. Altenberg (eds.), *English Corpus Linguistics* . (pp. 145-165) London: Longman.

Connor, U.
1987 Argumentative patterns in student essays: Cross cultural differences. In U. Connor &. R. Kaplan (eds.), *Writing across languages: Analysis of L2 text.* (pp. 57-71). Reading, MA: Addison Wesley.

Connor, U.
1996 *Contrastive Rhetoric.* Cambridge: CUP.

Connor, U. & Lauer, J.
1988 Cross-cultural variation in persuasive student writing. In A. Purves (ed.) *Writing across languages and cultures.* (pp. 138-195). Newbury Park, CA: Sage.

Cooper, J.

1979 *Think and Link: An Advanced Course in Reading and Writing.* Edward Arnold: London.

Cooper, M. & Holzman, M.

1985 Reply to Linda Flower & John Hayes. *College Composition & Communication.* 36 (1), 97-100.

Crismore, A.

1989 *Talking with readers: Metadiscourse as rhetorical act.* New York: Peter Lang.

Crismore, A. & Farnsworth, R.

1989 Mr. Darwin and his readers: Exploring interpersonal metadiscourse as a dimension of ethos. *Rhetoric Review*, 8, 1, 91-112.

Crismore, A., & Farnsworth, R.

1990 Metadiscourse in popular and professional science discourse. In W. Nash (ed.), *The Writing Scholar: Studies in Academic Discourse* (pp. 118-136). Newbury Park, Ca: Sage.

Crismore, A., Markkanen, R., & Steffensen, M.

1993 Metadiscourse in persuasive writing: A study of texts written by American and Finnish university students. *Written Communication*, 10 (1), 39-71.

Crismore, A., & Vande Kopple, W.

1988 Readers' learning from prose: The effects of hedges. *Written Communication*, 5 (2), 184-202.

Crookes, G

1986 Towards a validated analysis of scientific text structure. *Applied Linguistics, 7*, 57-70.

Crystal, D.

1995 In search of English: a traveler's guide. *ELT Journal, 49* (2), 107-121.

Crystal, D. & Davy, D.

1975 *Advanced conversational English.* London: Longman.

Currie, P. & Cray, E.

1987 *Strictly Academic: A Reading and Writing Text.* New York: Newbury House.

Davies, F.

1988 Designing a writing syllabus in English for academic purposes: Process and product. In P. Robinson (ed.), *Academic writing: Process and product. ELT Doc 129* . MET & British Council.

Day, R.
1988 *How to write and publish a scientific paper (3rd ed.)* Phoenix, NY: Oryx.

Day, R.
1992 *Scientific English: A guide for scientists and other professionals.* Phoenix, Arizona: Oryx.

Dillon, G.
1981 *Constructing texts, elements of a theory of a composition and style.* Bloomington: Indiana University Press.

Dillon, G., Coleman, L., Fahnestock, J. & Agar, M.
1985 Review of Leech 1983 *Language, 61*, 446-460.

Donovan, P.
1978 *Basic English for Science.* Oxford: OUP.

Dubois, B.
1982 The construction of noun phrases in biomedical journal articles. In J. Hoedt et. al. (eds.), *Pragmatics and LSP* (pp. 49-67). Copenhagen: Copenhagen School of Economics.

Dubois, B.
1987 "Something on the order of around forty to forty four": Imprecise numerical expressions in biomedical slide talks. *Language and Society, 16*, 527-541.

Dubois, B.
1988 Citation in biomedical journal articles. *English for Specific Purposes, 7*, 181-94.

Dudley-Evans, T. (ed.).
1983 *Papers on team teaching and syllabus design. Occasional papers 27.* Singapore: SEAMEO RELC.

Dudley-Evans, T.
1985 *Writing Laboratory Reports.* Nelson Wadsworth.

Dudley-Evans, T.
1986 Genre-analysis: an investigation of the introduction and discourse sections of MSc dissertations. In M. Coulthard (ed.), *Talking about text* (pp. 128-145). Birmingham: English Language Research.

Dudley-Evans, T.
1988 Recent developments in ESP: The trend to greater specialisation. In M. Tickoo (ed.), *ESP: State of the Art* Singapore: SEAMEO Regional Language Centre.

Dudley-Evans, T.

1991 Socialisation into the academic community: linguistic and stylistic expectations of a PhD thesis as revealed by supervisor comments. In P. Adams, B. Heaton & P. Howarth (eds.), *Socio-cultural issues in English for Academic Purposes.* (pp. 42-51) Review of ELT. Basingstoke: Modern English Publications/British Council.

Dudley-Evans, T.

1993 Variation in communication patterns between discourse communities: The case of highway engineering and plant biology. *Review of ELT. Language, learning and success: Studying through English,* 3 (1), 141-147.

Een, J.

1982 Tense usage in reporting research in geotechnical writing. *University of Minnesota Working Papers in ESL (2)* , 72-91.

Eggington, W.

1987 Written academic discourse in Korean: Implications for effective communication. In U. Connor &. R. Kaplan (eds.), *Writing across languages: Analysis of L2 text.* (pp. 153-168). Reading, MA: Addison-Wesley.

Ernst, T. B.

1984 *Towards an integrated theory of adverb position in English.* Indiana Linguistics Club.

Ewer, J. & Latorre, G.

1969 *A course in Basic Scientific English.* Harlow, Essex: Longman.

Fahnestock, J.

1986 Accommodating science: The rhetorical life of scientific facts. *Written Communication, 3* (3), 275-96.

Fairclough, N.

1992 *Discourse and social change.* Cambridge: Polity Press.

Fairclough, N.

1995 *Critical discourse analysis.* Harlow: Longman.

Faust, D.

1984 *The limits of scientific reasoning.* Minneapolis: University of Minnesota Press.

Fillmore, C.
1977 Topics in lexical semantics. In R. W. Cole (ed.), *Current issues in linguistic theory* . Bloomington: Indiana University Press.

Flowerdew, J.
1992 Definitions in science lectures. *Applied Linguistics, 13* (2), 202-221.

Flowerdew, J.
1993a Content-based language instruction in a tertiary setting. *English for Specific Purposes, 12* , 121-138.

Flowerdew, J.
1993b Variation across speech and writing in biology: A quantitative study. *Perspectives: Working papers of the department of English, City Polytechnic of Hong Kong., 5* (1), 75-88.

Flowerdew, J. & Miller, L.
1997 The teaching of academic listening comprehension and the question of authenticity. *English for specific Purposes.* 16 (1), 27-46.

Fraser, B.
1974 *Hedged performatives.* Bloomington, Ind.: Indiana University Linguistics Club (Mimeo).

Fraser, B.
1978 Acquiring social competence in a second language. *RELC Journal, 9* (2), 1-21.

Fraser, B.
1980 Conversational mitigation. *Journal of Pragmatics, 4* , 341-350.

Fraser, B.
1990 Perspectives on Politeness. *Journal of Pragmatics, 14* , 219-236.

Fraser, B. & Nolan, W.
1981 The association of deference with linguistic form. *International Journal of the Sociology of Language, 27* , 93-109.

Garfield, E.
1983 Talking science. *Nature, 303, p.* 354.

Gaston, J.
1971 Secretiveness and competition for priority in physics. *Minerva, 9* , 472-492.

Gazdar, G.
1979 *Pragmatics: Implicature, presupposition and logical form.* New York: Academic Press.

Gazdar, G.

1981 Speech act assignment. In AK Joshi, BL Webber & IA Sag (eds.), *Elements of Discourse Understanding* . Cambridge: CUP.

Geeraerts, D.

1993 Vagueness's puzzles, polysemy's vagaries. Cognitive Linguistics 4, 223-72.

Gilbert, G.

1976 The transformation of research findings into scientific knowledge. *Social Studies of Science, 6* , 281-306.

Gilbert, G., & Mulkay, M.

1984 *Opening Pandora's Box: A sociological Analysis of Scientific Discourse*. Cambridge: CUP.

Glendinning, E. & Mantell, H.

1983 *Write Ideas: An Intermediate Course in Writing Skills*. Harlow, Essex: Longman.

Goodwin, C. & Duranti, A.

1992 Rethinking context: an introduction. In A. Duranti & C. Goodwin (eds.) *Rethinking context: language as an interactional phenomenon* (pp. 1-37) Cambridge: CUP.

Gosden, H.

1992 Research writing and NNSs: From the editors. *Journal of Second Language Writing, 1* (2), 123-139.

Gosden, H.

1993 Discourse functions of subject in scientific research articles. *Applied Linguistics, 14* (1), 56-75.

Grabe, W. & Kaplan, R.

1996 *Theory and practice of writing*. Harlow: Longman.

Graetz, N.

1985 Teaching EFL students to extract structural information from abstracts. In J. M. Ulijn & A. K. Pugh (eds.), *Reading for professional purposes* (pp. 123-135). Leuven, Belgium: ACCO.

Granger, S.

1994 New insights into the learner lexicon: A preliminary report from the international corpus of English. In L. Flowerdew & K.K. Tong (eds.), *Entering text.* (pp. 102-113). Hong Kong: Hong Kong University of Science and Technology.

Greenbaum, S. & Quirk, R.
1990 *A student's grammar of the English language.* London: Longman.
Grice, H.
1975 Logic and conversation. In P. Cole &. J. Morgan (eds.), *Syntax and Semantics, Vol. 3: Speech Acts.* (pp. 41-58). New York: Academic Press.
Gross, A.
1984 Style and arrangement in scientific prose: Rules behind the rules. *Journal of Technical Writing and Communication, 14* , 241-253.
Gross, A.
1985 The form of the experimental paper: A realisation of the myth of induction. *Journal of Technical Writing and Communication, 15* (1), 15-26.
Gu, Y.
1990 Politeness phenomena in modern Chinese. *Journal of Pragmatics, 14* , 237-257.
Haack, S.
1978 *Philosophy of logic.* Cambridge: CUP.
Hagstrom, W. O.
1965 *The Scientific Community.* Carbondale: Southern Illinois University Press.
Hairston, M.
1981 *Successful writing, a rhetoric for advanced composition.* New York: W. W. Norton.
Halliday, M.A.K.
1970 Functional diversity in language as seen from a consideration of modality and mood in English. *Foundations of Language, 6* , 322-361.
Halliday, M.A.K.
1988 On the language of physical science. In M. Ghadessey (ed.), *Registers of Written English* (pp. 162-178). London: Pinter.
Halliday, M.A.K.
1994 *An introduction to functional grammar (2nd Ed).* London: Edward Arnold.
Halloran, S.
1984 The birth of molecular biology: An essay in the rhetorical criticism of scientific discourse. *Rhetoric Review, 3* (1), 70-83.

Hanania, E. & Akhtar, K.

1985 Verb form and rhetorical function in science writing: A study of MS theses in biology, chemistry and physics. *ESP Journal, 4* , 49-58.

Harder, B.

1984 Cultural attitudes in discourse analysis. *Canadian Journal of Linguistics, 29* , 115-130.

Harris, J.

1989 The idea of community in the study of writing. *College Composition and Communication, 40* , 11-22.

He, A.

1993 Exploring modality in institutional interactions: cases from academic counseling encounters. *Text, 13* (2), 503-528.

Hemeren, L.

1978 *On modality in English: a study of the semantics of the modals.* Lund, Sweden: CWK Gleerup.

Herrington, A.

1985 Writing in academic settings: A study of the contexts for writing in two college chemical engineering courses. *Research in the Teaching of English, 19* , 331-361.

Heslot, J.

1982 Tense and other indexical markers in the typology of scientific texts in English. In J. Hoedt et al (eds.), *Pragmatics and LSP.* Copenhagen: Copenhagen School of Economics.

Hill, S., Soppelsa, B. & West, G.

1982 Teaching ESL students to read and write experimental research papers. *TESOL Quarterly, 16* (3), 333-47.

Hinds, J.

1983 Contrastive rhetoric: Japanese & English. *Text, 3* (2), 183-195.

Hinds, J.

1987 Reader versus writer responsibility: A new typology. In U. Connor & R.B. Kaplan (eds.), *Writing across languages: Analysis of L2 text.* (pp. 141-152 Reading, Mass.: Addison Wesley.

Hinkel, E.

1994 Native and non-native speakers' pragmatic interpretations of English texts. *TESOL Quarterly, 28* (2), 353-376.

Hinkel, E.
1997 Indirectness in L1 and L2 academic writing. *Journal of Pragmatics*. 27: 361-386.

Hoey, M.
1983 *On the surface of discourse*. London: Allen & Unwin.

Holmes, J.
1982 Expressing doubt and certainty in English. *RELC Journal, 13* , 19-28.

Holmes, J.
1983 Speaking English with the appropriate degree of conviction. In C. Brumfit (ed.), *Learning and teaching languages for communication: Applied linguistics perspectives*. (pp. 100-121). London: British Assn. of Applied Linguistics.

Holmes, J.
1984 Modifying illocutionary force. *Journal of Pragmatics, 8* , 345-365.

Holmes, J.
1988a Paying compliments: A sex preferential positive politeness strategy. *Journal of Pragmatics, 12* , 445-465.

Holmes, J.
1988b Doubt and certainty in ESL textbooks. *Applied Linguistics, 9* 1), 20-44.

Holmes, J.
1990 Hedges and boosters in women's and men's speech. *Language and communication, 10* (3), 185-205.

Holmes, J.
1995 *Women, men and politeness*. London: Longman.

Hopkins, A. & Dudley-Evans, T.
1988 A genre-based investigation of the discussion sections in articles and dissertations. *English for specific purposes, 7* , 113-121.

House, J. & Kasper, G.
1981 Politeness markers in English and German. In F. Coulmas (ed.) *Conversational routine*. (pp. 157-185). The Hague: Mouton.

Howell, J. & Memering, D.
1986 *Brief Handbook for Writers*. Englewood Cliffs, NJ: Prentice Hall.

Hu, Z., Brown, D. & Brown, L.
1982 Some linguistic differences in the written English of Chinese and Australian students. *Language Learning and Communication*, 1, 39-49.

Hubler, A.
1983 *Understatement and hedges in English*. Amsterdam: John Benjamins.

Huddleston, R.
1971 *The sentence in written English: A syntactic study based on an analysis of scientific texts*. Cambridge: CUP.

Hudson, R.
1984 *Word grammar*. Oxford: Basil Blackwell.

Hunston, S.
1993 Evaluation and ideology in scientific writing. In M. Ghadessy (ed.), *Register analysis: theory and practice.*, (pp. 57-73 London: Pinter.

Hunston, S.
1994a Modal meanings in the structure of argumentation. In O. A. M. Brekke, T. Dahl & J. Myking (eds.), *9th European symposium on LSP* (pp. 621-28). University of Bergen.

Hunston, S.
1994b Evaluation and organisation in a sample of written academic discourse. In M. Coulthard. (ed.), *Advances in written text analysis*, (pp. 191-218). London: Routledge.

Hutchinson, T. & Waters, A.
1984 *Interface: English for Technical Communication*. Harlow, Essex: Longman.

Hyland, K.
1994 Hedging in academic writing and EAP textbooks. *English for Specific Purposes, 13* (3), 239-256.

Hyland, K.
1996 "I don't quite follow": Making sense of a modifier. *Language Awareness*. 5 (2): 91-109.

Hyland, K.
1997 Scientific claims and community values: articulating an academic culture. *Language and Communication*, 16, 1: 19-32.

Hyland, K.
1998 Exploring corporate rhetoric: metadiscourse in the Chairman's letter. *Journal of Business Communication*.

Hyland, K.
(In press a) Persuasion and context: the pragmatics of academic metadiscourse. *Journal of Pragmatics*.

Hyland, K.
(In press b) Talking to students: metadiscourse in introductory textbooks. *English for Specific Purposes*.

Hyland, K. & Hyland, F.

1992 Go for Gold: Integrating process and product in ESP. *English for Specific Purposes, 11* (3), 225-242.

Hyland, K. & Milton, J.

1997 Hedging in L1 and L2 student writing. *Journal of Second Language Writing.* 6, 2: 183-206.

Ide, S.

1989 Formal forms and discernment: Two neglected aspects of universals of linguistic politeness. *Multilingua, 8* , 223-248.

Intaraprawat, P. & Steffensen, M.

1995 The use of metadiscourse in good and poor ESL essays. *Journal of Second Language Writing*, 4, 3, 253-272.

Jacoby, S.

1987 References to other researchers in literary research articles. *ELR Journal, 1*, 33-78.

Jacoby, S., Leech, D. & Holten, C.

1995 A genre-based developmental writing course for undergraduate ESL science majors. In D. Belcher & G. Braine (eds.) *Academic writing in a second language: essays on research and pedagogy.* Ablex: New Jersey.

James, A.

1983 Compromisers in English: A cross-disciplinary approach to their interpersonal significance. *Journal of Pragmatics, 7* , 191-206.

Jenkins, S., Jordon, M. & Weiland, P.

1993 The role of writing in graduate engineering education: A survey of faculty beliefs and practices. *English for Specific Purposes, 12* , 51-67.

Johns, A.

1990 L1 composition theories: Implications for developing theories of L2 composition. In B. Kroll (ed.), *Second language writing* (pp. 24-36). Cambridge: CUP.

Johns, A.

1993 Written argumentation for real audiences: Suggestions for teacher research and classroom practice. *TESOL Quarterly, 27* (1), 75-90.

Johns, T. & King, P. (eds.).

1991 *Classroom concordancing.* Birmingham: Centre for English Language Studies.

Jordon, R.
1990 *Academic Writing Course*. London: Collins.
Judson, H.
1995 *The eighth day of creation: The makers of the revolution in biology*. Harmondsworth: Penguin Books.
Karttunen, L. & Peters, S.
1979 Conventional implicature. In C.-K. Oh & D.A. Dinneen (eds.), *Syntax and Semantics II: Presupposition*. New York: Academic Press.
Kasper, G. & Blum-Kulka, S. (eds.)
1993 *Interlanguage pragmatics*. New York: OUP.
Kasper, G. & Dahl, M.
1991 *Research Methods in Interlanguage pragmatics*. Hawaii: University of Hawaii Second Language Teaching and Curriculum Center.
Katriel, T.
1986 *Talking straight: 'Dugri' speech in Israeli 'Sabra' culture*. Cambridge: CUP.
Kennedy, G.
1979 Semantic priorities in English language teaching. *RELC Journal*, 10, 14-35.
Kennedy, G.
1987 Expressing temporal frequency in academic English. *TESOL Quarterly, 21* (1), 69-86.
Kennedy, G.
1993 Use and usefulness: Corpus analysis and the teaching of English. *TESOLANZ Journal, 1* , 7-18.
Kennedy, G.
1995 English as system and probability in academic and professional communication. In P. Bruthiaux, T. Boswood, B. Du-Babcock (eds.), *Explorations in English for professional communication* (pp. 139-154). Hong Kong: City University of Hong Kong Press.
Kennedy, M. & Smith, H.
1986 *Academic writing: Working with sources across the curriculum*. Englewood Cliffs, NJ: Prentice Hall.
Klinge, A.
1993 The English modal auxiliaries: form lexical semantics to utterance interpretation. *Journal of Linguistics, 29* , 315-357.

Klinge, A.
1995 On the linguistic interpretation of contractual modalities. *Journal of Pragmatics*. 23, 649-675.
Knorr-Cetina, K.
1981 *The manufacture of knowledge*. Oxford: Pergamon Press.
Kourany, J. A.
1987 *Scientific knowledge: basic issues in the philosophy of science*. Belmont, Ca.: Wadsworth.
Kroll, B. (ed.)
1990 *Second Language Writing: Research Insights for the Classroom*. Cambridge: CUP.
Kuhn, T.
1970 *The structure of scientific revolutions. 2nd ed.* Chicago: University of Chicago Press.
Kuhn, T.
1977 *The essential tension: Selected studies in scientific tradition and change*. Chicago: University of Chicago Press.
Kuo, C.-H.
1993 Problematic issues in EST materials development. *English for Specific Purposes, 12,* 171-181.
Labov, W.
1973 The boundaries of words and their meanings. In C.-J. Bailey & R. Shuy (eds.), *New ways of analysing variation in English*. (pp. 340-373). Washington: Georgetown University Press.
Lachowicz, D.
1981 On the use of the passive voice for objectivity, author responsibility and hedging in EST. *Science of Science, 2* (6), 105-115.
Lackstrom, J., Selinker, L. & Trimble, L.
1973 Technical rhetorical principles and grammatical choice. *TESOL Quarterly, 7* (2), 127-136.
Laib, N.
1985 Territoriality in rhetoric. *College English, 47* , 579-593.
Lakoff, G.
1972 Hedges: A study in meaning criteria and the logic of fuzzy concepts. *Chicago Linguistic Society Papers, 8* , 183-228.

Lakoff, G.

1982 Categories: an essay in cognitive linguistics. The Linguistics Society of Korea (ed.), *Linguistics in the morning calm*. Seoul: Hanshin Publishing Co.

Lakoff, G.

1987 *Women, fire and dangerous things*. London: University of Chicago Press.

Lakoff, R.

1972 The pragmatics of modality. In P. Peranteau, J. Levi & G. Phares (eds.), *Papers from the Eighth Regional Meeting* (pp. 229-46). Chicago: Chicago Linguistics Society.

Lakoff, R.

1989 The limits of politeness: Therapeutic and courtroom discourse. *Multilingua, 8*, 101-129.

Lannon, J.

1986 *The writing process, a concise rhetoric (2nd ed.* Boston: Little Brown.

Latour, B. & Woolgar, S.

1979 *Laboratory life: The social construction of scientific facts*. Beverly Hills: Sage.

Lee, D.

1992 *Competing discourses: Perspectives and ideology in language*. Harlow: Longman.

Leech, G.

1987 *Meaning and the English verb (2nd ed.)* London: Longman.

Leech, G.

1983 *Principles of pragmatics*. London: Longman.

Leech, G. N. & Coates, J.

1983 Semantic indeterminacy and the modals. In S. Greenbaum, G. Leech and J. Svartvik (eds.), *Studies in English linguistics*. (pp. 79-90). London: Longman.

Leggett, G., Mead, C. & Kramer, M.

1991 *Prentice Hall Handbook for Writers 11th ed.* Englewood Cliffs, NJ: Prentice Hall.

Leki, I. & Carson, J.

1997 "Completely different worlds": EAP and the writing experiences of ESL students in university courses. *TESOL Quarterly*, 31. 39-69.

Levinson, S.
1983 *Pragmatics*. Cambridge: CUP.
Lindsay, D.
1984 *A guide to scientific writing*. Melbourne: Longman.
Longino, H.
1990 *Science as social knowledge: Values and objectivity in scientific inquiry*. Princeton, New Jersey: Princeton University Press.
Low, G.
1996 Intensifiers and hedges in questionnaire items and the lexical invisibility hypothesis. *Applied Linguistics*. 17 (1): 1-37.
Lyon, A.
1992 Re-presenting communities: teaching turbulence. *Rhetoric Review*, 10: 279-290.
Lyons, J.
1977 *Semantics Vols. 1 & 2*. Cambridge: CUP.
Lyons, J.
1981 *Language, meaning and context*. London: Fontana.
Maher, J.
1986 The development of English as an international language of medicine. *Applied Linguistics, 7* (2), 206-218.
Malcolm, L.
1987 What rules govern tense usage in scientific articles ? *English for Specific Purposes, 6* , 31-44.
Mann, R. & Mann, P.
1989 *Essay Writing: Methods and Models*. Belmont, California: Wadsworth.
Marino, M.
1973 A feature analysis of the modal system of English. *Lingua, 32* , 309-323.
Markkanen, R. & Schroder, H.
1989 Hedging as a translation problem in scientific texts. In C. Lauren & M. Nordman (eds.), *Special language: from humans thinking to thinking machines*. (pp. 171-79). Clevedon: Multilingual Matters.
Martin, J.
1987 *Factual writing: exploring and challenging social reality*. Melbourne: Deakin University Press.

Matsumoto, Y.

1988 Reexamination of the universality of face: Politeness phenomena in Japanese. *Journal of Pragmatics, 12, 403-426* .

Mauranen, A.

1993 Contrastive ESP rhetoric: Metatext in Finnish-English Economics texts. *English for Specific Purposes, 12* , 3-22.

McCann, T.

1989 Student argumentative writing knowledge and ability at three grade levels. *Research in the Teaching of English, 23* 1), 62-76.

McEverdy, M. & Smith, P.

1990 *Plan, write and Rewrite: How to prepare assignments.* Melbourne: Nelson.

McEverdy, M. & Wyatt, P.

1990 *Assignment writing: Developing communication skills.* Melbourne: Nelson.

McKernan, J.

1988 *The writer's handbook.* New York: Holt Rinehart & Winston.

McKinlay, M. & Potter, J.

1987 Model discourse: Interpretive repertoires in scientists' conference talk. *Social Studies of Science, 17* , 443-63.

Mohan, B., & Lo, W. A.-Y.

1985 Academic writing and Chinese students: transfer and developmental factors. *TESOL Quarterly, 19* , 515-534.

Moore, B.

1980 *Reading and thinking in English. Vols. 1-4.* Oxford: OUP.

Morris, J.

1966 *Principles of scientific and technical writing.* New York: McGraw-Hill.

Muller, G.

1985 *The American college handbook of contemporary English.* New York: Harper & Row.

Murison-Bowie, S.

1993 *Oxford Microconcord users manual.* Oxford: OUP.

Myers, G.

1989 The pragmatics of politeness in scientific articles. *Applied Linguistics, 10* , 1-35.

Myers, G.
1990 *Writing biology: Texts in the social construction of scientific knowledge.* Madison: University of Wisconsin Press.

Myers, G.
1991 Lexical cohesion and specialised knowledge in science and popular science texts. *Discourse Processes, 14* , 1-26.

Myers, G.
1992 Textbooks and the sociology of scientific knowledge. *English for Specific Purposes, 11*, 3-17.

Myers, G.
1994 Narratives of science and nature in popularising molecular genetics. In M. Coulthard (ed.), *Advances in written text analysis.*, (pp. 179-190). London: Routledge.

Nash, W.
1990 Introduction: The stuff these people write. In W. Nash (ed.), *The Writing Scholar: Studies in Academic Discourse.* (pp. 8-30). Newbury Park, Ca.: Sage.

Neufeld, J.
1987 *A Textbook for Technical Communication.* Englewood Cliffs, NJ: Prentice Hall.

Nishimura, Y.
1993 Agentivity in cognitive grammar. In R.A. Geiger and B. Rudzke-Ostyn (eds.) *Conceptualisations and mental processing in language.* (pp. 487-530). Berlin: Mouton.

Nisbet, R. & Wilson, W.
1977 Telling more than we can know: verbal reports on mental processes. *Psychological Review*, 84, 231-259.

O'Connor, M. & Woodford, F.
1976 *Writing scientific papers in English.* Amsterdam: North Holland.

Oppenheim, A. N.
1992 *Questionnaire design, interviewing and attitude measurement (2nd ed.).* London: Pinter.

Oshima, A. & Hogue, A.
1981 *Writing Academic English.* Reading, Mass.: Addison-Wesley.

Oshima, A. & Hogue, A.
1988 *Introduction to Academic Writing.* Reading, Mass: Addison-Wesley.

Ota, A.
1963 *Tense and aspect of present day American English.* Tokyo: Kenkyu-sha.

Overington, M.
1977 The scientific community as audience: Toward a rhetorical analysis of science. *Philosophy and Rhetoric, 10* (3), 143-164.

Palmer, F.
1986 *Mood and modality.* Cambridge: CUP.

Palmer, F.
1990 *Modality and the English modals (2nd Ed* London: Longman.

Paltridge, B.
1993 Writing up research: A systemic functional perspective. *System, 21,* 175-192.

Paltridge, B.
1994 Genre analysis and the identification of textual boundaries. *Applied Linguistics, 15* (3), 288-299.

Paltridge, B.
1995 Working with genre: a pragmatic perspective. *Journal of pragmatics, 24,* 393-406.

Park, D.
1982 The meanings of "audience". *College English, 44* (3), 247-257.

Perfetti, C. A. & McCutchen, D.
1987 Schooled language competence: linguistic abilities in reading and writing. In S. Rosenberg (ed.), *Advances in Applied Psycholinguistics, Vol. 2.* (pp. 105-141). Cambridge: CUP.

Perkins, M.
1983 *Modal expressions in English.* London: Frances Pinter.

Pickard, V.
1993a Should we be teaching refutation? Concordanced evidence from the field of applied linguistics. In N. Bird, J. Harris & M. Ingham (eds.), *Language and content.* Hong Kong: Institute of Language in Education.

Pickard, V.
1993b Citing previous writers: What can we say instead of "say". Paper presented at *RELC Conference,* 1993. Singapore:

Pinch, T.

1985 Towards an analysis of scientific observation: the externality and evidential significance of observational reports in physics. *Social studies of science, 15* , 3-36.

Pindi, M. & Bloor, T.

1987 Playing safe with predictions: hedging, attribution and conditions in economic forecasting. In T. Bloor & J. Norrish (eds.) *Written Language.*(pp. 55-69) London: Centre for Information on Language Teaching and Research/British Assn. of Applied Linguistics.

Polanyi, M.

1964 *Personal Knowledge.* New York: Harper & Row.

Powell, M.

1985 Purposive vagueness: An evaluation dimension of vague quantifying expressions. *Journal of Linguistics, 21* , 31-50.

Prelli, L.

1989 *A rhetoric of science: Inventing scientific discourse.* Columbia: University of South Carolina Press.

Prince, E., Frader, J. & Bosk, C.

1982 On hedging in physician-physician discourse. In R. D. Pietro (ed.), *Linguistics and the professions* . (pp. 83-97). Hillsdale, NJ: Ablex.

Purves, A., & Purves, W.

1986 On the nature and formation of interpretive and rhetorical communities. In T. Postlethwaite (ed.), *International education research: Papers in honor of Torsten Hosen* . Oxford: Pergamon.

Quirk, R.

1965 Descriptive statement and serial relationship. *Language, 41* , 205-17.

Quirk, R., Greenbaum, S., Leech, G. & Svartvik, J.

1972 *A Grammar of Contemporary English.* Harlow, Essex: Longman.

Ramani, E.

1988 Developing a course in research writing for advanced ESP learners. In D. Chamberlain & R.J. Baumgardner (eds.), *ESP in the classroom: Practice and evaluation. ELT Doc 128.* MET & British Council.

Rawlins, J.

1980 What's so wrong with "In my opinion"? *College English, 41* , 670-674.

Redd-Boyd, T. & Slater, W.

1989 The effects of audience specification on under-graduates' attitudes, strategies and writing. *Research in the Teaching of English, 23*, 77-103.

Reid, J.

1988 *The process of composition (2nd ed.)*. Englewood Cliffs, NJ: Prentice Hall.

Richards, J.

1993 Beyond the textbook: The role of commercial materials in language teaching. *RELC Journal, 24* 1), 1-14.

Richards, K. & Skelton, J.

1991 How critical can you get? In P. Adams, B. Heaton & P. Howarth (eds.), *Socio-cultural issues in English for Academic Purposes.* Basingstoke: Modern English Publications/British Council.

Richards, R.

1988 Thesis/dissertation writing for EFL students: An ESP course design. *English for specific purposes, 7* , 171-180.

Richards, S.

1987 *Philosophy and sociology of science: An introduction (2nd ed.)*. Oxford: Blackwell.

Robberecht, P. & van Petegham, M.

1982 A functional model for the description of modality. In *Fifth International Conference on Contrastive Projects*. Jyvaskyla, Finland

Rorty, R.

1979 *Philosophy and the mirror of nature*. Princeton: Princeton University Press.

Rosch, E.

1975 Cognitive representations of semantic categories. *Journal of Experimental Psychology: General, 104* , 192-233.

Rose, K.

1994 On the validity of DCTs in non-Western contexts. *Applied Linguistics*, 15, 1-14.

Roth, A.

1989 *The research paper: process, form and content (6th ed.)* Belmont, Ca: Wadsworth.

Rounds, P.

1982 *Hedging in written academic discourse: Precision and flexibility*. The University of Michigan. Mimeo.

Royds-Irmak, D.

1975 *Beginning Scientific English (books 1 & 2)* Walton-on-Thames: Nelson.

Rudzka-Ostyn, B. (ed.).
1988 *Topics in cognitive linguistics*. Amsterdam: John Benjamins.

Salager-Meyer, F.
1994 Hedges and textual communicative function in medical English written discourse. *English for Specific Purposes, 13* (2), 149-70.

Scarcella, R. & Brunak, R.
1981 On speaking politely in a second language. *International Journal of the Sociology of Language, 27* , 59-75.

Schiffrin, D.
1980 Metatalk: organisational and evaluative brackets in discourse. *Sociological Inquiry: Language and social interaction*. 50: 199-236.

Schmidt, R.
1980 Review of 'Questions of politeness: Strategies in social interaction'. *RELC Journal, 11* , 100-114.

Schriver, K.
1992 Teaching writers to anticipate readers' needs. *Written Communication, 9* (2), 179-208.

Scollon, R. & Scollon, S.
1995 *Intercultural Communication*. Oxford: Blackwell.

SCI
1993 *1992 Journal Citation Reports (JCR)*. Philadelphia: Public Institute for Scientific Information.

Searle, J.
1976 A classification of illocutionary acts. *Language in Society, 5* , 1-23.

Searle, J.
1979 *Expression and meaning*. Cambridge: CUP.

Selzer, J.
1982 Review of "Style: Ten lessons in clarity and grace". *College Composition and Communication, 33* 1

Shapin, S.
1984 Pump and circumstance: Robert Boyle's literary technology. *Social Studies of Science, 14* , 481-520.

Sheldon, L.
1987 Introduction. In L. Sheldon (ed.), *ELT textbooks and materials: Problems in evaluation and development, ELT Doc 126*. MET & British Council.

Sheldon, L.
1988 Evaluating ELT textbooks and materials. *ELT Journal, 42* (4), 237-246.
Silva, T.
1993 Toward an understanding of the distinct nature of L2 wring: the ESL research and its implications. *TESOL Quarterly*. 27, 657-675.
Simpson, P.
1990 Modality in literary-critical discourse. In W. Nash (ed.), *The writing scholar: studies in academic discourse* (pp. 63-94). Newbury Park, Ca.: Sage.
Sinclair, J. M.
1991 *Corpus, concordance, collocation*. Oxford: OUP.
Skelton, J.
1988a The care and maintenance of hedges. *ELT Journal, 41* (1): 37-43.
Skelton, J.
1988b Comments in academic articles. In P. Grunwell (ed.), *Applied linguistics in society*. (pp. 98-108). London: CILT/BAAL.
Skelton, J.
1997 The representation of truth in academic medical writing. *Applied Linguistics*. 18 (2) 121-140.
Smagorinsky, P.
1994 Think-aloud protocol analysis: beyond the black box. In P. Smagorinsky (ed.) *Speaking about writing: reflections on research methodology*. (pp. 3-19). Thousand Oaks, Ca: Sage.
Smith, E. L.
1986 Achieving impact through the interpersonal component. In B. Couture (ed.), *Functional approaches to writing: Research perspectives* Norwood, NJ: Ablex.
Smith, H.
1985 *Readable writing, revising for style*. Belmont, Ca: Wadsworth.
Snow, M. & Brinton, D.
1988 The adjunct model of language instruction: An ideal framework. In S. Benesch (ed.), *Ending remediation: Linking ESL and content in higher education* . Washington, DC: TESOL.
Sötter, A.
1988 The second language learner and cultural transfer in narration. In A. Purves (ed.) *Writing across Languages and Cultures: Issues in Contrastive Rhetoric*. (pp. 177-205) Newbury Park, CA: Sage.

Sperber, D. & Wilson, D.
1986 *Relevance: Communication and cognition.* Oxford: Blackwell.

Stalnaker, R. C.
1991 Pragmatic Presuppositions. In S. Davis (eds.), *Pragmatics: A reader.* (pp. 471-481). New York: OUP.

Strunk, W. J. & White, E.
1959 *The Elements of Style.* New York: Macmillan.

Stubbe, M. & Holmes, J.
1995 You know, eh and other 'exasperating expressions': an analysis of social and stylistic variation in the use of pragmatic devices in a sample of New Zealand English. *Language and Communication, 15* (1), 63-88.

Stubbs, M. 1986 'A matter of prolonged fieldwork': Notes towards a modal grammar of English. *Applied Linguistics, 7* (1), 1-25.

Swales, J.
1980 ESP: The textbook problem. *ESP Journal, 1* (1), 11-23.

Swales, J.
1984 Research into the structure of introductions to journal articles and its application to the teaching of academic writing. In R. Williams, J. Swales & J. Kirkman (eds.), *Common Ground: Shared interests in ESP and Communication Studies.* (pp. 77-86). Oxford: Pergamon.

Swales, J.
1985 English language papers and authors' first language: Preliminary explorations. *Scientometrics, 8,* 91-101.

Swales, J.
1987 Utilising the literatures in teaching the research paper. *TESOL Quarterly, 21* (1), 41-68.

Swales, J.
1988 ESP and applied linguistics: Hopes for a brave new world. In M. Tickoo (eds.), *ESP: State of the Art* . Singapore: SEAMEO Regional Language Centre.

Swales, J.
1990 *Genre Analysis: English in academic and research settings.* Cambridge: CUP.

Swales, J.
1993 Genre and engagement. Revue Belge de Philogie et Histoire, 71, 3, 689-698.

Swales, J.

1996 Toward a textography of an academic site. Paper presented at the 11th AILA World Congress of Applied Linguistics, Jyvaskyla, Finland.

Swales, J. & Najjar, H.

1987 The writing of research article introductions. *Written Communication, 4*, 175-92.

Taylor, J. R.

1989 *Linguistic categorisation: Prototypes in linguistic theory.* Oxford: Clarendon Press.

Thomas, J.

1983 Cross-cultural pragmatic failure. *Applied Linguistics, 4* (2), 91-112.

Thomas, S. & Hawes, T.

1994 Reporting verbs in medical journal articles. *English for Specific Purposes, 13* (2), 129-148.

Thompson, D.

1993 Arguing for experimental "facts" in science: a study of research article results sections in biochemistry. *Written Communication.* 8 (1), 106-128.

Thompson, G. & Ye, Y.

1991 Evaluation of the reporting verbs used in academic papers. *Applied Linguistics, 12* , 365-82.

Toulmin, S.

1958 *The Uses of Argument.* Cambridge: CUP.

Trimble, L.

1985 *English for science and technology: A discourse approach.* Cambridge: CUP.

Turnbull, W. & Saxton, K.

1997 Modal expressions as facework in refusals to comply with requests: I think I should say 'no' right now. *Journal of Pragmatics.* 27, 145-181.

Turner, K.

1996 The principal principles of pragmatic inference: politeness. *Language Teaching* 29: 1-13.

Ungerer, F. & Schmid, H.

1996 *An introduction to cognitive linguistics.* Harlow: Longman.

Valero-Garces, C.
1996 Contrastive ESP rhetoric: Metatext in Spanish-English Economics texts. *English for Specific Purposes*. 15, 4: 279-294.
Van der Auwera, J.
1979 Pragmatic presupposition: Shared beliefs in a theory of irrefutable meaning. In O. Choon-Kyu & D.A. Dineen (eds.), *Syntax and Semantics Vol. 11: Presupposition*
Van Eemeren, F., & Grootendorst, R.
1984 *Speech acts in argumentative discussions*. Dordecht, Netherlands: Foris.
Van Frassen, B.
1980 *The scientific image*. Oxford: OUP.
Vande Kopple, W.
1985 Some exploratory discourse on metadiscourse. *College Composition and Communication, 36* , 82-93.
Ventola, E.
1992 Writing scientific English: overcoming cultural problems. *International Journal of Applied Linguistics, 2* (2), 191-220.
Wachtel, T.
1980 Pragmatic approximations. *Journal of Pragmatics, 4* , 201-212.
Watson, J.
1968 *The double helix*. Harmondsworth: Penguin.
Watson, J., & Crick, F.
1953 A structure for deoxyribose nucleic acid. *Nature, 171* (April 25), 737-8.
Weissberg, R.
1984 Given and new: paragraph development models for scientific English. *TESOL Quarterly, 18* , 485-500.
Weissberg, R. & Buker, S.
1990 *Writing up research: Experimental research report writing for students of English*. Englewood Cliffs, NJ: Prentice Hall.
West, G.
1980 That-nominal constructions in traditional rhetorical divisions of scientific research papers. *TESOL Quarterly, 14* , 483-9.
White, A.
1975 *Modal thinking*. Oxford: Blackwell.
Widdowson, H.
1979 *Explorations in applied linguistics 2*. Oxford: OUP.

Wierzbicka, A.
1985 Different cultures, different languages, different speech acts. *Journal of pragmatics, 9* , 145-178.
Wierzbicka, A.
1987 The semantics of modality. *Folia Linguistica, 21* , 25-43.
Williams, J.
1989 *Style: Ten lessons in clarity and grace (3rd ed.)*. Boston: Scott, Foresman.
Winkler, A. & McCuen, J.
1989 *Writing the research paper: A handbook (3rd ed.)*. Sydney: Harcourt Brace Jovanovich.
Winstanley, M.
1976 Assimilation into the literature of a critical advance in molecular biology. *Social studies of science*, 6, 545-549.
Witte, S.
1987 Pre-text and composing. *College Composition and Communication, 38:* 397-425.
Witte, S. & Cherry, R.
1994 Think-aloud protocols, protocol analysis, and research design. In Smagorinsky, P. (ed.) *Speaking about writing: reflections on research methodology*. (pp. 20-54). Thousand Oaks, Ca: Sage.
Wood, L. & Kroger, R.
1994 The analysis of facework in discourse: review and proposal. Journal of Language and Social Psychology. 13: 248-277.
Wu, M. H.
1992 Towards a contextual lexico-grammar: An application of concordance analysis in EST teaching. *RELC Journal, 23* (2), 18-34.
Yarber, R.
1985 *Writing for college*. Glenview, Ill.: Scott Foresman.
Yearley, S.
1981 Textual persuasion: the role of social accounting in the construction of scientific arguments. *Philosophy of the Social Sciences, 11* , 409-435.
Young, D.
1980 *The structure of English clauses*. London: Hutchinson.
Young, L.
1982 Inscrutability revisited. In J. Gumperz (eds.), *Language and social identity* (pp. 72-84). Cambridge: CUP.

Zadeh, L. A.

1972 A fuzzy-set-theoretic interpretation of linguistic hedges. *Journal of Cybernetics, 2* (3), 4-34.

Zappen, J.

1983 A rhetoric for research in sciences and technologies. In P. Anderson, R. Brockman, & C. Miller (eds.), *New essays in scientific and technical communication* (pp. 123- 38). Farmingdale, NY: Baywood.

Ziman, J.

1984 *An introduction to science studies*. Cambridge: CUP.

Zimmerman, F.

1989 *English for Science*. Englewood Cliffs, NJ: Prentice Hall.

Zinsser, W.

1976 *On writing well*. New York: Harper & Row.

Zuck, J. G. & Zuck, L. V.

1986 Hedging in news writing. In A.-M. Cornu, J. Van Parjis, M. Delahaye & L. Baten (eds.), *Beads or bracelets? How do we approach LSP, Selected papers from the fifth European symposium on LSP.* (pp. 172-180). Oxford: OUP.

Index

abstract rhetors, 123, 172, 186, 188, 254
academic disciplines, 10, 15, 19, 51, 96, 240, 241, 257
acceptability conditions, 91, 92, 94, 144, 251-253
accuracy-oriented hedges, 93, 138, 156, 160, 162-170, 173, 176, 185, 187, 191, 198, 201, 207-208, 210, 212, 214
adequacy conditions, 91, 92, 94, 162, 176, 246, 251-253
article structure
  introduction, 17, 25-27, 123, 154, 190-193, 201-202, 214
  methods, 17, 19, 25, 27-30, 123, 141, 153-155, 202-204, 214
  results, 17, 30-32, 122-123, 140, 153-154, 193-196, 205-214
  discussion, 17, 19, 25, 32-34, 123, 140, 149, 153-155, 196-200, 205-214, 246
attribute hedges, 93, 163-166, 169, 174, 180, 187, 195, 198, 207, 210, 244
Bartholomae, D., 217
Bazerman, C., 7, 15, 17, 19, 35, 57
Beaugrande, de R., 24, 25, 223
Becher, T., 69, 178
Berkenkotter, C., 15, 19, 21, 29, 33, 100, 216, 239
Bhatia, V., 97
biology, 11, 96, 97, 102, 166, 189, 196, 207
Bruffee, K., 82, 84
Butler, C., 10, 25, 48, 53, 54, 103, 107, 108, 109, 114, 116, 117, 155
categorical assertions, 3, 45, 65, 72, 85, 89, 91, 107, 109, 113, 120, 121, 125, 171, 215, 216
Channell, J., 9, 10, 41, 43, 54, 176, 256, 257
citation 8, 20, 57, 63-66, 74, 97, 154, 187, 201, 202, 212, 213, 236
Coates, J., 2, 3, 9, 10, 44, 46, 95, 103, 105-107, 110-113, 115, 116, 118, 155, 158, 161, 225, 247, 255
cohesion and coherence, 28, 82, 202, 246
concordance, 100, 233, 234
content-oriented hedges, 92, 93, 156, 158, 162-177, 185, 187, 190, 198, 251
context, 5, 11-16, 24, 27, 33-37, 38, 41, 47, 67, 82, 83, 100, 105, 111, 117, 125, 142, 144, 145, 160, 161, 170, 175, 181, 185, 187, 189, 193, 196, 199, 205, 209, 211, 215, 217, 222, 234, 235, 237-241, 244-247, 253-254, 258-260, 262
contrastive rhetoric, 221
corpora
  biology RA, 23, 29, 54, 55, 58, 66, 75, 96, 98, 100, 104, 108, 117, 123, 128-130, 134, 186, 226, 229, 247
  Brown, 5, 9, 44, 49, 62, 64, 67, 69, 94-96, 99, 100, 102-103, 107, 108, 120, 122, 127-128, 130, 149, 177, 225, 226, 228, 255

JDEST, 94-95, 102, 107-109, 115, 127, 128, 130, 149, 255
LOB, 94, 95, 96, 100, 102-108, 122, 127, 128, 130, 149, 255
Crismore, A., 6, 9, 10, 50-52, 220, 223
criticism, 137, 143, 179, 180, 193, 194, 200, 207, 209, 213, 230, 257
discourse
analysis, 44
academic, 8, 9, 10, 11, 19, 49, 77, 95, 100, 127, 162, 223, 224, 228, 245
conventions, 14, 16, 22, 38, 51, 83, 85, 177, 192, 220, 239, 242, 255, 260
scientific, 6-8, 12, 13-18, 23, 25, 35, 37-39, 49, 51, 53, 59, 61-64, 67-69, 76-79, 83, 85, 88, 89, 103, 105, 108, 111, 115-116, 123, 128-132, 149, 156, 157, 177, 185, 189, 216, 224, 237, 243-245, 250, 252, 253, 256, 258, 261
discourse community, 7, 12, 14, 18, 20, 21-24, 26, 29, 34-37, 57, 59, 64, 68, 69, 148, 157, 198, 200, 219, 245-258, 260-262
disjuncts, 135, 136, 137, 138, 139, 141, 165, 167, 186, 187
downtoners, 9, 135, 138, 141, 165, 186, 187, 198, 238
Dubois, B., 9, 10, 43, 54, 56, 133, 139, 256
epistemic adjectives, 3, 103, 118, 130-134, 141, 167, 191, 207
epistemic adverbs, 134-139, 148
epistemic lexical verbs, 3, 52, 60, 74, 100, 104, 119-129, 136, 137, 148, 150, 151, 153, 154, 173-174, 182-184, 193, 195, 196, 207, 213, 229, 230, 234, 237, 238
epistemic modality, 2, 3, 11, 12, 38, 42, 44-48, 51, 54, 62, 76, 88, 95, 100, 102-106, 119, 127, 131-134, 141, 152, 153, 166, 217-219, 225, 226, 228-231, 244, 248-249, 259, 263
epistemic nouns, 103, 104, 154
evidential verbs, 124-126, 182, 195, 199
experimental limitations, 4, 46, 99, 127, 141, 147, 154, 174, 176, 186, 195
Fairclough, N., 37, 157, 258
Flowerdew, J., 105, 122, 136, 240, 256
Fraser, B., 39, 40, 48, 177, 179, 200
fuzzy categories, 1, 70, 158-161, 170, 171, 174, 177, 186, 251, 254
Gilbert, N., 7, 14, 15, 16, 29, 34
Gosden, H., 8, 25, 54. 97
Halliday, M.A.K., 3, 5, 47, 48, 51, 100, 105, 124, 134, 136, 150, 160, 247, 248
harmonic combination, 118, 150-153, 207, 262
Hinds, J., 220
Holmes, J., 2, 3, 9, 10, 43, 44, 46, 49, 63, 96, 99, 103-105, 107, 108, 127, 130, 155, 175, 177, 190, 218, 223, 231, 237, 255
Huckin, T., 15, 19, 21, 29, 33, 100, 216
Hunston, S., 58, 78, 170
Hyland, K., 8, 9, 10, 49, 51, 217, 218, 220, 237, 255, 257, 261
inference, 106, 107, 115, 127
illocutionary force, 24, 37, 48, 50, 90, 220
indeterminacy, 100, 157-158, 214, 244, 247, 248, 262
inference, 89, 169, 215
judgement verbs, 120-124, 184, 195, 213
Kenndey, G., 95, 99, 100, 129, 130, 149, 177
knowledge claims, 12, 18, 25, 30, 31, 35, 38, 56-69, 89, 170, 171, 185, 220, 236, 239, 240, 248, 256, 260

In the PRAGMATICS AND BEYOND NEW SERIES the following titles have been published thus far or are scheduled for publication:

1. WALTER, Bettyruth: *The Jury Summation as Speech Genre: An Ethnographic Study of What it Means to Those who Use it.* Amsterdam/Philadelphia, 1988.
2. BARTON, Ellen: *Nonsentential Constituents: A Theory of Grammatical Structure and Pragmatic Interpretation.* Amsterdam/Philadelphia, 1990.
3. OLEKSY, Wieslaw (ed.): *Contrastive Pragmatics.* Amsterdam/Philadelphia, 1989.
4. RAFFLER-ENGEL, Walburga von (ed.): *Doctor-Patient Interaction.* Amsterdam/Philadelphia, 1989.
5. THELIN, Nils B. (ed.): *Verbal Aspect in Discourse.* Amsterdam/Philadelphia, 1990.
6. VERSCHUEREN, Jef (ed.): *Selected Papers from the 1987 International Pragmatics Conference. Vol. I: Pragmatics at Issue. Vol. II: Levels of Linguistic Adaptation. Vol. III: The Pragmatics of Intercultural and International Communication* (ed. with Jan Blommaert). Amsterdam/Philadelphia, 1991.
7. LINDENFELD, Jacqueline: *Speech and Sociability at French Urban Market Places.* Amsterdam/Philadelphia, 1990.
8. YOUNG, Lynne: *Language as Behaviour, Language as Code: A Study of Academic English.* Amsterdam/Philadelphia, 1990.
9. LUKE, Kang-Kwong: *Utterance Particles in Cantonese Conversation.* Amsterdam/Philadelphia, 1990.
10. MURRAY, Denise E.: *Conversation for Action. The computer terminal as medium of communication.* Amsterdam/Philadelphia, 1991.
11. LUONG, Hy V.: *Discursive Practices and Linguistic Meanings. The Vietnamese system of person reference.* Amsterdam/Philadelphia, 1990.
12. ABRAHAM, Werner (ed.): *Discourse Particles. Descriptive and theoretical investigations on the logical, syntactic and pragmatic properties of discourse particles in German.* Amsterdam/Philadelphia, 1991.
13. NUYTS, Jan, A. Machtelt BOLKESTEIN and Co VET (eds): *Layers and Levels of Representation in Language Theory: a functional view.* Amsterdam/Philadelphia, 1990.
14. SCHWARTZ, Ursula: *Young Children's Dyadic Pretend Play.* Amsterdam/Philadelphia, 1991.
15. KOMTER, Martha: *Conflict and Cooperation in Job Interviews.* Amsterdam/Philadelphia, 1991.
16. MANN, William C. and Sandra A. THOMPSON (eds): *Discourse Description: Diverse Linguistic Analyses of a Fund-Raising Text.* Amsterdam/Philadelphia, 1992.
17. PIÉRAUT-LE BONNIEC, Gilberte and Marlene DOLITSKY (eds): *Language Bases ... Discourse Bases.* Amsterdam/Philadelphia, 1991.
18. JOHNSTONE, Barbara: *Repetition in Arabic Discourse. Paradigms, syntagms and the ecology of language.* Amsterdam/Philadelphia, 1991.
19. BAKER, Carolyn D. and Allan LUKE (eds): *Towards a Critical Sociology of Reading Pedagogy. Papers of the XII World Congress on Reading.* Amsterdam/Philadelphia, 1991.
20. NUYTS, Jan: *Aspects of a Cognitive-Pragmatic Theory of Language. On cognition, functionalism, and grammar.* Amsterdam/Philadelphia, 1992.

21. SEARLE, John R. et al.: *(On) Searle on Conversation.* Compiled and introduced by Herman Parret and Jef Verschueren. Amsterdam/Philadelphia, 1992.
22. AUER, Peter and Aldo Di LUZIO (eds): *The Contextualization of Language.* Amsterdam/Philadelphia, 1992.
23. FORTESCUE, Michael, Peter HARDER and Lars KRISTOFFERSEN (eds): *Layered Structure and Reference in a Functional Perspective. Papers from the Functional Grammar Conference, Copenhagen, 1990.* Amsterdam/Philadelphia, 1992.
24. MAYNARD, Senko K.: *Discourse Modality: Subjectivity, Emotion and Voice in the Japanese Language.* Amsterdam/Philadelphia, 1993.
25. COUPER-KUHLEN, Elizabeth: *English Speech Rhythm. Form and function in everyday verbal interaction.* Amsterdam/Philadelphia, 1993.
26. STYGALL, Gail: Trial Language. *A study in differential discourse processing.* Amsterdam/Philadelphia, 1994.
27. SUTER, Hans Jürg: *The Wedding Report: A Prototypical Approach to the Study of Traditional Text Types.* Amsterdam/Philadelphia, 1993.
28. VAN DE WALLE, Lieve: *Pragmatics and Classical Sanskrit.* Amsterdam/Philadelphia, 1993.
29. BARSKY, Robert F.: *Constructing a Productive Other: Discourse theory and the convention refugee hearing.* Amsterdam/Philadelphia, 1994.
30. WORTHAM, Stanton E.F.: *Acting Out Participant Examples in the Classroom.* Amsterdam/Philadelphia, 1994.
31. WILDGEN, Wolfgang: *Process, Image and Meaning. A realistic model of the meanings of sentences and narrative texts.* Amsterdam/Philadelphia, 1994.
32. SHIBATANI, Masayoshi and Sandra A. THOMPSON (eds): *Essays in Semantics and Pragmatics.* Amsterdam/Philadelphia, 1995.
33. GOOSSENS, Louis, Paul PAUWELS, Brygida RUDZKA-OSTYN, Anne-Marie SIMON-VANDENBERGEN and Johan VANPARYS: *By Word of Mouth. Metaphor, metonymy and linguistic action in a cognitive perspective.* Amsterdam/Philadelphia, 1995.
34. BARBE, Katharina: Irony in Context. Amsterdam/Philadelphia, 1995.
35. JUCKER, Andreas H. (ed.): *Historical Pragmatics. Pragmatic developments in the history of English.* Amsterdam/Philadelphia, 1995.
36. CHILTON, Paul, Mikhail V. ILYIN and Jacob MEY: *Political Discourse in Transition in Eastern and Western Europe (1989-1991).* Amsterdam/Philadelphia, 1998.
37. CARSTON, Robyn, Nam SUN SONG and Seiji UCHIDA (eds): *Relevance Theory. Applications and implications.* Amsterdam/Philadelphia, 1998.
38. FRETHEIM, Thorstein and Jeanette K. GUNDEL (eds): *Reference and Referent Accessibility.* Amsterdam/Philadelphia, 1996.
39. HERRING, Susan (ed.): *Computer-Mediated Communication. Linguistic, social, and cross-cultural perspectives.* Amsterdam/Philadelphia, 1996.
40. DIAMOND, Julie: *Status and Power in Verbal Interaction. A study of discourse in a close-knit social network.* Amsterdam/Philadelphia, 1996.
41. VENTOLA, Eija and Anna MAURANEN, (eds): *Academic Writing. Intercultural and textual issues.* Amsterdam/Philadelphia, 1996.
42. WODAK, Ruth and Helga KOTTHOFF (eds): *Communicating Gender in Context.* Amsterdam/Philadelphia, 1997.

43. JANSSEN, Theo A.J.M. and Wim van der WURFF (eds): *Reported Speech. Forms and functions of the verb.* Amsterdam/Philadelphia, 1996.
44. BARGIELA-CHIAPPINI, Francesca and Sandra J. HARRIS: *Managing Language. The discourse of corporate meetings.* Amsterdam/Philadelphia, 1997.
45. PALTRIDGE, Brian: *Genre, Frames and Writing in Research Settings.* Amsterdam/Philadelphia, 1997.
46. GEORGAKOPOULOU, Alexandra: *Narrative Performances. A study of Modern Greek storytelling.* Amsterdam/Philadelphia, 1997.
47. CHESTERMAN, Andrew: *Contrastive Functional Analysis.* Amsterdam/Philadelphia, 1998.
48. KAMIO, Akio: *Territory of Information.* Amsterdam/Philadelphia, 1997.
49. KURZON, Dennis: *Discourse of Silence.* Amsterdam/Philadelphia, 1998.
50. GRENOBLE, Lenore: *Deixis and Information Packaging in Russian Discourse.* Amsterdam/Philadelphia, n.y.p.
51. BOULIMA, Jamila: *Negotiated Interaction in Target Language Classroom Discourse.* Amsterdam/Philadelphia, n.y.p.
52. GILLIS, Steven and Annick DE HOUWER (eds): *The Acquisition of Dutch.* Amsterdam/Philadelphia, n.y.p.
53. MOSEGAARD HANSEN, Maj-Britt: *The Function of Discourse Particles. A study with special reference to spoken standard French.* Amsterdam/Philadelphia, n.y.p.
54. HYLAND, Ken: *Hedging in Scientific Research Articles.* Amsterdam/Philadelphia, 1998.